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**ADVANCES IN APPLIED AND  
COMPUTATIONAL MATHEMATICS**



# **ADVANCES IN APPLIED AND COMPUTATIONAL MATHEMATICS**

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**Nova Science Publishers, Inc.**  
*New York*



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Library of Congress Cataloging-in-Publication Data

*Available upon request*

ISBN 1-60021-358-8

*Published by Nova Science Publishers, Inc. ✦ New York*

This work was supported by the ARO grant (DAAD 19-03-1-0375)

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## PREFACE

The 2005 Applied Mathematics Summer Workshop was held on the campus of the Delaware State University on August 18-20, 2005. It is the second of a series of annual event organized by the Applied Mathematics Research Center of the Delaware State University, funded by the Department of Defense (DoD). The 2005 Workshop brought together over 60 mathematicians, scientists and graduate students from the United States and abroad. The program includes invited lectures by distinguished speakers and contributed papers sessions.

The goal of the Workshop is to promote DoD's applied mathematics research interests among faculty and students from Historically Black Universities and Minority Institutions (HBCU/MI). During the Workshop, attendees also network, communicate, and exchange ideas and information on interdisciplinary applied mathematics research topics. They also discuss the trends and prospects of future research and have the opportunity to write joint research proposals.

The present book is the Proceedings of this 2005 Applied Mathematics Summer Workshop and consists of 22 carefully selected papers.

It is our great pleasure to acknowledge the financial support of the Department of Defense and the Delaware State University. We express our gratitude to Dr. Rajeev Parikh, Provost and Vice President of Academic Affairs for the Delaware State University for his encouragements and support during the preparation of the Workshop. We thank the members of the Organizing Committee and all the anonymous referees for their remarkable work.

The Editors:

Fengshan Liu, Delaware State University  
Zuhair Nashed, Central Florida University  
Gaston M. N'Guerekata, Morgan State University  
Dragoljub Pokrajac, Delaware State University  
Zhijun Qiao, University of Texas Pan American  
Xiquan Shi, Delaware State University  
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*Chapter 1*

# COMPUTATIONAL INVERSE MEDIUM SCATTERING AT FIXED FREQUENCY

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## Abstract

A continuation method is presented for solving the inverse medium scattering problem of the Helmholtz equation in  $\mathbb{R}^2$ . The algorithm requires only single-frequency scattering data. Using an initial guess from the Born approximation, each update is obtained via recursive linearization on the spatial frequency of a one-parameter family of plane waves by solving one forward and one adjoint problem of the Helmholtz equation.

## 1 Introduction

Consider the Helmholtz equation in two dimensions

$$\Delta\phi + k_0^2(1 + q(x))\phi = 0, \tag{1.1}$$

where  $\phi$  is the total field,  $k_0$  is the wavenumber, and  $q(x) > -1$ , which has a compact support, is the scatterer. The scatterer is illuminated by a one-parameter family of plane waves

$$\phi_0(x_1, x_2) = e^{i(\eta x_1 + k(\eta)x_2)},$$

---

\*E-mail address: bao@math.msu.edu, The research was supported in part by the NSF grants DMS 01-04001 and CCF-0514078, the ONR grant N000140210365, the National Science Foundation of China grant 10428105, and a special research grant from the KLA Tencor Foundation.

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where

$$k(\eta) = \begin{cases} \sqrt{k_0^2 - \eta^2} & \text{for } k_0 \geq |\eta|, \\ i\sqrt{\eta^2 - k_0^2} & \text{for } k_0 < |\eta|. \end{cases}$$

The number  $|\eta|$  is the spatial frequency.

The modes for which  $|\eta| \leq k_0$  correspond to propagating plane waves while the modes with  $|\eta| > k_0$  correspond to evanescent plane waves, which may be generated at the interface of two media by total internal reflection [5, 9]. These waves are oscillatory parallel to the  $x_1$  axis and decay exponentially along the  $x_2$  axis in the upper half plane  $\mathbb{R}_+^2 = \{(x_1, x_2) \in \mathbb{R}^2 : x_2 > 0\}$ . Evidently, such incident waves satisfy the homogeneous equation

$$\Delta\phi_0 + k_0^2\phi_0 = 0. \quad (1.2)$$

The total electric field  $\phi$  consists of the incident field  $\phi_0$  and the scattered field  $\psi$ :

$$\phi = \phi_0 + \psi.$$

It follows from the equations (1.1) and (1.2) that the scattered field satisfies

$$\Delta\psi + k_0^2(1 + q)\psi = -k_0^2q\phi_0. \quad (1.3)$$

Let  $D$  be a bounded domain in  $\mathbb{R}^2$  with boundary  $\partial D$ , which contains the compact support of the scatterer  $q(x)$ . Denote  $n$  the unit outward normal to  $\partial D$ . For simplicity, we employ the first order absorbing boundary condition [11] as

$$\frac{\partial\psi}{\partial n} - ik_0\psi = 0 \quad \text{on } \partial D, \quad (1.4)$$

The inverse medium scattering problem is to determine the scatterer  $q(x)$  from the measurements of near-field currents densities,  $\psi|_{\partial D}$ , given the incident field  $\phi_0$ . The inverse medium scattering problems arise naturally in diverse applications such as radar, sonar, geophysical exploration, medical imaging, and nondestructive testing [8]. There are two major difficulties associated with these inverse problems: the ill-posedness and the presence of many local minima. In this paper, we present an algorithm which overcomes the difficulties. Our algorithm requires single-frequency scattering data and the recursive linearization is obtained by a continuation method on the spatial frequency. It first solves a linear integral equation (Born approximation) at the largest spatial frequency. Updates are made by using the data at smaller spatial frequency sequentially. For each iteration, one forward and one adjoint problem of Helmholtz equation are solved. Two new computational examples are presented. We refer the reader to [4] for a complete description of the algorithm and related analysis. See also [2, 3, 6, 7] for related stable and efficient continuation methods for solving the two-dimensional Helmholtz equation and the three-dimensional Maxwell's equations in the case of full aperture data. A homotopy continuation method with limited aperture data may be found in [1].

## 2 Born Approximation

Rewrite (1.3) as

$$\Delta\psi + k_0^2\psi = -k_0^2q(\phi_0 + \psi). \quad (2.1)$$

Consider a test function  $\psi_0 = e^{ik_0x \cdot \vec{d}}$ ,  $\vec{d} = (\cos \theta, \sin \theta)$ ,  $\theta \in [0, 2\pi]$ . Hence  $\psi_0$  satisfies (1.2).

Multiplying the equation (2.1) by  $\psi_0$ , and integrating over  $D$  on both sides, we have

$$\int_D \psi_0 \Delta\psi dx + k_0^2 \int_D \psi_0 \psi dx = -k_0^2 \int_D q(\phi_0 + \psi) \psi_0 dx.$$

Integration by parts yields

$$\int_D \psi \Delta\psi_0 dx + \int_{\partial D} \left( \psi_0 \frac{\partial \psi}{\partial n} - \psi \frac{\partial \psi_0}{\partial n} \right) ds + k_0^2 \int_D \psi_0 \psi dx = -k_0^2 \int_D q(\phi_0 + \psi) \psi_0 dx.$$

We have by noting (1.2) and the boundary condition (1.4) that

$$\int_D q(\phi_0 + \psi) \psi_0 dx = \frac{1}{k_0^2} \int_{\partial D} \psi \left( \frac{\partial \psi_0}{\partial n} - ik_0 \psi_0 \right) ds.$$

Using the special form of the incident wave and the test function, we then get

$$\int_D q(x) e^{i(\eta + k_0 \cos \theta)x_1} e^{i(k(\eta) + k_0 \sin \theta)x_2} dx = \frac{i}{k_0} \int_{\partial D} \psi(n \cdot \vec{d} - 1) e^{ik_0x \cdot \vec{d}} ds - \int_D q\psi\psi_0 dx. \quad (2.2)$$

When the spatial frequency  $|\eta|$  is large, the scattered field is weak and the inverse scattering problem becomes essentially linear. See [4] for an energy estimate of the scattered field. Dropping the nonlinear (second) term of (2.2), we obtain the linearized integral equation

$$\int_D q(x) e^{i(\eta + k_0 \cos \theta)x_1} e^{(-\sqrt{\eta^2 - k_0^2} + ik_0 \sin \theta)x_2} dx = \frac{i}{k_0} \int_{\partial D} \psi(n \cdot \vec{d} - 1) e^{ik_0x \cdot \vec{d}} ds, \quad (2.3)$$

which is the Born approximation. In practice, the integral equation (2.3) is implemented by using Landweber iteration in order to reduce the computational cost and instability [10, 13].

When a medium is probed with an evanescent plane wave at a high spatial frequency, only a thin layer of the medium is penetrated. Corresponding to this exponentially decaying incident field, the scattered field measured on the boundary contains information of the medium in that thin layer. To accurately determine the medium, information at lower spatial frequencies of the evanescent plane waves is needed to illuminate the medium.

## 3 Recursive Linearization

As discussed in the previous section, when the spatial frequency  $|\eta|$  is large, the Born approximation allows a reconstruction of the thin layer for the true scatterer. Choose a large positive number  $\eta_{\max}$  and divide the interval  $[0, \eta_{\max}]$  into  $N$  subdivisions with the endpoints  $\{\eta_0, \eta_1, \dots, \eta_N\}$ , where  $\eta_0 = 0$ ,  $\eta_N = \eta_{\max}$ , and  $\eta_{i-1} < \eta_i$  for  $1 \leq i \leq N$ . We now describe a procedure that recursively determines  $q_\eta$  at  $\eta = \eta_N, \eta_{N-1}, \dots, \eta_0$ .



Suppose now that the scatterer  $q_{\tilde{\eta}}$  has been recovered at some  $\tilde{\eta} = \eta_{i+1}$  and that  $\eta = \eta_i$  is slightly less than  $\tilde{\eta}$ . We wish to determine  $q_\eta$ , or equivalently, to determine the perturbation

$$\delta q = q_\eta - q_{\tilde{\eta}}.$$

For the reconstructed scatterer  $q_{\tilde{\eta}}$ , we solve at the spatial frequency  $\eta$  the forward scattering problem

$$\Delta \tilde{\psi}^{(j,i)} + k_0^2(1 + q_{\tilde{\eta}})\tilde{\psi}^{(j,i)} = -k_0^2 q_{\tilde{\eta}} \phi_0^{(j,i)}, \quad (3.1)$$

$$\frac{\partial \tilde{\psi}^{(j,i)}}{\partial n} - ik_0 \tilde{\psi}^{(j,i)} = 0, \quad (3.2)$$

where the incident wave  $\phi_0^{(j,i)} = e^{i\eta_j x_1 + ik(\eta_j)x_2}$ ,  $|j| \geq i$ .

For the scatterer  $q_\eta$ , we have

$$\Delta \psi^{(j,i)} + k_0^2(1 + q_\eta)\psi^{(j,i)} = -k_0^2 q_\eta \phi_0^{(j,i)}, \quad (3.3)$$

$$\frac{\partial \psi^{(j,i)}}{\partial n} - ik_0 \psi^{(j,i)} = 0. \quad (3.4)$$

Subtracting (3.1), (3.2) from (3.3), (3.4) and omitting the second-order smallness in  $\delta q$  and in  $\delta \psi^{(j)} = \psi^{(j,i)} - \tilde{\psi}^{(j,i)}$ , we obtain

$$\Delta \delta \psi^{(j)} + k_0^2(1 + q_{\tilde{\eta}})\delta \psi^{(j)} = -k_0^2 \delta q (\phi_0^{(j,i)} + \tilde{\psi}^{(j,i)}), \quad (3.5)$$

$$\frac{\partial \delta \psi^{(j)}}{\partial n} - ik_0 \delta \psi^{(j)} = 0. \quad (3.6)$$

For the scatterer  $q_\eta$  and the incident wave  $\phi_0^{(j,i)}$ , we define the map  $S_j(q_\eta, \phi_0^{(j,i)})$  by

$$S_j(q_\eta, \phi_0^{(j,i)}) = \psi^{(j,i)},$$

where  $\psi^{(j,i)}$  is the scattering data corresponding to the incident wave  $\phi_0^{(j,i)}$ . Let  $\gamma$  be the trace operator to the boundary  $\partial D$ . Define the scattering map

$$M_j(q_\eta, \phi_0^{(j,i)}) = \gamma S_j(q_\eta, \phi_0^{(j,i)}).$$

For simplicity, denote  $M_j(q_\eta, \phi_0^{(j,i)})$  by  $M_j(q_\eta)$ . By the definition of the trace operator, we have

$$M_j(q_\eta) = \psi^{(j,i)}|_{\partial D}.$$

Let  $DM_j(q_{\tilde{\eta}})$  be the Fréchet derivative of  $M_j(q_\eta)$  and denote the residual operator by

$$R_j(q_{\tilde{\eta}}) = \psi^{(j,i)}|_{\partial D} - \tilde{\psi}^{(j,i)}|_{\partial D}.$$

It follows from [4] that

$$DM_j(q_{\tilde{\eta}})\delta q = R_j(q_{\tilde{\eta}}). \quad (3.7)$$

Similarly, in order to reduce the computation cost and instability, we consider the Landweber iteration of (3.7), which has the form

$$\delta q = \beta DM_j^*(q_{\tilde{\eta}})R_j(q_{\tilde{\eta}}) \quad \text{for all } |j| \geq i, \quad (3.8)$$

where  $\beta$  is a relaxation parameter and  $DM_j^*(q_{\tilde{\eta}})$  is the adjoint operator of  $DM_j(q_{\tilde{\eta}})$ .

In order to compute the correction  $\delta q$ , we need some efficient way to compute  $DM_j^*(q_{\tilde{\eta}})R_j(q_{\tilde{\eta}})$ , which is given by the following theorem. See [4] for the proof.

**Theorem 3.1.** *Given residual  $R_j(q_{\tilde{\eta}})$ , there exists a function  $\phi^{(j,i)}$  such that the adjoint Fréchet derivative  $DM_j^*(q_{\tilde{\eta}})$  satisfies*

$$[DM_j^*(q_{\tilde{\eta}})R_j(q_{\tilde{\eta}})](x) = k_0^2(\overline{\phi_0^{(j,i)}}(x) + \overline{\tilde{\psi}^{(j,i)}}(x))\phi^{(j,i)}(x), \quad (3.9)$$

where  $\phi_0^{(j,i)}$  is the incident wave and  $\tilde{\psi}^{(j,i)}$  is the solution of (3.1), (3.2) with the incident wave  $\phi_0^{(j,i)}$ .

Using this theorem, we can rewrite (3.8) as

$$\delta q = k_0^2\beta(\overline{\phi_0^{(j,i)}} + \overline{\tilde{\psi}^{(j,i)}})\phi^{(j,i)}. \quad (3.10)$$

So for each incident wave with a transverse part  $\eta_j$ , we have to solve one forward problem along with one adjoint problem for the Helmholtz equation. Since the adjoint problem has a similar variational form as the forward problem. Essentially, we need to compute two forward problems at each sweep. Once  $\delta q$  is determined,  $q_{\tilde{\eta}}$  is updated by  $q_{\tilde{\eta}} + \delta q$ . After completing sweeps with  $|\eta_j| \geq \eta$ , we get the reconstructed scatterer  $q_{\eta}$  at the spatial frequency  $\eta$ .

## 4 Numerical Experiments

In this section, we present two numerical examples to illustrate the performance of the algorithm. For test of stability, some relative random noise is added to the data, i.e. the electric field takes the form

$$\psi|_{\partial D} := (1 + \sigma \text{rand})\psi|_{\partial D}.$$

Here, rand gives uniformly distributed random numbers in  $[-1, 1]$  and  $\sigma$  is a noise level parameter taken to be 0.02 in our numerical experiments. The relaxation parameter  $\beta$  is taken to be 0.01.

Example 1. Reconstruct a scatterer defined by

$$q_1(x_1, x_2) = \begin{cases} 2.0 & \text{for } r \leq 0.6, \\ 0 & \text{for } r > 0.6, \end{cases}$$

inside the domain  $D = [-1, 1] \times [0, 2]$ , where  $r = \sqrt{x_1^2 + (x_2 - 1)^2}$ . See Figure 1 and Figure 2 for the surface and image views of the scatterer function. This example is used to examine the invalidity of Born approximation. In [12], the author derived an explicit error bound of the Born approximation for inverse scattering problem of the Helmholtz equation at fixed frequency. For the validity of the Born approximation, one needs a condition of the form

$$\rho k_0 \gamma(k_0) \sup_{|x| \leq \rho} |q(x)| < 1,$$

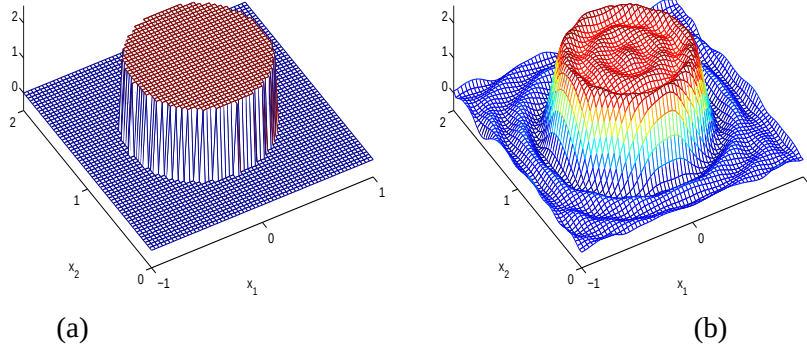


Figure 1: Reconstruction of  $q_1$ . (a) True scatterer; (b) Reconstruction.

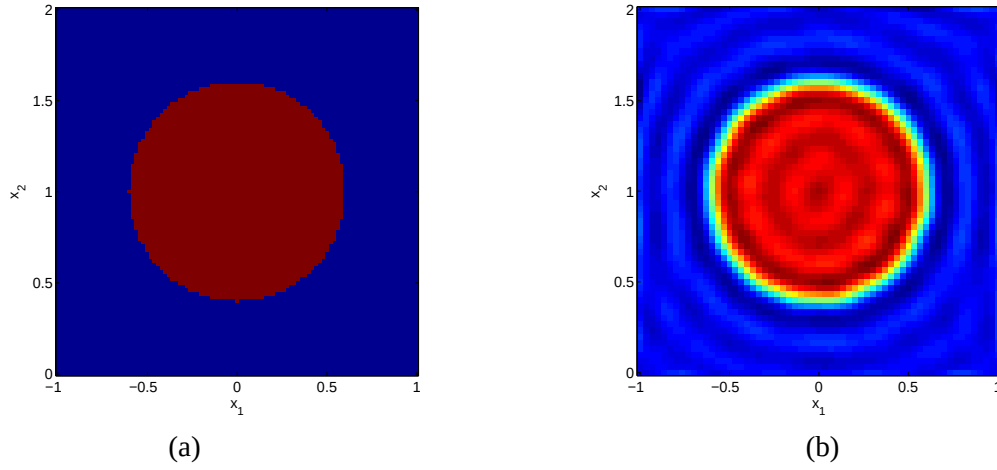


Figure 2: Image of reconstruction for  $q_1$ . (a) True scatterer; (b) Reconstruction.

where  $\rho$  is the radius of some region containing the compact support of the scatterer  $q$ ,  $k_0$  is the wavenumber, and  $\gamma$  is a positive constant, which depends on the wavenumber  $k_0$ . In the context of Example 1, these parameters are  $\rho = 0.6$ ,  $k_0 = 15.0$ ,  $\gamma = 0.63$ , and  $\sup_{|x| \leq \rho} |q(x)| = 2.0$ . It follows from simple calculation that

$$\rho k_0 \gamma(k_0) \sup_{|x| \leq \rho} |q(x)| = 11.34,$$

which is beyond the validity of Born approximation. Figure 3 gives the evolution of reconstruction horizontally across  $x_2 = 1.0$ . Due to the discontinuity of the given scatterer, the Gibbs phenomenon appears in the reconstructed scatterer.

Example 2. Reconstruct a scatterer defined by

$$q_2(x_1, x_2) = 0.5(1 + \cos(3\pi x_1)) \sin(2.5\pi x_2)$$

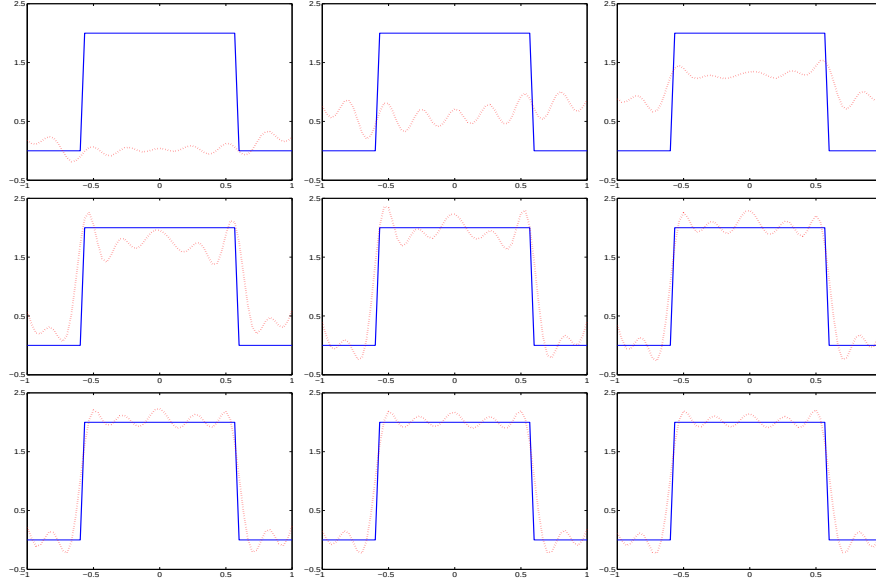


Figure 3: Evolution of slice for the reconstruction  $q_1$ . Solid curve: true scatterer; dotted curve: reconstruction. Top row from left to right: reconstruction at  $\eta = 14.45$ ; reconstruction at  $\eta = 13.60$ ; reconstruction at  $\eta = 12.75$ ; middle row from left to right: reconstruction at  $\eta = 10.20$ ; reconstruction at  $\eta = 8.50$ ; reconstruction at  $\eta = 6.80$ ; bottom row from left to right: reconstruction at  $\eta = 5.10$ ; reconstruction at  $\eta = 2.55$ ; reconstruction at  $\eta = 0.0$ .

inside the domain  $D_2 = [-1, 1] \times [0, 0.4]$ . This example is used to illustrate the resolution of the reconstruction using different wavenumbers. The  $x_1$ -transverse spatial frequency of  $q_2$  is  $3\pi$ , which accounts for the  $x_1$ -transverse wavelength about 0.67. Figure 4 shows the images of reconstructions using different wavenumbers  $k_0$  at  $\pi$ ,  $1.5\pi$ , and  $3\pi$ , corresponding to wavelengths of 2.0, 1.33, and 0.67, respectively. Figure 5 gives the slice of reconstructions at  $x_2 = 0.2$  using different wavenumbers. Figure 4 and Figure 5 present the effect of the wavenumber  $k_0$  on the result of reconstruction, which illustrates clearly that the inversion using a larger wavenumber  $k_0$  is better than that using a smaller one. This result may be explained by Heisenberg's uncertainty principle [6, 7].

## 5 Concluding remarks

We have presented a new continuation method with respect to the spatial frequency of a one-parameter family of plane waves. The recursive linearization algorithm is robust and efficient for solving the inverse medium scattering at fixed frequency. Finally, we point out some future directions along the line of this work. The first is concerned with the convergence analysis. Although our numerical experiments demonstrate the convergence and stability of the inversion algorithm, no rigorous mathematical result is available at present. Another important and interesting project is to investigate scattering problems in near-field optics since evanescent plane waves can only occur in the near-field zone. In the case of near-field optics, scattering problems are more appropriate to be formulated in the con-

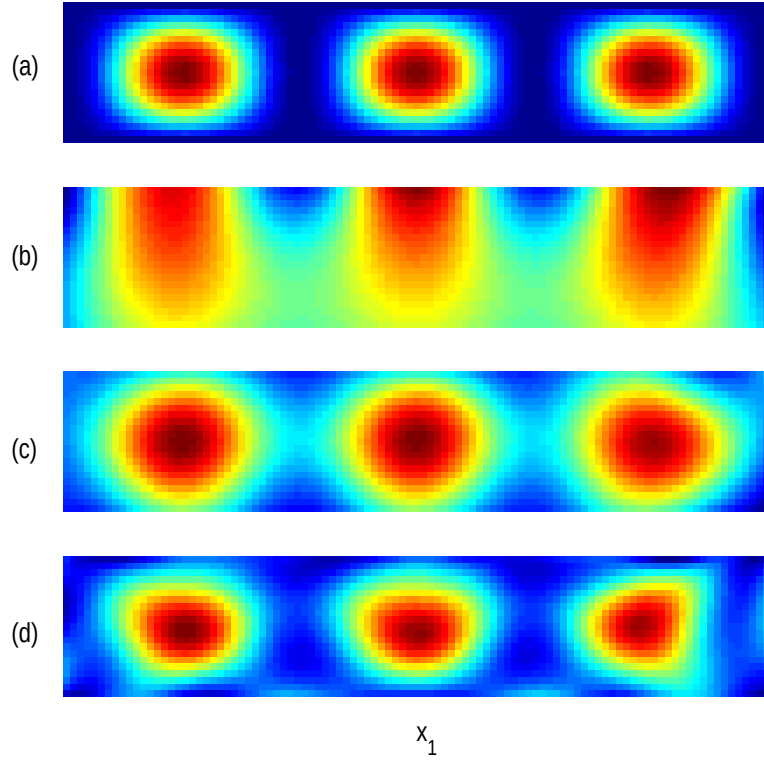


Figure 4: Image views of reconstructions for  $q_2$  with different wavenumbers. (a) true scatterer; (b) reconstruction using  $k_0 = \pi$ ; (c) reconstruction using  $k_0 = 1.5\pi$ ; (d) reconstruction using  $k_0 = 3\pi$ .

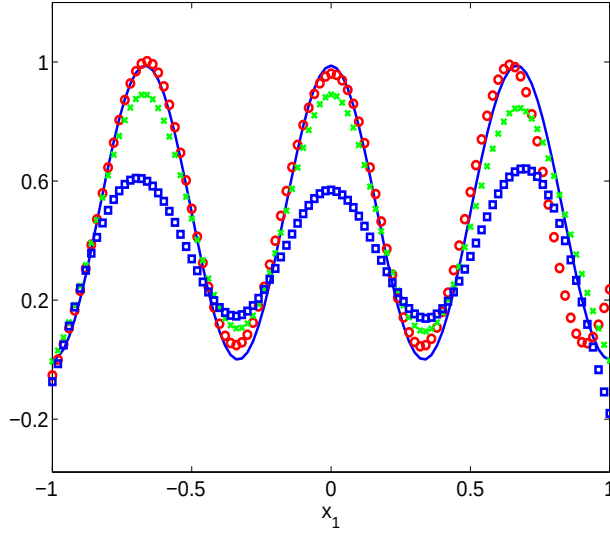


Figure 5: Slice of reconstructions for  $q_2$  with different wavenumbers. Solid curve: true scatterer;  $\square$ : reconstruction using  $k_0 = \pi$ ;  $\times$ : reconstruction using  $k_0 = 1.5\pi$ ;  $\circ$ : reconstruction using  $k_0 = 3\pi$ .

figuration of half-space instead of free space. We are currently attempting to extend the approach in this paper to more realistic models in the half-space geometry and will report the progress elsewhere.

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*Chapter 2*

## ASYMPTOTIC ANALYSIS OF DISPERSION-MANAGED VECTOR SOLITONS

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### Abstract

The higher order multiple-scale asymptotic analysis is carried out for the Gabitov-Turitsyn equation that governs the propagation of dispersion-managed solitons through birefringent fibers as well as multiple channels. The averaged equation, with the higher order terms, considerably improves the description of the soliton characteristics in such fibers.

**Key Words:** optical solitons, asymptotic analysis, Gabitov-Turitsyn equation

**AMS Subject Classification:** 35Q51; 35Q55; 37K10; 78A60

## 1 Introduction

The propagation of solitons through optical fibers has been a major area of research given its potential applicability in all optical communication systems. The field of telecommunications has undergone a substantial evolution in the last couple of decades due to the impressive progress in the development of optical fibers, optical amplifiers as well as transmitters and receivers. In a modern optical communication system, the transmission link is composed of optical fibers and amplifiers that replace the electrical regenerators. But the amplifiers introduce some noise and signal distortion that limit the system capacity. Presently the optical systems that show the best characteristics in terms of simplicity, cost and robustness against the degrading effects of a link are those based on intensity modulation with direct detection (IM-DD). Conventional IM-DD systems are based on non-return-to-zero (NRZ) format, but for transmission at higher data rate the return-to-zero (RZ) format is preferred.

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When the data rate is quite high, soliton transmission can be used. It allows the exploitation of the fiber capacity much more, but the NRZ signals offer very high potential especially in terms of simplicity [9].

There are limitations, however, on the performance of optical system due to several effects that are present in optical fibers and amplifiers. Signal propagation through optical fibers can be affected by group velocity dispersion (GVD), polarization mode dispersion (PMD) and the nonlinear effects. The chromatic dispersion that is essentially the GVD when waveguide dispersion is negligible, is a linear effect that introduces pulse broadening generates intersymbol interference. The PMD arises due the fact that optical fibers for telecommunications have two polarization modes, in spite of the fact that they are called monomode fibers. These modes have two different group velocities that induce pulse broadening depending on the input signal state of polarization. The transmission impairment due to PMD looks similar to that of the GVD. However, PMD is a random process as compared to the GVD that is a deterministic process. So PMD cannot be controlled at the receiver. Newly installed optical fibers have quite low values of PMD that is about  $0.1 \text{ ps}/\sqrt{\text{km}}$ .

The main nonlinear effects that arises in monomode fibers are the Brillouin scattering, Raman scattering and the Kerr effect. Brillouin is a backward scattering that arises from acoustic waves and can generate forward noise at the receiver. Raman scattering is a forward scattering from silica molecules. The Raman gain response is characterized by low gain and wide bandwidth namely about  $5 \text{ THz}$ . The Raman threshold in conventional fibers is of the order of  $500 \text{ mW}$  for copolarized pump and Stokes' wave (that is about  $1 \text{ W}$  for random polarization), thus making Raman effect negligible for a single channel signal. However, it becomes important for multichannel wavelength-division-multiplexed (WDM) signal due to an extremely wide band of wide gain curve.

The Kerr effect of nonlinearity is due to the dependence of the fiber refractive index on the field intensity. This effect mainly manifests as a new frequency when an optical signal propagates through a fiber. In a single channel the Kerr effect induces a spectral broadening and the phase of the signal is modulated according to its power profile. This effect is called self-phase modulation (SPM). The SPM-induced chirp combines with the linear chirp generated by the chromatic dispersion. If the fiber dispersion coefficient is positive namely in the normal dispersion regime, linear and nonlinear chirps have the same sign while in the anomalous dispersion regime they are of opposite signs. In the former case, pulse broadening is enhanced by SPM while in the later case it is reduced. In the anomalous dispersion case the Kerr nonlinearity induces a chirp that can compensate the degradation induced by GVD. Such a compensation is total if soliton signals are used.

If multichannel WDM signals are considered, the Kerr effect can be more degrading since it induces nonlinear cross-talk among the channels that is known as the cross-phase modulation (XPM). In addition WDM generates new frequencies called the Four-Wave mixing (FWM). The other issue in the WDM system is the collision-induced timing jitter that is introduced due to the collision of solitons in different channels. The XPM causes further nonlinear chirp that interacts with the fiber GVD as in the case of SPM. The FWM is a parametric interaction among waves satisfying a particular relationship called phase-matching that lead to power transfer among different channels.

To limit the FWM effect in a WDM it is preferable to operate with a local high GVD that is periodically compensated by devices having an opposite sign of GVD. One such

device is a simple optical fiber with opportune GVD and the method is commonly known as the dispersion-management. With this approach the accumulated GVD can be very low and at the same time FWM effect is strongly limited. Through dispersion-management it is possible to achieve highest capacity for both RZ as well as NRZ signals. In that case the overall link dispersion has to be kept very close to zero, while a small amount of chromatic anomalous dispersion is useful for the efficient propagation of a soliton signal. It has been demonstrated that with soliton signals, the dispersion-management is very useful since it reduces collision induced timing jitter [3] and also the pulse interactions. It thus permits the achievement of higher capacities as compared to the link having constant chromatic dispersion.

## 2 Governing Equations

The relevant equation is the Nonlinear Schrödinger's Equation (NLSE) with damping and periodic amplification [1, 7] that is given in the dimensionless form as

$$iu_z + \frac{D(z)}{2}u_{tt} + |u|^2u = -i\Gamma u + i \left[ e^{\Gamma z_a} - 1 \right] \sum_{n=1}^N \delta(z - nz_a)u \quad (2.1)$$

Here,  $\Gamma$  is the normalized loss coefficient,  $z_a$  is the normalized characteristic amplifier spacing and  $z$  and  $t$  represent the normalized propagation distance and the normalized time, respectively, expressed in the usual nondimensional units.

Also,  $D(z)$  is used to model strong dispersion-management. The fiber dispersion  $D(z)$  into two components namely a path-averaged constant value  $\delta_a$  and a term representing the large rapid variation due to large local values of the dispersion [11, 12]. Thus,

$$D(z) = \delta_a + \frac{1}{z_a} \Delta(\zeta) \quad (2.2)$$

where  $\zeta = z/z_a$ . The function  $\Delta(\zeta)$  is taken to have average zero (namely  $\langle \Delta \rangle = 0$ ), so that the path-averaged dispersion  $\langle D \rangle = \delta_a$ . The proportionality factor in front of  $\Delta(\zeta)$  is chosen so that both  $\delta_a$  and  $\Delta(\zeta)$  are quantities of order one. In practical situations, dispersion-management is often performed by concatenating together two or more sections of given length of a fiber with different values of fiber dispersion. In the special case of a two-step map it is convenient to write the dispersion map as a periodic extension of [12]

$$\Delta(\zeta) = \begin{cases} \Delta_1 & : 0 \leq |\zeta| < \frac{\theta}{2} \\ \Delta_2 & : \frac{\theta}{2} \leq |\zeta| < \frac{1}{2} \end{cases} \quad (2.3)$$

where  $\Delta_1$  and  $\Delta_2$  are given by

$$\Delta_1 = \frac{2s}{\theta} \quad (2.4)$$

$$\Delta_2 = -\frac{2s}{1-\theta} \quad (2.5)$$

with the map strength  $s$  defined as

$$s = \frac{\theta\Delta_1 - (1 - \theta)\Delta_2}{4} \quad (2.6)$$

Conversely,

$$s = \frac{\Delta_1\Delta_2}{4(\Delta_2 - \Delta_1)} \quad (2.7)$$

and

$$\theta = \frac{\Delta_2}{\Delta_2 - \Delta_1} \quad (2.8)$$

Now, taking into account the loss and amplification cycles by looking for a solution of (1) of the form  $u(z, t) = A(z)q(z, t)$  for real  $A$  and letting  $A$  satisfy

$$A_z + \Gamma A - [e^{\Gamma z_a} - 1] \sum_{n=1}^N \delta(z - nz_a) A = 0 \quad (2.9)$$

it can be shown that (1) transforms to

$$iq_z + \frac{D(z)}{2} q_{tt} + g(z)|q|^2 q = 0 \quad (2.10)$$

where

$$g(z) = A^2(z) = a_0^2 e^{-2\Gamma(z - nz_a)} \quad (2.11)$$

for  $z \in [nz_a, (n+1)z_a)$  and  $n > 0$  and also

$$a_0 = \left[ \frac{2\Gamma z_a}{1 - e^{-2\Gamma z_a}} \right]^{\frac{1}{2}} \quad (2.12)$$

so that  $\langle g(z) \rangle = 1$  over each amplification period [12]. Equation (10) governs the propagation of a dispersion-managed soliton through a polarization preserved optical fiber with damping and periodic amplification [14, 18, 19, 20].

### 3 Birefringent Fibers

A single mode fiber supports two degenerate modes that are polarized in two orthogonal directions. Under ideal conditions of perfect cylindrical geometry and isotropic material, a mode excited with its polarization in one direction would not couple with the mode in the orthogonal direction. However, small deviations from the cylindrical geometry or small fluctuations in material anisotropy result in a mixing of the two polarization states and the mode degeneracy is broken. Thus, the mode propagation constant becomes slightly different for the modes polarized in orthogonal directions. This property is referred to as modal birefringence [16]. Birefringence can also be introduced artificially in optical fibers.

The propagation of solitons in birefringent nonlinear fibers has attracted much attention in recent years. It has potential applications in optical communications and optical logic devices. The equations that describe the pulse propagation through these fibers was originally derived by Menyuk [7]. They can be solved approximately in certain special cases only. The localized pulse evolution in a birefringent fiber has been studied analytically, numerically and experimentally [16] on the basis of a simplified chirp-free model without oscillating terms under the assumptions that the two polarizations exhibit different group velocities. The equations that describe the pulse propagation in birefringent fibers are of the following dimensionless form:

$$i(u_z + \delta u_t) + \beta u + \frac{D(z)}{2} u_{tt} + g(z) (|u|^2 + \alpha |v|^2) u + \gamma v^2 u^* = 0 \quad (3.13)$$

$$i(v_z - \delta v_t) + \beta v + \frac{D(z)}{2} v_{tt} + g(z) (|v|^2 + \alpha |u|^2) v + \gamma u^2 v^* = 0 \quad (3.14)$$

Equations (13) and (14) are known as the Dispersion Managed Vector Nonlinear Schrodinger's Equation (DM-VNLSE). Here,  $u$  and  $v$  are slowly varying envelopes of the two linearly polarized components of the field along the  $x$  and  $y$  axis. Also,  $\delta$  is the group velocity mismatch between the two polarization components and is called the birefringence parameter,  $\beta$  corresponds to the difference between the propagation constants,  $\alpha$  is the cross-phase modulation (XPM) coefficient and  $\gamma$  is the coefficient of the coherent energy coupling (four-wave mixing) term. These equations are, in general, not integrable. However, they can be solved analytically for certain specific cases only [10, 16].

In this paper, the terms with  $\delta$  will be neglected as  $\delta \leq 10^{-3}$  [13]. Also, neglecting  $\beta$  and the four wave mixing terms given by the coefficient of  $\gamma$ , gives

$$iu_z + \frac{D(z)}{2} u_{tt} + g(z) (|u|^2 + \alpha |v|^2) u = 0 \quad (3.15)$$

$$iv_z + \frac{D(z)}{2} v_{tt} + g(z) (|v|^2 + \alpha |u|^2) v = 0 \quad (3.16)$$

Equations (15) and (16) are now going to be studied using the method of multiple-scale perturbation since there is no inverse scattering solution to them.

### 3.1 Integrals of Motion

The DM-VNLSE has only a couple of conserved quantities namely the energy ( $E$ ) and the linear momentum ( $M$ ) of the pulses that are respectively given by

$$E = \int_{-\infty}^{\infty} (|u|^2 + |v|^2) dt \quad (3.17)$$

$$M = \frac{i}{2} D(z) \int_{-\infty}^{\infty} (u^* u_t - u u_t^* + v^* v_t - v v_t^*) dt \quad (3.18)$$

The Hamiltonian ( $H$ ) which is given by

$$H = \frac{1}{2} \int_{-\infty}^{\infty} \left[ \frac{D(z)}{2} (|u_t|^2 + |v_t|^2) - \beta (|u|^2 - |v|^2) - \frac{g(z)}{2} (|u|^4 + |v|^4) - i \frac{\delta}{2} (u^* u_t - u u_t^* + v^* v_t - v v_t^*) - \alpha |u|^2 |v|^2 - \frac{1}{2} (1 - \alpha) (u^2 v^{*2} + v^2 u^{*2}) \right] dt \quad (3.19)$$

is however not a constant of motion, in general unless  $D(z)$  and  $g(z)$  are constants. For the reduced set of equation given by (15) and (16), the Hamiltonian is

$$H = \int_{-\infty}^{\infty} \left[ \frac{D(z)}{2} (|u_t|^2 + |v_t|^2) - \frac{g(z)}{2} (|u|^4 + |v|^4) - \alpha |u|^2 |v|^2 - \frac{1}{2} (1 - \alpha) (u^2 v^{*2} + v^2 u^{*2}) \right] dt \quad (3.20)$$

The existence of a Hamiltonian implies that (15) and (16) can be written as

$$i \frac{\partial u}{\partial z} = \frac{\delta H}{\delta u^*} \quad (3.21)$$

and

$$i \frac{\partial v}{\partial z} = \frac{\delta H}{\delta v^*} \quad (3.22)$$

This defines a Hamiltonian dynamical system on an infinite-dimensional phase space of two complex functions  $u$  and  $v$  that decrease to zero at infinity and can be analysed using the theory of Hamiltonian system.

### 3.2 Asymptotic Analysis

Equations (15) and (16) contains both large and rapidly varying terms. To obtain the asymptotic behaviour the fast and slow  $z$  scales are introduced as

$$\zeta = \frac{z}{z_a} \quad (3.23)$$

and

$$Z = z \quad (3.24)$$

The fields  $u$  and  $v$  are expanded in powers of  $z_a$  as

$$u(\zeta, Z, t) = u^{(0)}(\zeta, Z, t) + z_a u^{(1)}(\zeta, Z, t) + z_a^2 u^{(2)}(\zeta, Z, t) + \dots \quad (3.25)$$

$$v(\zeta, Z, t) = v^{(0)}(\zeta, Z, t) + z_a v^{(1)}(\zeta, Z, t) + z_a^2 v^{(2)}(\zeta, Z, t) + \dots \quad (3.26)$$

Equating coefficients of like powers of  $z_a$  gives

$$O\left(\frac{1}{z_a}\right): \quad i \frac{\partial u^{(0)}}{\partial \zeta} + \frac{\Delta(\zeta)}{2} \frac{\partial^2 u^{(0)}}{\partial t^2} = 0 \quad (3.27)$$

$$O\left(\frac{1}{z_a}\right) : \quad i\frac{\partial v^{(0)}}{\partial \zeta} + \frac{\Delta(\zeta)}{2} \frac{\partial^2 v^{(0)}}{\partial t^2} = 0 \quad (3.28)$$

$$\begin{aligned} O(1) : \quad & i\frac{\partial u^{(1)}}{\partial \zeta} + \frac{\Delta(\zeta)}{2} \frac{\partial^2 u^{(1)}}{\partial t^2} \\ & + \left\{ i\frac{\partial u^{(0)}}{\partial Z} + \frac{\delta_a}{2} \frac{\partial^2 u^{(0)}}{\partial t^2} + g(z) \left( |u^{(0)}|^2 + \alpha |v^{(0)}|^2 \right) u^{(0)} \right\} = 0 \end{aligned} \quad (3.29)$$

$$\begin{aligned} O(1) : \quad & i\frac{\partial v^{(1)}}{\partial \zeta} + \frac{\Delta(\zeta)}{2} \frac{\partial^2 v^{(1)}}{\partial t^2} \\ & + \left\{ i\frac{\partial v^{(0)}}{\partial Z} + \frac{\delta_a}{2} \frac{\partial^2 v^{(0)}}{\partial t^2} + g(z) \left( |v^{(0)}|^2 + \alpha |u^{(0)}|^2 \right) v^{(0)} \right\} = 0 \end{aligned} \quad (3.30)$$

$$\begin{aligned} O(z_a) : \quad & i\frac{\partial u^{(2)}}{\partial \zeta} + \frac{\Delta(\zeta)}{2} \frac{\partial^2 u^{(2)}}{\partial t^2} \\ & + \left\{ i\frac{\partial u^{(1)}}{\partial Z} + \frac{\delta_a}{2} \frac{\partial^2 u^{(1)}}{\partial t^2} + g(z) \left[ 2|u^{(0)}|^2 u^{(1)} + \left(u^{(0)}\right)^2 \left(u^{(1)}\right)^* \right. \right. \\ & \left. \left. + \alpha \left\{ 2|v^{(0)}|^2 v^{(1)} + \left(v^{(0)}\right)^2 \left(v^{(1)}\right)^* \right\} \right] \right\} \end{aligned} \quad (3.31)$$

$$\begin{aligned} O(z_a) : \quad & i\frac{\partial v^{(2)}}{\partial \zeta} + \frac{\Delta(\zeta)}{2} \frac{\partial^2 v^{(2)}}{\partial t^2} \\ & + \left\{ i\frac{\partial v^{(1)}}{\partial Z} + \frac{\delta_a}{2} \frac{\partial^2 v^{(1)}}{\partial t^2} + g(z) \left[ 2|v^{(0)}|^2 v^{(1)} + \left(v^{(0)}\right)^2 \left(v^{(1)}\right)^* \right. \right. \\ & \left. \left. + \alpha \left\{ 2|u^{(0)}|^2 u^{(1)} + \left(u^{(0)}\right)^2 \left(u^{(1)}\right)^* \right\} \right] \right\} \end{aligned} \quad (3.32)$$

Now the Fourier transform and its inverse are respectively defined as

$$\hat{f}(\omega) = \mathbf{F}[f] \equiv \int_{-\infty}^{\infty} f(t) e^{i\omega t} dt \quad (3.33)$$

$$f(t) = \mathbf{F}^{-1}[\hat{f}] \equiv \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{f}(\omega) e^{-i\omega t} d\omega \quad (3.34)$$

At  $O(1/z_a)$  equations (27) and (28), respectively, in the Fourier domain are given by

$$i\frac{\partial \hat{u}^{(0)}}{\partial \zeta} - \frac{\omega^2}{2} \Delta(\zeta) \hat{u}^{(0)} = 0 \quad (3.35)$$

and

$$i \frac{\partial \hat{v}^{(0)}}{\partial \zeta} - \frac{\omega^2}{2} \Delta(\zeta) \hat{v}^{(0)} = 0 \quad (3.36)$$

whose respective solutions are

$$\hat{u}^{(0)}(\zeta, Z, \omega) = \hat{U}_0(Z, \omega) e^{-\frac{i\omega^2}{2} C(\zeta)} \quad (3.37)$$

and

$$\hat{v}^{(0)}(\zeta, Z, \omega) = \hat{V}_0(Z, \omega) e^{-\frac{i\omega^2}{2} C(\zeta)} \quad (3.38)$$

where

$$\hat{U}_0(Z, \omega) = \hat{u}^{(0)}(0, Z, \omega) \quad (3.39)$$

$$\hat{V}_0(Z, \omega) = \hat{v}^{(0)}(0, Z, \omega) \quad (3.40)$$

and

$$C(\zeta) = \int_0^\zeta \Delta(\zeta') d\zeta' \quad (3.41)$$

At  $O(1)$ , equations (27) and (28) are solved in the Fourier domain by substituting the respective solutions given by (35) and (36) for  $u^{(0)}$  and  $v^{(0)}$  in (27) and (28). This gives

$$\begin{aligned} & i \frac{\partial \hat{u}^{(1)}}{\partial \zeta} - \frac{\omega^2}{2} \Delta(\zeta) \hat{u}^{(1)} \\ &= -e^{-\frac{i\omega^2}{2} C(\zeta)} \left( \frac{\partial \hat{U}_0}{\partial Z} - \frac{\omega^2}{2} \delta_a \hat{U}_0 \right) - g(\zeta) \int_{-\infty}^{\infty} \left( |u^{(0)}|^2 + \alpha |v^{(0)}|^2 \right) u^{(0)} e^{i\omega t} dt \end{aligned} \quad (3.42)$$

and

$$\begin{aligned} & i \frac{\partial \hat{v}^{(1)}}{\partial \zeta} - \frac{\omega^2}{2} \Delta(\zeta) \hat{v}^{(1)} \\ &= -e^{-\frac{i\omega^2}{2} C(\zeta)} \left( \frac{\partial \hat{V}_0}{\partial Z} - \frac{\omega^2}{2} \delta_a \hat{V}_0 \right) - g(\zeta) \int_{-\infty}^{\infty} \left( |v^{(0)}|^2 + \alpha |u^{(0)}|^2 \right) v^{(0)} e^{i\omega t} dt \end{aligned} \quad (3.43)$$

Equations (42) and (43) are inhomogeneous equations for  $\hat{u}^{(1)}$  and  $\hat{v}^{(1)}$  respectively with the homogenous parts having the same structures as in (35) and (36) respectively. For the non-secularity conditions of  $\hat{u}^{(1)}$  and  $\hat{v}^{(1)}$ , it is necessary that the forcing terms be orthogonal to the adjoint solutions of (35) and (36) respectively, a condition that is commonly

known as Fredholm's Alternative (FA). This gives the conditions for  $\hat{U}_0(Z, \omega)$  and  $\hat{V}_0(Z, \omega)$  respectively as

$$\begin{aligned} \frac{\partial \hat{U}_0}{\partial Z} - \frac{\omega^2}{2} \delta_a \hat{U}_0 \\ + \int_0^1 \int_{-\infty}^{\infty} e^{\frac{i\omega^2}{2} C(\zeta)} g(\zeta) \left( |u^{(0)}|^2 + \alpha |v^{(0)}|^2 \right) u^{(0)} e^{i\omega t} dt d\zeta = 0 \end{aligned} \quad (3.44)$$

and

$$\begin{aligned} \frac{\partial \hat{V}_0}{\partial Z} - \frac{\omega^2}{2} \delta_a \hat{V}_0 \\ + \int_0^1 \int_{-\infty}^{\infty} e^{\frac{i\omega^2}{2} C(\zeta)} g(\zeta) \left( |v^{(0)}|^2 + \alpha |u^{(0)}|^2 \right) v^{(0)} e^{i\omega t} dt d\zeta = 0 \end{aligned} \quad (3.45)$$

Equations (44) and (45) can be respectively simplified to

$$\begin{aligned} \frac{\partial \hat{U}_0}{\partial Z} - \frac{\omega^2}{2} \delta_a \hat{U}_0 \\ + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} r_0(\omega_1 \omega_2) \hat{U}_0(Z, \omega_1 + \omega_2) \left[ \hat{U}_0(Z, \omega + \omega_1) \hat{U}_0(Z, \omega + \omega_1 + \omega_2) \right. \\ \left. + \alpha \hat{V}_0(Z, \omega + \omega_1) \hat{V}_0(Z, \omega + \omega_1 + \omega_2) \right] d\omega_1 d\omega_2 = 0 \end{aligned} \quad (3.46)$$

and

$$\begin{aligned} \frac{\partial \hat{V}_0}{\partial Z} - \frac{\omega^2}{2} \delta_a \hat{V}_0 \\ + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} r_0(\omega_1 \omega_2) \hat{V}_0(Z, \omega_1 + \omega_2) \left[ \hat{V}_0(Z, \omega + \omega_1) \hat{V}_0(Z, \omega + \omega_1 + \omega_2) \right. \\ \left. + \alpha \hat{U}_0(Z, \omega + \omega_1) \hat{U}_0(Z, \omega + \omega_1 + \omega_2) \right] d\omega_1 d\omega_2 = 0 \end{aligned} \quad (3.47)$$

where, the kernel  $r_0(x)$  is given by

$$r_0(x) = \frac{1}{(2\pi)^2} \int_0^1 g(\zeta) e^{ixC(\zeta)} d\zeta \quad (3.48)$$

Equations (46) and (47) are commonly known as the Gabitov-Turitsyn equations (GTE) for the propagation of solitons through a birefringent fiber.

Equations (42) and (43) will now be solved to obtain  $u^{(1)}(\zeta, Z, t)$  and  $v^{(1)}(\zeta, Z, t)$ . Substituting  $\hat{U}_0$  and  $\hat{V}_0$  into the right side of equations (42) and (43) respectively and using the pairs (37)-(38) and (44)-(45) gives

$$\begin{aligned} \frac{\partial}{\partial \zeta} \left[ i \hat{u}^{(1)} e^{\frac{i\omega^2}{2} C(\zeta)} \right] \\ = \int_0^1 \int_{-\infty}^{\infty} g(\zeta) e^{\frac{i\omega^2}{2} C(\zeta)} \left( |u^{(0)}|^2 + \alpha |v^{(0)}|^2 \right) u^{(0)} e^{i\omega t} dt d\zeta \\ - g(\zeta) e^{\frac{i\omega^2}{2} C(\zeta)} \int_{-\infty}^{\infty} \left( |u^{(0)}|^2 + \alpha |v^{(0)}|^2 \right) u^{(0)} e^{i\omega t} dt \end{aligned} \quad (3.49)$$



and

$$\begin{aligned}
& \frac{\partial}{\partial \zeta} \left[ i\hat{v}^{(1)} e^{\frac{i\omega^2}{2}C(\zeta)} \right] \\
&= \int_0^1 \int_{-\infty}^{\infty} g(\zeta) e^{\frac{i\omega^2}{2}C(\zeta)} \left( |v^{(0)}|^2 + \alpha |u^{(0)}|^2 \right) v^{(0)} e^{i\omega t} dt d\zeta \\
&\quad - g(\zeta) e^{\frac{i\omega^2}{2}C(\zeta)} \int_{-\infty}^{\infty} \left( |v^{(0)}|^2 + \alpha |u^{(0)}|^2 \right) v^{(0)} e^{i\omega t} dt
\end{aligned} \tag{3.50}$$

Integration of equations (49) and (50) yields

$$\begin{aligned}
& i\hat{u}^{(1)} e^{\frac{i\omega^2}{2}C(\zeta)} = \hat{U}_1(Z, \omega) \\
& + \zeta \int_0^1 \int_{-\infty}^{\infty} g(\zeta) e^{\frac{i\omega^2}{2}C(\zeta)} \left( |u^{(0)}|^2 + \alpha |v^{(0)}|^2 \right) u^{(0)} e^{i\omega t} dt d\zeta \\
& - \int_0^\zeta \int_{-\infty}^{\infty} g(\zeta') e^{\frac{i\omega^2}{2}C(\zeta')} \left( |u^{(0)}|^2 + \alpha |v^{(0)}|^2 \right) u^{(0)} e^{i\omega t} dt d\zeta'
\end{aligned} \tag{3.51}$$

and

$$\begin{aligned}
& i\hat{v}^{(1)} e^{\frac{i\omega^2}{2}C(\zeta)} = \hat{V}_1(Z, \omega) \\
& + \zeta \int_0^1 \int_{-\infty}^{\infty} g(\zeta) e^{\frac{i\omega^2}{2}C(\zeta)} \left( |v^{(0)}|^2 + \alpha |u^{(0)}|^2 \right) v^{(0)} e^{i\omega t} dt d\zeta \\
& - \int_0^\zeta \int_{-\infty}^{\infty} g(\zeta') e^{\frac{i\omega^2}{2}C(\zeta')} \left( |v^{(0)}|^2 + \alpha |u^{(0)}|^2 \right) v^{(0)} e^{i\omega t} dt d\zeta'
\end{aligned} \tag{3.52}$$

where

$$\hat{U}_1(Z, \omega) = i\hat{u}^{(1)}(0, Z, \omega) e^{\frac{i\omega^2}{2}C(0)} \tag{3.53}$$

and

$$\hat{V}_1(Z, \omega) = i\hat{v}^{(1)}(0, Z, \omega) e^{\frac{i\omega^2}{2}C(0)} \tag{3.54}$$

Also,  $\hat{U}_1(Z, \omega)$  and  $\hat{V}_1(Z, \omega)$  are so chosen that

$$\int_0^1 i\hat{u}^{(1)} e^{\frac{i\omega^2}{2}C(\zeta)} d\zeta = 0 \tag{3.55}$$

and

$$\int_0^1 i\hat{v}^{(1)} e^{\frac{i\omega^2}{2}C(\zeta)} d\zeta = 0 \tag{3.56}$$

which are going to be useful relations at subsequent orders. Applying (55) and (56) to (51) and (52) respectively gives

$$\begin{aligned}
& \hat{U}_1(Z, \omega) = \\
& \int_0^1 \int_0^\zeta \int_{-\infty}^{\infty} g(\zeta') e^{\frac{i\omega^2}{2}C(\zeta')} \left( |u^{(0)}|^2 + \alpha |v^{(0)}|^2 \right) u^{(0)} e^{i\omega t} dt d\zeta' d\zeta \\
& - \frac{1}{2} \int_0^1 \int_{-\infty}^{\infty} g(\zeta) e^{\frac{i\omega^2}{2}C(\zeta)} \left( |u^{(0)}|^2 + \alpha |v^{(0)}|^2 \right) u^{(0)} e^{i\omega t} dt d\zeta
\end{aligned} \tag{3.57}$$

and

$$\begin{aligned}\hat{V}_1(Z, \omega) = & \int_0^1 \int_0^\zeta \int_{-\infty}^\infty g(\zeta') e^{\frac{i\omega^2}{2}C(\zeta')} \left( |v^{(0)}|^2 + \alpha |u^{(0)}|^2 \right) v^{(0)} e^{i\omega t} dt d\zeta' d\zeta \\ & - \frac{1}{2} \int_0^1 \int_{-\infty}^\infty g(\zeta) e^{\frac{i\omega^2}{2}C(\zeta)} \left( |v^{(0)}|^2 + \alpha |u^{(0)}|^2 \right) v^{(0)} e^{i\omega t} dt d\zeta\end{aligned}\quad (3.58)$$

Now, equations (51) and (52), by virtue of (57) and (58) can be respectively written as

$$\begin{aligned}\hat{u}^{(1)}(\zeta, Z, \omega) = & ie^{\frac{i\omega^2}{2}C(\zeta)} \left[ \int_0^\zeta \int_{-\infty}^\infty g(\zeta') e^{\frac{i\omega^2}{2}C(\zeta')} \left( |u^{(0)}|^2 + \alpha |v^{(0)}|^2 \right) u^{(0)} e^{i\omega t} dt d\zeta' \right. \\ & - \int_0^1 \int_0^\zeta \int_{-\infty}^\infty g(\zeta') e^{\frac{i\omega^2}{2}C(\zeta')} \left( |u^{(0)}|^2 + \alpha |v^{(0)}|^2 \right) u^{(0)} e^{i\omega t} dt d\zeta' d\zeta \\ & \left. - \left( \zeta - \frac{1}{2} \right) \int_0^1 \int_{-\infty}^\infty g(\zeta) e^{\frac{i\omega^2}{2}C(\zeta)} \left( |u^{(0)}|^2 + \alpha |v^{(0)}|^2 \right) u^{(0)} e^{i\omega t} dt d\zeta \right]\end{aligned}\quad (3.59)$$

and

$$\begin{aligned}\hat{v}^{(1)}(\zeta, Z, \omega) = & ie^{\frac{i\omega^2}{2}C(\zeta)} \left[ \int_0^\zeta \int_{-\infty}^\infty g(\zeta') e^{\frac{i\omega^2}{2}C(\zeta')} \left( |v^{(0)}|^2 + \alpha |u^{(0)}|^2 \right) v^{(0)} e^{i\omega t} dt d\zeta' \right. \\ & - \int_0^1 \int_0^\zeta \int_{-\infty}^\infty g(\zeta') e^{\frac{i\omega^2}{2}C(\zeta')} \left( |v^{(0)}|^2 + \alpha |u^{(0)}|^2 \right) v^{(0)} e^{i\omega t} dt d\zeta' d\zeta \\ & \left. - \left( \zeta - \frac{1}{2} \right) \int_0^1 \int_{-\infty}^\infty g(\zeta) e^{\frac{i\omega^2}{2}C(\zeta)} \left( |v^{(0)}|^2 + \alpha |u^{(0)}|^2 \right) v^{(0)} e^{i\omega t} dt d\zeta \right]\end{aligned}\quad (3.60)$$

The pair (59) and (60) can now be respectively further written as

$$\begin{aligned}\hat{u}^{(1)}(\zeta, Z, \omega) = & ie^{-\frac{i\omega^2}{2}C(\zeta)} \left[ \int_{-\infty}^\infty \int_{-\infty}^\infty \hat{U}_0^*(\omega + \Omega_1 + \Omega_2) \left\{ \hat{U}_0(\omega + \Omega_1) \hat{U}_0(\omega + \Omega_2) \right. \right. \\ & \left. \left. + \alpha \hat{V}_0(\omega + \Omega_1) \hat{V}_0(\omega + \Omega_2) \right\} \right. \\ & \left\{ \int_0^\zeta g(\zeta') e^{i\Omega_1\Omega_2C(\zeta')} d\zeta' - \int_0^1 \int_0^\zeta g(\zeta') e^{i\Omega_1\Omega_2C(\zeta')} d\zeta' d\zeta \right. \\ & \left. \left. - \left( \zeta - \frac{1}{2} \right) \int_0^1 g(\zeta) e^{i\Omega_1\Omega_2C(\zeta)} d\zeta \right\} d\Omega_1 d\Omega_2 \right]\end{aligned}\quad (3.61)$$

and

$$\begin{aligned}
\hat{v}^{(1)}(\zeta, Z, \omega) = & i e^{-\frac{i\omega^2}{2}C(\zeta)} \left[ \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \hat{V}_0^*(\omega + \Omega_1 + \Omega_2) \left\{ \hat{V}_0(\omega + \Omega_1) \hat{V}_0(\omega + \Omega_2) \right. \right. \\
& + \alpha \hat{U}_0(\omega + \Omega_1) \hat{U}_0(\omega + \Omega_2) \left. \right\} \\
& \left\{ \int_0^\zeta g(\zeta') e^{i\Omega_1 \Omega_2 C(\zeta')} d\zeta' - \int_0^1 \int_0^\zeta g(\zeta') e^{i\Omega_1 \Omega_2 C(\zeta')} d\zeta' d\zeta \right. \\
& \left. \left. - \left( \zeta - \frac{1}{2} \right) \int_0^1 g(\zeta) e^{i\Omega_1 \Omega_2 C(\zeta)} d\zeta \right\} d\Omega_1 \Omega_2 \right] \quad (3.62)
\end{aligned}$$

Thus, at  $O(z_a)$ ,

$$\hat{u}(\zeta, Z, \omega) = \hat{u}^{(0)}(\zeta, Z, \omega) + z_a \hat{u}^{(1)}(\zeta, Z, \omega) \quad (3.63)$$

and

$$\hat{v}(\zeta, Z, \omega) = \hat{v}^{(0)}(\zeta, Z, \omega) + z_a \hat{v}^{(1)}(\zeta, Z, \omega) \quad (3.64)$$

Moving on to the next order at  $O(z_a^2)$ , one can note that the GTE given by (46) and (47) are allowed to have an additional term of  $O(z_a)$  such as

$$\begin{aligned}
& \frac{\partial \hat{U}_0}{\partial Z} - \frac{\omega^2}{2} \delta_a \hat{U}_0 \\
& + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} r_0(\omega_1 \omega_2) \hat{U}_0(Z, \omega_1 + \omega_2) \left[ \hat{U}_0(Z, \omega + \omega_1) \hat{U}_0(Z, \omega + \omega_1 + \omega_2) \right. \\
& \left. + \alpha \hat{V}_0(Z, \omega + \omega_1) \hat{V}_0(Z, \omega + \omega_1 + \omega_2) \right] d\omega_1 d\omega_2 = z_a \hat{n}_1(Z, \omega) + O(z_a^2) \quad (3.65)
\end{aligned}$$

and

$$\begin{aligned}
& \frac{\partial \hat{V}_0}{\partial Z} - \frac{\omega^2}{2} \delta_a \hat{V}_0 \\
& + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} r_0(\omega_1 \omega_2) \hat{V}_0(Z, \omega_1 + \omega_2) \left[ \hat{V}_0(Z, \omega + \omega_1) \hat{V}_0(Z, \omega + \omega_1 + \omega_2) \right. \\
& \left. + \alpha \hat{U}_0(Z, \omega + \omega_1) \hat{U}_0(Z, \omega + \omega_1 + \omega_2) \right] d\omega_1 d\omega_2 = z_a \hat{n}_2(Z, \omega) + O(z_a^2) \quad (3.66)
\end{aligned}$$

The higher order corrections  $\hat{n}_1$  and  $\hat{n}_2$  can be obtained from suitable non-secular conditions at  $O(z_a^2)$  in (31) and (32) respectively. Now, equations (31) and (32), in the Fourier domain, respectively are

$$\begin{aligned}
& \frac{\partial}{\partial \zeta} \left[ i \hat{u}^{(2)} e^{\frac{i\omega^2}{2}C(\zeta)} \right] + \hat{n}_1 + e^{\frac{i\omega^2}{2}C(\zeta)} \left( i \frac{\partial \hat{u}^{(1)}}{\partial Z} - \frac{\omega^2}{2} \delta_a \hat{u}^{(1)} \right) + e^{\frac{i\omega^2}{2}C(\zeta)} g(\zeta) \\
& \int_{-\infty}^{\infty} \left[ 2 |u^{(0)}|^2 u^{(1)} + \left( u^{(0)} \right)^2 u^{(1)*} + \alpha \left\{ 2 |v^{(0)}|^2 v^{(1)} + \left( v^{(0)} \right)^2 v^{(1)*} \right\} \right] e^{i\omega t} dt = 0 \quad (3.67)
\end{aligned}$$

and

$$\begin{aligned} \frac{\partial}{\partial \zeta} \left[ i \hat{v}^{(2)} e^{\frac{i\omega^2}{2} C(\zeta)} \right] + \hat{n}_2 + e^{\frac{i\omega^2}{2} C(\zeta)} \left( i \frac{\partial \hat{v}^{(1)}}{\partial Z} - \frac{\omega^2}{2} \delta_a \hat{v}^{(1)} \right) + e^{\frac{i\omega^2}{2} C(\zeta)} g(\zeta) \\ \int_{-\infty}^{\infty} \left[ 2 \left| v^{(0)} \right|^2 v^{(1)} + \left( v^{(0)} \right)^2 v^{(1)*} + \alpha \left\{ 2 \left| u^{(0)} \right|^2 u^{(1)} + \left( u^{(0)} \right)^2 u^{(1)*} \right\} \right] e^{i\omega t} dt = 0 \end{aligned} \quad (3.68)$$

But, again (55) and (56) gives

$$\int_0^1 \hat{u}^{(1)} e^{\frac{i\omega^2}{2} C(\zeta)} d\zeta = 0 \quad (3.69)$$

and

$$\int_0^1 \hat{v}^{(1)} e^{\frac{i\omega^2}{2} C(\zeta)} d\zeta = 0 \quad (3.70)$$

Applying the non-secularity conditions (69) and (70) to (67) and (68) respectively gives

$$\begin{aligned} \hat{n}_1 = - \int_0^1 \int_{-\infty}^{\infty} e^{\frac{i\omega^2}{2} C(\zeta)} g(\zeta) \\ \left[ 2 \left| u^{(0)} \right|^2 u^{(1)} + \left( u^{(0)} \right)^2 u^{(1)*} + \alpha \left\{ 2 \left| v^{(0)} \right|^2 v^{(1)} + \left( v^{(0)} \right)^2 v^{(1)*} \right\} \right] e^{i\omega t} dt d\zeta \end{aligned} \quad (3.71)$$

and

$$\begin{aligned} \hat{n}_2 = - \int_0^1 \int_{-\infty}^{\infty} e^{\frac{i\omega^2}{2} C(\zeta)} g(\zeta) \\ \left[ 2 \left| v^{(0)} \right|^2 v^{(1)} + \left( v^{(0)} \right)^2 v^{(1)*} + \alpha \left\{ 2 \left| u^{(0)} \right|^2 u^{(1)} + \left( u^{(0)} \right)^2 u^{(1)*} \right\} \right] e^{i\omega t} dt d\zeta \end{aligned} \quad (3.72)$$

Using the pairs (37)-(38) and (59)-(60), equations (71) and (72) can respectively be written

as

$$\begin{aligned}
\hat{n}_1 = & \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} r_1(\omega_1 \omega_2, \Omega_1 \Omega_2) \\
& \left[ \left\{ 2\hat{U}_0(\omega + \omega_1) \hat{U}_0^*(\omega + \omega_1 + \omega_2) \hat{U}_0(\omega + \omega_2 + \Omega_1) \right. \right. \\
& \hat{U}_0(\omega + \omega_2 + \Omega_2) \hat{U}_0^*(\omega + \omega_2 + \Omega_1 + \Omega_2) \\
& - \hat{U}_0(\omega + \omega_1) \hat{U}_0(\omega + \omega_2) \hat{U}_0^*(\omega + \omega_1 + \omega_2 + \Omega_1) \\
& \left. \hat{U}_0^*(\omega + \omega_1 + \omega_2 - \Omega_2) \hat{U}_0^*(\omega + \omega_1 + \omega_2 + \Omega_1 - \Omega_2) \right\} \\
& + \alpha \left\{ 2\hat{V}_0(\omega + \omega_1) \hat{V}_0^*(\omega + \omega_1 + \omega_2) \hat{V}_0(\omega + \omega_2 + \Omega_1) \right. \\
& \hat{V}_0(\omega + \omega_2 + \Omega_2) \hat{V}_0^*(\omega + \omega_2 + \Omega_1 + \Omega_2) \\
& - \hat{V}_0(\omega + \omega_1) \hat{V}_0(\omega + \omega_2) \hat{V}_0^*(\omega + \omega_1 + \omega_2 + \Omega_1) \\
& \left. \left. \hat{V}_0^*(\omega + \omega_1 + \omega_2 - \Omega_2) \hat{V}_0^*(\omega + \omega_1 + \omega_2 + \Omega_1 - \Omega_2) \right\} \right] d\omega_1 d\omega_2 d\Omega_1 d\Omega_2
\end{aligned} \tag{3.73}$$

and

$$\begin{aligned}
\hat{n}_2 = & \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} r_1(\omega_1 \omega_2, \Omega_1 \Omega_2) \\
& \left[ \left\{ 2\hat{V}_0(\omega + \omega_1) \hat{V}_0^*(\omega + \omega_1 + \omega_2) \hat{V}_0(\omega + \omega_2 + \Omega_1) \right. \right. \\
& \hat{V}_0(\omega + \omega_2 + \Omega_2) \hat{V}_0^*(\omega + \omega_2 + \Omega_1 + \Omega_2) \\
& - \hat{V}_0(\omega + \omega_1) \hat{V}_0(\omega + \omega_2) \hat{V}_0^*(\omega + \omega_1 + \omega_2 + \Omega_1) \\
& \left. \hat{V}_0^*(\omega + \omega_1 + \omega_2 - \Omega_2) \hat{V}_0^*(\omega + \omega_1 + \omega_2 + \Omega_1 - \Omega_2) \right\} \\
& + \alpha \left\{ 2\hat{U}_0(\omega + \omega_1) \hat{U}_0^*(\omega + \omega_1 + \omega_2) \hat{U}_0(\omega + \omega_2 + \Omega_1) \right. \\
& \hat{U}_0(\omega + \omega_2 + \Omega_2) \hat{U}_0^*(\omega + \omega_2 + \Omega_1 + \Omega_2) \\
& - \hat{U}_0(\omega + \omega_1) \hat{U}_0(\omega + \omega_2) \hat{U}_0^*(\omega + \omega_1 + \omega_2 + \Omega_1) \\
& \left. \left. \hat{U}_0^*(\omega + \omega_1 + \omega_2 - \Omega_2) \hat{U}_0^*(\omega + \omega_1 + \omega_2 + \Omega_1 - \Omega_2) \right\} \right] d\omega_1 d\omega_2 d\Omega_1 d\Omega_2
\end{aligned} \tag{3.74}$$

where, the kernel  $r_1(x, y)$  is given by

$$\begin{aligned}
r_1(x, y) = & \frac{1}{(2\pi)^4} \int_0^1 \int_0^\zeta g(\zeta) g(\zeta') e^{i(xC(\zeta) + yC(\zeta'))} d\zeta d\zeta' \\
& - \left[ \int_0^1 g(\zeta) e^{ixC(\zeta)} d\zeta \right] \left[ \int_0^\zeta g(\zeta') e^{ixC(\zeta')} d\zeta' \right] \\
& - \left[ \int_0^1 \left( \zeta - \frac{1}{2} \right) g(\zeta) e^{ixC(\zeta)} d\zeta \right] \left[ \int_0^1 g(\zeta) e^{iyC(\zeta)} d\zeta \right]
\end{aligned} \tag{3.75}$$

Equations (65) and (66) represent the higher order GTE (HO-GTE) for the propagation of solitons through birefringent optical fibers.

## 4 Multiple Channels

The successful design of low-loss dispersion-shifted and dispersion-flattened optical fibers with low dispersion over relatively large wavelength range can be used to reduce or completely eliminate the group velocity mismatch for the multi-channel WDM systems resulting in the desirable simultaneous arrival of time aligned bit pulses, thus creating a new class of bit-parallel wavelength links that is used in high speed single fiber computer buses. In spite of the intrinsically small value of the nonlinearity-induced change in the refractive index of fused silica, nonlinear effects in optical fibers cannot be ignored even at relatively low powers. In particular, in WDM systems with simultaneous transmission of pulses of different wavelengths, the cross-phase modulation (XPM) effects needs to be taken into account. Although the XPM will not cause the energy to be exchanged among the different wavelengths, it will lead to the interaction of pulses and thus the pulse positions and shapes gets altered significantly. The multi-channel WDM transmission of co-propagating wave envelopes in a nonlinear optical fiber, including the XPM effect, can be modeled [8] by the following  $N$ -coupled NLSE in the dimensionless form

$$iq_z^{(l)} + \frac{D(z)}{2}q_{tt}^{(l)} + g(z) \left\{ |q^{(l)}|^2 + \sum_{m \neq l}^N \alpha_{lm} |q^{(m)}|^2 \right\} q^{(l)} = 0 \quad (4.76)$$

where  $1 \leq l \leq N$ . Equation (76) is the  $N$ -dimensional vector NLSE and is the model for bit-parallel WDM soliton transmission. Here  $\alpha_{lm}$  are known as the XPM coefficients. It is well known [7] that the straightforward use of this system for description of WDM transmission could potentially give incorrect results. However, this model can be applied to describe WDM transmission for dispersion flattened fibers, the dispersion of which weakly depends on the operating wavelength.

Another important medium in which the model given by (76) arises is the photorefractive medium [8]. In the case of incoherent beam propagation in a biased photorefractive crystal, which is a noninstantaneous nonlinear media, the diffraction behaviour of that incoherent beam is to be treated somewhat differently. The diffraction behaviour of an incoherent beam can be effectively described by the sum of the intensity contributions from all its coherent components. Then the governing equation of  $N$  self-trapped mutually incoherent wave packets in such a media is given by (76).

### 4.1 Integrals of Motion

Equation (76) has at least two integrals of motion and they are the energy ( $E$ ) and the linear momentum ( $M$ ) that are respectively given by

$$E = \sum_{l=1}^N \int_{-\infty}^{\infty} |q_l|^2 dt \quad (4.77)$$

and

$$M = \frac{i}{2} D(z) \sum_{l=1}^N \int_{-\infty}^{\infty} \left( q_l^* \frac{\partial q_l}{\partial t} - q_l \frac{\partial q_l^*}{\partial t} \right) dt \quad (4.78)$$

The Hamiltonian ( $H$ ) given by

$$H = \frac{1}{2} \sum_{l=1}^N \int_{-\infty}^{\infty} \left\{ D(z) \left| \frac{\partial q_l}{\partial t} \right|^2 - g(z) \sum_{m \neq l}^N \alpha_{lm} |q_l|^2 |q_m|^2 \right\} dt \quad (4.79)$$

is, however, not a conserved quantity unless, in addition to  $D(z)$  and  $g(z)$  being constants, the matrix of XPM coefficients  $\Lambda = (\alpha_{ij})_{N \times N}$  is a symmetric matrix namely  $\alpha_{ij} = \alpha_{ji}$  for  $1 \leq i, j \leq N$ . Thus, for a birefringent fiber the matrix should be of the form

$$\Lambda = \begin{bmatrix} 0 & \alpha_{12} \\ \alpha_{12} & 0 \end{bmatrix} \quad (4.80)$$

while for a triple channeled fiber

$$\Lambda = \begin{bmatrix} 0 & \alpha_{12} & \alpha_{13} \\ \alpha_{12} & 0 & \alpha_{23} \\ \alpha_{13} & \alpha_{23} & 0 \end{bmatrix} \quad (4.81)$$

and so on. The existence of a Hamiltonian implies that (76) can be written as

$$i \frac{\partial q_l}{\partial z} = \frac{\delta H}{\delta q_l^*} \quad (4.82)$$

## 4.2 Asymptotic Analysis

The fields  $q_l$  are expanded in powers of  $z_a$  as

$$q_l(\zeta, Z, t) = q_l^{(0)}(\zeta, Z, t) + z_a q_l^{(1)}(\zeta, Z, t) + z_a^2 q_l^{(2)}(\zeta, Z, t) + \dots \quad (4.83)$$

Equating coefficients of like powers of  $z_a$  gives

$$O\left(\frac{1}{z_a}\right): \quad i \frac{\partial q_l^{(0)}}{\partial \zeta} + \frac{\Delta(\zeta)}{2} \frac{\partial^2 q_l^{(0)}}{\partial t^2} = 0 \quad (4.84)$$

$$\begin{aligned} O(1): \quad & i \frac{\partial q_l^{(1)}}{\partial \zeta} + \frac{\Delta(\zeta)}{2} \frac{\partial^2 q_l^{(1)}}{\partial t^2} \\ & + \left\{ i \frac{\partial q_l^{(0)}}{\partial Z} + \frac{\delta_a}{2} \frac{\partial^2 q_l^{(0)}}{\partial t^2} + g(z) \left( |q_l^{(0)}|^2 + \sum_{m \neq l}^N \alpha_{lm} |q_m^{(0)}|^2 \right) q_l^{(0)} \right\} = 0 \end{aligned} \quad (4.85)$$

$$\begin{aligned} O(z_a): \quad & i \frac{\partial q_l^{(2)}}{\partial \zeta} + \frac{\Delta(\zeta)}{2} \frac{\partial^2 q_l^{(2)}}{\partial t^2} \\ & + \left\{ i \frac{\partial q_l^{(1)}}{\partial Z} + \frac{\delta_a}{2} \frac{\partial^2 q_l^{(1)}}{\partial t^2} + g(z) \left[ 2 |q_l^{(1)}|^2 q_l^{(1)} + (q_l^{(1)})^2 q_l^{(1)*} \right. \right. \\ & \left. \left. + \sum_{m \neq l}^N \alpha_{lm} \left\{ 2 |q_m^{(1)}|^2 q_m^{(1)} + (q_m^{(1)})^2 q_m^{(1)*} \right\} \right] \right\} = 0 \end{aligned} \quad (4.86)$$

At  $O(1/z_a)$  equation (84), in the Fourier domain, is given by

$$i \frac{\partial \hat{q}_l^{(0)}}{\partial \zeta} - \frac{\omega^2}{2} \Delta(\zeta) \hat{q}_l^{(0)} = 0 \quad (4.87)$$

whose solution is

$$\hat{q}_l^{(0)}(\zeta, Z, \omega) = \hat{Q}_l^{(0)}(Z, \omega) e^{-\frac{i\omega^2}{2} C(\zeta)} \quad (4.88)$$

where

$$\hat{Q}_l^{(0)}(Z, \omega) = \hat{q}_l^{(0)}(0, Z, \omega) \quad (4.89)$$

At  $O(1)$ , equation (85) is solved in the Fourier domain by substituting the solution given by (88) into (85). This gives

$$\begin{aligned} i \frac{\partial \hat{q}_l^{(1)}}{\partial \zeta} - \frac{\omega^2}{2} \Delta(\zeta) \hat{q}_l^{(1)} \\ = -e^{-\frac{i\omega^2}{2} C(\zeta)} \left( \frac{\partial \hat{Q}_l^{(0)}}{\partial Z} - \frac{\omega^2}{2} \delta_a \hat{Q}_l^{(0)} \right) - g(\zeta) \int_{-\infty}^{\infty} \left( |q_l^{(0)}|^2 + \sum_{m \neq l}^N \alpha_{lm} |q_m^{(0)}|^2 \right) q_l^{(0)} e^{i\omega t} dt \end{aligned} \quad (4.90)$$

Equation (90) is an inhomogeneous equations for  $\hat{q}_l^{(1)}$  and with the homogenous parts having the same structure as in (84). For the non-secularity condition of  $\hat{q}_l^{(1)}$ , FA gives the condition on  $\hat{Q}_l^{(0)}(Z, \omega)$  as

$$\begin{aligned} \frac{\partial \hat{Q}_l^{(0)}}{\partial Z} - \frac{\omega^2}{2} \delta_a \hat{Q}_l^{(0)} \\ + \int_0^1 \int_{-\infty}^{\infty} e^{\frac{i\omega^2}{2} C(\zeta)} g(\zeta) \left( |q_l^{(0)}|^2 + \sum_{m \neq l}^N \alpha_{lm} |q_m^{(0)}|^2 \right) q_m^{(0)} e^{i\omega t} dt d\zeta = 0 \end{aligned} \quad (4.91)$$

Equation (91) can be simplified to

$$\begin{aligned} \frac{\partial \hat{Q}_l^{(0)}}{\partial Z} - \frac{\omega^2}{2} \delta_a \hat{Q}_l^{(0)} \\ + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} r_0(\omega_1 \omega_2) \hat{Q}_l^{(0)}(Z, \omega_1 + \omega_2) \left[ \hat{Q}_l^{(0)}(Z, \omega + \omega_1) \hat{Q}_l^{(0)}(Z, \omega + \omega_1 + \omega_2) \right. \\ \left. + \sum_{m \neq l}^N \alpha_{lm} \hat{Q}_m^{(0)}(Z, \omega + \omega_1) \hat{Q}_m^{(0)}(Z, \omega + \omega_1 + \omega_2) \right] d\omega_1 d\omega_2 = 0 \end{aligned} \quad (4.92)$$

Equation (92) is commonly known as GTE for the propagation of solitons through multiple channels [7]. Equation (85) will now be solved to obtain  $\hat{q}_l^{(1)}(\zeta, Z, t)$ . Substituting  $\hat{Q}_l^{(0)}$



into the right side of equation (90) and using (87) and (91) gives

$$\begin{aligned}
& \frac{\partial}{\partial \zeta} \left[ i\hat{q}_l^{(1)} e^{\frac{i\omega^2}{2}C(\zeta)} \right] \\
&= \int_0^1 \int_{-\infty}^{\infty} g(\zeta) e^{\frac{i\omega^2}{2}C(\zeta)} \left( |q_l^{(0)}|^2 + \sum_{m \neq l}^N \alpha_{lm} |q_m^{(0)}|^2 \right) q_l^{(0)} e^{i\omega t} dt d\zeta \\
&- g(\zeta) e^{\frac{i\omega^2}{2}C(\zeta)} \int_{-\infty}^{\infty} \left( |q_l^{(0)}|^2 + \sum_{m \neq l}^N \alpha_{lm} |q_m^{(0)}|^2 \right) q_l^{(0)} e^{i\omega t} dt \quad (4.93)
\end{aligned}$$

which integrates to

$$\begin{aligned}
i\hat{q}_l^{(1)} e^{\frac{i\omega^2}{2}C(\zeta)} &= \hat{Q}_l^{(1)}(Z, \omega) \\
&+ \zeta \int_0^1 \int_{-\infty}^{\infty} g(\zeta) e^{\frac{i\omega^2}{2}C(\zeta)} \left( |q_l^{(0)}|^2 + \sum_{m \neq l}^N \alpha_{lm} |q_m^{(0)}|^2 \right) q_l^{(0)} e^{i\omega t} dt d\zeta \\
&- \int_0^\zeta \int_{-\infty}^{\infty} g(\zeta') e^{\frac{i\omega^2}{2}C(\zeta')} \left( |q_l^{(0)}|^2 + \sum_{m \neq l}^N \alpha_{lm} |q_m^{(0)}|^2 \right) q_l^{(0)} e^{i\omega t} dt d\zeta' \quad (4.94)
\end{aligned}$$

where

$$\hat{Q}_l^{(1)}(Z, \omega) = i\hat{q}_l^{(1)}(0, Z, \omega) e^{\frac{i\omega^2}{2}C(0)} \quad (4.95)$$

Also,  $\hat{Q}_l^{(1)}(Z, \omega)$  is so chosen that

$$\int_0^1 i\hat{q}_l^{(1)} e^{\frac{i\omega^2}{2}C(\zeta)} d\zeta = 0 \quad (4.96)$$

which is going to be an useful relation for subsequent orders. Applying (96) to (94) gives

$$\begin{aligned}
\hat{Q}_l^{(1)}(Z, \omega) &= \\
&\int_0^1 \int_0^\zeta \int_{-\infty}^{\infty} g(\zeta') e^{\frac{i\omega^2}{2}C(\zeta')} \left( |q_l^{(0)}|^2 + \sum_{m \neq l}^N \alpha_{lm} |q_m^{(0)}|^2 \right) q_l^{(0)} e^{i\omega t} dt d\zeta' d\zeta \\
&- \frac{1}{2} \int_0^1 \int_{-\infty}^{\infty} g(\zeta) e^{\frac{i\omega^2}{2}C(\zeta)} \left( |q_l^{(0)}|^2 + \sum_{m \neq l}^N \alpha_{lm} |q_m^{(0)}|^2 \right) q_l^{(0)} e^{i\omega t} dt d\zeta \quad (4.97)
\end{aligned}$$

Now, (94), by virtue of (97), can be written as

$$\begin{aligned} \hat{q}_l^{(1)}(\zeta, Z, \omega) = & i e^{\frac{i\omega^2}{2}C(\zeta)} \left[ \int_0^\zeta \int_{-\infty}^\infty g(\zeta') e^{\frac{i\omega^2}{2}C(\zeta')} \left( |q_l^{(0)}|^2 + \sum_{m \neq l}^N \alpha_{lm} |q_m^{(0)}|^2 \right) q_l^{(0)} e^{i\omega t} dt d\zeta' \right. \\ & - \int_0^1 \int_0^\zeta \int_{-\infty}^\infty g(\zeta') e^{\frac{i\omega^2}{2}C(\zeta')} \left( |q_m^{(0)}|^2 + \sum_{m \neq l}^N \alpha_{lm} |q_m^{(0)}|^2 \right) q_l^{(0)} e^{i\omega t} dt d\zeta' d\zeta \\ & \left. - \left( \zeta - \frac{1}{2} \right) \int_0^1 \int_{-\infty}^\infty g(\zeta) e^{\frac{i\omega^2}{2}C(\zeta)} \left( |q_l^{(0)}|^2 + \sum_{m \neq l}^N \alpha_{lm} |q_m^{(0)}|^2 \right) q_l^{(0)} e^{i\omega t} dt d\zeta \right] \end{aligned} \quad (4.98)$$

which can be further rewritten as

$$\begin{aligned} \hat{q}_l^{(1)}(\zeta, Z, \omega) = & i e^{-\frac{i\omega^2}{2}C(\zeta)} \left[ \int_{-\infty}^\infty \int_{-\infty}^\infty \hat{Q}_l^{(0)*}(\omega + \Omega_1 + \Omega_2) \left\{ \hat{Q}_l^{(0)}(\omega + \Omega_1) \hat{Q}_l^{(0)}(\omega + \Omega_2) \right. \right. \\ & \left. \left. + \sum_{m \neq l}^N \alpha_{lm} \hat{Q}_l^{(0)}(\omega + \Omega_1) \hat{Q}_m^{(0)}(\omega + \Omega_2) \right\} \right. \\ & \left\{ \int_0^\zeta g(\zeta') e^{i\Omega_1 \Omega_2 C(\zeta')} d\zeta' - \int_0^1 \int_0^\zeta g(\zeta') e^{i\Omega_1 \Omega_2 C(\zeta')} d\zeta' d\zeta \right. \\ & \left. \left. - \left( \zeta - \frac{1}{2} \right) \int_0^1 g(\zeta) e^{i\Omega_1 \Omega_2 C(\zeta)} d\zeta \right\} d\Omega_1 d\Omega_2 \right] \end{aligned} \quad (4.99)$$

Thus, at  $O(z_a)$ ,

$$\hat{q}_l(\zeta, Z, \omega) = \hat{q}_l^{(0)}(\zeta, Z, \omega) + z_a \hat{q}_l^{(1)}(\zeta, Z, \omega) \quad (4.100)$$

Moving on to the next order at  $O(z_a^2)$ , one can note that the GTE given by (92) is allowed to have an additional term of  $O(z_a)$  as

$$\begin{aligned} \frac{\partial \hat{Q}_l^{(0)}}{\partial Z} - \frac{\omega^2}{2} \delta_a \hat{Q}_l^{(0)} & + \int_{-\infty}^\infty \int_{-\infty}^\infty r_0(\omega_1 \omega_2) \hat{Q}_l^{(0)}(Z, \omega_1 + \omega_2) \left[ \hat{Q}_l^{(0)}(Z, \omega + \omega_1) \hat{Q}_l^{(0)}(Z, \omega + \omega_1 + \omega_2) \right. \\ & \left. + \sum_{m \neq l}^N \alpha_{lm} \hat{Q}_m^{(0)}(Z, \omega + \omega_1) \hat{Q}_m^{(0)}(Z, \omega + \omega_1 + \omega_2) \right] d\omega_1 d\omega_2 = z_a \hat{n}_l(Z, \omega) + O(z_a^2) \end{aligned} \quad (4.101)$$

The higher order correction  $\hat{n}_l$  can be obtained from suitable a non-secular conditions at  $O(z_a^2)$  in (86). Now, equation (86), in the Fourier domain, is

$$\begin{aligned} & \frac{\partial}{\partial \zeta} \left[ i \hat{q}_l^{(2)} e^{\frac{i\omega^2}{2} C(\zeta)} \right] + \hat{n}_l + e^{\frac{i\omega^2}{2} C(\zeta)} \left( i \frac{\partial \hat{q}_l^{(1)}}{\partial Z} - \frac{\omega^2}{2} \delta_a \hat{q}_l^{(1)} \right) \\ & + e^{\frac{i\omega^2}{2} C(\zeta)} g(\zeta) \int_{-\infty}^{\infty} \left[ 2 \left| q_l^{(0)} \right|^2 q_l^{(1)} + \left( q_l^{(0)} \right)^2 q_l^{(1)*} \right. \\ & \left. + \sum_{m \neq l}^N \alpha_{lm} \left\{ 2 \left| q_m^{(0)} \right|^2 q_m^{(1)} + \left( q_m^{(0)} \right)^2 q_m^{(1)*} \right\} \right] e^{i\omega t} dt = 0 \end{aligned} \quad (4.102)$$

But, again (96) gives

$$\int_0^1 \hat{q}_l^{(1)} e^{\frac{i\omega^2}{2} C(\zeta)} d\zeta = 0 \quad (4.103)$$

Applying the non-secularity condition (103) to (102) gives

$$\begin{aligned} \hat{n}_l = & - \int_0^1 \int_{-\infty}^{\infty} e^{\frac{i\omega^2}{2} C(\zeta)} g(\zeta) \\ & \left[ 2 \left| q_l^{(0)} \right|^2 q_l^{(1)} + \left( q_l^{(0)} \right)^2 q_l^{(1)*} + \sum_{m \neq l}^N \alpha_{lm} \left\{ 2 \left| q_m^{(0)} \right|^2 q_m^{(1)} + \left( q_m^{(0)} \right)^2 q_m^{(1)*} \right\} \right] e^{i\omega t} dt d\zeta \end{aligned} \quad (4.104)$$

Using (88) and (98), equation (104) can be written as

$$\begin{aligned} \hat{n}_l = & \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} r_1(\omega_1 \omega_2, \Omega_1 \Omega_2) \\ & \left[ \left\{ 2 \hat{Q}_l^{(0)}(\omega + \omega_1) \hat{Q}_l^{(0)*}(\omega + \omega_1 + \omega_2) \hat{Q}_l^{(0)}(\omega + \omega_2 + \Omega_1) \right. \right. \\ & \hat{Q}_l^{(0)}(\omega + \omega_2 + \Omega_2) \hat{Q}_l^{(0)*}(\omega + \omega_2 + \Omega_1 + \Omega_2) \\ & - \hat{Q}_l^{(0)}(\omega + \omega_1) \hat{Q}_l^{(0)}(\omega + \omega_2) \hat{Q}_l^{(0)*}(\omega + \omega_1 + \omega_2 + \Omega_1) \\ & \left. \hat{Q}_l^{(0)*}(\omega + \omega_1 + \omega_2 - \Omega_2) \hat{Q}_l^{(0)*}(\omega + \omega_1 + \omega_2 + \Omega_1 - \Omega_2) \right\} \\ & + \sum_{m \neq l}^N \alpha_{lm} \left\{ 2 \hat{Q}_m^{(0)}(\omega + \omega_1) \hat{Q}_m^{(0)*}(\omega + \omega_1 + \omega_2) \hat{Q}_m^{(0)}(\omega + \omega_2 + \Omega_1) \right. \\ & \hat{Q}_m^{(0)}(\omega + \omega_2 + \Omega_2) \hat{Q}_m^{(0)*}(\omega + \omega_2 + \Omega_1 + \Omega_2) \\ & - \hat{Q}_m^{(0)}(\omega + \omega_1) \hat{Q}_m^{(0)}(\omega + \omega_2) \hat{Q}_m^{(0)*}(\omega + \omega_1 + \omega_2 + \Omega_1) \\ & \left. \left. \hat{Q}_m^{(0)*}(\omega + \omega_1 + \omega_2 - \Omega_2) \hat{Q}_m^{(0)*}(\omega + \omega_1 + \omega_2 + \Omega_1 - \Omega_2) \right\} \right] d\omega_1 d\omega_2 d\Omega_1 d\Omega_2 \end{aligned} \quad (4.105)$$

Equation (105) represents the HO-GTE for the propagation of solitons through multiple channels.

## 5 Properties of the Kernel

The HO-GTE, for different types of fibers, are the fundamental equations that govern the evolution of optical pulses for a strong dispersion-managed soliton systems corresponding to the frequency and time domain respectively. In these GT equations, all the fast variations and large quantities are removed and so they contain only slowly varying quantities of order one. These equations are not limited to the case  $\delta_a > 0$ , however, they are also applicable to the case of pulse dynamics with zero or normal values of average dispersion. If the fiber dispersion is constant namely if  $\Delta(\zeta) = 0$ , then  $C(\zeta) = 0$  and so  $r_0(x) = 1/(2\pi)^2$ . The kernels  $r_0(x)$  and  $r_1(x, y)$  are now going to be studied in the following two cases

### 5.1 Lossless Case

For the lossless case, namely when  $g(\zeta) = 1$ , the kernels  $r_0(x)$  and  $r_1(x, y)$  for a two-step map defined in (3) take very simple forms namely

$$r_0(x) = \frac{1}{(2\pi)^2} \frac{\sin(sx)}{sx} \quad (5.106)$$

$$r_1(x, y) = \frac{i(2\theta - 1)}{2s^3 x^2 y^2 (x + y)} \left[ sxy \{y \cos(sx) \sin(sy) - x \cos(sy) \sin(sx)\} + (x^2 - y^2) \sin(sx) \sin(sy) \right] \quad (5.107)$$

It can be seen that  $\theta$  appears in  $r_1$  but not in  $r_0$ . This means that the leading order of HO-GTE is independent of  $\theta$ . Equation (107) also shows that  $r_1(x, y)$  vanishes at  $\theta = 1/2$  so that the leading order GTE is valid for long distances  $O(1/z_a)$  if the positive and negative dispersion lengths of the fiber are the same. It can also be observed that

$$\lim_{s \rightarrow 0} r_0(x) = \frac{1}{(2\pi)^2} \quad (5.108)$$

$$\lim_{s \rightarrow 0} r_1(x, y) = 0 \quad (5.109)$$

This shows that the higher-order GTE reduces to the ideal NLSE as the map strength approaches zero.

### 5.2 Lossy Case

For the lossy case, namely when  $g(\zeta) \neq 1$ , the kernel  $r_0(x)$  depends on the relative position of the amplifier with respect to the dispersion map. The two step map given by  $\Delta(\zeta)$  in (3) is considered and  $\zeta_a$  is defined to represent the position of the amplifier within the dispersion map. So  $|\zeta_a| < 1/2$  and  $\zeta_a = 0$  means that the amplifier is placed at the mid point of the anomalous fiber segment. The function  $g(\zeta)$  given by (11) can then be written as

$$g(\zeta) = \frac{2Ge^{2G}}{\sinh(2G)} e^{-4G(\zeta - n\zeta_a)} \quad (5.110)$$

for  $\zeta_a + n \leq \zeta < \zeta_a + n + 1$  where  $G = \Gamma z_a/2$ . The kernel  $r_0(x)$  in the lossy case is computed in a similar method as in the lossless case. If  $|\zeta_a| < \theta/2$ , namely the amplifier is located in the anomalous fiber segment, the resulting expression for kernel is

$$r_0(x) = \frac{1}{(2\pi)^2} \frac{Ge^{iC_0x}}{(sx + 2iG\theta)(sx - 2iG(1 - \theta))} \left[ e^{G(4\zeta_a - 2\theta + 1)} sx \frac{\sin(sx - 2iG(1 - \theta))}{\sinh(2G)} + i\theta e^{i(4\zeta_a - 2\theta + 1)\frac{sx}{2\theta}} (sx - 2iG(1 - \theta)) \right] \quad (5.111)$$

In (111), unlike the lossless case, the kernel  $r_0(x)$  is complex and is explicitly dependent on the parameters  $\theta$ ,  $\Gamma$ ,  $z_a$  and  $\zeta_a$  in a nontrivial way. However, one still gets

$$\lim_{s \rightarrow 0} r_0(x) = \frac{1}{(2\pi)^2} \quad (5.112)$$

and moreover

$$\lim_{G \rightarrow 0} r_0(x) = \frac{1}{(2\pi)^2} \frac{\sin(sx)}{sx} \quad (5.113)$$

which means that as  $z_a \rightarrow 0$  (106) is recovered. For the particular case  $\theta = 1/2$ ,  $\zeta_a = 0$  which corresponds to fiber segments of equal length with amplifiers placed at the middle of the anomalous fiber segment, the kernel modifies to

$$r_0(x) = \frac{1}{(2\pi)^2} \frac{G}{x^2 s^2 + G^2} \left[ sx \frac{\sin(sx)}{\sinh(G)} + isx \left\{ 1 - \frac{\cos(sx)}{\cosh(G)} \right\} + G \right] \quad (5.114)$$

Also for  $g(\zeta) \neq 1$ , (75) gives

$$\lim_{s \rightarrow 0} r_1(x, y) = 0 \quad (5.115)$$

Thus, even in the lossy case, HO-GTE reduces to the case of ideal NLSE.

## 6 Conclusions

In this paper, the dynamics of vector optical solitons, propagating through optical fibers, with strong dispersion-management was studied. The birefringent fibers as well as multiple channels are considered. The technique that was used is the multiple-scale perturbation expansion. By using this technique the pulses in the Fourier domain was decomposed into a slowly evolving amplitude and a rapid phase that describes the chirp of the pulse. The fast phase is calculated explicitly that is driven by the large variations of the dispersion about the average. The amplitude evolution is described by the nonlocal evolution equations that is the HO-GTE.

These equations can be used to study the propagation of solitons with higher order accuracy, namely accuracy to  $O(z_a^2)$ . Also, the dynamics of quasilinear pulses [2, 3, 8], in optical fibers, can also be studied with greater accuracy. HO-GTE can also be used to study

the four-wave mixing, timing and amplitude jitter and ghost pulses [5] for optical fibers, with better estimates and further accuracy than that was already obtained before. Better yet, HO-GTE can be used to study the detailed asymptotic properties governing the long-scale dynamics of optical pulses. It needs to be noted here that the derivation of the HO-GTE is valid for any arbitrary dispersion map  $D(z)$  and with the general effects of damping and periodic amplification  $g(z)$ .

Although the HO-GTE is useful for studying the structure and properties, it is inconvenient for numerical computations because of the presence of the four-fold integrals that are given in the  $O(z_a)$  terms of the HO-GTE. In the case of polarization preserving fibers, there was some numerical analysis done with special solutions of the HO-GTE and bi-solitons, tri-solitons and quartic-solitons was observed [4, 16, 17].

In future, one can extend this study to include the perturbation terms, for example, filters, higher order dispersion, Raman scattering, self-steepening just to name a few. Also, it is possible to take a look at the GTE and HO-GTE in the context of other laws of nonlinearity like parabolic law, saturable law and others.

## Acknowledgment

This research was fully supported by NSF Grant No: HRD-970668 and the support is very thankfully appreciated.

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*Chapter 3*

## PARTITION OF UNITY FINITE ELEMENT METHOD IMPLEMENTATION FOR POISSON EQUATION

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### Abstract

Partition of Unity Finite Element Method (PUFEM) is a very powerful tool to deal overlapping grids. It is flexible and keeps the global continuity. In this paper, we consider PUFEM for Poisson equation for minimal overlapping grids. We present details of the implementation of Poisson equation in 2D for two overlapping domains using triangular meshes.

**Key Words:** Partition of Unity, FEM, Poisson Equation, Overlapping Grids

**AMS Subject Classification:** 43A60

## 1 Introduction

The main idea of overlapping grids is to divide a physical domain into overlapping regions which can accommodate smooth, simple, easily generated grids. Objects with complex geometry can be divided into simple domains and then meshes for each of them are generated separately. Refinement of the grids in one or part of domains can be done without affecting other domains. Furthermore, overlapping grids are suited to parallelization.

The study of finite element method applied to overlapping grids is done mainly in the framework of mortar method or Lagrange multiplier [5, 6, 8]. Partition of unity method [1] is used implicitly or explicitly to develop so-called generalized finite element methods.

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The study for the development of conforming finite element methods for overlapping and nonmatching grid can be found in [2]. Bacuta (et.al) study partition of unity method on nonmatching grids for the stokes problem [3]. Parallel partition of unity method is considered by Holst [4]. In this paper, we focus on the implementation aspects of PUFEM for Poisson equation and numerical consequences.

The rest of the paper is organized as following. In Section 2, we shall give a review of the standard finite element method for Poisson equation and various implementation aspects for it. In Section 3, we present the theoretic results of partition of unity method of Melenk and Babuska [1] and Huang and Xu's approach for overlapping and nonmatching grid [2]. In Section 4, we deal with the implementation of PUFEM for Poisson equation in two dimensions with minimal overlapping region. Numerical results are also presented. Finally in Section 5, we make some conclusions of the implementation and problems that might be interesting.

## 2 Standard FEM for Poisson Equation

The problem we deal with is the following Poisson equation

$$-\Delta u = f \quad \text{in } \Omega, \quad (2.1)$$

$$u = 0 \quad \text{on } \partial\Omega. \quad (2.2)$$

where  $\Omega = (0, 1) \times (0, 1)$ . Its weak formulation is to find  $u \in H_0^1(\Omega)$  such that

$$a(u, v) = (f, v) \quad \forall v \in H_0^1(\Omega)$$

where

$$a(u, v) = \int \nabla u \cdot \nabla v dx.$$

The corresponding finite element method formulation can be expressed as: Find  $u_h \in V_h$  such that

$$a(u_h, v_h) = (f, v_h) \quad \forall v_h \in V_h$$

where  $V_h$  is a finite dimension subspace of  $H_0^1(\Omega)$ . For elliptic problems of the second order, the error bounds can be found in [7].

In the implementation, we discretize the domain by uniform triangles. Figure 1 shows the generated mesh and numbering.

For overlapping grids, we would like to generate meshes on the overlapping domains separately. Figure 2 shows the meshes on two overlapping domains for the unit square and their combination.

**Remark 2.1.** *In the implementation of the PUFEM, we need to know if a triangle is in the overlapping region or not. It is not efficient to loop through all the triangles. When we generate the mesh in each subdomain, we put the triangles in the overlapping region together sequentially.*

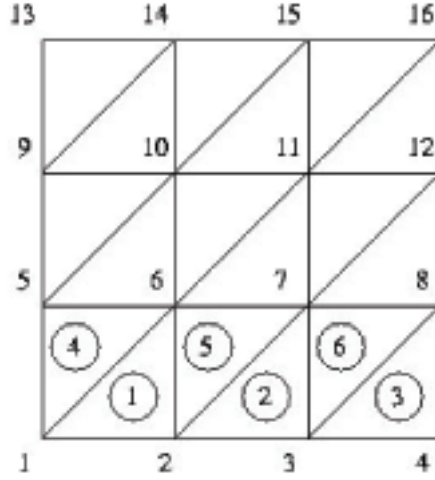


Figure 1: Mesh generated for the unit square.

We use the standard finite element method to solve the following model problem for the purpose of later comparison.

$$-\Delta u = 2(x + y) - 2(x^2 + y^2) \quad \text{in } \Omega \quad (2.3)$$

$$u = 0 \quad \text{on } \partial\Omega \quad (2.4)$$

where  $\Omega = [0, 1] \times [0, 1]$ . The exact solution is

$$u(x, y) = xy(1 - x)(1 - y).$$

Table 1 is the  $L_\infty$  norm of the error  $u - u_h$  for different values of  $h$ . Actually it is the maximum of all values evaluated at grid points.

 Table 1: Numerical results of FEM for Poisson Equation ( $u = xy(1 - x)(1 - y)$ ).

| Element Diameter | $L_\infty$ Error      |
|------------------|-----------------------|
| 1/10             | 4.873229035610716e-04 |
| 1/20             | 1.225445153886506e-04 |
| 1/40             | 3.068129255456853e-05 |
| 1/80             | 7.673154667112159e-06 |
| 1/160            | 1.918465776740153e-06 |

Other than the  $L_\infty$  norm, we may also look at the  $H^1$  semi-norm. Let  $u$  be the solution of the continuous problem and  $u_h$  be the solution of the discrete problem, then

$$\begin{aligned}
 a(u - u_h, u - u_h) &= a(u, u - u_h) - a(u_h, u - u_h) \\
 &= a(u, u) - a(u, u_h) \\
 &= a(u, u) - (f, u_h).
 \end{aligned}$$

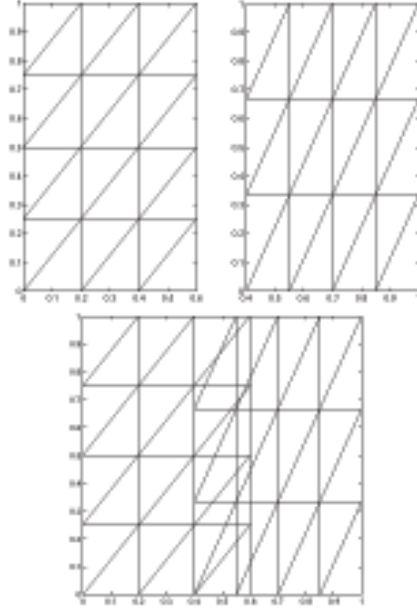


Figure 2: Two overlapping meshes and their combination.

where we have already used the fact that  $a(u_h, u - u_h) = 0$ . Hence if we know  $u_h$  and the exact solution  $u$ , we can use above formula to calculate the  $H^1$  semi-norm. For our first model problem,  $u = xy(1 - x)(1 - y)$ , we have  $a(u, u) = \frac{1}{45}$ . We use a accurate quadrature formula to evaluate  $(f, u_h)$ . Table 2 shows the  $H^1$  semi-norm of  $u - u_h$  for the model problem.

Table 2: Numerical results of FEM for Poisson equation ( $u = xy(1 - x)(1 - y)$ ). The first column is the element size. The second column is the  $H^1$  semi-norm of  $u - u_h$ .

| <i>Element Size</i> | $H^1$ Semi-norm of the Error. |
|---------------------|-------------------------------|
| 1/10                | 0.02420557358585              |
| 1/20                | 0.01215431899870              |
| 1/40                | 0.00608364175173              |
| 1/80                | 0.00304263245469              |
| 1/160               | 0.00152141771552              |

### 3 PUFEM for Overlapping Grids

#### 3.1 Basic Theory of PUFEM

We present the preliminaries of the PUFEM which can be found in detail in [1].

**Definition 3.1.** Let  $\Omega \in R^n$  be an open set,  $\{\Omega_i\}$  be an open cover of  $\Omega$  satisfying a pointwise overlap condition:

$$\exists M \in N \quad \forall x \in \Omega \quad \text{card}\{i | x \in \Omega_i\} \leq M.$$

Let  $\{\phi_i\}$  be a Lipschitz partition of unity subordinate to the cover  $\{\Omega_i\}$  satisfying

$$\begin{aligned} \text{supp}\phi_i &\subset \text{closure}(\Omega_i) \quad \forall i, \\ \sum_i \phi_i &\equiv 1 \quad \text{on } \Omega, \\ \|\phi_i\|_{L^\infty(R^n)} &\leq C_\infty, \\ \|\nabla\phi_i\|_{L^\infty(R^n)} &\leq \frac{C_G}{\text{diam}\Omega_i}, \end{aligned}$$

where  $C_\infty, C_G$  are two constants. Then  $\{\phi_i\}$  is called a  $(M, C_\infty, C_G)$  partition of unity subordinate to the cover  $\{\Omega_i\}$ . The partition of unity  $\{\phi_i\}$  is said to be of degree  $m \in N_0$  if  $\{\phi_i\} \subset C^m(R^n)$ . The covering sets  $\{\Omega_i\}$  are called patches.

**Definition 3.2.** Let  $\{\Omega_i\}$  be an open cover of  $\Omega \subset R^n$  and let  $\{\phi_i\}$  be a  $(M, C_\infty, C_G)$  partition of unity subordinate to  $\{\Omega_i\}$ . Let  $V_i \subset H^1(\Omega_i \cap \Omega)$  be given. Then the space

$$V := \sum_i \phi_i V_i = \left\{ \sum_i \phi_i v_i | v_i \in V_i \right\} \subset H^1(\Omega)$$

is called the PUFEM space. The PUFEM space  $V$  is said to be of degree  $m$  if  $V \subset C^m(\Omega)$ . The space  $V_i$  are referred to as the local approximation spaces.

**Theorem 3.3.** Let  $\Omega \in R^n$  be given. Let  $\{\phi_i\}, \{\Omega_i\}$  and  $\{V_i\}$  be as in the definitions above. Let  $u \in H^1(\Omega)$  be the function to be approximated. Assume that the local approximation spaces  $V_i$  have the following approximation properties: On each patch  $\Omega_i \cap \Omega$ ,  $u$  can be approximated by a function  $v_i \in V_i$  such that

$$\begin{aligned} \|u - v_i\|_{L^2(\Omega_i \cap \Omega)} &\leq \epsilon_1(i), \\ \|\nabla(u - v_i)\|_{L^2(\Omega_i \cap \Omega)} &\leq \epsilon_2(i). \end{aligned}$$

Then the function

$$u_{ap} = \sum_i \phi_i v_i \in V \subset H^1(\Omega)$$

satisfies

$$\begin{aligned} \|u - u_{ap}\|_{L^2(\Omega)} &\leq \sqrt{M} C_\infty \left( \sum_i \epsilon_1^2(i) \right)^{1/2}, \\ \|\nabla(u - u_{ap})\|_{L^2(\Omega)} &\leq \sqrt{2M} \left( \sum_i \left( \frac{C_G}{\text{diam}\Omega_i} \right)^2 \epsilon_1^2(i) + C_\infty^2 \epsilon_2^2(i) \right)^{1/2}. \end{aligned}$$

### 3.2 Huang and Xu's Approach

Let each  $\Omega_i$  is partitioned by quasi-uniform triangulation  $\tau^{h_i}$  of maximal mesh size  $h_i$ . With each triangulation  $\tau^{h_i}$ , associate a finite element spaces  $V_i \subset H^r(\Omega_i)$ . Let  $u \in H^r(\Omega)$ , and let  $m_i \geq 1$  denote the additional degree of smoothness of  $u$  on  $\Omega_i$ . Assume optimal approximation property on subdomains:

For any  $u \in H^{m_i+r}(\Omega_i)$ , there exist a  $v_h \in V_i$  such that

$$\sum_{k=0}^r h_i^k |u - v_h|_{k,\Omega_i} \leq ch_i^{m_i+r} \|u\|_{m_i+r,\Omega_i}.$$

Also assume that

$$|\nabla^k \phi_i| \leq cd_i^{-k}, 1 \leq k \leq r$$

where  $d_i$  is the minimal overlapping size of  $\Omega_i$  with its neighboring subdomains.

**Theorem 3.4.** (Huang and Xu) *If the overlapping size  $d_i \geq ch_i$ , then for  $0 \leq k \leq r$ ,*

$$\inf_{v_h \in V} \|u - v_h\|_{k,\Omega} \leq C \sum_{i=1}^p h_i^{m_i+r-k} \|u\|_{m_i+r,\Omega_i},$$

for any  $u \in H^r(\Omega) \cap \bigcap_{i=1}^p H^{m_i+r}(\Omega_i)$ .

For  $u \in H^2(\Omega)$  and  $H^1$  conforming finite element space, we have

$$\begin{aligned} \inf_{v_h \in V} \|u - v_h\|_{0,\Omega} &\leq C \sum_{i=1}^p h_i^2 \|u\|_{2,\Omega_i}, \\ \inf_{v_h \in V} \|u - v_h\|_{1,\Omega} &\leq C \sum_{i=1}^p h_i^1 \|u\|_{2,\Omega_i}, \end{aligned}$$

where we set  $k = 0$  and  $k = 1$  in the above theorem and  $m_i = 1, r = 1$ .

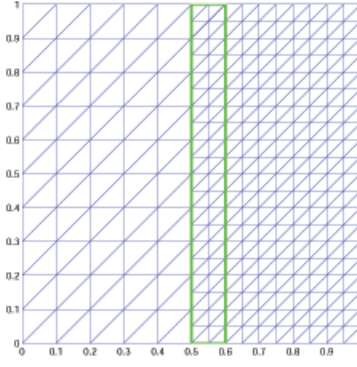
### 3.3 Two Overlapping Domains

We consider a simple case for overlapping grids. Let  $\Omega = (0, 1) \times (0, 1)$ ,  $\Omega_1 = (0, 0.6) \times (0, 1)$  and  $\Omega_2 = (0.5, 1) \times (0, 1)$ . The overlapping region is  $\Omega_o = (0.5, 0.6) \times (0, 1)$ . We generate meshes on  $\Omega_1$  and  $\Omega_2$  and end up with an overlapping mesh on  $\Omega$  (Figure 3). Notice that the meshes in  $\Omega_o$  overlap. To be precise,  $\{\Omega_1, \Omega_2\}$  is an open cover of  $\Omega$  satisfying a point wise overlap condition (See [1]). Let

$$\phi_1 = \begin{cases} 1 & 0 \leq x \leq 0.5, & 0 \leq y \leq 1, \\ \frac{0.6-x}{0.6-0.5} & 0.5 < x < 0.6, & 0 \leq y \leq 1, \\ 0 & 0.6 \leq x \leq 1, & 0 \leq y \leq 1, \end{cases} \quad (3.1)$$

and

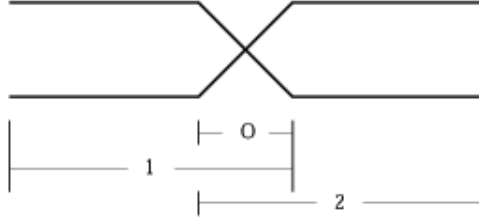
$$\phi_2 = \begin{cases} 0 & 0 \leq x \leq 0.5, & 0 \leq y \leq 1, \\ \frac{x-0.5}{0.6-0.5} & 0.5 < x < 0.6, & 0 \leq y \leq 1, \\ 1 & 0.6 \leq x \leq 1, & 0 \leq y \leq 1. \end{cases} \quad (3.2)$$


 Figure 3: Overlapping Meshes in  $\Omega$ .

Then  $\{\phi_1, \phi_2\}$  is a Lipschitz partition of unity subordinate to the cover  $\{\Omega_1, \Omega_2\}$ , i.e.,

$$\phi_1 + \phi_2 = 1, \quad 0 \leq \phi_1, \phi_2 \leq 1, \quad \|\nabla \phi_i\|_{L^\infty} \leq 1/d$$

and  $\phi_i \equiv 1, i = 1, 2$  on  $\Omega_i \setminus \Omega_o$ , and  $\phi_i \equiv 0$  on  $\Omega_j, j \neq i$ . In the numerical tests, the mesh


 Figure 4: Partition of unity functions  $\phi_1$  and  $\phi_2$  in  $x$ -direction.

in each domain will be refined and the width of the overlapping region also decreases. We set  $\Omega_1 = (0.5, 0.5 + h_1) \times (0, 1)$ , where  $h_1$  is the width of one element of the mesh for  $\Omega_1$ , and  $\Omega_2 = (0.5, 1) \times (0, 1)$ . Hence the width of the overlapping region  $\Omega_o$  is  $h_1$ .

Let  $V_1$  and  $V_2$  be the local approximation spaces corresponding to  $\Omega_1$  and  $\Omega_2$ . Then the PUFEM space is given by

$$V := \sum_{i=1,2} \phi_i V_i = \left\{ \sum_{i=1,2} \phi_i v_i \mid v_i \in V_i \right\} \subset H^1(\Omega).$$

Thus  $V$  is a conforming subspaces of  $H^1(\Omega)$ . The discrete problem is to find  $u_h$  of the form

$$u_h = \phi_1 v_i + \phi_2 w_j$$

where  $v_i \in V_1$  and  $w_j \in V_2$ , such that

$$a(u_h, v_h) = (f, v_h) \quad \forall \quad v_h \in V.$$

To numerically solve the above discrete problem we need to find a basis for  $V$ . The next theorem is of crucial importance in our paper.

**Theorem 3.5.** *Suppose we have the regular triangulation meshes for  $\Omega_1$  and  $\Omega_2$ , which are defined as before and the overlapping region is a strip-type rectangle domain. Suppose that  $V_k, k = 1, 2$  are linear finite element spaces with base functions  $\{v_i\}_{i \in I}$  and  $\{w_j\}_{j \in J}$ . Then the set*

$$\{\phi_1 v_i, \phi_2 w_j\}_{i \in I, j \in J}$$

*forms a basis for the PUFEM space  $V$ .*

*Proof.* Since  $\phi_1 \equiv 1, \phi_2 \equiv 0$  in domain  $\Omega_1 \setminus \Omega_o$  and  $\phi_2 \equiv 1, \phi_1 \equiv 0$  in domain  $\Omega_2 \setminus \Omega_o$ , the linear independence is obvious. We will show the linear independence in the overlapping region.

Suppose

$$\phi_1 \sum_i \alpha_i v_i + \phi_2 \sum_j \beta_j w_j = 0.$$

Write  $\phi_2 = 1 - \phi_1$  to obtain

$$\phi_1 \left( \sum_i \alpha_i v_i - \sum_j \beta_j w_j \right) = - \sum_j \beta_j w_j$$

where the right hand side is piece-wise linear at most. Since  $\phi_1$  is linear, we can conclude that

$$\sum_i \alpha_i v_i - \sum_j \beta_j w_j$$

is constant in  $\Omega_o$ . Using the zero boundary condition, the above expression must be zero. Then we have  $-\sum_j \beta_j w_j = 0$  and  $\beta_j = 0$  for all  $j$  because of the linear independence of  $w_j$ 's. Thus  $\sum_i \alpha_i v_i = 0$  and we conclude that  $\alpha_i = 0$  for all  $i$  because of the linear independence of  $v_i$ 's. Hence  $\{\phi_1 v_i, \phi_2 w_j\}_{i \in I, j \in J}$  are linearly independent and forms a basis for  $V$ . □

**Remark 3.6.** *To show the linear independence of the base functions, we use the zero boundary condition. If  $\phi_1$  and  $\phi_2$  are not linear in the overlapping region, the linear independence can be shown without the application of the boundary condition.*

## 4 Implementation and Numerical Results

### 4.1 Meshing

We generate the mesh on  $\Omega_1$  and  $\Omega_2$  and then combine the meshes together to obtain the mesh on  $\Omega$ . Since  $\Omega_1$  and  $\Omega_2$  overlap, we need to be deliberate when we generate meshes. To simplify coding, we make the following assumptions:

1. The boundary of  $\Omega_o$  aligns with the two meshes on  $\Omega_1$  and  $\Omega_2$ .

2. In the overlapping region, a triangle in the finer mesh (smaller in size) is contained entirely in a triangle in the coarser mesh. In other words, it can be thought as the refinement of the coarser mesh in the overlapping region.

## 4.2 Stiffness Matrix

As usual, we would like to setup the local stiffness matrix for every element  $e$  and then assemble the global stiffness matrix. If the element  $e$  is inside the non-overlapping regions,  $\Omega_1 \setminus \Omega_o$  or  $\Omega_2 \setminus \Omega_o$ , the setup of the local stiffness matrix is as usual. But if the element  $e$  is inside the overlapping region, we have two cases.

**Case 1:** Suppose  $e$  is a triangle in the fine mesh, then  $e$  is contained in a triangle  $e'$  in the coarse mesh (see Figure 5). We will have two kinds of entries in the local stiffness matrix  $K_e$ . The first kind is due to the  $e - e$  connection:

$$a(\phi_2\psi_i, \phi_2\psi_j)|_e, \quad i, j = 1, 2, 3$$

where  $\psi_i$  and  $\psi_j$  are nodal base functions on  $e$ . The second kind is due to the  $e - e'$  connection:

$$a(\phi_2\psi_i, \phi_1\varphi_j)|_e, \quad i, j = 1, 2, 3$$

where  $\phi_i, i = 1, 2, 3$  are nodal base functions on  $e$  and  $\varphi_j, j = 1, 2, 3$  are nodal base functions on  $e'$ . Hence in this case, the local stiffness matrix has 18 entries.

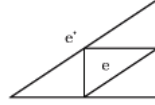


Figure 5: The figure shows that the relations of the triangles in the overlapping region.  $e'$  is the large triangle in the coarse mesh in  $\Omega_1$ , which contains four small triangles in the fine mesh of  $\Omega_2$ .  $e$  is one of the small triangles in  $e'$ .

**Case 2:** Suppose  $e'$  is a triangle in the coarse mesh, then it contains a couple of fine triangles, denoted by  $e_k, k = 1, \dots, N_{e'}$ . The local stiffness matrix  $K_e$  also contains two kinds of entries. The first kind is due to the  $e' - e'$  connection:

$$a(\phi_1\varphi_i, \phi_1\varphi_j)|_{e'}, \quad i, j = 1, 2, 3.$$

where  $\varphi_i$  and  $\varphi_j$  are nodal base functions on  $e'$ . The second kind is due to the  $e' - e_k, k = 1, \dots, N_{e'}$  connection:

$$a(\phi_1\varphi_i, \phi_2\psi_j)|_{e'}, \quad i, j = 1, 2, 3$$

where  $\varphi_i, i = 1, 2, 3$  are nodal base functions on  $e'$  and  $\psi_j, j = 1, 2, 3$  are nodal base functions on  $e_k$ . Hence in this case, the local stiffness matrix has  $9 + 9 \times N_{e'}$  entries.

Note that by the symmetry of the bilinear form  $a(\cdot, \cdot)$ , we need to compute either case 1 or case 2. In our implementation, we compute case 1.

**Remark 4.1.** The setup of the right hand side of the linear system is similar to the above situation for the stiffness matrix.



### 4.3 Numerical Result

Let  $\Omega$  be the unit square and  $\Omega_1 = (0, 0.6) \times [0, 1]$  and  $\Omega_2 = (0.5, 1) \times (0, 1)$ . The mesh is exactly like that in Figure 3. The solution of the model problem by PUFEM is plotted (Figure 6). Since in the overlapping region, the value of the function is given by

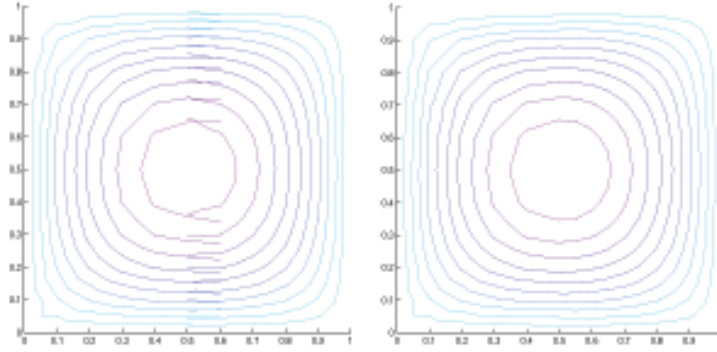


Figure 6: The figure on the left hand side is the solution of the problem. The figure on the right hand side is the contour plot of solution after adjustment. On  $\Omega_1$ ,  $h = 1/10$ . On  $\Omega_2$ ,  $h = 1/20$ .

$$\alpha_i \phi_1 v_i + \beta_j \phi_2 w_j,$$

we need to combine the value in the overlapping region to get the solution. We can see the solution is smoother in the right part of the plot than that in the left part of the plot where the mesh are coarser.

We decrease the mesh size in both  $\Omega_1$  and  $\Omega_2$ . We also decrease the width of the overlapping region. Hence we will have  $\Omega_1 = (0, 0.5 + h_1) \times (0, 1)$  where  $h_1$  is the width of one grid in  $\Omega$ . We can get a series of the results and calculate the  $H^1$  semi-norm as in the standard FEM. The  $H^1$  semi-norm of the error is evaluated (Table 3).

Table 3: Numerical results of PUFEM for Poisson equation ( $u = xy(1 - x)(1 - y)$ ). The first column is the element size in  $\Omega_1$  and the second is the element size in  $\Omega_2$ . The third column is the  $H^1$  semi-norm of  $u - u_h$ .

| Element Size in $\Omega_1$ | Element Size in $\Omega_2$ | $H^1$ Semi-norm of the Error |
|----------------------------|----------------------------|------------------------------|
| 1/10                       | 1/20                       | 0.01821113459601             |
| 1/20                       | 1/40                       | 0.00865699798935             |
| 1/40                       | 1/80                       | 0.00380050384524             |
| 1/80                       | 1/160                      | 0.00120187622777             |

Compare with Table 2, the results verifies the error estimate.

We also look at the condition number of the stiffness matrix. We first calculate the condition number of the stiffness matrix for the standard FEM (Table 4). We see that the condition number is of  $O(h^{-2})$ .

Table 4: Condition number of the stiffness matrix for standard finite element method.

| Element Size in $\Omega$ | Condition number of the stiffness matrix. |
|--------------------------|-------------------------------------------|
| 1/10                     | 58.4787                                   |
| 1/20                     | 235.2855                                  |
| 1/40                     | 942.5293                                  |
| 1/80                     | 3.7715e+03                                |

The condition number of the stiffness matrix for PUFEM is given in Table 5. We see that we end up with much larger condition numbers. But fortunately, it is still of  $O(h^{-2})$ .

Table 5: Condition number of the stiffness matrix for PUFEM.

| $h_1$ in $\Omega_1$ | $h_2$ in $\Omega_2$ | Condition number of the stiffness matrix. |
|---------------------|---------------------|-------------------------------------------|
| 1/10                | 1/20                | 4.2017e+03                                |
| 1/20                | 1/40                | 1.6950e+04                                |
| 1/40                | 1/80                | 6.8245e+04                                |
| 1/80                | 1/160               | 2.7403e+05                                |

## 5 Conclusions and Future Work

In this paper, we implement PUFEM for Poisson equation in two dimensions. Numerical results verify the error estimates for PUFEM. We also show the condition number of the stiffness matrix for PUFEM is of  $O(h^{-2})$  numerically.

We are considering to implement smooth partition of unity functions in the overlapping regions and study how it affects the stiffness matrix. Implementation of PUFEM for other equations, such as Helmholtz's Equation, on a more complex geometry is definitely more interesting.

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*Chapter 4*

**INVESTIGATION OF THE HETEROGENEOUS  
PROBLEMS OF THE ELASTICITY WITH COUPLED  
BOUNDARY FINITE ELEMENT SCHEMES**

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**Abstract**

This paper presents a numerical approach of elasticity problems for compound structures. It is based on a combination of the linear elasticity theory and Timoshenko's shells theory for elasticity problems. Subdomains of the compound structure, which are described by different theories, are joined by special interface boundary conditions. Variational statements are formulated and their properties are investigated. A numerical solution of the problem is performed by coupled direct boundary element and finite element methods, and domain decomposition method. Corresponding discretizations and asymptotic error estimates are presented. The obtained results of numerical simulation demonstrate the effectiveness of the proposed techniques.

**Key Words:** Heterogeneous models, compound structures, finite element method, boundary element method  
**AMS Subject Classification:** 65N30, 65N38, 65N55.

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## 1 Introduction

Coupled boundary and finite element approximations is used within the framework of an algorithm that utilizes the advantages of both boundary and finite elements approximations. This approach became a powerful tool for solving various scientific and engineering problems. In early 1980s, there existed prevailing schemes that led to the formation of a mutual matrix of system of linear algebraic equations. Recently, domain decomposition method became most wide spread technique of coupling both methods. Theoretical bases of such approaches, are formulated in papers [3] and [4]. The aim of this work is to use Timoshenko's shell elements for thin walled of domain and three-dimensional equations of the theory of elasticity for the subdomain where the shell theory no longer applies. To achieve this goal, the advantages of boundary element method (BEM) and finite element method (FEM) are exploited. We propose a numerical analysis approach of compound structures that is based on heterogeneous mathematical and numerical models. This approach utilizes the equations of the elasticity theory, the equations of the Timoshenko shell theory, and numerical solution of the model by coupling BEM and FEM. Boundary-value formulation and variational formulation for the considered heterogeneous mathematical model are proposed. Properties of the operators of the considered heterogeneous mathematical model are investigated. The existence and uniqueness of weak solutions of the heterogeneous model are proved. Several approaches to the analysis of compound structures are known in the literature. Among them we emphasize the approach developed in [2]. It consists of modelling the effect of the thin walled structure elements with special generalized boundary conditions that leads to a nonclassical boundary-value problem. The numerical solution of the obtained nonclassical boundary value problem encounters significant difficulties. A new class of mathematical models called D-adaptive mathematical models was proposed in [6] and [12]. These models are described by the equations of the elasticity theory and the equations of asymptotic shell theory that are joined by special nonclassical boundary conditions. The numerical analysis of compound structures on the basis of such approach also encounters some difficulties. They are caused by the presence of high order derivatives in the prescribed equations. In this paper, as in [5] and [8]-[11], we consider the numerical analysis of compound structures that is based on heterogeneous mathematical models. The stress-strain state of the massive part of the structure is described by the equations of the elasticity theory. The stress-strain state of the shell part of the structure is described by the equations of the Timoshenko's shell theory. In order to obtain numerical solutions of the variational problems corresponding to heterogeneous models we employ coupled Galerkin's direct boundary element for elasticity problem and finite element method for the shell theory problem as in [10] and [11], and a domain decomposition method. The method proposed in [1] and [4] is used to prove the existence and uniqueness of the heterogeneous numerical solution of the heterogeneous mathematical model.

## 2 Statement of the Heterogeneous Model

Let us consider a domain  $\Omega = \Omega_1 \cup \Omega_2^*$ ,  $\bar{\Omega}_1 \cap \bar{\Omega}_2^* = \emptyset$ , (Fig.1) with Lipschitz continuous boundary  $\partial\Omega$ , where  $\Omega_1$  and  $\Omega_2^*$  are arbitrary connected domains of  $\mathbf{R}^3$ . Assume that the domain  $\Omega_1$  is bounded by Lipschitz boundary  $\Gamma_1$ , and the domain  $\Omega_2^*$  is bounded by two

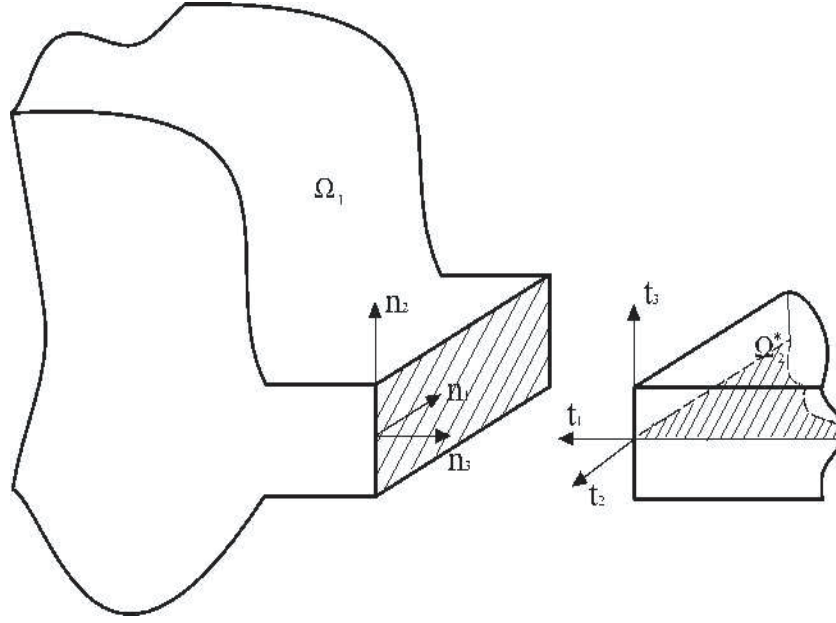


Figure 1: An elastic body with thin shell.

parallel planes with distance  $h$  between them and a cylindrical lateral surface, perpendicular to these planes. Let's assume, that the size  $h$  of the domain  $\Omega_2^*$  considerably smaller, than other characteristic sizes of this domain. Between parallel planes on identical distance from them, the placed middle plane  $\Omega_2$ ,  $\Gamma_2 = \Gamma_2^{(1)} \cup \Gamma_2^{(2)} \cup \Gamma_2^{(3)}$  and  $\Gamma_2^{(1)} \cap \Gamma_2^{(2)} \cap \Gamma_2^{(3)} = \emptyset$  where  $\Gamma_2^{(1)}, \Gamma_2^{(2)}, \Gamma_2^{(3)}$  - are piecewise smooth curves. Let's  $x_1, x_2, x_3$  be the Cartesian coordinates system of the elastic body  $\Omega_1$ , and we shall define  $\bar{n}_1, \bar{n}_2, \bar{n}_3$  - are an orthogonal right triple of unit-length vectors on  $\Gamma_1$ , where  $\bar{n}_3$  is an outer normal to boundary  $\Gamma_1$ . On a middle plane  $a_1, a_2, a_3$  the Cartesian system of coordinates, a direction  $a_3$  coincides with a normal to a middle plane. In points of border of a middle plane we shall define  $\bar{t}_1, \bar{t}_2$  a pair of orthogonal unit vectors, where  $\bar{t}_1$  - an outer normal to the boundary,  $\bar{t}_2$  - the tangent vector corresponding to a positive direction of  $\Omega_2$  along curve  $\Gamma_2$ . We shall consider, that the part of boundary  $\Gamma_1^{(3)}$  is common for domains  $\Omega_1$  and  $\Omega_2^*$  and  $\Gamma_1^{(3)} = \Gamma_2^{(3)} \times [-\frac{h}{2}, \frac{h}{2}]$  are also carried out  $\bar{t}_1 = -\bar{n}_3, \bar{t}_2 = -\bar{n}_1$ .

The stress-strain state of the elastic body in the domain  $\Omega_1$  is described by the equations of the linear elasticity theory [13] (summation over repeated indices)

$$\frac{\partial \sigma_{ij}}{\partial x_j} + f_i = 0, \quad i, j = 1, 2, 3, \quad x \in \Omega \subset \mathbf{R}^3, \quad (2.1)$$

the stress-strain state of the domain  $\Omega_2^*$  is described by the equations of the Timoshenko's shells theory [13]

$$\begin{aligned}
\frac{\partial T_{kl}}{\partial \alpha_l} + p_k &= 0, \\
\frac{\partial Q_k}{\partial \alpha_k} + p_3 &= 0, \\
\frac{\partial M_{kl}}{\partial \alpha_l} - Q_k + m_k &= 0, \quad k, l = 1, 2, \quad x \in \Omega_2 \subset \mathbf{R}^2,
\end{aligned} \tag{2.2}$$

where  $\sigma_{ij}$  are the components of the stress tensor;  $f_i$  are the components of the body force vector;  $T_{kl}$ ,  $M_{kl}$ ,  $Q_k$  are the normal forces, the shear forces, and the bending moments, respectively;  $p_i$ ,  $m_k$  are the components of surface loads reduced to the middle plane  $\Omega_2$ . It is known [13], that:

$$\begin{aligned}
p_i &= \sigma_{i3}^+ + \sigma_{i3}^- + \rho_i, \quad i = 1, 2, 3; \\
m_k &= 0.5h (\sigma_{k3}^+ - \sigma_{k3}^-) + \mu_k, \quad k = 1, 2;
\end{aligned}$$

where  $\sigma_{i3}^+$ ,  $\sigma_{i3}^-$  are the normal components of the tractions on the top and bottom surfaces  $\alpha_3 = \pm \frac{h}{2}$  of the shell;  $\rho_i$ ,  $\mu_k$  are the components of body forces and moments reduced to the middle plane

$$\rho_i = \int_{-h/2}^{h/2} q_i d\alpha_3, \quad \mu_k = \int_{-h/2}^{h/2} q_k \alpha_3 d\alpha_3;$$

and  $q_i$  are the components of the body force vector in the middle plane. The components of the stress tensor  $\sigma_{ij}$  are given by

$$\sigma_{ij} = C_{ijkl} \varepsilon_{kl}, \quad i, j, k, l = 1, 2, 3; \tag{2.3}$$

$C_{ijkl}$  are the elasticity constants, which in the case of isotropic domain are given by

$$\begin{aligned}
C_{iiii} &= \frac{E(1-\nu)}{(1+\nu)(1-2\nu)}, \quad C_{iikk} = \frac{E\nu}{(1+\nu)(1-2\nu)}, \\
C_{ikik} &= \frac{E}{2(1+\nu)}, \quad i, k = 1, 2, 3; \quad i \neq k;
\end{aligned}$$

where  $E$  is Young's modulus and  $\nu$  is Poisson's ratio of the elastic body.

The forces and moments in  $\Omega_2$  in terms of deformations  $\varepsilon_{kl}$ ,  $\kappa_{kl}$  by means of physical law of Timoshenko's theory are

$$\begin{aligned}
T_{kk} &= B(\varepsilon_{kk} + \nu\varepsilon_{ll}), \quad T_{kl} = B\frac{1-\nu}{2}\varepsilon_{kl}, \quad Q_k = G\varepsilon_{k3}, \\
M_{kk} &= D(\kappa_{kk} + \nu\kappa_{ll}), \quad M_{kl} = D\frac{1-\nu}{2}\kappa_{kl}, \quad k, l = 1, 2; \quad k \neq l.
\end{aligned} \tag{2.4}$$

where the constants  $B, D, G$  in the case of isotropic material are given by

$$B = \frac{Eh}{1-\nu^2}, \quad D = \frac{Eh^3}{12(1-\nu)}, \quad G = \frac{5Eh(1+\nu)}{12}.$$

Let  $u_i (i = 1, 2, 3)$  be the components of a vector of displacements of the elastic body in  $x_1, x_2, x_3$ —system of coordinates,  $v_i$  be the displacements of points of the middle plane in the direction of axes  $\alpha_i$ , and  $\gamma_l (l = 1, 2)$  are the angles of rotation of the normal to the middle surface in a direction of axes  $\alpha_l$ .

The following Cauchy relations hold

$$e_{ij} = \frac{1}{2} \left( \frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right), \quad i, j = 1, 2, 3 \quad (2.5)$$

$$\begin{aligned} \varepsilon_{kl} &= \frac{1}{2} \left( \frac{\partial v_k}{\partial \alpha_l} + \frac{\partial v_l}{\partial \alpha_k} \right), \quad \varepsilon_{k3} = \frac{\partial v_3}{\partial \alpha_k} + \gamma_k, \\ \kappa_{kl} &= \frac{1}{2} \left( \frac{\partial \gamma_k}{\partial \alpha_l} + \frac{\partial \gamma_l}{\partial \alpha_k} \right), \quad k, l = 1, 2 \end{aligned} \quad (2.6)$$

Typical boundary conditions for the subdomain  $\Omega_1$  are the following:

$$u_{ni} = g, \quad i = 1, 2, 3, \quad x \in \Gamma_1^{(1)}; \quad (2.7)$$

$$\sigma_{ni,n3} = t, \quad i = 1, 2, 3, \quad x \in \Gamma_1^{(2)}; \quad (2.8)$$

where  $u_{ni} = u_k n_{ik}$ ,  $\sigma_{ni,n3} = \sigma_{kl} n_{ik} n_{3l}$ ,  $(k, l = 1, 2, 3)$ ,  $n_{ij} = \cos(n_i, x_j)$ —are the direction cosines of the triple  $\bar{n}_j$ . The boundary conditions on the boundary of  $\Omega_2$  are

$$v_{tk} = g_k, \quad v_3 = g_3, \quad \gamma_{tk} = 0, \quad k = 1, 2, \quad x \in \Gamma_2^{(1)}; \quad (2.9)$$

$$T_{tk,tl} = 0, \quad Q_{ti} = 0, \quad M_{tk,tl} = 0, \quad i = 1, \quad k = 1, 2, \quad x \in \Gamma_2^{(2)}; \quad (2.10)$$

where  $v_{tk} = v_l t_{kl}$ ,  $\gamma_{tk} = \gamma_l t_{kl}$ ,  $k, l = 1, 2$ ,  $T_{tk,tl} = T_{ij} t_{ki} t_{lj}$ ,  $Q_{ti} = Q_l t_{il}$ ,  $M_{tk,tl} = M_{ij} t_{ki} t_{lj}$ ,  $i, j = 1, 2$ , and  $t_{kl} = \cos(t_k, \alpha_l)$  are the direction cosines of coordinate system  $\alpha_l$ .

Kinematic and static condition are defined at the coupling boundary  $\Gamma_1^{(3)} = \Gamma_2^{(3)} \times [-h/2, h/2]$  [9]:

$$v_1 + \alpha_3 \gamma_1 = -u_{n3}, \quad v_2 + \alpha_3 \gamma_2 = -u_{n1}, \quad v_3 = u_{n2}; \quad (2.11)$$

$$\begin{aligned} \int_{-h/2}^{h/2} \sigma_{n3,n3} d\alpha_3 &= T_{t1,t1}, \quad \int_{-h/2}^{h/2} \sigma_{n1,n3} d\alpha_3 = T_{t2,t1} \\ \int_{-h/2}^{h/2} \sigma_{n2,n3} d\alpha_3 &= -Q_t, \\ \int_{-h/2}^{h/2} \sigma_{n3,n3} \alpha_3 d\alpha_3 &= M_{t1,t1}, \quad \int_{-h/2}^{h/2} \sigma_{n1,n3} \alpha_3 d\alpha_3 = M_{t1,t2}. \end{aligned} \quad (2.12)$$



Thus, the system of partial differential equations (2.1),( 2.2 ) with the boundary conditions (2.7)-(2.10 )and interface boundary conditions (2.11),(2.12) and physical and geometrical relationships(2.3)-(2.6) is referred to as the heterogeneous mathematical model of an elastic body and Timoshenko's shell.

### 3 Variational Statement

Let us consider the special case of zero right hand sides in boundary conditions ( 2.7), ( 2.9). In operator notation, we can formulate the boundary value problem for the heterogeneous mathematical model as

$$\mathbf{AZ} = \mathbf{f}, \quad \mathbf{f} \in H \quad (3.13)$$

where  $H = [L_2(\Omega_1)]^3 \times [L_2(\Omega_2)]^5$ ,  $\mathbf{Z} = (u_1, u_2, u_3, v_1, v_2, v_3, \gamma_1, \gamma_2)$ ,  
 $\mathbf{f} = (f_1, f_2, f_3, \sigma_{13}^+ + \rho_1, \sigma_{23}^+ + \rho_2, \sigma_{33}^+ + \rho_3, \frac{h}{2}\sigma_{13}^+ + \mu_1, \frac{h}{2}\sigma_{23}^+ + \mu_2)$ . The operator in (3.13) is defined on the set

$$D_A = \{u_i, v_i, \gamma_k : i = 1, 2, 3; k = 1, 2; u_i \in [W_2^2(\Omega_1)]^3, v_i \in [W_2^2(\Omega_2)]^3; \gamma_k \in [W_2^2(\Omega_2)]^2; \text{ with conditions (2.7), (2.9), (2.11)}\}. \quad (3.14)$$

Let us define the scalar product of vector functions  $u, \tilde{u}$  over the lineal set  $D_A$  as

$$(u, \tilde{u}) = \int_{\Omega_1} u_i \tilde{u}_i d\Omega_1 + \int_{\Omega_2} (u_i \tilde{u}_i + \gamma_k \tilde{\gamma}_k) d\Omega_2. \quad (3.15)$$

The following lemma holds [8]:

**Lemma 3.1** *The operator of the problem (3.13) is symmetric in space  $H$  with the scalar product (3.15).*

The following theorem holds also [8].

**Theorem 3.2** *The operator of problem (3.13) is positive definite on  $D_A$ , i.e.*

$$(AZ, Z) \geq C_1^2 \|Z\|^2, \quad (3.16)$$

where  $C_1 \in R$ , and  $C_1 > 0$ ,  $\|Z\|^2 = \int_{\Omega_1} u_i u_i d\Omega_1 + \int_{\Omega_2} (v_i v_i + \gamma_k \gamma_k) d\Omega_2$ ,  $i = 1, 2, 3$ ;  $k = 1, 2$ , and

$$\begin{aligned} (AZ, \tilde{Z}) = & \int_{\Omega_1} \sigma_{ij}(u_1, u_2, u_3) e_{ij}(\tilde{v}_1, \tilde{v}_2, \tilde{v}_3) d\Omega_1 + \\ & + \int_{\Omega_2} (T_{kl}(v_1, v_2) \varepsilon_{kl}(\tilde{v}_1, \tilde{v}_2) + Q_{kl}(v_3, \gamma_1, \gamma_2) \varepsilon_{k3}(\tilde{v}_3, \tilde{\gamma}_1, \tilde{\gamma}_2) + \end{aligned} \quad (3.17)$$

$$+M_{kl}(\gamma_1, \gamma_2)\kappa_{kl}(\tilde{\gamma}_1, \tilde{\gamma}_2))d\Omega_2, \quad i, j = 1, 2, 3; k, l = 1, 2$$

Existence and uniqueness of the weak solutions of the problem (3.13) follow from the Lax-Milgram theorem [7].

Further we will present weak formulation of heterogeneous numerical model in which we will utilize DBEM in  $\Omega_1$  and FEM in the middle plane of  $\Omega_2$ , in the same manner as in [4]. Let us define the given domain as subsets  $\Omega_B$  and  $\Omega_F$  with  $\Omega = \Omega_F \cup \Omega_B \cup \Gamma_C$ , where  $\Gamma_C = \Gamma_1^{(3)} = \Gamma_2^{(3)} \times [-\frac{h}{2}, \frac{h}{2}] = \Omega_F \cap \Omega_B$  is the global coupling boundary in which we have boundary conditions (2.11), (2.12). The subset  $\Omega_B$  describes the boundary element geometry while  $\Omega_F$  denotes the finite element one. Let us denote  $u_F := v$ , if  $x \in \Omega_F = \Omega_2$  and  $u_B := u$ , if  $x \in \Omega_B = \Omega_1$ .

We assume the boundary condition are

$$u = g, \quad \text{if } x \in \Gamma_D, \quad \text{where } \Gamma_D = \Gamma_1^{(D)} \cup \Gamma_2^{(D)}, \quad (3.18)$$

$$p = h, \quad \text{if } x \in \Gamma_N, \quad \text{where } \Gamma_N = \Gamma_1^{(N)} \cup \Gamma_2^{(N)}, \quad (3.19)$$

The vectors  $g$  and  $h$  are prescribed boundary-displacement and boundary-stress respectively, and  $p_i = \sigma_{ij}n_{ij}$ ,  $p(\xi) = p[u(\xi)] := \lambda(\operatorname{div} u) + 2\mu\frac{\partial u}{\partial n} + \mu \times \operatorname{rot} u$  boundary tractions. For every  $u \in H^1(\Omega_F)$ , the trace  $\bar{u} \in H^{\frac{1}{2}}(\partial\Omega_F)$ , and  $p \in H^{-\frac{1}{2}}(\partial\Omega_F)$ . Corresponding duality pairing given by

$$\langle p, u \rangle = \int_{\partial\Omega_B} (p, u) d\Gamma, \quad (3.20)$$

where  $(\cdot, \cdot)$  is the  $R^n$ -scalar product. We introduce the subspace

$$H_0^{-\frac{1}{2}}(\partial\Omega_B) := \left\{ \chi \in H^{-\frac{1}{2}}(\partial\Omega_B) : \langle \chi, 1 \rangle = 0 \wedge \langle \chi, r \rangle = 0 \right\}, \quad (3.21)$$

where  $r$  denotes the distance vector to the origin.

Let us rewrite the bilinear form (3.17) as

$$(AZ, \tilde{Z}) = a_B(u, \tilde{u}) + a_F(v, \tilde{v}), \quad (3.22)$$

where

$$a_B(u, \tilde{u}) = \int_{\Omega_1} \sigma_{ij}(u) \varepsilon_{ij}(\tilde{u}) d\Omega_1, \quad (3.23)$$

and

$$a_F(v, \tilde{v}) = \int_{\Omega_2^*} (T_{kl}(v) \varepsilon_{kl}(\tilde{v}) + Q_k(v) \varepsilon_{k3}(\tilde{v}) + M_{kl}(v) k_{kl}(\tilde{v})) d\Omega_2. \quad (3.24)$$

The energy test space corresponding to  $\Omega_F$  is denoted by

$$H_D^1(\Omega_F) := \{v_F \in H^1(\Omega_F) : v_F = 0, \text{ if } x \in \Gamma_{FD}\}, \quad (3.25)$$

If  $\Gamma_{FD} \neq \emptyset$ , then the bilinear form (3.24) is  $H_D^1(\Omega_F)$ –elliptic [9], i.e.

$$\exists \alpha_0 > 0, \text{ such that } a_F(v, v) \geq \alpha_0 \|\mathbf{v}\|_{H^1(\Omega_F)}^2, \text{ for } \forall u \in H^1(\Omega_F). \quad (3.26)$$

We will denote  $\Gamma_{BD} := \partial\Omega_B \cap \Gamma_D$ ,  $\Gamma_{BN} := \partial\Omega_B \cap \Gamma_N$ ,  $\Gamma_{FD} := \partial\Omega_F \cap \Gamma_D$ ,  $\Gamma_{FN} := \partial\Omega_F \cap \Gamma_N$ . Let us apply Green's formula to (3.23)

$$a_B(u, \tilde{u}) = \int_{\Omega_1} \sigma_{ij}(u) \varepsilon_{ij}(\tilde{u}) d\Omega_1 = \int_{\partial\Omega_1} p_j(u) \tilde{u}_j d\Gamma - \int_{\Omega_1} \sigma_{ij,i}(u) \tilde{u}_j d\Omega_1. \quad (3.27)$$

If there are no body forces in  $\Omega_B$ , we have

$$a_B(u, u_1) = \int_{\partial\Omega_1} (p(u), u_1) d\Gamma. \quad (3.28)$$

We introduce the mortar functions

$$\tilde{u} \in H^{\frac{1}{2}}(\partial\Omega_B) := \left\{ \tilde{w} = w, x \in \partial\Omega_B : w \in H^1(\Omega) \text{ and } \|\tilde{w}\|_{H^{\frac{1}{2}}(\partial\Omega_B)} := \inf \|w\|_{H^1(\Omega)} \right\}, \quad (3.29)$$

and product space of pairs of restrictions

$$\mathfrak{R} = \left\{ (w_F, \tilde{w}) \in H^1(\Omega_F) \times H^{\frac{1}{2}}(\partial\Omega_B) : \tilde{w} = w_F \text{ as means in (2.11) if } x \in \Gamma_C \right\}, \quad (3.30)$$

equipped with the norm

$$\|(w_F, \tilde{w})\|_{\mathfrak{R}} := \inf \left\{ \|w\|_{H^1(\Omega)} : w \in H^1(\Omega), w = w_F, x \in \Omega_F, \text{ and } w = \tilde{w}, x \in \partial\Omega_B \right\}.$$

The test function space we denoted by

$$\mathfrak{R}_D = \left\{ (v_F, \tilde{v}) \in \mathfrak{R} : v_F \in H_D^1(\Omega_F), \tilde{v} = w, x \in \Gamma_B, w \in H^1(\Omega) \text{ and } \tilde{v} = 0, x \in \Gamma_{DB} \right\}.$$

The variational problem of the heterogeneous numerical model now reads: Find

$$((u_F, \tilde{u}), u_B, p_B) \in \mathfrak{R} \times H^{\frac{1}{2}}(\Gamma_B) \times H_0^{-\frac{1}{2}}(\Gamma_B),$$

with

$$(u_F, \tilde{u}) \in \mathfrak{R} \text{ and } u_F = g, \text{ if } x \in \Gamma_{DF}, \tilde{u} = g, \text{ if } x \in \Gamma_{DB}, \quad (3.31)$$

such that the global equations

$$a_{\Omega_F}(u_F, v_F) + \langle p_B, \tilde{v} \rangle_{\Gamma_B} = \int_{\Gamma_{FN}} h v_F d\Gamma + \int_{\Gamma_{BN}} p \tilde{v} dS, \quad (3.32)$$

for every  $(v_F, \tilde{v}) \in \mathfrak{R}_D$ ,

the weak coupling condition  $\int_{\Gamma_B} (\tilde{u} - u_B) \chi_B d\Gamma = 0$  for every  $\chi_B = p[u_B] \in H_0^{-\frac{1}{2}}(\Gamma_B)$ , are satisfied, together with the boundary integral equations

$$\left\langle \left( \frac{1}{2}I + K \right) u_B, \chi_B \right\rangle_{\Gamma_B} = \langle V p_B, \chi_B \rangle_{\Gamma_B}, \text{ for every } \chi_B \in H_0^{-\frac{1}{2}}(\Gamma_B). \quad (3.33)$$

In equation (3.33)  $V$  and  $K$  are the boundary integral operators of elastic single and double layer potentials respectively, defined by

$$V p(x) = \int_{\Gamma_B} E(x, y) p(y) dS_y,$$

$$K u(x) = p.v. \int_{\Gamma_B} (T_y [E(x, y)])^T u(y) dS_y,$$

with the fundamental solution

$$E(x, y) = \frac{\lambda + 3\mu}{4(n-1)\pi(\lambda + 2\mu)\mu} \left\{ e(x, y)I + \frac{(\lambda + \mu)(x - y)(y - x)^T}{\lambda + 3\mu |y - x|^n} \right\},$$

where

$$e(x, y) = \begin{cases} -\log |y - x|, & \text{for } n = 2, \\ |y - x|^{-1}, & \text{for } n = 3, \end{cases}.$$

$p.v.$  stands for the Cauchy principal value integral.

In the case  $\Gamma_B \neq \emptyset$ , the boundary integral operator  $V$  is a continuous mapping

$$V : H^{\frac{1}{2}}(\partial\Omega_B) \rightarrow H^{-\frac{1}{2}}(\partial\Omega_B)$$

and it is  $H^{-\frac{1}{2}}(\partial\Omega_B)$ -elliptic for the case  $n = 3$ , i.e.

$$\langle V\sigma, \sigma \rangle \geq \alpha_0 \|\sigma\|_{H^{\frac{1}{2}}(\partial\Omega_B)}^2 \text{ for every } \sigma \in H^{-\frac{1}{2}}(\partial\Omega_B).$$

The singular integral operator  $K$  defines a continuous mapping

$$K : H^{\frac{1}{2}}(\partial\Omega_B) \rightarrow H^{\frac{1}{2}}(\partial\Omega_B).$$

Hence, the boundary integral equation (3.33) defines the Dirichlet-Neumann mapping

$$S : H^{\frac{1}{2}}(\partial\Omega_B) \ni u_B \rightarrow p_B = S u_B \in H^{-\frac{1}{2}}(\partial\Omega_B),$$

given by

$$S = V^{-1} \left( \frac{1}{2}I + K \right), \text{ on } \partial\Omega_B.$$

It is well-known [1],[4] that  $S$  is a Steklov-Poincaré operator which is symmetric and  $H_0^{\frac{1}{2}}(\partial\Omega_B)$ -elliptic.

Hence,  $S$ —admits a right-inverse  $U$  on  $H_0^{-\frac{1}{2}}(\partial\Omega_B)$ ,

$$\begin{aligned} SUP &= p \quad \text{for every } p \in H_0^{-\frac{1}{2}}(\partial\Omega_B), \text{ and} \\ USu - u &= u_R = a + b \times r, \text{ for every } u \in H_0^{\frac{1}{2}}(\partial\Omega_B), \text{ and } a, b \in R^n. \end{aligned}$$

$U$  is called Poincare-Steklov operator, and

$$U : H^{\frac{1}{2}}(\Gamma_{BD}) \times H^{-\frac{1}{2}}(\Gamma_{BN}) \ni (g, p^*) \mapsto u \in H^{\frac{1}{2}}(\Gamma_B),$$

where  $u$  is the solution of boundary value problem of the theory of elasticity:

$$Lu_B = 0, \quad x \in \Omega_B,$$

$$u_B = g, \quad x \in \Gamma_{BD},$$

$$pu_B = p^*, \quad x \in \Gamma_{BN},$$

where

$$p^* = \begin{cases} p_B, & \text{if } x \in \Gamma_B \setminus \Gamma_N \\ t, & \text{if } x \in \Gamma_B \cap \Gamma_N \end{cases}.$$

Now, we formulate the variational problem (3.31)-(3.33) with use the definition of the Steklov-Poincare operator and their generalizations as:

Find  $(u_F, \tilde{u}) \in \mathfrak{R}$  and  $p_B \in H^{-\frac{1}{2}}(\Gamma_B)$ , which satisfy boundary condition (3.31) and such weak equations.

$$a_{\Omega_F}(u_F, v_F) + \langle p_B, \tilde{v} \rangle_{\Gamma_B} = \int_{\Gamma_{FN}} h v_F d\Gamma + \int_{\Gamma_{BN}} S^*(0, h) \tilde{v} dS, \quad (3.34)$$

for every  $(v_F, \tilde{v}) \in \mathfrak{R}_D$ , and

$$\int_{\Gamma_B} (\tilde{u} - U)\chi_B d\Gamma = 0, \text{ for every } \chi_B \in H^{-\frac{1}{2}}(\Gamma_B). \quad (3.35)$$

In the same way, as well as in [4] by utilizing the existence of the solution of the problem (3.13) we have the following existence and uniqueness results.

**Theorem 3.3** *The bilinear form defined by (3.34) and (3.35) is continuous and  $(\mathfrak{R} \times H^{-\frac{1}{2}}(\Gamma_B))$ —elliptic, i.e. there exists  $\alpha_0 > 0$  such that*

$$a_{\Omega_F}(v_F, v_F) + \langle S^*(v_B, 0), v_B \rangle_{\Gamma_B} \geq \alpha_0 \left\{ \|v_F\|_{H^1(\Omega_F)}^2 + \|v_B\|_{H^{\frac{1}{2}}(\Omega_B)}^2 \right\}, \quad (3.36)$$

with

$$\int_{\Gamma_B} (\tilde{v} - v_B) S^*(v_B, 0) d\Gamma = 0.$$

For the solution of the variational equation (3.32), (3.33) we use the heterogeneous numerical scheme. It is based on two families of meshes of the same type of finite elements with parameters of meshwidth  $H$  for the subdomain  $\Omega_F$  and  $h$  ( $h \leq H$ ) for the subdomain  $\Omega_B$ . Let us consider a conforming finite element space  $H_H^1(\Omega_F) \subset H^1(\Omega_F)$  with the approximation degree  $d \geq 2$  on a triangulation  $\{\tau_l\}_{l=1}^N$  of middle plane which satisfy the condition  $\text{diam}(\tau_l) \leq cH$  for all  $l = 1, \dots, N$ . On the surface of  $\Omega_B$  we introduce a family of finite-dimensional subspaces of continuous functions  $B_h^{\frac{1}{2}}(\partial\Omega_B) \subset H^{\frac{1}{2}}(\partial\Omega_B)$  (we use the same finite element functions). The elements in  $B_h^{\frac{1}{2}}(\partial\Omega_B)$  are used as mortar elements for global coupling. The pairs  $(v_F^H, \tilde{v}^H) \in H_H^1(\Omega_F) \times B_h^{\frac{1}{2}}(\partial\Omega_B)$ , which satisfy  $v_F^H = \tilde{v}^H$  in the sense of (2.11) for  $x \in \Gamma_C$ , define the finite-dimensional subspace of 'global' approximations  $\mathfrak{R}_H \subset \mathfrak{R}$ .

We consider a family of finite element spaces with  $H \rightarrow 0$  for the asymptotic analysis [4]. For this aim we require:

Approximation property: for every  $v \in H^\mu(\Omega)$ , with  $1 \leq \mu \leq d$ , there exists  $(v_F^H, \tilde{v}^H) \in \mathfrak{R}_H$ , such that

$$\|v - v_F^H\|_{H^1(\Omega_F)} + \|\tilde{v} - \tilde{v}^H\|_{H^{\frac{1}{2}}(\partial\Omega_B)} \leq cH^{\mu-1} \|v\|_{H^\mu(\Omega)}. \quad (3.37)$$

Inverse assumption:

With some  $\delta \in (0, \frac{1}{2})$  and for every  $\tilde{w} \in \mathfrak{R}_H$  we have on  $\partial\Omega_B$ ,

$$\|\tilde{w}_H\|_{H^{\frac{1}{2}+\delta}(\Omega_B)} \leq cH^{-\delta} \|\tilde{w}\|_{H^{\frac{1}{2}}(\partial\Omega_B)}. \quad (3.38)$$

Here, the norms on  $\partial\Omega_B$  are defined by

$$\|\tilde{w}_H\|_{H^{\frac{1}{2}+\delta}(\Omega_B)} = \inf \left\{ \|v\|_{H^{\frac{1}{2}+\delta}(\partial\Omega)} \in H^{1+\delta}(\Omega) \wedge v = \tilde{w} \text{ for } x \in \partial\Omega_B \right\}.$$

Similar to FEM analysis, the boundary elements are assumed to provide the approximation property and the inverse assumption.

We may introduce the global finite element space on  $\Omega_F \cup \partial\Omega_B$  as follows:

$$\mathfrak{R}_H = \text{span} \{ \varphi_1, \varphi_2, \varphi_3, \dots, \varphi_N \},$$

The discretized version of the similar approximation BEM - FEM for the problem of elasticity was very carefully investigated in [4]. Using the estimations of BEM approximations obtained in [4] for problems of the theory of elasticity, and FEM approximations for problem of Timoshenko's shell [9], we obtain results of asymptotic error estimates for Galerkin's approximations of the solution of the heterogeneous problem.

**Theorem 3.4** *Under assumptions (3.37), (3.38) there exist positive constants  $c_0, c > 0$  such that for all  $h < c_0H$  we have uniform stability estimate*

$$\left\{ \|u_F^H\|_{H^1(\Omega_F)}^2 + \|\tilde{u}_H\|_{H^{\frac{1}{2}}(\partial\Omega_B)}^2 \right\}^{\frac{1}{2}} \leq c \left( \|t\|_{H^{-\frac{1}{2}}(\Gamma_N)} + \|g\|_{H^{\frac{1}{2}}(\Gamma_D)} \right) \leq \|u\|_{H^1(\Omega)}.$$

This theorem guarantees the existence and uniqueness of the numerical solution.

**Theorem 3.5** *There exist positive constants  $c_1, c_2, c_3 > 0$  that don't depend on  $h$  and  $H$  such that we have the asymptotic estimate*

$$\begin{aligned} & \|u - u^H\|_{\mathfrak{R}} + \|pu - \chi^h\|_{H^{-\frac{1}{2}}(\partial\Omega_B)} + \|u - u^h\|_{H^{\frac{1}{2}}(\partial\Omega_B)} \leq \\ & c_1 \inf_{v^H \in \mathfrak{R}_H} \|u - v^H\|_{\mathfrak{R}} + c_2 \inf_{\mu^h \in \mathfrak{R}_h^{\frac{1}{2}}} \|u - \mu^h\|_{H^{\frac{1}{2}}(\partial\Omega_B)} + \\ & c_3 \inf_{\chi^h \in \mathfrak{R}_h^{-\frac{1}{2}}} \|pu - \chi^h\|_{H^{-\frac{1}{2}}(\partial\Omega_B)} \leq \\ & c'_1 H^{t-1} \|u\|_{H^t(\Omega)} + c'_2 h^{s'-\frac{1}{2}} \|u\|_{H^{s'+\frac{1}{2}}(\Omega)} + c'_3 h^{s+\frac{1}{2}} \|u\|_{H^{s+\frac{3}{2}}(\Omega)}, \end{aligned}$$

where  $1 \leq t \leq d$ ,  $\frac{1}{2} \leq s' \leq d'$ ,  $-\frac{1}{2} \leq s \leq d''$ ,  $s' = s + 1$ ,  $d' = d'' + 1$ .

## 4 Numerical Implementation

Here we use one of the domain decomposition techniques called iteration by subdomain, that is based on an iterative solution of a sequence of mixed discretized subproblems in  $\Omega_F$  and  $\Omega_B$  [8]. The elasticity problem in  $\Omega_B$  is solved by 3D FEM and the dimensionally reduced Timoshenko's shell elasticity problem in  $\Omega_F$  is solved by 2D FEM. Advantages of this technique are (a) the discretized problems on subdomain can be solved by standard single-domain finite element solvers, (b) no preconditioning is required, (c) interface boundary conditions are easily imposed.

In numerical approximation of the solution we use isoparametric 8-node serendipity finite elements in  $\Omega_B$  and in  $\Omega_F$ . The discretized version of (3.32), (3.33) obtained by a Galerkin procedure is as follows:

$$\begin{bmatrix} \mathbf{K}_{BB} & \mathbf{K}_{BC} & \mathbf{0} \\ \mathbf{K}_{BC}^T & \mathbf{K}_{CC} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{K}_{FF} \end{bmatrix} \begin{bmatrix} \mathbf{x}_B \\ \mathbf{x}_C \\ \mathbf{x}_F \end{bmatrix} = \begin{bmatrix} \mathbf{b}_B \\ \mathbf{b}_C \\ \mathbf{b}_F \end{bmatrix}, \quad (4.39)$$

where  $\mathbf{x}_B$  is the vector of degrees of freedom associated with  $\Gamma_B$ ,  $\mathbf{x}_C$  is the vector of degrees of freedom associated with  $\Gamma_C$ , and  $\mathbf{x}_F$  is the vector of degrees of freedom associated with  $\Omega_F$ . Therefore, the system (4.39) must be supplemented with the discretized boundary conditions on the interface boundary (2.11):

$$\mathbf{x}_C = \mathbf{T}\mathbf{x}_F. \quad (4.40)$$

Let  $\{\beta_k\}_{k=1}^{\infty}$  be a sequence of positive relaxation parameters, and let  $\mathbf{x}_F^{(0)}$  be given, e.g.  $\mathbf{x}_F^{(0)} = \mathbf{0}$ . Then the iteration by subdomain algorithm for solving the coupled system (4.39), (4.40) is as follows:

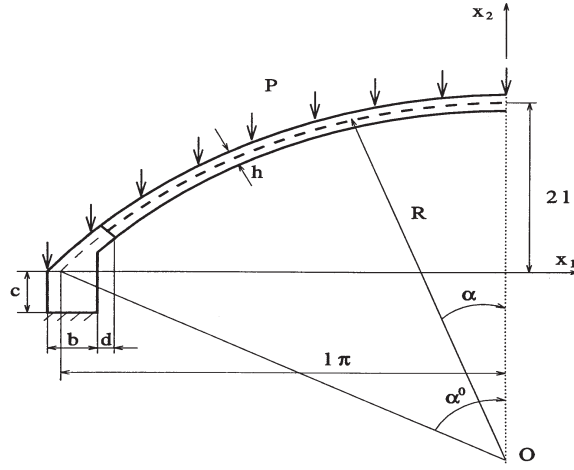


Figure 2: Cross section of a cylindrical shell supported by a massive edge beam.

$$\mathbf{x}_C^{(k+1)} = \mathbf{T} \mathbf{x}_F^{(k)}, \quad (4.41)$$

$$\mathbf{K}_{BB} \mathbf{x}_B^{(k+1)} = \mathbf{b}_B - \mathbf{K}_{BC} \mathbf{x}_C^{(k+1)}, \quad (4.42)$$

$$\mathbf{K}_{FF} \mathbf{x}_F^{(k+\frac{1}{2})} = \mathbf{b}_F + \mathbf{S}_1 \mathbf{x}_B^{(k+1)} + \mathbf{S}_2 \mathbf{x}_C^{(k+1)}, \quad (4.43)$$

$$\mathbf{x}_F^{(k+1)} = \mathbf{x}_F^{(k+\frac{1}{2})} + \beta_k \left( \mathbf{x}_F^{(k+\frac{1}{2})} - \mathbf{x}_F^{(k+1)} \right), \quad (4.44)$$

Stop if

$$\left\| \mathbf{x}_F^{(k+1)} - \mathbf{x}_F^{(k)} \right\| + \left\| \mathbf{x}_C^{(k+1)} - \mathbf{x}_C^{(k)} \right\| + \left\| \mathbf{x}_B^{(k+1)} - \mathbf{x}_B^{(k)} \right\| \leq \varepsilon \quad (4.45)$$

(or any other reasonable criterion),

otherwise go back to (4.41).

Where  $\mathbf{S}_1 \mathbf{x}_B^{(k+1)} + \mathbf{S}_2 \mathbf{x}_C^{(k+1)}$  represents the contribution to the right-hand side of the discretized problem in  $\Omega_F$  resulting from the natural interface boundary conditions (2.12). Convergence of numerical solution can be greatly accelerated by a judicious choice of a relaxation parameter at subdomain interface  $\Gamma_C$ .

## 5 Numerical Example

Let us consider the plain strain problem for the one-parameter family cylinder shells-roof. Fig.2 shows the cross-section of a cylindrical shell, supported by a massive edge beam. The



shell is subjected to a uniformly distributed load  $p$  parallel to the axes  $x_2$ . Because of symmetry of the problem, only one half of cross-section needs to be discretized. It consists of two parts: the massive part and the thin-walled part (shell). The cross-section of the massive part is treated as a plane strain problem. The thin part is described by Timoshenko's shell theory. The following types of curves are considered for the middle curves of the shell:

circular segment

$$x_1 = R \cos \alpha_1, \quad x_2 = R \sin \alpha_1 + c_1, \quad c_1 = -\frac{\pi^2 - 4}{4};$$

chain curve

$$x_1 = R \cos \alpha_1, \quad x_2 = -\frac{a}{2} \left( e^{\frac{x_1}{a}} + e^{-\frac{x_1}{a}} \right) + c_2, \quad a = \frac{\pi l}{k}, \quad c_2 = 2l + a, \quad k = 1.143187;$$

and quadratic parabola

$$x_1 = R \cos \alpha_1, \quad x_2 = -\frac{2}{\pi^2 l} x_1^2 + 2l.$$

The following terms are the same for all types of middle curves:

$$l = \text{const}, \quad R = \frac{4 + \pi^2}{4} l, \quad \alpha_1^0 \leq \alpha_1 \leq \frac{\pi}{2},$$

$$\alpha_1^0 = \arccos \frac{4\pi}{4 + \pi^2}.$$

The chosen parametrization of the middle curves leads to the same horizontal ( $l\pi$ ) and vertical ( $2l$ ) dimension of the shell structures for the three different middle curves. It is worth mention that the shapes of the cross-section of the shells differ weakly.

We utilize for numerical experiments the following parameter values:  $l = 8m$ ,  $h/R = 0.1$ ,  $c = 0.5m$ ,  $b = 0.5m$ ,  $d = 0.1m$ ,  $E = 2 \cdot 10^{10} N/m^2$ ,  $\nu = 0.1$ ,  $p = 1N/m^2$ .

The boundary of the massive fragment of the roof is approximated by the linear boundary elements both for the displacements and the tractions. The displacements and the angle of the normal to the middle curve of the structure rotation are approximated by the quadratic isoparametric finite elements.

The results confirm that the stressed state of the given types of shells is sensitive to small changes of the middle surface of the shell. Fig. 3 shows the curves of the normal stresses  $\sigma_{\alpha\alpha} (N/m^2)$  on the upper part of the roofs surface as a function of the natural parameter  $s$ . We introduce the following notations: the continuous curve represents the results, which are obtained for the roof with the parabolic section of shell, the dotted line curve represents the results, which are obtained for the roof with the circle section of shell, and the dash-dot line curve represents the results, which are obtained for the roof with the chain section of shell.

The analysis of the obtained results shows that the lowest bending stress among the above mentioned types has the shell with chain curve section. It is worth noticing, that the forms of the shells cross section differ weakly. These results confirm the fact, that the

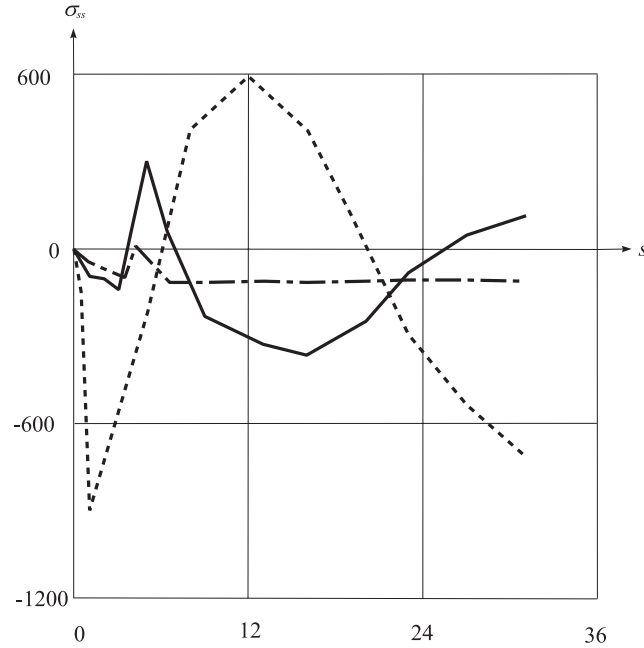


Figure 3: Normal stresses on upper surface of the structure.

stressed state of shell is rather sensitive to the form of the shells middle surface. Therefore, the middle surface change can be effective to control the stressed state of shell.

The effect of the use of the combined model for this example is: it is impossible to analyze the structure with the equations of the shell theory, because the massive part is present. An analysis of the structure by the 2D FEM on the basis of a plane strain model of the elasticity theory (i.e. without using shell theory) would require a significantly greater amount of data processing.

## 6 Conclusion

The considered heterogeneous models are an effective tool for analysis of compound structures. They allow us to employ the most suitable mathematical model and numerical method in each substructure. One type of isoparametric approximations is used for both parts of the domain. This method can be easily extended to problems with nonlinear field in a thin part of construction. The use of lower-dimensional models entails a significant reduction of preprocessing, processing and postprocessing time without significant losses of accuracy.

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*Chapter 5*

## ON RELAXED BOUNDARY SMOOTHING SPLINES FOR NONPARAMETRIC REGRESSION

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### Abstract

We study the relaxed boundary splines of OEHLERT (1992) for the nonparametric regression problem with deterministic quasi-uniform interior designs. For integer  $m \geq 1$ , we show that the relaxed boundary splines of order  $2m$  realize the optimal convergence rates for the expected  $L^2$  error when the regression function belongs to a Sobolev space of order  $k$  with  $m \leq k \leq 2m$ . The analysis is based on the identification of the relevant weighted Sobolev spaces as reproducing kernel Hilbert spaces (but not in the usual Bayesian view of smoothing splines).

**Key Words:** Nonparametric function estimation, maximum penalized likelihood estimation, kernel density estimation, smoothing splines **AMS Subject Classification:** Primary 62G08, Secondary 41A15.

## 1 Introduction

In this paper, we study the relaxed boundary spline estimator of OEHLERT [13] for the nonparametric regression problem. Relaxed boundary splines are an attempt at avoiding the problematic boundary behavior of the usual smoothing spline estimators when the degree of smoothness of the regression function has been underestimated. This difficulty arises as follows. Consider the nonparametric regression problem with deterministic design,

$$y_{in} = f_o(x_{in}) + d_{in}, \quad i = 1, 2, \dots, n, \quad (1.1)$$

where the  $x_{in}$  are quasi-uniform interior design points (for the precise definition, see §3) in a bounded interval, which we take to be  $(0, 1)$ . By way of example, take  $x_{in} = t_{in}$ ,

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where

$$t_{in} = \frac{i - 12}{n}, \quad i = 1, 2, \dots, n. \quad (1.2)$$

The  $1 \times n$  noise vector  $d_n = (d_{1,n}, d_{2,n}, \dots, d_{in}, \dots, d_{n,n})$  satisfies

$$\mathbb{E}[d_n] = 0, \quad \mathbb{E}[d_n d_n^T] = \sigma^2 I, \quad (1.3)$$

with  $\sigma^2$  unknown. The unknown regression function  $f_o$  is assumed to be a smooth function, in the sense that for some integer  $m \geq 1$ ,

$$f_o \in W^{m,2}(0, 1), \quad (1.4)$$

where, for  $a < b$ ,  $W^{m,2}(a, b)$  is the Sobolev space of order  $m$  on the interval  $(a, b)$ ,

$$W^{m,2}(a, b) = \left\{ f \in C^{m-1}[a, b] \mid 1 f^{(m-1)} \text{ absolutely continuous } f^{(m)} \in L^2(a, b) \right\}, \quad (1.5)$$

see, e.g., ADAMS and FOURNIER [1]. Here,  $f^{(m)}$  denotes the  $m$ -th derivative of  $f$ .

The goal is to estimate  $f_o$  given the data  $y_{1,n}, y_{2,n}, \dots, y_{in}, \dots, y_{n,n}$ . The (polynomial) smoothing spline estimator of order  $2m$  (or degree  $2m - 1$ ) is defined as the solution  $f = f^{nh}$  of

$$2 \text{ minimize } 1n \sum_{i=1}^n |f(x_{in}) - y_{in}|^2 + h^{2m} \|f^{(m)}\|^2 \text{ subject to } f \in W^{m,2}(0, 1), \quad (1.6)$$

where  $\|\cdot\|$  denotes the  $L^2(0, 1)$  norm. Typically, one takes  $m = 2$ , which corresponds to cubic spline smoothing. It is well-known that for  $n > m$ , the solution exists and is unique, see, e.g., EUBANK [8] and WAHBA [18]. For more on nonparametric regression in general, see GYÖRFY *et al.* [10] and the forthcoming [7].

The estimation problem (1.6) has all the characteristics of a regularized ill-posed problem, with  $h$  (or  $h^{2m}$ ) being the regularization or smoothing parameter. If the smoothing parameter  $h$  tends to  $\infty$ , then the solution  $f^{nh}$  must satisfy  $\|(f^{nh})^{(m)}\| \rightarrow 0$ , so that  $f^{nh}$  tends to a polynomial of order  $\leq m$ . Indeed, one shows that  $f^{nh} \rightarrow f^{n,\infty}$ , where  $p = f^{n,\infty}$  is the solution to

$$2 \text{ minimize } 1n \sum_{i=1}^n |p(x_{in}) - y_{in}|^2 \text{ subject to } p \text{ is a polynomial of order } \leq m. \quad (1.7)$$

Surely, if  $m = 2$ , or generally, if  $m$  is small and fixed, this is not a good estimator. (If  $m$  is allowed to increase with  $n$ , good estimators result, but that is not the subject of this paper. See COX [5].)

At the other end of the spectrum, when  $h \rightarrow 0$ , there is no “penalty” on the size of the  $m$ -th derivative of the estimator, and one shows that  $f^{nh} \rightarrow f^{n,0}$ , the spline interpolator of the data, i.e., the solution of the problem

$$2 \text{ minimize } \|f^{(m)}\|^2 \text{ subject to } f \in W^{m,2}(0, 1), f(x_{in}) = y_{in}, i = 1, 2, \dots, n. \quad (1.8)$$

So, the optimal  $h$  is somewhere between 0 and  $\infty$ . In fact, one shows under the conditions (1.1) through (1.4), that with  $f_h = \mathbb{E}[f^{nh}]$  for deterministic  $h$ ,

$$1\mathbb{E}[\|f^{nh} - f_h\|^2] = \mathcal{O}((nh)^{-1}), \quad nh \rightarrow \infty, \|f_h - f_o\|^2 = \mathcal{O}(h^{2m}), \quad h \rightarrow 0. \quad (1.9)$$

Then, since the expected  $L^2$  error decomposes into squared bias and variance,

$$\mathbb{E}[\|f^{nh} - f_o\|^2] = \|f_h - f_o\|^2 + \mathbb{E}[\|f^{nh} - f_h\|^2],$$

we have the asymptotic behavior of the expected  $L^2$  error,

$$\mathbb{E}[\|f^{nh} - f_h\|^2] = \mathcal{O}((nh)^{-1} + h^{2m}), \quad h \rightarrow 0, \quad nh \rightarrow \infty. \quad (1.10)$$

It follows that the asymptotically optimal  $h$  satisfies  $h \asymp n^{-1/(2m+1)}$ , and the optimal error is

$$\mathbb{E}[\|f^{nh} - f_h\|^2] = \mathcal{O}(n^{-2m/(2m+1)}). \quad (1.11)$$

Here, for sequences  $\{a_n\}_n$  and  $\{b_n\}_n$ , we write  $a_n \asymp b_n$  to mean that  $a_n/b_n$  and  $b_n/a_n$  are bounded for  $n \rightarrow \infty$ . If  $f_o \in W^{m,2}(0,1)$ , the bound (1.11) is the best possible, not only for smoothing splines, but for any regression estimator, as shown by STONE [17] and SPECKMAN [16]. What is striking about it is that no matter how smooth  $f_o$  is (in the scale of Sobolev spaces), one never achieves the  $n^{-1}$  rate of convergence.

We now come to the crux of the matter: The spline estimation problem (1.6) requires knowledge of the degree of smoothness of the regression function  $f_o$  as embodied in (1.4). In particular, if  $f_o$  is smoother than the selected value of  $m$ , e.g., in the most extreme case,

$$f_o \in W^{2m,2}(0,1), \quad (1.12)$$

then the bound (1.11) still cannot be improved upon. Indeed, RICE and ROSENBLATT [14] have shown that there exist functions  $b_{\ell,h}(x)$ ,  $\ell = m, \dots, 2m-1$ , such that for  $f_o \in W^{2m,2}(0,1)$ , the bias of the smoothing spline estimator (1.6) may be written as

$$f_h(x) - f_o(x) = [BC_h(f_o)](x) + e_h(x), \quad x \in [0,1], \quad (1.13)$$

where

$$\|e_h\|^2 = \mathcal{O}(h^{4m}), \quad (1.14)$$

and

$$[BC_h(f)](x) = \sum_{\ell=m}^{2m-1} h^\ell \left\{ f_o^{(\ell)}(0) b_{\ell,h}(x) + f_o^{(\ell)}(1) (-1)^\ell b_{\ell,h}(1-x) \right\}. \quad (1.15)$$

Thus, if  $f_o$  satisfies the *natural boundary conditions* associated with the spline smoothing problem (1.6),

$$f_o^{(\ell)}(0) = f_o^{(\ell)}(1) = 0, \quad \ell = m, \dots, 2m-1, \quad (1.16)$$

then we get for the squared bias,

$$\|f_h - f_o\|^2 = \mathcal{O}(h^{4m}), \quad (1.17)$$

and one obtains for  $h \asymp n^{-1/(4m+1)}$  the optimal error bound

$$\mathbb{E}[\|f^{nh} - f_h\|^2] = \mathcal{O}(n^{-4m/(4m+1)}). \quad (1.18)$$

REMARK 1.1 (a) The case (1.12) is the most extreme case for which interesting things happen to the spline smoother (1.6). Any more smoothness does not yield better rates than (1.18). (b) The above needs to be modified if the natural boundary conditions (1.16) are only fulfilled for  $\ell = m, \dots, m+k$ , with  $0 \leq k < m-1$ . We omit the details.

Computable approximations to the expression  $BC_h(f_o)$  are called boundary corrections. There is essentially only one way to go about obtaining them, to wit via the *bias reduction principle* of EUBANK and SPECKMAN [9], see HUANG [12]. An alternative is to avoid the boundary behavior (1.13) all together, and consider the relaxed boundary splines of OEHLERT [13]. They are defined as follows. For  $m \geq 1$ , define the spaces  $\mathcal{W}_m$  of functions on  $(0, 1)$  by

$$\mathcal{W}_m = \left\{ f \mid \forall \delta : 0 < \delta < 1/2, f \in W^{m,2}(\delta, 1-\delta) \mid f|_{\mathcal{W}_m} < \infty \right\}, \quad (1.19)$$

where

$$\|f\|_{\mathcal{W}_m} \text{ def} = \left\{ \int_0^1 \{x(1-x)\}^m |f^{(m)}(x)|^2 dx \right\}^{1/2}. \quad (1.20)$$

Note that the statement “ $f \in W^{m,2}(\delta, 1-\delta)$ ” must be interpreted as “the restriction of  $f$  to  $(\delta, 1-\delta)$  belongs to  $\dots$ ”.

Now, the relaxed boundary spline estimator of order  $2m$  is defined as the solution  $f = \varphi^{nh}$  to

$$2 \text{ minimize } \ln \sum_{i=1}^n |f(x_{in}) - y_{in}|^2 + h^{2m} \|f\|_{\mathcal{W}_m}^2 \text{ subject to } f \in \mathcal{W}_m. \quad (1.21)$$

The idea is that this estimator should have the correct bias behavior (1.17) and that its variance should still be governed by (1.9), albeit with a larger constant. The variance part requires a fair amount of work, but it can “easily” be seen that the bias part works as advertised. Note that the natural boundary conditions for the relaxed boundary spline problem are

$$D^\ell \left( \{x(1-x)\}^m D^m f \right) \text{ vanishes at } x = 0, 1 \text{ for } \ell = m, \dots, 2m-1, \quad (1.22)$$

where  $D$  denotes differentiation with respect to  $x$ , and that these conditions are automatically satisfied by  $f = f_o$  if  $f_o \in W^{2m,2}(0, 1)$ . OEHLERT [13] proves (1.18) for the relaxed boundary spline estimator for  $m \geq 4$ . In this paper, we conduct a more precise analysis and show that it works for  $m \geq 1$ . Moreover, the condition  $f_o \in W^{2m,2}(0, 1)$  can be replaced by the somewhat weaker condition

$$f_o \in \mathcal{W}_{2m}. \quad (1.23)$$

The main result is as follows. (The notion of quasi-uniform interior designs is defined in § 3. For now, it suffices to say that the uniform design (1.2) is indeed quasi-uniform interior.)

MAIN THEOREM. *Let  $m \geq 1$ . Under the assumptions (1.1), (1.3), and (1.23), for any quasi-uniform interior design, the relaxed boundary spline estimator  $\varphi^{nh}$ , defined as the solution of (1.21), exists and is unique, and satisfies*

$$\mathbb{E}[\|\varphi^{nh} - f_o\|^2] = \mathcal{O}(n^{-4m/(4m+1)}),$$

provided  $h \asymp n^{-1/(4m+1)}$ , deterministically.

In the remainder of the paper we prove the Main Theorem. To that end, in § 2, we study the relevant Hilbert spaces, and show that they are *reproducing kernel Hilbert spaces*. In § 3, we give the formal definition of quasi-uniform interior designs and prove a quadrature result for these designs. Finally, in § 4, the Main Theorem proper is proved.

## 2 The Reproducing Kernel Hilbert Space Setting

The first step in studying the relaxed boundary spline estimator (1.21) is whether the estimator actually exists. The crucial question concerns the continuity of the point evaluation functionals  $f \mapsto f(x_{in})$  in the  $\mathcal{W}_m$ -topology. This is (also) the key to establishing convergence rates for the relaxed boundary spline estimator.

We introduce the following innerproducts on the spaces  $\mathcal{W}_m$ , depending on the (smoothing) parameter  $h$ ,

$$\begin{aligned} \langle f, g \rangle_{\mathcal{W}_m} &= \int_0^1 \{x(1-x)\}^m f^{(m)}(x) g^{(m)}(x) dx, \\ \langle\langle f, g \rangle\rangle_{h, \mathcal{W}_m} &= \langle f, g \rangle + h^{2m} \langle f, g \rangle_{\mathcal{W}_m}, \end{aligned} \quad (2.1)$$

where  $\langle \cdot, \cdot \rangle$  denotes the standard innerproduct on  $L^2(0, 1)$ , and the norms

$$\|f\|_{h, \mathcal{W}_m} = \left\{ \langle\langle f, f \rangle\rangle_{h, \mathcal{W}_m} \right\}^{1/2}. \quad (2.2)$$

(Note that  $\langle \cdot, \cdot \rangle_{\mathcal{W}_m}$  is only a semi-innerproduct, but never mind.) In what follows, we write  $\log x(1-x)$  for  $\log\{x(1-x)\}$ .

**Lemma 2.1** *There exists a constant  $c$  such that for all  $h$ ,  $0 < h < 12$ , for all  $x \in (0, 1)$  and all  $f \in \mathcal{W}_1$ ,  $|f(x)| \leq c h^{-1} |\log x(1-x)|^{1/2} \|f\|_{h, \mathcal{W}_1}$ , a*  
 *$|f(x)| \leq 2 h^{-1/2} |x(1-x)|^{-1/4} \|f\|_{h, \mathcal{W}_1}$ . b*

**Lemma 2.2** *For all integers  $m \geq 1$ , there exists constants  $c_m$  such that*

$$\|f\|_{h, \mathcal{W}_m} \leq c_m \|f\|_{h, \mathcal{W}_{m+1}}$$

for all  $h$ ,  $0 < h < 1$ , and all  $f \in \mathcal{W}_{m+1}$ .

**REMARK 2.3** *Note that the functions  $f(x) = |\log x(1-x)|^{1/2-\varepsilon}$  belong to  $\mathcal{W}_1$  for  $\varepsilon > 0$ , but not for  $\varepsilon = 0$ . Thus, in this sense, the given rate in Lemma 2.1(a) is sharp. Likewise, in part (b), the rate  $h^{-1/2}$  is sharp, otherwise one would get better rates than (1.11), i.e., better than optimal. Unfortunately, the authors are unable to get sharp rates in  $h$  and  $x$  simultaneously.*



The proofs of the two lemmas are drastically different. For the first one, elementary considerations suffice. For the second one, the use of Legendre polynomials, which already featured prominently in OEHLERT [13], seems to be the most direct.

*Proof of Lemma 2.1.* Let  $f \in \mathcal{W}_1$ , let  $0 < x \leq 12$  and let  $0 < \lambda < 14$ . Since  $f$  is continuous on the interval  $(x, x + \lambda)$ , there exists a  $y$  in this interval such that

$$|f(y)| = \lambda^{-1/2} \left\{ \int_x^{x+\lambda} |f(x)|^2 dx \right\}^{1/2},$$

so that  $|f(y)| \leq \lambda^{-1/2} \|f\|$ . Also,

$$\begin{aligned} |f(x) - f(y)| &= \\ &= \left| \int_x^y f'(t) dt \right| \\ &\leq \left| \int_x^y \frac{dt}{t(1-t)} \right|^{1/2} \left| \int_x^y t(1-t) |f'(t)|^2 dt \right|^{1/2} \\ &\leq w^{1/2} \|f\|_{\mathcal{W}_1}, \end{aligned} \tag{2.3}$$

in which

$$w = \int_x^{x+\lambda} \{t(1-t)\}^{-1} dt. \tag{2.4}$$

Then,

$$|f(x)| \leq |f(y)| + |f(x) - f(y)| \leq \lambda^{-1/2} \|f\| + w^{1/2} \|f\|_{\mathcal{W}_1},$$

and we get with Cauchy-Schwarz that

$$|f(x)| \leq W^{1/2} \|f\|_{h, \mathcal{W}_1}, \tag{2.5}$$

with

$$W = \lambda^{-1} + h^{-2} w. \tag{2.6}$$

So all that remains is to suitably bound  $W$ .

Bounding  $\{t(1-t)\}^{-1}$  on the interval  $(x, x + \lambda) \subset (x, 34)$  by  $4x^{-1}$  yields

$$w \leq 4 \int_x^{x+\lambda} \frac{dt}{t} = 4 \log(1 + \lambda/x) \leq 4\lambda/x. \tag{2.7}$$

This gives

$$W \leq \lambda^{-1} + 4\lambda/(xh^2).$$

Now, take

$$\lambda = 12 h x^{1/2}. \tag{2.8}$$

Note that for  $h < 12$ , this gives  $\lambda < 14$ . Then,  $W \leq 4 h^{-1} x^{-1/2}$ . It follows trivially that

$$W \leq 4 h^{-1} \{x(1-x)\}^{-1/2}. \tag{2.9}$$

Now, the above holds for  $x \in (0, 12]$ . By replacing  $x$  by  $1 - x$ , we see that (2.9) holds for all  $x \in (0, 1)$ . Together with (2.5), this proves part (b).

For part (a), from the first part of (2.7), we use

$$w \leq 4 \log(1 + \lambda/x).$$

Now, take

$$\lambda = \gamma h^2 |\log x(1 - x)|^{-1},$$

with  $\gamma = \log 4$ . Then  $\lambda \leq h^2 < 14$ , so the bound on  $w$  applies. Also, it implies that

$$1 + \frac{\lambda}{x} \leq 1 + \frac{1}{x} = \frac{x+1}{x} \leq \frac{1}{x(1-x)},$$

and so

$$w \leq 4 \log(1 + \lambda/x) \leq 4 |\log x(1 - x)|.$$

Substituting this into (2.6), we obtain that

$$W \leq \{\gamma^{-1} + 4\} h^{-2} |\log x(1 - x)|.$$

Again, the above holds for all  $x \in (0, 12]$ . By replacing  $x$  by  $1 - x$ , we see that it holds for all  $x \in (0, 1)$ . Part (a) follows.

The proof of Lemma (2.2) is based on the availability of an orthogonal basis for  $\mathcal{W}_m$ , viz. the shifted Legendre polynomials. OEHLERT [13] already made extensive use of this. First, we define the shifted Legendre polynomials,

$$Q_k(x) = (2k+1)^{1/2} P_k(2x-1), \quad k \geq 0, \quad (2.10)$$

where  $P_k$  is the standard Legendre polynomial of exact degree  $k$  on  $[-1, 1]$ . They satisfy the orthogonality relations  $2 \langle Q_k, Q_\ell \rangle = \begin{cases} 2 & \text{if } k = \ell, \\ 0 & \text{otherwise} \end{cases}$ .

$\langle Q_k, Q_\ell \rangle_{\mathcal{W}_m} = \begin{cases} 2 & \text{if } k \neq \ell, \\ 0 & \text{if } k = \ell, \end{cases} \frac{4^m (k+m)!}{(k-m)!}$ , Note that the last integral vanishes (also) for  $k = \ell < m$ . The polynomials satisfy the pointwise bounds,

$$|Q_k(x)| \leq (2k+1)^{1/2}, \quad (2.11)$$

for all  $0 \leq x \leq 1$ , and  $k \geq 0$ ,

$$\leq c_k [x(1-x)]^{1/4}, \quad \text{for all } 0 < x < 1, \text{ and } k \geq 1,$$

with  $c_k = 4 \{(2k+1)/(\pi k)\}^{1/2} \leq 4 \{3/\pi\}^{1/2}$ . It follows that  $Q_k, k \geq 0$ , is an orthonormal basis for  $L^2(0, 1)$ , and an orthogonal basis for  $\mathcal{W}_m$ : For all  $h > 0$  and all  $m \geq 1$ ,

$$\langle\langle Q_k, Q_\ell \rangle\rangle_{h, \mathcal{W}_m} = \begin{cases} 21 + (2h)^{2m} \lambda_{k,m}, & \text{if } k = \ell, \\ 0, & \text{for } k \neq \ell, \end{cases} \quad (2.12)$$

where

$$\lambda_{k,m} = \frac{(k+m)!}{(k-m)!}. \quad (2.13)$$

Note that  $\lambda_{k,m} = 0$  for  $k < m$ , and that there exist constants  $c_m$  such that

$$(c_m)^{-1} k^{2m} \leq \lambda_{k,m} \leq c_m k^{2m}, \quad k \geq m \geq 1. \quad (2.14)$$

(For all of this and more, see SANSONE [15].) The following lemma is then obvious.

**Lemma 2.4** Let  $m \geq 1$ . For  $f \in \mathcal{W}_m$ , define  $\hat{f}_k = \langle f, Q_k \rangle$ ,  $k \geq 0$ . Then,

$$f(x) = \sum_{k=0}^{\infty} \hat{f}_k Q_k(x), \quad x \in [0, 1],$$

where the series converges in  $L^2(0, 1)$ , and

$$\|f\|_{h, \mathcal{W}_m}^2 = \sum_{k=0}^{\infty} (1 + (2h)^{2m} \lambda_{k,m}) |\hat{f}_k|^2 < \infty.$$

The above suggests the equivalent norms  $\|\cdot\|_{h, \mathcal{W}_m}$  for  $\mathcal{W}_m$ ,

$$\|f\|_{h, \mathcal{W}_m}^2 = \sum_{k=0}^{\infty} (1 + (2hk)^{2m}) |\hat{f}_k|^2, \quad (2.15)$$

The following lemma states the equivalence in precise terms, after which the proof of Lemma 2.2 is straight forward.

**Lemma 2.5** For  $m \geq 1$ , there exist constants  $\gamma_m$  such that

$$(\gamma_m)^{-1} \|f\|_{h, \mathcal{W}_m} \leq \|f\|_{h, \mathcal{W}_m} \leq \gamma_m \|f\|_{h, \mathcal{W}_m}$$

for all  $h$ ,  $0 < h < 1$ , and all  $f \in \mathcal{W}_m$ .

*Proof.* Let  $f \in \mathcal{W}_m$ , and use the expression of Lemma 2.4 for the norm  $\|f\|_{h, \mathcal{W}_m}$ . In view of (2.14), the upperbound  $\lambda_{k,m} \leq c_m k^{2m}$  holds for all  $k$ , so that

$$\|f\|_{h, \mathcal{W}_m}^2 \leq c_m \sum_{k=0}^{\infty} (1 + (2hk)^{2m}) |\hat{f}_k|^2 = c_m \|f\|_{h, \mathcal{W}_m}^2.$$

Of course, for the lower bound we have for  $k = 0, 1, \dots, m-1$ , and all  $h$ ,  $0 < h < 1$ ,

$$\frac{1 + (2hk)^{2m}}{1 + (2m)^{2m}} \leq 1 = 1 + (2h)^{2m} \lambda_{k,m},$$

and so, with

$$\alpha_m = \min\{1, (c_m)^{-1}, (1 + (2m)^{2m})^{-1}\},$$

we get

$$\alpha_m (1 + (2kh)^{2m}) \leq 1 + (2h)^{2m} \lambda_{k,m} \quad \text{for all } k \geq 0,$$

and consequently,

$$\alpha_m \|f\|_{h, \mathcal{W}_m}^2 \leq \|f\|_{h, \mathcal{W}_m}^2.$$

The equivalence of the norms uniformly in  $h$ , follows.

*Proof of Lemma 2.2.* Let  $m \geq 1$ , and  $f \in \mathcal{W}_{m+1}$ . Hölder's inequality implies that

$$\sum_{k=0}^{\infty} (1 + (2hk)^{2m}) |\hat{f}_k|^2 \leq \left\{ \sum_{k=0}^{\infty} |\hat{f}_k|^2 \right\}^{1-\theta} \left\{ \sum_{k=0}^{\infty} (1 + (2hk)^{2m})^{1/\theta} |\hat{f}_k|^2 \right\}^{\theta},$$

where  $\theta = m/(m+1)$ . Now, since

$$\sup_{h>0, k \geq 0} \frac{(1 + (2hk)^{2m})^{(m+1)/m}}{1 + (2hk)^{2m+2}} \leq \sup_{t>0} \frac{(1 + t^m)^{(m+1)/m}}{1 + t^{m+1}} \leq c < \infty,$$

we may conclude that (with the same  $c$ )

$$\|f\|_{h, \mathcal{W}_m} \leq c^\theta \|f\|^\theta \|f\|^{1-\theta}_{h, \mathcal{W}_{m+1}} \leq c^\theta \|f\|_{h, \mathcal{W}_{m+1}},$$

and of course,  $c^\theta < c$ . Lemma 2.2 follows.

Now, let us take stock. Lemmas (2.1) and (2.2) show that the point evaluation functionals  $f \mapsto f(x)$  for each  $x \in (0, 1)$  are continuous on  $\mathcal{W}_m$  ( $m \geq 1$ ). Precisely, for  $m \geq 1$ , there exists a constant  $c_m$ , such that for all  $x \in (0, 1)$  and all  $f \in \mathcal{W}_m$ ,

$$|f(x)| \leq c_m h^{-1/2} [x(1-x)]^{-1/4} \|f\|_{h, \mathcal{W}_m}, \quad |f(x)| \leq c_m h^{-1} |\log x(1-x)|^{1/2} \|f\|_{h, \mathcal{W}_m}. \quad (2.16)$$

Consequently, there exists elements  $\mathfrak{R}_{mhx} \in \mathcal{W}_m$  such that for all  $f \in \mathcal{W}_m$  and all  $x \in (0, 1)$ ,

$$f(x) = \langle f, \mathfrak{R}_{mhx} \rangle_{h, \mathcal{W}_m}. \quad (2.17)$$

Applying this to  $\mathfrak{R}_{mhx}(y)$ ,  $y \in (0, 1)$ , shows that  $\mathfrak{R}_{mhx}(y) = \mathfrak{R}_{mhy}(x)$ . It is customary to define the *reproducing kernel*  $\mathfrak{R}_{mh}(x, y)$ ,  $x, y \in (0, 1)$ , by

$$\mathfrak{R}_{mh}(x, y) = \mathfrak{R}_{mhx}(y) = \mathfrak{R}_{mhy}(x). \quad (2.18)$$

The above says that for  $m \geq 1$ , the spaces  $\mathcal{W}_m$  with the innerproducts  $\langle \cdot, \cdot \rangle_{h, \mathcal{W}_m}$ , are reproducing kernel Hilbert spaces with reproducing kernels  $\mathfrak{R}_{mh}$ . From (2.16), one obtains the bounds  $1 \|\mathfrak{R}_{mh}(x, \cdot)\|_{h, \mathcal{W}_m} \leq c h^{-1/2} |x(1-x)|^{-1/4}$ ,  $\|\mathfrak{R}_{mh}(x, \cdot)\|_{h, \mathcal{W}_m} \leq c h^{-1} |\log x(1-x)|^{-1/2}$ . For more on reproducing kernel Hilbert spaces, see ARONSZAJN [3].

**REMARK 2.6** *In probability and statistics, the usual way reproducing kernel Hilbert spaces arise is by constructing the Hilbert space from the reproducing kernel, which is given as the covariance operator of some stochastic process. See, e.g., DOLPH and WOODBURY [6]. In nonparametric regression, this comprises the Bayesian view of splines smoothing. See, e.g., WAHBA [18]. In the approach of this paper and [7], this does not seem useful.*

Finally, we need an integrations-by-parts trick, except that we prove it by way of expansions in terms of Legendre polynomials.

**Lemma 2.7** *Let  $m \geq 1$ . There exists a constant  $c_m$  such that for  $f \in \mathcal{W}_m$  and  $g \in \mathcal{W}_{2m}$ ,*

$$\langle f, g \rangle_{\mathcal{W}_m} \leq c \|f\| \|g\|_{1, \mathcal{W}_{2m}}.$$

*Proof.* Expand  $f$  and  $g$  in terms of the shifted Legendre polynomials, with convergence in  $L^2$ ,

$$f(x) = \sum_{k \geq 0} \hat{f}_k Q_k(x) \quad , \quad g(x) = \sum_{k \geq 0} \hat{g}_k Q_k(x) \quad ,$$

where  $\hat{f}_k = \langle f, Q_k \rangle$ , and likewise for  $\hat{g}_k$ . Then, see (2.15),

$$\|f\|^2 = \sum_{k \geq 0} |\hat{f}_k|^2 < \infty \quad , \quad \|g\|_{1, \mathcal{W}_{2m}}^2 = \sum_{k \geq 0} (1 + (2k)^{4m}) |\hat{g}_k|^2 < \infty \quad ,$$

and, from (2.12)-(2.13),  $\langle f, g \rangle_{\mathcal{W}_m} = \sum_{k \geq m} 2^{2m} \lambda_{k,m} \hat{f}_k \hat{g}_k$ . Then (2.14) implies that

$$\langle f, g \rangle_{\mathcal{W}_m} \leq c_m \sum_{k \geq m} (2k)^{2m} |\hat{f}_k| |\hat{g}_k| \quad ,$$

and so, with Cauchy-Schwarz,

$$\langle f, g \rangle_{\mathcal{W}_m} \leq c_m \left\{ \sum_{k \geq m} |\hat{f}_k|^2 \right\}^{1/2} \left\{ \sum_{k \geq m} (2k)^{4m} |\hat{g}_k|^2 \right\}^{1/2} \leq c_m \|f\| \|g\|_{1, \mathcal{W}_{2m}}$$

By the equivalence of the norms, see Lemma 2.5, the proof is complete (with a different constant  $c_m$ ).

**REMARK 2.8** *The reason we called Lemma 2.7 an integration by parts trick is that  $m$  times integration by parts gives*

$$\int_0^1 \{x(1-x)\}^m f^{(m)}(x) g^{(m)}(x) dx = \int_0^1 f(x) (-D)^m \left\{ \{x(1-x)\}^m g^{(m)}(x) \right\} dx$$

with  $D$  denoting differentiation with respect to  $x$ , provided the boundary terms vanish. [They do, but it requires bounds on  $f^{(k)}(x)$  in terms of the  $\mathcal{W}_m$  norm of  $f$ , similar to Lemma 2.1, and the realization that

$$\left| (-D)^k \left\{ \{x(1-x)\}^m f^{(m)}(x) \right\} \right| \leq c \{x(1-x)\}^{m-k} \quad ,$$

with the constant depending on  $L^2$  norms of  $g^{(\ell)}$ ,  $\ell = m, \dots, m+k$ .] Then, Cauchy-Schwarz gives the upperbound

$$\langle f, g \rangle_{\mathcal{W}_m} \leq \|f\| \left\{ \int_0^1 \left| D^m \left\{ \{x(1-x)\}^m g^{(m)}(x) \right\} \right|^2 dx \right\}^{1/2} \quad ,$$

and it requires some work to see that the last factor may be bounded in terms of  $\|g\|_{1, \mathcal{W}_{2m}}$ . These details are hidden in the expansions in terms of Legendre polynomials.

### 3 Quadrature with Quasi-Uniform Interior Designs

In this section, we introduce a (sufficient) condition on the designs so that the “design” sum of squares and squared norms

$$1/n \sum_{i=1}^n |f(x_{in})|^2 \quad \text{and} \quad \int_0^1 |f(x)|^2 dx$$

are comparable on  $\mathcal{W}_m$ .

**Definition 3.1** The design  $x_{in}$ ,  $i = 1, 2, \dots, n$ , is quasi-uniform interior if there exist positive constants  $c$  and  $\alpha$  such that

$$c n^{-1} \leq x_{in} \leq 1 - c n^{-1}, \quad i = 1, 2, \dots, n,$$

and for all  $f \in \mathcal{W}_1$ ,

$$\left| 1n \sum_{i=1}^n |f(x_{in})|^2 - \|f\|^2 \right| \leq c \frac{(\log n)^\alpha}{n} \|f\|_{1, \mathcal{W}_1}^2.$$

**Lemma 3.2** The design (1.2) is quasi-uniform interior.

*Proof.* In this proof, let  $a = t_{1,n} = 1/(2n)$  and  $b = t_{n,n} = 1 - 1/(2n)$ . We write

$$2 \sum_{i=1}^n |f(t_{in})|^2 = \sum_{i=1}^{n-1} |f(t_{in})|^2 + \sum_{i=2}^n |f(t_{in})|^2 + \text{rem}, \quad (3.1)$$

with

$$\text{rem} = |f(t_{1,n})|^2 + |f(t_{n,n})|^2. \quad (3.2)$$

From Lemma 2.1(a),

$$\text{rem} \leq c \log n \|f\|_{1, \mathcal{W}_1}^2. \quad (3.3)$$

We proceed with the first sum on the right of (3.1), which we interpret as Euler's compound quadrature formula. Hence, see ATKINSON [4],

$$\left| 1n - 1 \sum_{i=1}^{n-1} |f(t_{in})|^2 - \int_a^b |f(t)|^2 dt \right| \leq 1n - 1 \int_a^b |\{f^2(t)\}'| dt. \quad (3.4)$$

Now, by Lemma 2.1 and Cauchy-Schwarz,

$$\begin{aligned} & \int_a^b |\{f^2(t)\}'| dt \\ &= 2 \int_a^b |f(t) f'(t)| dt \\ &\leq 2c \|f\|_{1, \mathcal{W}_1} \int_a^b |\log t(1-t)|^{1/2} |f'(t)| dt \\ &\leq 2c \sqrt{\kappa_n} \|f\|_{1, \mathcal{W}_1} \|f\|_{\mathcal{W}_1}, \end{aligned} \quad (3.5)$$

where

$$\kappa_n = \int_a^b \frac{|\log t(1-t)|}{t(1-t)} dt. \quad (3.6)$$

Then, obviously,

$$\int_a^b |\{f^2(t)\}'| dt \leq 2c \sqrt{\kappa_n} \|f\|_{1, \mathcal{W}_1}^2, \quad (3.7)$$

It is straight forward to show that  $\kappa_n \leq c |\log n|^2$  for a suitable constant  $c$ , so

$$\left| 1n - 1 \sum_{i=1}^{n-1} |f(x_{in})|^2 - \int_a^b |f(t)|^2 dt \right| \leq c n^{-1} \log n \|f\|_{1, \mathcal{W}_1}^2. \quad (3.8)$$

Finally, for the appropriate constants  $c$  and  $c_1$ ,

$$\int_0^a |f(x)|^2 dx \leq c \|f\|_{1, \mathcal{W}_m}^2 \int_0^a |\log x(1-x)| dx \leq \frac{c_1 \log n}{n} \|f\|_{1, \mathcal{W}_1}^2. \quad (3.9)$$

A similar inequality holds for the integral over  $(b, 1)$ . So, on the left of (3.8), we may replace the integral over the interval  $(a, b)$  by the integral over  $(0, 1)$ .

Since the other sum in (3.1) may be treated in the same way, the lemma follows.

**REMARK 3.3** *The quadrature lemma suggests that it may be fruitful to study the semi-continuous analogue of (1.21),*

$$2 \text{ minimize } \|f\|^2 - 2n \sum_{i=1}^n y_{in} f(x_{in}) + h^{2m} |f|_{\mathcal{W}_m}^2 \text{ subject to } f \in \mathcal{W}_m.$$

*The solution exists if the design does not include the endpoints, and is given by*

$$\hat{f}(t) = 1n \sum_{i=1}^n y_{in} \mathfrak{R}_{mh}(x_{in}, t), \quad t \in (0, 1).$$

*This estimator should be an interesting object of study.*

The quadrature result also leads to the asymptotic existence of the relaxed boundary spline estimator.

**Lemma 3.4** *Let  $m \geq 1$ , and let  $x_{in}$ ,  $i = 1, 2, \dots, n$ , be a quasi-uniform interior design. Then, for all  $n$  large enough, the solution of the relaxed boundary spline smoothing problem (1.21) exists and is unique.*

*Proof.* Let  $m \geq 1$ . Note that for quasi-uniform interior designs, for suitably chosen positive constants  $c_1$  and  $c_2$ , and for all  $n$  large enough,

$$\begin{aligned} 1n \sum_{i=1}^n |f(x_{in})|^2 & \geq \{c - c_2 n^{-1} \log n\} \|f\|^2 - c_2 n^{-1} \log n |f|_{\mathcal{W}_m}^2 \\ & \geq c_1 \|f\|^2 - c_2 n^{-1} \log n |f|_{\mathcal{W}_m}^2. \end{aligned} \quad (3.10)$$

It follows that

$$1n \sum_{i=1}^n |f(x_{in})|^2 + \gamma_{nh} h^{2m} |f|_{\mathcal{W}_m}^2 \geq c_1 \{ \|f\|^2 + h^{2m} |f|_{\mathcal{W}_m}^2 \}, \quad (3.11)$$

where  $\gamma_{nh} = c_2 (nh^{2m})^{-1} \log n + c_1$  is positive. Consequently, with  $\kappa_{nh} = 1 \vee \gamma_{nh}$ , from (3.11),

$$\kappa_{nh} \left\{ 1n \sum_{i=1}^n |f(x_{in})|^2 + h^{2m} |f|_{\mathcal{W}_m}^2 \right\} \geq c_1 \|f\|_{h, \mathcal{W}_m}^2. \quad (3.12)$$

This inequality says that for  $n$  large enough, the objective function in (1.21) is strongly convex on  $\mathcal{W}_m$ , and so the solution of (1.21) exists and is unique, see, e.g., HOLMES [11].

**REMARK 3.5** *In Lemma 3.4, it would seem that the condition “for all  $n$  large enough” can be replaced by “ $n \geq m$  and all the design points are distinct”, but we shall not pursue this further.*

## 4 Random Sums

The reproducing kernel Hilbert space setting has a nice application to bounding random sums. The first lemma decouples the random noise from (possible) randomness of  $f$ . The second lemma bounds the decoupled random sum.

**Lemma 4.1** *Let  $m \geq 1$ , and let  $x_{in}, i = 1, 2, \dots, n$ , be a quasi-uniform interior design. Then, for all  $f \in \mathcal{W}_m$ , and all noise vectors  $d_n$ ,*

$$\left| 1n \sum_{i=1}^n d_{in} f(x_{in}) \right| \leq \|f\|_{h, \mathcal{W}_m} \|1n \sum_{i=1}^n d_{in} \mathfrak{R}_{mh}(x_{in}, \cdot)\|_{h, \mathcal{W}_m}.$$

**Lemma 4.2** *Let  $m \geq 1$ , and let  $x_{in}, i = 1, 2, \dots, n$ , be a quasi-uniform interior design. If the noise vector  $d_n$  satisfies (1.3), there exists a constant  $c$  depending only the order  $m$  and the design, such that for all  $h$  and all  $n$ ,*

$$\mathbb{E} \left[ \left\| 1n \sum_{i=1}^n d_{in} \mathfrak{R}_{mh}(x_{in}, \cdot) \right\|_{h, \mathcal{W}_m}^2 \right] \leq c (nh)^{-1}.$$

*Proofs.* With the representation (2.17),

$$1n \sum_{i=1}^n d_{in} f(x_{in}) = \langle f, \mathfrak{S}^{nh} \rangle_{h, \mathcal{W}_m}, \quad (4.1)$$

where

$$\mathfrak{S}^{nh}(t) = 1n \sum_{i=1}^n d_{in} \mathfrak{R}_{mh}(x_{in}, \cdot), \quad (4.2)$$

and Lemma 4.1 follows from Cauchy-Schwarz.

To prove Lemma 4.2, note that

$$\|\mathfrak{S}^{nh}\|_{h, \mathcal{W}_m}^2 = n^{-2} \sum_{i,j=1}^n d_{in} d_{jn} \langle \mathfrak{R}_{mh}(x_{in}, \cdot), \mathfrak{R}_{mh}(x_{jn}, \cdot) \rangle_{h, \mathcal{W}_m}, \quad (4.3)$$

Since  $\mathbb{E}[d_{in} d_{jn}] = \sigma^2$  for  $i = j$ , and  $= 0$  otherwise, then

$$\mathbb{E} \left[ \|\mathfrak{S}^{nh}\|_{h, \mathcal{W}_m}^2 \right] = \sigma^2 n^{-2} \sum_{i=1}^n \|\mathfrak{R}_{mh}(x_{in}, \cdot)\|_{h, \mathcal{W}_m}^2. \quad (4.4)$$

To bound the right hand side of (4.4), we start from (2),

$$n^{-2} \sum_{i=1}^n \|\mathfrak{R}_{mh}(x_{in}, \cdot)\|_{h, \mathcal{W}_m}^2 \leq c (nh)^{-1} \cdot 1n \sum_{i=1}^n |\psi(x_{in})|^2, \quad (4.5)$$

where  $\psi(t) = \{t(1-t)\}^{-1/4}$ . We now use the fact that the  $x_{in}$  form a quasi-uniform interior design, but first, we need to truncate  $\psi$  on  $(0, x_{1,n})$  and  $(x_{1,n}, 1)$ ,

$$\begin{aligned} \psi_n(t) = & \quad (4.6) \\ & \begin{cases} \{t(1-t)\}^{-1/4}, & x_{1,n} \leq t \leq x_{n,n}, \\ \{x_{1,n}(1-x_{1,n})\}^{-1/4}, & t < x_{1,n}, \\ \{x_{n,n}(1-x_{n,n})\}^{-1/4}, & t > x_{n,n}. \end{cases} \end{aligned}$$



(4.8)

Replacing  $\psi$  by  $\psi_n$  does not alter the expression on the right of (4.5). Then, from the quasi-uniformity of the design,

$$1n \sum_{i=1}^n |\psi_n(x_{in})|^2 \leq \|\psi_n\|^2 + c n^{-1} \log n \|\psi_n\|_{1, \mathcal{W}_1}^2. \quad (4.9)$$

One verifies that

$$\|\psi_n\|^2 \leq \int_0^1 \{x(1-x)\}^{-1/2} dx = \pi,$$

and that

$$|\psi_n|_{\mathcal{W}_1}^2 = \int_{x_{1,n}}^{x_{n,n}} x(1-x) \left| \frac{d}{dx} \{x(1-x)\}^{-1/4} \right|^2 dx. \quad (4.10)$$

Now, the integrand can be bounded by

$$const \cdot x(1-x) |x^{-5/4} + (1-x)^{-5/4}|^2 \leq otherconst \cdot \{x^{-3/2} + (1-x)^{-3/2}\},$$

so that, for suitable constants  $C$  and  $C_1$ ,

$$|\psi_n|_{\mathcal{W}_1}^2 \leq C \{x_{1,n}\}^{-1/2} + C \{1 - x_{n,n}\}^{-1/2} \leq C_1 n^{1/2}. \quad (4.11)$$

From (4.9), we then get

$$\sup_{n \geq 2} 1n \sum_{i=1}^n |\psi_n(x_{in})|^2 \leq \sup_{n \geq 2} \{\pi + C_1 n^{-1/2} \log n\} \leq C_2. \quad (4.12)$$

With (4.5), the right hand side of (4.4) may then be bounded by  $c(nh)^{-1}$ .

**Corollary 4.3** *Under the conditions of Lemma 4.1,*

$$\mathbb{E} \left[ \left| 1n \sum_{i=1}^n d_{in} f(x_{in}) \right| \right] \leq c(nh)^{-1/2} \left\{ \mathbb{E}[\|f\|_{h, \mathcal{W}_m}^2] \right\}^{1/2}.$$

*Proof.* Since

$$1n \sum_{i=1}^n d_{in} f(x_{in}) = \langle f, \mathfrak{S}^{nh} \rangle_{h, \mathcal{W}_m},$$

the corollary follows from the Cauchy-Schwarz inequality in the form

$$\mathbb{E} \left[ \left| \langle f, \mathfrak{S}^{nh} \rangle_{h, \mathcal{W}_m} \right| \right] \leq \left\{ \mathbb{E}[\|f\|_{h, \mathcal{W}_m}^2] \right\}^{1/2} \left\{ \mathbb{E}[\|\mathfrak{S}^{nh}\|_{h, \mathcal{W}_m}^2] \right\}^{1/2},$$

and the bound from Lemma 4.2.

## 5 Convergence Rates for the Relaxed Boundary Spline Estimator

We now come to the proof of the Main Theorem. It is useful to introduce the abbreviation

$$RLS(f) \text{ def } = 1n \sum_{i=1}^n |f(x_{in}) - y_{in}|^2 + h^{2m} |f|_{\mathcal{W}_m}^2, \quad (5.1)$$

so that the relaxed boundary spline estimation problem (1.21) reads

$$\text{minimize } RLS(f) \quad \text{subject to } f \in \mathcal{W}_m. \quad (5.2)$$

The first step is to describe the behavior of  $RLS(f)$  around its minimum.

**Lemma 5.1** *Let  $m \geq 1$ , and let  $f = \varphi^{nh}$  be the relaxed boundary spline estimator of (1.21). Then,  $\varepsilon \equiv f_o - \varphi^{nh}$  satisfies*

$$1n \sum_{i=1}^n |\varepsilon(x_{in})|^2 + h^{2m} |\varepsilon|_{\mathcal{W}_m}^2 = 1n \sum_{i=1}^n d_{in} \varepsilon(x_{in}) + h^{2m} \langle f_o, \varepsilon \rangle_{\mathcal{W}_m}. \quad (5.3)$$

*Proof.* Using the Taylor expansion of  $RLS(f)$  with  $f = f_o$  around  $f = \varphi^{nh}$ , one obtains,

$$RLS(f_o) = RLS(\varphi^{nh}) + \text{“linear term”} + 1n \sum_{i=1}^n |\varepsilon(x_{in})|^2 + h^{2m} |\varepsilon|_{\mathcal{W}_m}^2 \quad (5.4)$$

where

$$\text{“linear term”} = 2n \sum_{i=1}^n \varepsilon(x_{in}) (\varphi^{nh}(x_{in}) - y_{in}) + 2h^{2m} \langle \varepsilon, \varphi^{nh} \rangle_{\mathcal{W}_m}.$$

Of course, the “linear term” is the Gateaux variation of  $RLS(f)$  at  $\varphi^{nh}$  in the direction  $\varepsilon$ . Since  $RLS(f)$  is convex,  $\varphi^{nh}$  solves (5.2) if and only if this Gateaux variation vanishes for all  $\varepsilon$ , see, e.g., HOLMES [11]. So, in (5.4), the “linear term” may be omitted, and yields

$$1n \sum_{i=1}^n |\varepsilon(x_{in})|^2 + h^{2m} |\varepsilon|_{\mathcal{W}_m}^2 = RLS(f_o) - RLS(\varphi^{nh}). \quad (5.5)$$

Now, expand again  $RLS(\varphi^{nh}) = RLS(f_o) + 1n \sum_{i=1}^n |\varepsilon(x_{in})|^2 + h^{2m} |\varepsilon|_{\mathcal{W}_m}^2 - 2n \sum_{i=1}^n \varepsilon(x_{in}) (f_o(x_{in}) - y_{in}) - 2h^{2m} \langle \varepsilon, f_o \rangle_{\mathcal{W}_m}$ , and observe that  $f_o(x_{in}) - y_{in} = d_{in}$ . Then, substitute into (5.5). After moving the quadratic terms to the left and division by 2, the lemma follows.

*Proof of the Main Theorem.* The starting point is the inequality (5.3), and all we have to do is quote the appropriate results. From Definition 3.1, we conclude that

$$1n \sum_{i=1}^n |\varepsilon(x_{in})|^2 + h^{2m} |\varepsilon|_{\mathcal{W}_m}^2 \geq \eta_n \|\varepsilon\|^2 + \theta_n h^{2m} |\varepsilon|_{\mathcal{W}_m}^2, \quad (5.6)$$

where  $\eta_n = 1 - c n^{-1} (\log n)^\alpha$  and

$$\theta_n = 1 - c (nh^{2m})^{-1} (\log n)^\alpha = 1 - \mathcal{O}(n^{-(2m+1)/(4m+1)} (\log n)^\alpha).$$

Let  $\zeta_n = \eta_n \vee \theta_n$ . Then, the right hand side of (5.6) is bounded from below by  $\zeta_n \|\varepsilon\|_{h, \mathcal{W}_m}^2$ , and (5.6) implies

$$1n \sum_{i=1}^n |\varepsilon(x_{in})|^2 + h^{2m} \|\varepsilon\|_{\mathcal{W}_m}^2 \geq \zeta_n \|\varepsilon\|_{h, \mathcal{W}_m}^2. \quad (5.7)$$

Note that  $\zeta_n \rightarrow 1$ . Substituting (5.7) into (5.3) gives

$$\zeta_n \|\varepsilon\|_{h, \mathcal{W}_m}^2 \leq 1n \sum_{i=1}^n d_{in} \varepsilon(x_{in}) + h^{2m} \langle f_o, \varepsilon \rangle_{\mathcal{W}_m}. \quad (5.8)$$

Lemma 4.1 gives a nice bound for the first term on the right of (5.8), so that

$$\zeta_n \|\varepsilon\|_{h, \mathcal{W}_m}^2 \leq \|\varepsilon\|_{h, \mathcal{W}_m} \|\mathfrak{S}^{nh}\|_{h, \mathcal{W}_m} + h^{2m} \langle f_o, \varepsilon \rangle_{\mathcal{W}_m}. \quad (5.9)$$

For the last term on the right, we have of course by plain Cauchy-Schwarz that

$$h^{2m} \langle f_o, \varepsilon \rangle_{\mathcal{W}_m} \leq h^{2m} \|f_o\|_{\mathcal{W}_m} \|\varepsilon\|_{\mathcal{W}_m} \leq h^m \|f_o\|_{\mathcal{W}_m} \|\varepsilon\|_{h, \mathcal{W}_m}$$

but this loses a factor  $h^m$ . Instead, the integration-by-parts trick of Lemma 2.7 gives for  $f_o \in \mathcal{W}_{2m}$ ,

$$h^{2m} \langle f_o, \varepsilon \rangle_{\mathcal{W}_m} \leq c h^{2m} \|\varepsilon\| \|\| f_o \|_{1, \mathcal{W}_{2m}}, \quad (5.10)$$

so the factor  $h^{2m}$  survives as is. Now, the inequality (5.8) results in

$$\zeta_n \|\varepsilon\|_{h, \mathcal{W}_m}^2 \leq \|\varepsilon\|_{h, \mathcal{W}_m} \left\{ \|\mathfrak{S}^{nh}\|_{h, \mathcal{W}_m} + c h^{2m} \|f_o\|_{1, \mathcal{W}_m} \right\}, \quad (5.11)$$

and it follows that  $\zeta_n^2 \|\varepsilon\|_{h, \mathcal{W}_m}^2 \leq \left\{ \|\mathfrak{S}^{nh}\|_{h, \mathcal{W}_m} + c h^{2m} \|f_o\|_{1, \mathcal{W}_m} \right\}^2 \leq 2 \|\mathfrak{S}^{nh}\|_{h, \mathcal{W}_m}^2 + 2 c^2 h^{4m} \|f_o\|_{1, \mathcal{W}_m}^2$ . Upon taking expectations, see Lemma 4.2, everything follows.

**REMARK 5.2** *The idea of the proof of the Main Theorem (minus the last expectation step) is well known in the analysis of ill-posed problems, see, e.g., ARCANGELI [2].*

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*Chapter 6*

# **A NEW ALGEBRAIC STRUCTURE APPROPRIATE FOR FINDING THE RELIABILITY OF ORGANIZATIONAL SYSTEMS**

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## **Abstract**

A new algebraic structure called a reliability algebra is defined which can be used to compute the reliability of organizational systems with independently functioning components outside the realm of probability. Sufficient conditions are given for a reliability algebra to be a lattice and a Boolean Algebra.

**Key Words:** Ordered Monoids, Reliability Algebra, Reliability of Systems and Boolean Algebra

**AMS Subject Classification:** Primary 60K99, 06F05 and 06A11.

## **1 Introduction**

We will discuss an algebraic system which can be used to compute the reliability of an organizational system outside the realm of probability and its relationship to other better known algebraic systems including lattices and Boolean algebras.

Organizational systems consisting of components which are interrelated in certain ways are discussed in many probability texts, so we will give only a brief summary on reliability of systems.

Two adjacent components of a system can be placed in series or in parallel. Following are systems of two components  $c_1$  and  $c_2$  placed in series (figure 1) and parallel (figure 2).

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Figure 1: Geometric representation for a two component series system

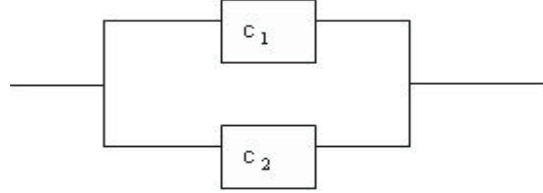


Figure 2: Geometric representation for a two component parallel system

In a series system, the entire system functions properly if and only if all components in the system functions properly and in a parallel system, the entire system functions properly if and only if at least one component in the system functions properly.

The reliability of an organizational system is the probability that the system functions properly for a specific time interval.

Consider an organizational system consisting of  $n$  components  $c_1, \dots, c_n$ . Suppose components  $c_i$ ,  $1 \leq i \leq n$ , are either functioning or not functioning for a specific time interval and events

$E_i$  = the event that component  $c_i$  is functioning properly,  $1 \leq i \leq n$ , are well defined and independent events.

The reliability,  $r$ , of a system consisting of components  $c_1$  and  $c_2$  placed in series (see figure 1) is given by

$$r = P(E_1 \cap E_2)$$

$$r = P(E_1) \bullet P(E_2) \quad (1.1)$$

since  $E_1$  and  $E_2$  are independent.

The reliability,  $r$ , of a system consisting components  $c_1$  and  $c_2$  placed in parallel (see Figure 2) is given by

$$r = P(E_1 \cup E_2)$$

$$r = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

$$r = P(E_1) + P(E_2) - P(E_1) \bullet P(E_2) \quad (1.2)$$

*Example 1.* Consider the system consisting of six components shown below on figure 3.

The reliability of the system is

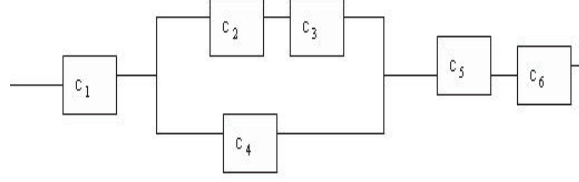


Figure 3: Geometric representation of a system of both series and parallel components

$$r = P(E_1 \cap ((E_2 \cap E_3) \cup E_4) \cap E_5 \cap E_6)$$

If the reliabilities of components  $r_1, r_2, r_3, r_4, r_5$  and  $r_6$  are  $\frac{1}{4}, \frac{1}{5}, \frac{1}{3}, \frac{2}{5}, \frac{1}{2}$  and  $\frac{2}{3}$ , respectively, then the reliability of the system is

$$r = P(E_1 \cap ((E_2 \cap E_3) \cup E_4) \cap E_5 \cap E_6)$$

$$r = P(E_1) \bullet P(E_5) \bullet P(E_6) \bullet [P(E_2 \cap E_3) + P(E_4) + P(E_2 \cap E_3 \cap E_4)]$$

$$r = P(E_1) \bullet P(E_5) \bullet P(E_6) \bullet [P(E_2) \bullet P(E_3) + P(E_4) - P(E_2) \bullet P(E_3) \bullet P(E_4)]$$

$$r = \frac{1}{4} \bullet \frac{1}{2} \bullet \frac{2}{3} \bullet [\frac{1}{5} \bullet \frac{1}{3} + \frac{2}{5} - \frac{1}{5} \bullet \frac{1}{3} \bullet \frac{2}{5}]$$

$$r = \frac{1}{12} \bullet [\frac{5}{75} + \frac{30}{75} - \frac{2}{75}]$$

$$r = \frac{33}{900}$$

$$r = \frac{11}{300}$$

## 2 Reliability Algebra

We will now look at an algebraic structure which is appropriate for computing the reliability of an organizational system with independent functioning components, i.e., the events  $E_i, i = 1, \dots, n$  defined earlier are independent events.

As a general algebraic structure, consider a commutative monoid  $(M, \bullet)$  with identity element  $e$  on which is defined a unary operation  $\overline{(\ )}$  satisfying

$$\overline{\overline{a}} = a \text{ for all } a \in M. \quad (2.1)$$

A unary operation  $\overline{(\ )}$  satisfying the condition  $\overline{\overline{a}} = a$  for all  $a \in M$ , is commonly called an involution. One property that holds for  $\overline{(\ )}$  is

$$\overline{\overline{a}} = \overline{b} \text{ if and only if } a = b \quad (2.2)$$



An example of an involution in a group  $M$  is the inverse operation  $^{-1}$ , since

$$(a^{-1})^{-1} = a, \text{ for all } a \in M.$$

One can use the involution  $\overline{(\ )}$  to define a second binary operation  $\oplus$  on  $M$  by the relation

$$a \oplus b = \overline{\overline{a} \bullet \overline{b}}, \text{ for all } a, b \in M \quad (2.3)$$

By (2.1) and (2.2) it follows that  $\overline{(\ )} : (M, \bullet) \rightarrow (M, \oplus)$  is bijective.

By (2.3) and (2.1) it follows that  $\overline{(\ )}$  is a homomorphism. So we have:

**Theorem 1** *If  $\overline{(\ )}$  is an involution on monoid  $(M, \bullet)$  and binary operation  $\oplus$  on  $M$  is defined by  $a \oplus b = \overline{\overline{a} \bullet \overline{b}}$ , for all  $a, b \in M$ , then,  $\overline{(\ )} : (M, \bullet) \rightarrow (M, \oplus)$  is an isomorphism.*

**Corollary**  $(M, \oplus)$  is a monoid with identity element  $\overline{e}$ .

It is possible for the two operations  $\bullet$  and  $\oplus$  to be identical. As an example, if  $M$  is a commutative group and the unary operation  $\overline{(\ )}$  is the inverse operator  $^{-1}$ .

**Definition 1** *A reliability algebra is a set  $M$  together with two binary operations  $\bullet$  and  $\oplus$  and a unary operation  $\overline{(\ )}$  satisfying*

$R_1$   $(M, \bullet)$  is a commutative monoid

$R_2$   $\overline{\overline{a}} = a$  for  $a \in M$

$R_3$   $a \oplus b = \overline{\overline{a} \bullet \overline{b}}$  for all  $a, b \in M$  (De Morgan's Law)

$R_2$  and  $R_3$  define binary operation  $\oplus$  in terms of the operations  $\bullet$  and  $\overline{(\ )}$  by formula (2.3). Later, we will mention other properties that reliability algebras may possess and its relation to other known algebraic structures.

Following is an example of a reliability algebra which is appropriate for computing the reliability of organization systems with independently functioning components.

**Example 2.** Let  $M$  be the closed interval  $[0,1]$ ,  $\bullet$  be the multiplication operation on  $M$  and  $\overline{(\ )}$  be the unary operation defined on  $M$  by  $\overline{a} = 1 - a$ . Clearly  $(M, \bullet)$  is a commutative monoid with identity 1 and  $\overline{(\ )}$  satisfies

$$\begin{aligned} \overline{\overline{a}} &= a \text{ and } \oplus \text{ is defined by } a \oplus b = \overline{\overline{a} \bullet \overline{b}} \\ &= 1 - (1 - a) \bullet (1 - b) \\ a \oplus b &= a + b - a \bullet b \end{aligned} \quad (2.4)$$

We have motivated the following definitions which give the same result one would obtain using probability.

**Definition 2** *The reliability of two components or subsystems in series is  $r = r_1 \bullet r_2$ , where  $r_1$  and  $r_2$  are the reliabilities of the components or subsystems.*

**Definition 3** *The reliability of two components or subsystems in parallel is  $r = r_1 \oplus r_2$ , where  $r_1$  and  $r_2$  are the reliability of the two components or subsystems.*

*Example 3.* Using the specified reliability algebra, we can compute the reliability of the system of example 1 as follows. For this example, it is easier to express all operations in terms of multiplication and subtraction from 1.

In our reliability algebra, the reliability of Example 1 can be expressed as

$$r = \frac{1}{4} \bullet ((\frac{1}{5} \bullet \frac{1}{3}) \oplus \frac{2}{5}) \bullet \frac{1}{2} \bullet \frac{2}{3}$$

$$r = \frac{1}{4} \bullet \overline{\overline{\frac{1}{5} \bullet \frac{1}{3} \bullet \frac{2}{5}}} \bullet \frac{1}{2} \bullet \frac{2}{3}$$

$$r = \frac{1}{4} \bullet \frac{14}{15} \bullet \frac{3}{5} \bullet \frac{1}{3}$$

$$r = \frac{1}{4} \bullet \frac{33}{75} \bullet \frac{1}{3}$$

$$r = \frac{11}{300}$$

*Example 4.* Find an expression for the reliability of the following system where we assume  $c_i$  has reliability  $r_i$ ,  $i = 1, \dots, 5$ , respectively.

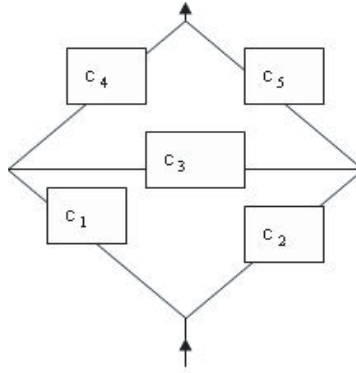


Figure 4: Geometric representation of a bridged system

One finds that  $r = r_1 \bullet r_4 \oplus r_1 \bullet r_3 \bullet r_5 \oplus r_2 \bullet r_5 \oplus r_2 \bullet r_3 \bullet r_4$ . So the addition of the bridge  $c_3$  to the system increases the reliability of the system by  $r_1 \bullet r_3 \bullet r_5 \oplus r_2 \bullet r_3 \bullet r_4$  in terms of the operation  $\oplus$ . One can argue that the natural operations to assess reliability of organizational systems are  $\bullet$  and  $\oplus$  since they are closed operations in  $[0, 1]$  in contrast to ordinary addition  $+$ .

### 3 Results on Reliability Algebras

Due to De Morgan's Law's, reliability algebra satisfies the duality principle.

**Theorem 2 (Duality principle)** *If  $s$  is an equality in a reliability algebra involving  $\bullet$  and  $\oplus$ , then the dual statement obtained by interchanging  $\bullet$  and  $\oplus$  throughout  $s$  is an equality.*

Following are properties which reliability algebras may possess.

$B_1$  For all  $a \in M$ ,  $a \bullet \bar{a} = \bar{e}$  (complemented)

- B<sub>2</sub> For all  $a \in M$ ,  $\bar{e} \bullet a = \bar{e}$  ( $\bar{e}$  is absorbing).  
 B<sub>3</sub> For all  $a \in M$ ,  $a \bullet a = a$  (idempotent)  
 B<sub>4</sub> For all  $a, b \in M$ ,  $a \bullet \overline{a \bullet b} = a$  or  $a \bullet (a \oplus \bar{b}) = a$  (absorption)  
 B<sub>5</sub> If there exists  $a \in M$  such that  $a \bullet x = a \bullet y$  and  $\bar{a} \bullet x = \bar{a} \bullet y$  then,  $x = y$  (cancellation)  
 B<sub>6</sub> For all  $a, b$  and  $c \in G$ ,  $a \bullet (b \oplus c) = (a \bullet b) \oplus (a \bullet c)$  (distributive)

We will discuss some better known algebraic systems including lattices and Boolean algebras. A lattice is a set  $M$  together with two binary operations  $\wedge$  and  $\vee$  defined on  $M$  which is closed, associative, commutative and satisfy:

- L<sub>1</sub> For all  $a \in M$ ,  $a \wedge a = a$  and  $a \vee a = a$  (idempotent)  
 L<sub>2</sub> For all  $a, b \in M$   $a \wedge (a \vee b) = a$  and  $a \vee (a \wedge b) = a$  (absorption)

Due to the Duality principle, it is clear that in a reliability algebra  $M$ , statements L<sub>1</sub> and L<sub>2</sub> follow from B<sub>3</sub> and B<sub>4</sub>.

**Theorem 3** *A reliability algebra with each member idempotent under operation  $\bullet$  (B<sub>3</sub>) and satisfying the absorption law (B<sub>4</sub>) is a lattice.*

In a reliability algebra  $M$  with operation  $\bullet$  one can define partial ordering on  $M$  if various properties listed above are satisfied. Their existences and properties of the partial ordering can be used to prove further properties of the reliability algebra. Properties a partial may possess include congruence with respect to operation  $\bullet$  and ordering reversing as given below under O<sub>1</sub> and O<sub>2</sub>.

O<sub>1</sub> For all  $a, b$  and  $c \in M$ , if  $a \geq b$  then  $a \bullet c \geq b \bullet c$  (congruent with respect operation  $\bullet$ ).

O<sub>2</sub> For all  $a, b \in M$ , if  $a > b$  then  $\bar{b} > \bar{a}$  (order reversing).

**Theorem 4** *Let  $M$  be a reliability algebra with operation  $\bullet$  in which each element is idempotent (B<sub>3</sub>), then there exists a partial ordering  $\geq$  on  $M$  which is congruent with respect operation  $\bullet$  and having  $e$  as the largest member in  $M$ . Furthermore, if  $M$  satisfies  $\bar{e} \bullet a = \bar{e}$  (B<sub>2</sub>) for all  $a \in M$ , then  $\bar{e}$  is the smallest member in  $M$  and if  $M$  satisfies  $a \bullet \overline{a \bullet b} = a$  (B<sub>4</sub>) for all  $a, b \in M$ , then  $\geq$  is order reversing (O<sub>2</sub>).*

Defined relation  $\geq$  by:

$$a \geq b \text{ if and only if } a \bullet b = b \quad (3.1)$$

$\geq$  is reflective since  $a \geq a$  is equivalent to  $a \bullet a = a$ , which holds by B<sub>3</sub>

$\geq$  is anti-symmetric and transitive by associativity and commutativity of  $\bullet$ . For congruence, assume  $a \geq b$  and  $c \in M$ ,

So

$$\begin{aligned} a \bullet b &= b \\ a \bullet b \bullet c &= b \bullet c \\ a \bullet c \bullet b \bullet c &= b \bullet c \text{ by } B_3 \\ \therefore a \bullet c &\geq b \bullet c \end{aligned}$$

Since  $e$  is the identity for  $M$ ,  $e \bullet a = a$  for all  $a \in M$ , which is equivalent to  $e \geq a$  for all  $a \in M$ .

By  $\bar{e} \bullet a = \bar{e}$  ( $B_2$ ),  $a \geq \bar{e}$  for all  $a \in M$  and  $\bar{e}$  is the smallest member in  $M$ .

To show  $O_2$ , let  $a \geq b$ , then

$$\begin{aligned} a \bullet b &= b \\ \bar{b} \bullet \bar{a} &= \overline{a \bullet b} = \bar{a} \text{ by } B_4 \\ \therefore \bar{b} &\geq \bar{a} \end{aligned}$$

There is a converse to the above lemma in the following sense.

**Theorem 5** *Let  $M$  be a reliability algebra with operation  $\bullet$  and with every member idempotent ( $B_3$ ). If  $\geq$  is any partial ordering with largest element  $e$  and which is congruent with respect to  $\bullet$  ( $O_1$ ) and order reversing ( $O_2$ ), then  $M$  is absorbing ( $B_4$ ).*

To show ( $B_4$ ), let  $a, b \in M$

$$\begin{aligned} e &\geq \overline{\bar{a} \bullet \bar{b}} \text{ since } e \text{ is the largest member in } M \\ a \bullet e &\geq a \bullet \overline{\bar{a} \bullet \bar{b}} \text{ by } O_1 \\ \text{So } a &\geq a \bullet \overline{\bar{a} \bullet \bar{b}} \end{aligned}$$

$$\begin{aligned} \text{Likewise, } \bar{a} &\geq \bar{a} \bullet b \text{ follows from } e \geq b \text{ and } O_1 \\ \text{So } \overline{\bar{a} \bullet b} &\geq \bar{a} \text{ by } O_2 \\ a \bullet \overline{\bar{a} \bullet b} &\geq a \bullet a \text{ by } O_1 \\ a \bullet \overline{\bar{a} \bullet b} &\geq a \text{ by } B_3 \\ \therefore a \bullet \overline{\bar{a} \bullet b} &= a \end{aligned}$$

A Boolean algebra is a lattice  $M$  with two binary operations  $\bullet$  and  $\oplus$ , a unary operation  $\bar{()}$  together with two elements  $e$  and  $\bar{e}$  satisfying for all  $a, b$  and  $c \in M$

$$\begin{array}{ll} B_1 & a \bullet \bar{a} = \bar{e} & B'_1 & a \oplus \bar{a} = e \text{ (complemented)} \\ B_2 & \bar{e} \bullet a = \bar{e} \text{ (}\bar{e} \text{ is the smallest member)} & B'_2 & e \oplus a = e \text{ (}e \text{ is the largest member)} \\ B_6 & a \bullet (b \oplus c) = (a \bullet b) \oplus (a \bullet c) & B'_6 & a \oplus (b \bullet c) = (a \oplus b) \bullet (a \oplus c) \end{array}$$

(distributive)

Due to the duality principle, to show that a reliability algebra  $M$  is a Boolean algebra, one needs only show that  $B_1, B_2, B_3, B_4$ , and  $B_6$  holds.

One requirement for a reliability algebra  $M$  to be a Boolean algebra is that every element  $a \in M$  have an unique compliment.

Definition 4 A compliment of a member  $a$  in a reliability algebra  $M$  is a member  $a'$  in  $M$  such that

$$a' \bullet a = \bar{e} \text{ and } a' \oplus a = e \quad (\bar{a'} \bullet \bar{a} = \bar{e})$$

Assuming  $B_1$  holds in  $M$ , then

$$a' = \bar{a} \text{ satisfies } a' \bullet a = \bar{e} \text{ and } \bar{a'} \bullet \bar{a} = \bar{e} \text{ by } R_3 \text{ and } B_2$$

Hence  $\bar{a}$  is a compliment of  $a$ .

$\bar{a}$  is a unique compliment of  $a$  if,

$$\text{for } b \in M, b \bullet a = \bar{e} \text{ and } \bar{b} \bullet \bar{a} = \bar{e} \text{ if and only if } b = \bar{a} \quad (3.2)$$

(3.2) implies the condition:

$$\text{for all } a, b \in M \text{ if } \bar{b} \bullet a = \bar{e} \text{ and } b \bullet \bar{a} = \bar{e} \text{ then } b = a \quad (3.3)$$

This suggest that in a reliability algebra having a unique compliments, one can define a relation  $\geq$  on  $M$  by

$$a \geq b \text{ if and only if } \bar{a} \bullet b = \bar{e} \quad (3.4)$$

Statement (3.4) is equivalent to condition that  $\geq$  is anti-symmetric.

One has the following statement about unique compliments.

Theorem 6 If  $M$  is a reliability algebra in which every element is idempotent ( $B_3$ ),  $\bar{e} \bullet a = \bar{e}$  ( $B_2$ ),  $a \bullet \bar{a} = \bar{e}$  holds for all  $a \in M$  ( $B_1$ ) and  $M$ , satisfies the cancellation law ( $B_5$ ), then every member  $a$  in  $M$  has unique compliment  $\bar{a}$ .

Suppose  $b$  is a member in  $M$  such that

$$b \bullet a = \bar{e} \text{ and } \bar{b} \bullet \bar{a} = \bar{e}$$

Using  $b \bullet a = \bar{e}$  and  $\bar{b} \bullet \bar{a} = \bar{e}$  we will first show that  $\bar{b} \bullet a = a$

This follows by  $B_1$ ,  $B_2$ ,  $B_3$  and  $B_5$ .

$$\begin{aligned} b \bullet \bar{b} \bullet a &= \bar{e} \bullet a \text{ by } B_1 \quad \text{and} \quad b \bullet a = \bar{e} \text{ by assumption} \\ &= \bar{e} \text{ by } B_2 \end{aligned}$$

$$\bar{b} \bullet \bar{b} \bullet a = \bar{b} \bullet a \text{ by } B_3 \quad \text{and} \quad \bar{b} \bullet a = \bar{b} \bullet a$$

$$\text{So} \quad \bar{b} \bullet a = a \text{ by } B_5$$

Next we show that  $\bar{b} = a$

$$a \bullet \bar{b} = a \quad \text{and} \quad a \bullet a = a \text{ by } B_3$$

$$\bar{a} \bullet \bar{b} = \bar{e} \quad \text{and} \quad \bar{a} \bullet a = \bar{e} \text{ by } B_1$$

$$\text{So} \quad \bar{b} = a \text{ by } B_5$$

$$\therefore \quad b = \bar{a} \quad \text{and} \quad \bar{a} \text{ is the unique compliment of } a.$$

Theorem 7 If  $M$  is a reliability algebra in which every element is idempotent ( $B_3$ ),  $\bar{e} \bullet a = \bar{e}$  ( $B_2$ ),  $a \bullet \bar{a} = \bar{e}$  holds for all  $a \in M$  ( $B_1$ ) and  $M$  satisfies the cancellation law ( $B_5$ ), then there exists a partial ordering  $\geq$  which is congruent with respect to  $\bullet$  ( $O_1$ ), order reversing ( $O_2$ ), has largest element  $e$  and smallest element  $\bar{e}$ .

Let relation  $\geq$  on reliability algebra  $M$  be given by  $a \geq b$  if and only if  $\bar{a} \bullet b = \bar{e}$

$\geq$  is reflective by ( $B_1$ )

$\geq$  is anti-symmetric by Theorem 6 and statement (3.3)

To show  $\geq$  is transitive, assume  $a \geq b$  and  $b \geq c$ .

$$\text{So} \quad \bar{a} \bullet b = \bar{e} \text{ and } \bar{b} \bullet c = \bar{e}$$

We need to show that  $\bar{a} \bullet c = \bar{e}$

$$\begin{aligned} b \bullet \bar{e} &= \bar{e} \text{ by } B_2 \quad \text{and} \\ b \bullet \bar{a} \bullet c &= \bar{e} \bullet c \text{ by assumption} \\ &= \bar{e} \text{ by } B_2 \quad \text{and} \\ b \bullet \bar{e} &= \bar{e} \text{ by } B_2 \quad \text{and} \\ \bar{b} \bullet \bar{e} &= \bar{e} \quad \text{and} \\ \bar{b} \bullet \bar{a} \bullet c &= \bar{a} \bullet \bar{e} \text{ by assumption} \\ &= \bar{e} \text{ by } B_2 \\ \bar{b} \bullet \bar{e} &= \bar{e} \quad \text{and} \\ \bar{a} \bullet c &= \bar{e} \text{ by } B_5 \end{aligned}$$

$$\therefore \quad \geq \text{ is transitive}$$

To show that  $\geq$  is congruent, let  $a \geq b$  and  $c \in M$ .

We will first show that  $b \bullet c = a \bullet b \bullet c$ .

$$\begin{aligned} a \bullet b \bullet c &= a \bullet b \bullet c \quad \text{and} \\ a \bullet a \bullet b \bullet c &= a \bullet b \bullet c \text{ by } B_3 \\ \bar{a} \bullet b \bullet c &= \bar{e} \bullet c \text{ by assumption} \\ &= \bar{e} \text{ by } B_2 \quad \text{and} \\ \bar{a} \bullet a \bullet b \bullet c &= \bar{e} \bullet b \bullet c \text{ by } B_1 \quad \text{and} \\ &= \bar{e} \text{ by } B_2 \\ \therefore \quad b \bullet c &= a \bullet b \bullet c \text{ by } B_5 \end{aligned}$$

By  $b \bullet c = a \bullet b \bullet c$ ,

$$\begin{aligned} \overline{a \bullet c} \bullet b \bullet c &= \overline{a \bullet c} \bullet a \bullet b \bullet c \\ &= \bar{e} \bullet b \text{ by } B_1 \\ &= \bar{e} \text{ by } B_2 \\ \therefore \quad a \bullet c &\geq b \bullet c \text{ and } \geq \text{ is congruent.} \end{aligned}$$

To show  $\geq$  is an ordering reversing, assume  $a \geq b$

$$\begin{aligned}\bar{a} \bullet b &= \bar{e} \\ \bar{\bar{b}} \bullet \bar{a} &= \bar{e} \\ \therefore \quad \bar{b} &\geq \bar{a} \text{ and } \geq \text{ is an ordering reversing.}\end{aligned}$$

That  $e \geq a$  and  $a \geq \bar{e}$  holds for all  $a \in M$  follows from  $\bar{e} \bullet a = \bar{e}$  and  $a \bullet \bar{e} = \bar{e}$  for all  $a \in M$  ( $B_2$ ). So  $e$  is the largest and  $\bar{e}$  is the smallest member in  $M$ .

We now intend to show that reliability algebra  $M$  satisfying: every element is idempotent ( $B_3$ ),  $\bar{e} \bullet a = \bar{e}$  ( $B_2$ ),  $a \bullet \bar{a} = \bar{e}$  holds for all  $a \in M$  ( $B_1$ ) and the cancellation law ( $B_5$ ) is a Boolean algebra.

Lemma 1 If  $M$  is a reliability algebra satisfying  $B_1$ ,  $B_2$ ,  $B_3$  and  $B_5$  then  $B_4$  holds.

By Theorem 7,  $M$  has defined on it a partial ordering  $\geq$  which is congruent, order reversing and has largest member  $e$ . By Theorem 5,  $\geq$  satisfies  $B_4$ .

Lemma 2 If  $M$  is a reliability algebra satisfying  $B_1$ ,  $B_2$ ,  $B_3$  and  $B_5$ , then

$$a \bullet \overline{a \bullet b} = a \bullet \bar{b} \text{ for all } a, b \in M$$

Using the partial ordering  $\geq$  on  $M$  given by  $a \geq b$  if and only if  $\bar{a} \bullet b = \bar{e}$  as defined in Theorem 7, we have  $e \geq \overline{a \bullet b}$  by Theorem 7. So

$$\begin{aligned}a \bullet e &\geq a \bullet \overline{a \bullet b} \text{ by } O_1 \\ a &\geq a \bullet \overline{a \bullet b}\end{aligned}$$

To show that  $\bar{b} \geq a \bullet \overline{a \bullet b}$ , note that  $b \bullet a \bullet \overline{a \bullet b} = \bar{e}$  by  $B_1$ . So

$$\bar{b} \geq a \bullet \overline{a \bullet b}$$

By transitivity and congruence of  $\geq$ ,

$$\begin{aligned}a \bullet \bar{b} &\geq a \bullet \overline{a \bullet b} \bullet a \bullet \overline{a \bullet b}. \\ \text{Hence } a \bullet \bar{b} &\geq a \bullet \overline{a \bullet b} \text{ by } B_3 \\ \text{By } b &\geq a \bullet b \\ \text{So } \overline{a \bullet b} &\geq \bar{b} \text{ by } O_2 \\ \text{Hence } a \bullet \overline{a \bullet b} &\geq a \bullet \bar{b} \text{ by } O_1 \\ \therefore a \bullet \overline{a \bullet b} &= a \bullet \bar{b}\end{aligned}$$

Lemma 3 If  $M$  is a reliability algebra satisfying  $B_1$ ,  $B_2$ ,  $B_3$  and  $B_5$  then  $B_6$  holds.

A statement of the distributive law ( $B_6$ ) in terms of  $\bullet$  is:  $a \bullet \bar{b} \bullet \bar{c} = \overline{a \bullet b \bullet a \bullet c}$ .

$$\begin{aligned}
(a \bullet b) \bullet a \bullet \overline{\overline{b} \bullet \overline{c}} &= a \bullet a \bullet b \bullet \overline{\overline{b} \bullet \overline{c}} \\
&= a \bullet a \bullet b \text{ by Lemma 1} \\
&= a \bullet b \text{ by } B_3 \\
(a \bullet b) \bullet \overline{\overline{a \bullet b \bullet a \bullet c}} &= a \bullet b \text{ by Lemma 1} \\
(\overline{a \bullet b}) \bullet a \bullet \overline{\overline{b} \bullet \overline{c}} &= a \bullet \overline{b} \bullet \overline{\overline{b} \bullet \overline{c}} \text{ by Lemma 2} \\
&= a \bullet \overline{b} \bullet c \text{ by Lemma 1} \\
(\overline{a \bullet b}) \bullet \overline{\overline{a \bullet b \bullet a \bullet c}} &= \overline{a \bullet b} \bullet a \bullet c \text{ by Lemma 2} \\
&= a \bullet \overline{b} \bullet c \text{ by Lemma 2} \\
\therefore a \bullet \overline{\overline{b} \bullet \overline{c}} &= \overline{\overline{a \bullet b \bullet a \bullet c}} \text{ by } B_5
\end{aligned}$$

**Theorem 8** *If  $M$  is a reliability algebra in which every element is idempotent ( $B_3$ ),  $\overline{e} \bullet a = \overline{e}$  ( $B_2$ ),  $a \bullet \overline{a} = \overline{e}$  holds for all  $a \in M$  ( $B_1$ ) and  $M$  satisfies the cancellation law ( $B_5$ ), then  $M$  is a Boolean algebra.*

By Theorem 5,  $B_1$ ,  $B_2$ ,  $B_3$  and  $B_5$  imply  $B_4$ . By Lemma 3,  $B_1$ ,  $B_2$ ,  $B_3$  and  $B_5$  imply  $B_6$ . Therefore  $M$  is a Boolean algebra.

We can restate Theorem 8 in the following form.

**Theorem 9** *Let  $M$  be a commutative monoid with binary operation  $\bullet$  and unary operation  $\overline{(\ )}$  which satisfies:*

1.  $\overline{\overline{a}} = a$  for all  $a \in M$
2.  $\overline{e} \bullet a = \overline{e}$  for all  $a \in M$ , where  $e$  is the identity member in  $M$
3.  $a \bullet \overline{a} = \overline{e}$  for all  $a \in M$
4.  $a \bullet a = a$  for all  $a \in M$
5. If there exists  $a \in M$  such that  $a \bullet x = a \bullet y$  and  $\overline{a} \bullet x = \overline{a} \bullet y$  then  $x = y$ .

Then  $(M, \bullet)$  generates a Boolean algebra with second operation  $\oplus$  on  $M$  defined by

$$a \oplus b = \overline{\overline{a} \bullet \overline{b}} \text{ for all } a, b \in M$$

## 4 Another Example of a Reliability Algebra

The original Reliability algebra we introduced  $[0,1]$  satisfies only properties (1), (2) and (5) of Theorem 8. It is not a lattice.

Following is an example of a Reliability algebra which satisfies all properties required to generate Boolean algebra except for the cancellation law. It is, however, a lattice.

**Example 4.** Let  $V$  be any finite dimensional vector space,  $L(V)$  the set of all subspaces of  $V$ ,  $\bullet$  the intersection operation and  $\perp$  the orthogonal complement operation in  $L(V)$ .  $L(V)$  is a monoid under operation  $\bullet$  with identity member  $V$  and the following also holds in  $L(V)$ .

1.  $(A^\perp)^\perp = A$  for all  $A \in L(V)$ .
2.  $V^\perp \bullet A = V^\perp$  for all  $A \in L(V)$  since  $V^\perp = \{0\}$ .



$$3. A \bullet A^\perp = V^\perp \text{ for all } A \in L(V).$$

$$4. A \bullet A = A \text{ for all } A \in L(V).$$

$$A \oplus B = (A^\perp \bullet B^\perp)^\perp \text{ can be shown to be the following.}$$

$$A \oplus B = A + B, \text{ the sum of the two vector subspaces } A \text{ and } B.$$

The Reliability algebra  $L(V)$  does not satisfy the cancellation law, so it is not a Boolean algebra. It is also easy to find subspaces of  $V$  which do not satisfy the distributive law.

$L(V)$  can be generalized to be the set of all closed subspaces of a Hilbert space  $V$ .

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## Chapter 7

# STABILIZATION VIA PROJECTION\*

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### Abstract

The solution of unstable linear inverse problems inevitably involves the evaluation of an unbounded operator. In this note we investigate a stable approximate evaluation scheme for closed unbounded linear Hilbert space operators that is based on projection onto finite-dimensional subspaces with a “duality twist”.

**Key Words:** inverse problem, ill-posed problem, unbounded operators, stabilization.

**AMS Subject Classification:** Primary 65J20, 47A52; Secondary 47A58.

## 1 Introduction

Myriad linear inverse problems in mathematical physics and technology may be posed as finding a solution  $f$  of a linear operator equation of the first kind

$$Kf = g \tag{1.1}$$

(see e.g. [11], [6], [2], [10]). In technical applications the operator  $K$  typically represents the action of an apparatus “point spread” function that produces an imperfect representation  $g$  of the desired signal  $f$  (the unattainable ideal of perfect measurement would correspond to the identity operator, for example, an integral operator with a “delta function” kernel). In inverse problems in mathematical physics, the operator  $K$  often models some evolutionary process that transforms a detail-rich “object”  $f$  into a much smoother “image”  $g$ . In either context,  $K$  is typically a highly smoothing operator, which in the conventional setting of operators acting on Hilbert space translates into saying that  $K$  is a *compact* linear operator.

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\*This paper is dedicated to the people of Texas in recognition of their extraordinary generosity to the victims of hurricane Katrina.

<sup>†</sup>E-mail address: groetsch@uc.edu; This work was supported in part by the Charles Phelps Taft Research Center. I am indebted to Martin Hanke of the University of Mainz for a helpful conversation.

If  $K : H_1 \rightarrow H_2$  is a bounded linear operator from a Hilbert space  $H_1$  into a Hilbert space  $H_2$ , then the generally recognized solution of (1.1) is the minimal norm least squares solution, that is the vector  $f \in N(K)^\perp$  satisfying

$$K^* K f = K^* g$$

where  $K^*$  is the adjoint of the operator  $K$  and  $N(K)^\perp$  is the orthogonal complement of  $N(K)$ , the nullspace of  $K$ . Such a least squares solution exists if and only if  $g \in R(K) + R(K)^\perp$ , where  $R(K)$  is the range of the operator  $K$ , and the minimal norm least squares solution is always unique when it exists. The operator

$$K^\dagger : \mathcal{D}(K^\dagger) := R(K) + R(K)^\perp \rightarrow N(K)^\perp$$

which maps  $g$  to the minimal norm least squares solution  $K^\dagger g$  is called the Moore-Penrose generalized inverse of  $K$ . The Moore-Penrose generalized inverse is a closed, densely defined linear operator which is bounded if and only if  $R(K)$  is closed (see e.g. [6], [2]). For compact operators  $K$  (e.g., integral operators with square integrable kernels acting on a space of square integrable functions), the range  $R(K)$  is closed if and only if  $R(K)$  is finite dimensional, which is very seldom the case in models of linear inverse problems. For such operators  $K$  the Moore-Penrose generalized inverse  $K^\dagger$  is unbounded, leading to the inevitable consequence that the solution process  $g \mapsto K^\dagger g$  is *unstable*, that is, vanishingly small perturbations in  $g$  can lead to unbounded perturbations in  $K^\dagger g$ .

In some linear inverse problems the forward operator may be inverted thereby giving the solution of the inverse problem explicitly as the value of some linear operator  $L$  applied to the “data”  $g$ :

$$f = Lg \tag{1.2}$$

(for example, in the abstract formulation  $L = K^\dagger$ ). Such is the case for the inversion of Abel and other integral transforms (see, e.g., [13]) and for explicit inversion formulae for various tomographic transforms (see, e.g., [3]). For ill-posed inverse problems the considerations of the first paragraph suggest that the operator  $L$  will be unbounded and consequently the evaluation of the operator will be an unstable process.

In this paper we focus on the stability problem, namely we isolate the problem of approximately evaluating a closed unbounded operator in such a manner that each individual approximation is stable with respect to perturbations in the data. Given an unbounded densely defined linear operator  $L : \mathcal{D}(L) \subseteq H_2 \rightarrow H_1$  and a vector  $g \in \mathcal{D}(L)$  the problem is to construct a stable approximation to  $Lg$  given approximate data  $g^\delta$  satisfying  $\|g - g^\delta\| \leq \delta$ , where  $\delta$  is a given error bound for the approximate data. We note that it will not do to take  $Lg^\delta$  as an approximation to  $Lg$ , for it may happen that  $g^\delta \notin \mathcal{D}(L)$ . Furthermore, even if  $g^\delta \in \mathcal{D}(L)$  for all  $\delta$ , one can not be assured that  $Lg^\delta \rightarrow Lg$  as  $\delta \rightarrow 0$ , as  $L$  is unbounded, and hence discontinuous. We study certain stabilized approximations to  $Lg$  of the form  $L_m g^\delta$  where  $L_m$  is a bounded, indeed a finite rank, operator obtained from  $L$  by a projection process. Since the operator  $L_m$  is bounded it follows that the mapping  $g^\delta \mapsto L_m g^\delta$  gives a stable approximation to  $Lg$ . The final ingredients in the analysis are an error-dependent rank choice  $m = m(\delta)$  and the identification of conditions that insure  $L_{m(\delta)} g^\delta \rightarrow Lg$  as  $\delta \rightarrow 0$ . This will involve relating the error level  $\delta$  to the approximation properties of the finite dimensional approximation subspaces. Before proceeding with

the development we present an example model problem and review some known results on projective approximations for the problem (1).

## 2 An Illustrative Example

We illustrate the two abstract formulations of the direct problem (1) and the inverse problem (2) on an elementary model problem for the one-dimensional heat equation. Suppose  $f$  is a stationary source function in the heat equation

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + f(x), \quad 0 < x < \pi, \quad 0 < t$$

with boundary conditions

$$u(0, t) = u(\pi, t) = 0,$$

and initial condition

$$u(x, 0) = 0 \quad \text{for } 0 \leq x \leq \pi.$$

The source term  $f$  will then drive the temperature to a distribution

$$g(x) = u(x, 1)$$

at the later time  $t = 1$ . We consider the inverse problem of determining the source term  $f$  given the later temperature distribution  $g$ .

It is easy to see that a purely spatial particular solution of the nonhomogeneous partial differential equation, which also satisfies the boundary conditions is

$$\begin{aligned} y(x) &= \frac{x}{\pi} \int_0^\pi \int_0^s f(\eta) d\eta ds - \int_0^x \int_0^s f(\eta) d\eta ds \\ &= \int_0^\pi h(x, \eta) f(\eta) d\eta = Hf(x) \end{aligned}$$

where

$$h(x, \eta) = \frac{x(\pi - \eta)}{\pi} - (x - \eta)_+$$

and

$$(x - \eta)_+ = \begin{cases} x - \eta & \text{if } 0 \leq \eta \leq x \\ 0 & \text{if } x \leq \eta \leq \pi \end{cases}$$

Note that the kernel  $h$  generates a compact linear operator  $H$  on  $L^2[0, \pi]$ .

The general solution of the nonhomogeneous heat problem has the form

$$u(x, t) = \sum_{n=1}^{\infty} b_n \exp(-n^2 t) \sin nx + y(x).$$

where the initial condition requires that

$$b_n = \frac{-2}{\pi} \int_0^\pi y(s) \sin ns ds, \quad n = 1, 2, \dots \quad (2.1)$$

and hence

$$g(x) = u(x, 1) = \int_0^\pi m(x, s)y(s)ds + y(x) = (M + I)y(x)$$

where

$$m(x, s) = -\frac{2}{\pi} \sum_{n=1}^{\infty} e^{-n^2} \sin nx \sin ns,$$

and  $M$  is the compact operator generated by the kernel  $m(x, s)$  and  $I$  is the identity operator. It then follows that the inverse problem of determining the forcing function  $f$  takes the form

$$Kf = g$$

where the operator  $K := MH + H$  is compact.

Note that the function  $y$  defined above satisfies  $y'' = -f$ , and hence applying integration by parts to (2.1) leads to

$$-n^2 b_n = \frac{2}{\pi} \int_0^\pi f(s) \sin ns ds.$$

Therefore,

$$f(x) = \sum_{n=1}^{\infty} (-n^2 b_n) \sin nx. \quad (2.2)$$

However,

$$\begin{aligned} g(x) = u(x, 1) &= \sum_{n=1}^{\infty} b_n e^{-n^2} \sin nx + y(x) \\ &= \sum_{n=1}^{\infty} \left\{ b_n e^{-n^2} + \frac{2}{\pi} \int_0^\pi y(s) \sin ns ds \right\} \sin nx \\ &= \sum_{n=1}^{\infty} (b_n e^{-n^2} - b_n) \sin nx, \quad \text{by (2.1),} \end{aligned}$$

and hence

$$b_n(e^{-n^2} - 1) = \frac{2}{\pi} \int_0^\pi g(s) \sin ns ds.$$

By (2.2) one is then led to the explicit representation

$$f(x) = Lg(x) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{n^2}{1 - e^{-n^2}} \sin nx \int_0^\pi g(s) \sin ns ds.$$

That is,  $f = Lg$ , where  $L$  is the linear operator on  $L^2[0, \pi]$  with domain

$$\mathcal{D}(L) = \left\{ g : \sum_{m=1}^{\infty} m^4 a_m^2 < \infty, \quad a_m = \frac{2}{\pi} \int_0^\pi g(s) \sin ms ds \right\}.$$

We note that  $\mathcal{D}(L)$  is dense in  $L^2[0, \pi]$  since all finite linear combinations of the eigenfunctions  $\varphi_n(s) = \sin ns$  are contained in  $\mathcal{D}(L)$ . Moreover,  $L$  is unbounded since  $\|\varphi_n\|^2 = \pi/2$  while

$$\|L\varphi_n\|^2 = \frac{\pi n^2}{2(1 - e^{-n^2})} \rightarrow \infty$$

as  $n \rightarrow \infty$ .

### 3 Projective Regularization

In this section we briefly review some results on approximating a solution of (1) by use of finite rank operators. The analysis of regular approximations to the solution of (1) obtained by solving the corresponding problem on finite dimensional subspaces goes back to Natterer (although he points out that the idea is “keineswegs neu”, see [14]; [16] contains another early suggestion along these lines). Suppose

$$U_1 \subseteq U_2 \subseteq \dots \subseteq U_m \subseteq H_1$$

are finite-dimensional subspaces of  $H_1$  satisfying

$$\overline{\bigcup_{m=1}^{\infty} U_m} = H_1.$$

Let  $K_m = K|_{U_m}$  be the restriction of  $K$  to  $U_m$  and define  $f_m = K_m^\dagger g$  where  $g \in H_2$ . Since  $K_m$  has finite rank, the operator  $K_m^\dagger$  is bounded and hence the computation of  $f_m$  is stable with respect to perturbations in  $g$ . However, Seidman [18] has shown by example that, even in the case of error free data  $g \in \mathcal{D}(K^\dagger)$ , it does not necessarily follow that  $f_m \rightarrow K^\dagger g$  as  $m \rightarrow \infty$ . In [4] it is shown that a necessary and sufficient condition for the convergence of  $f_m$  to  $K^\dagger g$  for each  $g \in \mathcal{D}(K^\dagger)$  is that the sequence  $\{\|R_m\|\}$  be uniformly bounded, where  $R_m$  is the (generally non-orthogonal) projector defined by  $R_m = K_m^\dagger Q_m K$ , and  $Q_m$  is the orthogonal projector of  $H_2$  onto  $R(K_m)$  (see also [12] and [8] for other results along these lines). Under these conditions, if the subspace index  $m = m(\delta)$  satisfies  $\|K_{m(\delta)}^\dagger\| \delta \rightarrow 0$  as  $\delta \rightarrow 0$ , then the finite rank approximations are *regular* in the sense that

$$f_{m(\delta)} = K_{m(\delta)}^\dagger g^\delta \rightarrow f = K^\dagger g \quad \text{as } \delta \rightarrow 0.$$

In 1976 Seidman [17] pointed out that a convergent finite rank scheme may be obtained by choosing the finite-dimensional subspaces in the form  $U_m = K^*(V_m)$ , where  $V_m$  are finite-dimensional subspaces of  $N(K^*)^\perp$ . The approximations take the form

$$f_m = \tilde{K}_m^\dagger r_m g$$

where  $r_m : H_2 \rightarrow \mathbb{R}^m$  is the operator defined by

$$(r_m g)_j = \langle g, u_j \rangle \quad j = 1, \dots, m$$

where  $\{u_1, \dots, u_m\}$  is a basis for  $U_m$  and  $\tilde{K}_m := r_m K$ . Engl [1] proved a regularity theorem for this method which has become known as the dual least squares method (see [2], p. 67).

In the next section we concentrate on the *stabilization* problem, that is, the problem of producing computable stable approximations to  $Lg$  where  $L$  is a given closed unbounded linear operator corresponding, for example, to solving an inverse problem in the form (2). In particular we review some known results on projective stabilization and we present a stability result that is closely akin to the dual least squares regularization approach.

## 4 Projective Stabilization

Suppose  $L : \mathcal{D}(L) \subseteq H_2 \rightarrow H_1$  is a closed densely defined unbounded linear operator. In [7] approximations to  $Lg$ , where  $g \in \mathcal{D}(L)$ , of the form  $LQ_m g^\delta$  are investigated. Here  $g^\delta \in H_2$  is an approximation to  $g$  satisfying  $\|g - g^\delta\| \leq \delta$  and  $Q_m$  is a (not necessarily orthogonal) projector onto a finite-dimensional subspace  $W_m$  of  $\mathcal{D}(L) \cap N(L)^\perp$ . It is assumed that the subspaces satisfy

$$W_1 \subseteq W_2 \subseteq \dots \subseteq W_m \quad \text{and} \quad H_2 = \overline{\bigcup_{m=1}^{\infty} W_m}$$

and that  $LQ_m g \rightarrow Lg$  as  $m \rightarrow \infty$  for each  $g \in \mathcal{D}(L)$ . Such is the case, for example, if  $L$  is self-adjoint and possesses a complete system of eigenfunctions and  $Q_m$  is the orthogonal projector onto the  $m$ th eigenspace. Another example is provided by the first derivative operator defined on the space of  $C^2$  functions where  $Q_m$  is the cubic spline interpolation operator (see e.g., [15]). The condition

$$\text{dist}(g, W_m) \|LQ_m\| \rightarrow 0 \quad \text{as} \quad m \rightarrow \infty,$$

which is satisfied in each of the examples just cited, is also assumed. The basic stability result established in [7] is:

If  $m = m(\delta) \rightarrow \infty$  and  $\|LQ_{m(\delta)}\| \delta \rightarrow 0$  as  $\delta \rightarrow 0$ , then  $LQ_{m(\delta)} g^\delta \rightarrow Lg$ .

An a posteriori method for choosing the subspace index  $m(\delta)$  is also investigated in [7]. Namely, given  $\tau > 1$  and  $\gamma_m \geq \|I - Q_m\|$  it is shown that there is a first index  $m = m(\delta)$  satisfying

$$\|g^\delta - Q_m g^\delta\| \leq \tau \gamma_m \delta$$

and for this choice

$$LQ_{m(\delta)} g^\delta \rightarrow Lg \quad \text{as} \quad \delta \rightarrow 0.$$

Also, if  $g \in \mathcal{D}((L^*L)^{\nu+1})$  for some  $\nu > 0$  and if  $L^*L$  has compact resolvent, then

$$\|LQ_{m(\delta)} g^\delta - Lg\| = O(\delta^{(2\nu+1)/(2\nu+2)}).$$

We now treat a different projection scheme which relies on choosing basis functions in a special subspace. This method relies on projection, with a 'duality twist', onto a finite-dimensional subspace of  $\mathcal{D}(L^*)$ , the domain of the adjoint operator. In the previous method stabilization was achieved, under hypotheses that may be difficult to verify, by projecting the *data* onto a finite-dimensional subspace of the operator domain and then applying the operator. The method here in a sense reverses this procedure: the *value* of the operator is projected onto a finite-dimensional subspace of the domain of the adjoint and a weak form of the equations that characterize this projection is used to extend the technique to data that are not necessarily in the domain of the operator.

Since  $L : \mathcal{D}(L) \subseteq H_2 \rightarrow H_1$  is densely defined, the adjoint operator  $L^*$  has a domain  $\mathcal{D}(L^*)$  which is dense in  $H_1$ . Suppose that for each positive integer  $m$ , the set

$$\{l_1^{(m)}, l_2^{(m)}, \dots, l_{n(m)}^{(m)}\}$$

consists of linearly independent vectors in  $\mathcal{D}(L^*)$  and let

$$V_m = \text{span}\{l_1^{(m)}, \dots, l_{n(m)}^{(m)}\}.$$

Denote the orthogonal projector of  $H_1$  onto  $V_m$  by  $P_m$ . We suppose that the finite dimensional subspaces ultimately fill  $H_1$  in the sense that

$$\overline{\bigcup_{m=1}^{\infty} V_m} = H_1$$

and so,  $P_m y \rightarrow y$  as  $m \rightarrow \infty$  for each  $y \in H_1$ . In particular,  $P_m Lg \rightarrow Lg$  as  $m \rightarrow \infty$  for each  $g \in \mathcal{D}(L)$ . The projection  $f_m := P_m Lg$  is characterized by the conditions  $f_m \in V_m$  and

$$\langle f_m, l_i^{(m)} \rangle = \langle Lg, l_i^{(m)} \rangle = \langle g, L^* l_i^{(m)} \rangle, \quad i = 1, \dots, n(m). \quad (4.1)$$

Note that while the formulation  $f_m = P_m Lg$  requires that  $g \in \mathcal{D}(L)$ , the conditions

$$\langle f_m, l_i^{(m)} \rangle = \langle g, L^* l_i^{(m)} \rangle, \quad i = 1, \dots, n(m)$$

have meaning for *any*  $g \in H_1$ . The approximation  $f_m$  has another operational characterization. Define operators  $B_m : H_2 \rightarrow \mathbb{R}^{n(m)}$  and  $I_m : \mathbb{R}^{n(m)} \rightarrow H_1$  by

$$B_m z = \begin{bmatrix} \langle z, L^* l_1^{(m)} \rangle \\ \vdots \\ \langle z, L^* l_{n(m)}^{(m)} \rangle \end{bmatrix}$$

and

$$I_m \mathbf{c} = \sum_{j=1}^{n(m)} c_j l_j^{(m)},$$

respectively, and let  $G_m : \mathbb{R}^{n(m)} \rightarrow \mathbb{R}^{n(m)}$  be the linear operator whose matrix representation relative to the standard basis is the Grammian matrix

$$[G_m]_{ij} = [\langle l_i^{n(m)}, l_j^{n(m)} \rangle].$$

If

$$f_m = \sum_{j=1}^{n(m)} c_j l_j^{(m)} = I_m \mathbf{c},$$

then by (4.1),  $G_m \mathbf{c} = B_m g$  and hence  $f_m = L_m g$ , where  $L_m := I_m G_m^{-1} B_m$ . Note that the operators  $L_m : H_2 \rightarrow H_1$  are bounded and defined on all of  $H_2$ . Evaluating these operators is therefore a stable process.

We now show that the operators  $L_m$  are stabilizers of the unbounded operator  $L$  when the index  $m$ , which acts as a stabilization parameter, is suitably matched with the approximating subspace  $V_m$  and the error level  $\delta$  in the data. The stabilization relies on relating the error level to the subspace  $V_m$  by way of the smallest eigenvalue  $\lambda_m$  of the matrix  $G_m$  and the parameter  $\|B_m\|$ .



**Theorem 4.1** *If  $n(m) = n(m(\delta)) \rightarrow \infty$  and  $\delta \|B_{m(\delta)}\| / \sqrt{\lambda_{m(\delta)}} \rightarrow 0$ , then*

$$\|Lg - L_{m(\delta)}g^\delta\| \rightarrow 0$$

as  $\delta \rightarrow 0$ , for each  $g \in \mathcal{D}(L)$ .

**Proof:** First note that

$$\|Lg - L_{m(\delta)}g\| = \|Lg - P_{m(\delta)}Lg\| \rightarrow 0, \quad \text{as } \delta \rightarrow 0. \quad (4.2)$$

It remains only to estimate  $\|L_m\| = \|I_m G_m^{-1} B_m\|$ . Since

$$L_m g = \sum_{j=1}^{n(m)} c_j l_j^{(m)}$$

where  $G_m \mathbf{c} = B_m g$  we have

$$\begin{aligned} \|L_m g\|^2 &= \langle I_m \mathbf{c}, I_m \mathbf{c} \rangle = \sum_{i,j} c_i c_j \langle l_i^{(m)}, l_j^{(m)} \rangle = \mathbf{c}^T G_m \mathbf{c} \\ &= (G_m^{-1} B_m g)^T B_m g \leq \|G_m^{-1}\| \|B_m\|^2 \|g\|^2 \end{aligned}$$

and hence

$$\|L_m\| \leq \sqrt{\|G_m^{-1}\| \|B_m\|} \quad (4.3)$$

Furthermore, by a well-known property of positive definite matrices,  $\|G_m^{-1}\| = \lambda_m^{-1}$ , where  $\lambda_m$  is the smallest eigenvalue of  $G_m$ . We then have the stability estimate

$$\|L_m g - L_m g^\delta\| \leq \delta \|L_m\| \leq \delta \|B_m\| / \sqrt{\lambda_m}$$

which with (4.2) proves the result.

The fact that  $\|(I - P_m)Lg\| \rightarrow 0$  as  $m \rightarrow \infty$  used in (4.2) above can be expressed in a more quantitative form given additional hypotheses on  $L$  and  $g$ . In fact, suppose that  $g$  possesses additional smoothness in the sense that  $g \in \mathcal{D}(L^*L)$ . Suppose further that  $L^{*\dagger}$  is compact. Let  $w = L^*Lg$ . Then it is easy to see that

$$Lg = L^{*\dagger} L^* Lg = L^{*\dagger} w$$

and hence

$$\|(I - P_m)Lg\| \leq \gamma_m \|w\|$$

where  $\gamma_m = \|(I - P_m)L^{*\dagger}\| \rightarrow 0$  as  $m \rightarrow \infty$ , since  $L^{*\dagger}$  is compact. Therefore  $\gamma_m$  provides a rate of convergence for  $\|(I - P_m)Lg\|$  in this case.

We now illustrate the above result for the inverse heat problem introduced earlier. Let  $l_k(s) = \sin ks$ , then the orthogonal projector of  $L^2[0, \pi]$  onto

$$V_m = \text{span}\{l_1, \dots, l_m\}$$

is

$$P_m \phi = \frac{2}{\pi} \sum_{n=1}^{\infty} \langle \phi, l_j \rangle l_j.$$

Also,  $G_m = \frac{\pi}{2}I$ , where  $I$  is the identity operator on  $\mathbb{R}^m$  and hence  $\lambda_m = \pi/2$ . Since  $L^*l_k = \frac{k^2}{1 - e^{-k^2}}l_k$ , we have

$$\|B_m g\|^2 = \sum_{k=1}^m \left( \frac{k^2}{1 - e^{-k^2}} \right)^2 |\langle g, l_k \rangle|^2 \leq O(m^4) \|g\|^2$$

and hence  $\|B_m\| = O(m^2)$ . Finally, since

$$Lg - P_m Lg = \frac{2}{\pi} \sum_{n>m} \frac{n^2}{1 - e^{-n^2}} \langle g, l_n \rangle l_n,$$

if we assume greater smoothness on  $g$  in the form  $g \in H_0^r[0, \pi]$  for  $r > 2$ , we have

$$\|Lg - P_m Lg\|^2 \leq C \sum_{n>m} (n^4 n^{-2r}) n^{2r} |\langle g, l_n \rangle|^2 \leq C m^{4-2r} \|g\|_{H^r}^2.$$

If  $g^\delta \in L^2[0, 1]$  satisfies  $\|g - g^\delta\| \leq \delta$  and  $g \in H_0^{2+\nu}[0, \pi]$  for some  $\nu > 0$ , then a choice of cut-off level of the form  $m \sim \delta^{\frac{-1}{2(\nu+1)}}$  yields (on setting  $r = 2 + \nu$ )

$$\begin{aligned} \|Lg - L_m g^\delta\| &\leq \|Lg - L_{m(\delta)} g\| + \|L_{m(\delta)} g - L_{m(\delta)} g^\delta\| \\ &= O(m^{-2\nu}) + \delta O(m^2) \\ &= O(\delta^{\frac{\nu}{\nu+1}}) \end{aligned}$$

and hence an order of approximation arbitrarily near to the optimal order  $O(\delta)$  is achievable in principle for sufficiently smooth data. On the other hand, the Tikhonov-Morozov stabilizer can not achieve a rate better than  $O(\delta^{2/3})$  regardless of the order of smoothness of the true data [9].

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**Chapter 8****FAST SCAN CONVERSION ALGORITHM FOR CIRCLES*****Kam Kong\****Department of Computer Information Science  
Delaware State University, Dover, DE 19901**Abstract**

In this article, we present a very fast approximate scan conversion algorithm applicable to certain differentiable curves. Under favorable circumstances, the result of the algorithm is very close (not more than 1 pixel point-wisely) to the result generated by exact but slower algorithms. This fast approximate algorithm can be used for situation such as rubber-banding during interactive object manipulation on screen, where speed is more important than accuracy. We illustrate this approximate algorithm with circles. The result is a very fast and almost exact (the error is not more than 1 pixel point-wisely anywhere) scan conversion algorithm for circles.

**1 An Approximate Algorithm**

Scan-conversion algorithm for a curve computes the coordinates of the pixels that lie on or near the curve imposed on a 2D raster grid. In principle, we would like the sequence of pixels to lie as close to the curve as possible. In this article, we will consider differentiable curves for which the infinitesimal equation  $\Delta y = (dy/dx)\Delta x$  may be used as a good approximation for small increment  $\Delta x$ .

To trace the curve  $y = f(x)$  over the range  $[a, b]$  where  $a$  and  $b$  are integers: We start by setting  $x = a$ , and keep increasing the  $x$ -coordinate by the fixed amount  $\Delta x = 1$  until we reach  $x = b$ . During the process, we accumulate  $\Delta y$  in a variable `dySum` computed using the approximation  $\Delta y = (dy/dx)\Delta x$ . Whenever `dySum` is greater than 1, we increase the  $y$ -coordinate by the integral amount of `dySum`, and decrease `dySum` by the same amount (and thus 'resetting' `dySum`) at the same time. Similarly, whenever `dySum` is less than  $-1$ , we decrease the  $y$ -coordinate by the integral amount of `dySum`, and increase `dySum` by that

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amount at the same time. The procedure can be summarized by the following pseudocode ( $f_1$  is the derivative of  $f$ ):

```

1  // algorithm A
2  void draw(int a, int b) {
3      int y = (int)f(a);
4      double dySum = 0;
5      for (int x=a; x<=b; x++) {
6          plot(x, y);
7          dySum += f1(x, y);
8          if (dySumN >= 1) {
9              while (dySum >= 1) {
10                 y++;
11                 plot(x, y);
12                 dySum--;
13             }
14         } else if (dySumN <= -1) {
15             while (dySum <= -1) {
16                 y--;
17                 plot(x, y);
18                 dySum++;
19             }
20         }
21     }
22 }
```

It is clear that the validity of the algorithm depends on the validity of the approximation  $\Delta y = (dy/dx)\Delta x$ . In general, the error tends to be larger (i) if the absolute value of the derivative is large, or/and (ii) if we are far away from the starting point. We will not perform the mathematics of error analysis in its generality. Instead, we will demonstrate that the above approximate algorithm, when applied to certain kinds of curves, provides us a very fast and very good approximate scan conversion algorithm.

## 2 Favorable Conditions

We will now focus our discussion on how algorithm A can be coded more efficiently under favorable conditions.

*Favorable condition FC1:* If it is known that (such as when the absolute value of the derivative is small)  $dySumN$  is never more than twice the value of  $dySumD$  over the same  $x$ , then the 'while loops' in the above code can be replaced by a single loop (i.e., we can remove lines 9, 13, 15 and 19).

*Favorable condition FC2:* If it is known that the derivative is always positive over the whole range  $[a, b]$ , then the 'else branch' (lines 14-22) can be removed from the code. Similarly, if the derivative is always negative, then the 'then branch' can be removed.

*Favorable condition FC3:* For certain curves, the derivative  $dy/dx$  is a fractional expression  $g(x, y)/h(y)$ , where  $g$  is a continuous function in  $x$  and  $y$ ; and  $h$  is a never zero continue function in  $y$ . By a simple change of signs of  $g$  and  $h$ , we may further assume that  $h$  is always positive over the range  $[a, b]$ . To avoid division, we store  $dySum$  into two separate variables: the numerator in the variable  $dySumN$ , and the denominator in the variable  $dySumD$  so that  $dySum = dySumN/dySumD$  where  $dySumD$  is always positive. It is easy to see that

$$\frac{dySumN}{dySumD} > 1 \iff dySumN > dySumD$$

$$\frac{dySumN}{dySumD} < -1 \iff dySumN < -dySumD$$

Each time when  $\Delta x$  is increased by 1, we increase  $dySumN$  by  $g(x, y)$ . Note that we do not have to change  $dySumD$  because the denominator  $h(y)$  is independent of  $x$ . As we move along the curve, we also check the value of  $dySumN$  and  $dySumD$  constantly. Whenever  $dySumN > dySumD$ , we increase  $y$  and decrease  $dySumN$  by  $dySumD$  (thus effectively subtracting  $dySum$  by 1) at the same time. Similarly, whenever  $dySumN < -dySumD$ , we decrease  $y$  and increase  $dySumN$  by  $dySumD$  (thus effectively adding  $dySum$  by 1) at the same time. After  $|dySum|$  is reduced to less than 1, we re-normalize  $dySumN$  and  $dySumD$  using the new current  $y$ -coordinate:

Since

$$\frac{dySumN_{old}}{dySumD_{old}} = \frac{dySumN_{new}}{dySumD_{new}}$$

Hence:

$$dySumN_{new} = (dySumN_{old}) \frac{dySumD_{new}}{dySumD_{old}}$$

Finally, in terms of the variables  $dySumN$  and  $dySumD$ :

$$dySumN = (dySumN) \frac{updated\_dySumD}{dySumD},$$

where  $dySumN$  on the left hand side is the variable  $dySumN$  evaluated at updated position; and  $updated\_dySumD$  is the variable  $dySumD$  evaluated at updated position. This observation is utilized in the re-normalizing of  $dySumN$  and  $dySumD$  below (see lines 15, 16, 23 and 24).

Note that, using the popular C/Java programming language notation, the last formula is written as follows:

$$dySumN *= updated\_dySumD / dySumD.$$

```

1 // algorithm B
2 void draw(int a, int b) {
3     int y = (int)f(a);
4     double dySumN = 0;    // numerator of accumulated dy
```

```

5      double dySumD = h(y); // denominator of accumulated dy
6      for (x=a; x<=b; x++) {
7          plot(x, y);
8          dySumN += g(x, y);
9          if (dySumN >= dySumD) {
10             while (dySumN >= dySumD) {
11                 y++;
12                 plot(x, y);
13                 dySumN -= dySumD;
14             }
15             dySumN *= h(y)/dySumD;
16             dySumD = h(y);
17         } else if (dySumN <= -dySumD) {
18             while (dySumN <= -dySumD) {
19                 y--;
20                 plot(x, y);
21                 dySumN += dySumD;
22             }
23             dySumN *= h(y)/dySumD;
24             dySumD = h(y);
25         }
26     }
27 }

```

*Favorable condition FC31:* Under the favorable condition FC3, it may happen that the functions  $g$  and  $h$  are polynomials. Under such conditions, the variables `dySumN` and `dySumD` can be declared integer types. Hence the whole algorithm can be performed with integer arithmetic which tends to be faster than floating point arithmetic on most performs.

### 3 The Circles

We will now concentrate on the circles. Consider a circle of radius  $r$  centered at the origin:  $x^2 + y^2 = r^2$ . We will assume that the radius is a positive integer. Let  $A = (0, r)$  be the North pole, and  $B = (r/\sqrt{2}, r/\sqrt{2})$  be the point  $45^\circ$  to the East of  $A$ . Due to the 8-way (w.r.t orthogonal and diagonal axes) symmetry of circles, it is enough to concentrate on drawing the octant arc  $AB$  only. Our strategy is to start at  $A$ , where  $x = 0$ , and keep increasing the  $x$ -coordinate until we reach  $B$ , where  $x = y = r/\sqrt{2}$ . Since the derivative  $dy/dx$  is zero at the starting point  $A$ , and decreases monotonously until it reaches  $-1$  at the destination point  $B$ , the circles satisfy the favorable condition FC1.

The derivative  $\frac{dy}{dx} = -\frac{x}{y} \leq 0$  on the octant arc  $AB$ . Hence circles satisfy favorable condition FC2. Now the derivative can be expressed as a rational function:  $\frac{dy}{dx} = -\frac{x}{y} = -\frac{g(x, y)}{h(y)}$ , where  $g(x, y) = x$  and  $h(y) = y > 0$  are polynomials. Hence circles satisfy

favorable condition FC3 and FC31. We will denote the integral variables `dySumN` and `dySumD` by `N` and `D` below.

The range of  $x$  can be specified in the following two equivalent conditions: (i) from  $x = 0$  to  $x = r/\sqrt{2}$ , or (ii) from  $x = 0$  to  $x = y$ . It is obvious that the second condition is more convenient for our purpose. Hence we replace the 'for loop' (see line 6 of algorithm B above) with a 'while loop' (see line 19 of circle *algorithm C* below).

```

1  // plot 8 symmetric points on the circle
2  void plot8(int x, int y) {
3      plot( x,  y);
4      plot(-x, y);
5      plot( x, -y);
6      plot(-x, -y);
7      plot( y,  x);
8      plot( y, -x);
9      plot(-y,  x);
10     plot(-y, -x);
11 }
12
13 // algorithm C
14 void draw(int r) {
15     int x = 0;
16     int y = r;
17     int N = 0;
18     int D = r;
19     while (x <= y) {
20         plot8(x, y);
21         x++;
22         N += -x;
23         if (N <= -D) {
24             y--;
25             N += D;
26             N *= y/D;
27             D = y;
28         }
29     }
30 }
```

We will now make a bold assumption which will allow us to simplify the circle *algorithm C* further. By profiling the *algorithm C*, one can conclude that the effect of line 26 is to reduce `N` by 1 (most of the time). Hence we will replace line 26 by `N--`. It is also easy to see that such simplification renders the variable `D` useless. Hence our final version of the circle algorithm is as follows:



```
1  // algorithm D
2  void draw(int r) {
3      int x = 0;
4      int y = r;
5      int n = 0;
6      while (x <= y) {
7          plot8(x, y);
8          x++;
9          n += -x;
10         if (n <= -y) {
11             y--;
12             n += y-1;
13         }
14     }
15 }
```

## 4 Error Analysis and the Justification

There are several well-known scan conversion for circles. The following are the most common ones: the Midpoint Circle Algorithms, and the Bresenham's Circle Algorithm.

```
1  // floating midpoint circle algorithm
2  void draw(int r) {
3      int x = 0;
4      int y = r;
5      double n = 5.0/4.0-r;
6      plot8(x, y);
7      while (x < y) {
8          if (n < 0) {
9              n += 2.0*x+3.0;
10         } else {
11             n += 2.0*(x-y)+5.0;
12             y--;
13         }
14         x++;
15         plot8(x, y);
16     }
17 }
```

```
1  // integer midpoint circle algorithm
2  void draw(int r) {
3      int x = 0;
```

```
4      int y = r;
5      int n = 1-r;
6      plot8(x, y);
7      while (x < y) {
8          if (n < 0) {
9              n += 2*x+3;
10         } else {
11             n += 2*(x-y)+5;
12             y--;
13         }
14         x++;
15         plot8(x, y);
16     }
17 }
```

```
1  // second-order midpoint circle algorithm
2  void draw(int r) {
3      int x = 0;
4      int y = r;
5      int n = 1-r;
6      int dE = 3;
7      int dSE = 5-2*r;
8      plot8(x, y);
9      while (x < y) {
10         if (n < 0) {
11             n += dE;
12             dE += 2;
13             dSE += 2;
14         } else {
15             n += dSE;
16             dE += 2;
17             dSE += 4;
18             y--;
19         }
20         x++;
21         plot8(x, y);
22     }
23 }
```

```
1  // Bresenham's circle algorithm
2  void draw(int r) {
3      int x = 0;
4      int y = r;
```

```

5      int n = 3-2*r;
6      while (x <= y) {
7          plot8(x, y);
8          if (n <= 0) {
9              n += 4*x+6;
10         } else {
11             n += 4*(x-y)+10;
12             y--;
13         }
14         x++;
15     }
16 }

```

It should be noted that the above algorithms all involve multiplication, and involve more than one statements in the then and/or else branches inside the while loops. The *algorithm D* does not require multiplication/division. The while loop is very tight. It is clear that *algorithm D* is faster. But how accurate is the *algorithm D*? Since the algorithm is deterministic, and our primary goal is for drawing circles on computer screens of common sizes. We can therefore answer this question by actually testing the algorithm for all integral radius from  $r = 0$  to 5000 over a 45-degree-octant. For any fixed radius, the test consists of calculating all the  $y$ -coordinates of the pixels on the octant over  $x = 0$  to  $x = r/\sqrt{2}$  using both the exact method  $y = \sqrt{r^2 - x^2}$  and the proposed algorithm. The error at  $x$  is the vertical distance between two corresponding points on the exact octant and the approximate octant. The sign of the error changes over the octant

We find that the algorithm yields exact circles for  $r = 0, 1, 2, 3, 5, 6$ , and 10. For other circles, the error is never larger than 1 pixel anywhere on the octant. The average error varies from circles to circles, it is close to 0.1 pixel for most circles.

As mentioned before, the algorithm depends on the approximation  $\Delta y = (dy/dx)\Delta x$ . Recall the Taylor Series, which is applicable to the infinitely differentiable curve  $y = f(x) = \sqrt{r^2 - x^2}$  over the octant.

$$f(x) = f(c) + f'(c)(x - c) + f''(\zeta)(x - c)^2/2,$$

for some  $\zeta$  that depends on  $x$  and lies between  $c$  and  $x$ . In terms of  $\Delta x$  of  $\Delta y$ , the above formula can be written as:

$$\Delta y = f'(x)\Delta x + f''(\zeta)(\Delta x)^2/2.$$

for some  $\zeta$  that depends on  $x$  and lies between  $x$  and  $x + \Delta x$ . It is easy to show that  $f'(x) = -x/y$  and  $f''(x) = -r^2/y^2$  over the octant. Hence for  $\Delta x = 1$ :

$$\Delta y = -\frac{x}{y}(\Delta x) - \frac{r^2}{\mu^2}(\Delta x)^2 = -\frac{x}{y} - \frac{r^2}{\mu^2},$$

where  $\mu$  is between  $f(x + 1)$  and  $f(x)$  since  $f$  is monotonous decreasing over the octant. Based on the above equation, we arrive at the following conclusion:

The first approximation  $\Delta y = -\frac{x}{y}$  is an over-estimation: We need to subtract a little bit more (by the undetermined amount  $\frac{r^2}{\mu^2}$ ) to get the correct  $\Delta y$ .

Based on the above observation, it is natural to see what happen if we modify line 12 of *algorithm D* to  $n+=y-2$ . The algorithm tends to yield better approximations: The modified version will still generate exact circles for integral radius  $r = 0, 1, 2, 3, 5, 6$ , and 10. The maximum error is 1 pixel for almost all circles, and is 2 pixels for the rest of the circles (the first ones are  $r = 332, 431, 501, 571, \dots$ ). By actual enumeration, it can be verified that there are 0.4% of such circles of integral radius in the range  $r = [0, 500]$ ; 1.3% in the range  $[0, 1000]$ ; about 4% in the range  $[0, 5000]$ . The average error varies. It is close to 0.01 pixels for most circles (it tends to be smaller for larger circles). We will not pursue this issue any further, as incorporating such observation in the estimate of  $\Delta y$  would probably result a more complicated algorithm. Such action would defect the purpose of this article: We are investigating how to obtain a fast approximate algorithm to scan convert circles.

In order to compare the effect of  $K$  in line 13 of algorithm D, we calculate the average errors and plot the results on the same graph. The errors are calculated as follows:

For each integral  $x$ -coordinate between 0 and  $r/\sqrt{2}$ , we calculate the exact and approximate integral  $y$ -coordinates; the error at  $x$  is then defined by:

$$\text{ErrY}(x) = \text{ApproxY}(x) - \text{RoundedExactY}(x).$$

If the exact  $y$ -coordinate is half-way between two integers, we will round the exact  $y$  towards the approximate  $y$ . The sign of the error  $\text{ErrY}$  changes over the range of  $x$ . The following statistics provide two different measures of the fitness of the approximation:

Average signed error:

$$\text{AvgSigErr} = \text{sum\_of\_ErrY} / \text{range\_of\_x}.$$

Average absolute error:

$$\text{AvgAbsErr} = \text{sum\_of\_absolute\_value\_of\_ErrY} / \text{range\_of\_x}.$$

Below are the graphs of  $\text{AvgSigErr}$  and  $\text{AvgAbsErr}$  for  $K=2..-4$  over the range  $[0,5000]$ . Both graphs have the same scale. It should be pointed out that the best choice of  $K$  depends on the choice of error measure. In general, portions of the approximate circle rendered by algorithm D will overlap with the exact circle, portions will be inside the exact circle (under-shoot error), the remaining portions will be outside the exact circle (over-shoot error). Thus  $K=0$  is the best choice if we want the approximate circles to be as close to the exact circles as possible. On the other hand, if we allow over-shoot and under-shoot to cancel, then  $K=-2$  is the best choice.

We conclude that *algorithm D* (with  $K=0, 1$ , or  $-2$ ) is a very good approximation for circles of integral radius in the range  $[0,5000]$ . The error is never more than 1 pixel anywhere. For  $K=-2$ , the average signed error is in the order of 0.001 pixels. Therefore the algorithm can be used in situations such as rubber-banding in interactive selection of objects on the screen, where speed is more important than accuracy.

Since *algorithm D* is faster and the error is never more than 1 pixel anywhere. It can be used in situations such as rubber-banding in interactive selection of objects on the screen, where speed is more important than accuracy.

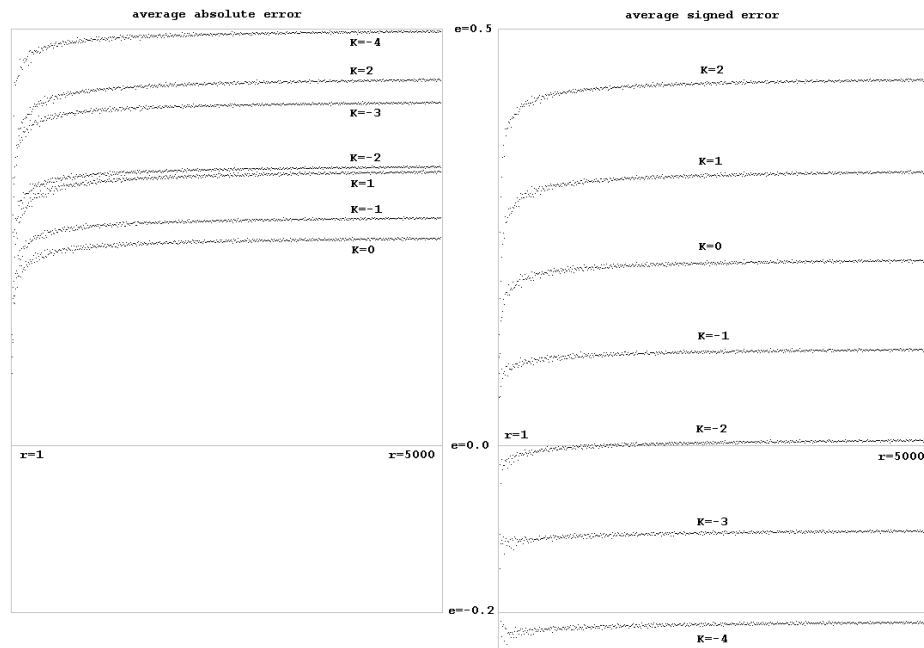


Figure 1:

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## Chapter 9

# INVERTING A MATRIX USING NEWTON'S METHOD

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### Abstract

The reciprocal of a number can be computed by Newton's Method. It has been shown by Ben Israel and later by Pan & Reif that the Newton's method can be used to invert a non-singular matrix provided a suitable initial approximate is chosen. In this paper, it is shown that a better initial approximate can be chosen when the eigenvalues of the matrix are positive. In general, the experiments reveal that the number of iterations required with the new approximate is fifty percent less compared to using the initial approximates suggested by those authors.

**Key Words:** Newton's Method.

**AMS Subject Classification:** Primary 43A60.

## 1 Introduction

Newton's method

$$X_{n+1} = 2X_n - X_n A X_n$$

converges to the inverse of a non-singular matrix  $A$ , provided a suitable initial approximate  $X_0$  is chosen. This problem has been solved by Ben-Israel [1] and later by Pan and Reif [2]. In this note we give a better initial approximate when the matrix  $A$  has real positive eigenvalues. With this new initial approximate the number of iterations required is almost half compared to that of the initial approximate suggested in [1] and [2]. Let  $A$  be a real matrix with positive eigenvalues. Let  $\{X_n\}$  denote the successive approximates given by the Newton's method with an initial approximate  $X_0$ . The spectral radius of a matrix  $A$  is

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denoted by  $\rho(A)$ . In the next section, we prove a lemma and the main theorem. In the last section we give results of experiments and symmetric and non-symmetric matrices using initial approximate suggested by Pan & Reif and us.

## 2 Main Result

First we prove the following lemma:

**Lemma 2.1.** *Let  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n > 0$  be the absolute values of the eigenvalues of a given matrix  $A$ . If*

$$p(x) = \frac{2}{\lambda_1^2 + \lambda_n^2} x^2 - 1$$

and

$$q(x) = \frac{2}{\tilde{\mu}_1 + \mu_n} (\lambda_1 + \lambda_n - x)x - 1$$

are polynomials where  $\mu_n = \lambda_1 \lambda_n$  and  $\tilde{\mu}_1$  is the maximum of the polynomial  $(\lambda_1 + \lambda_n - x)x$  on the interval  $[\lambda_n, \lambda_1]$  then  $\rho_p = \rho(p(A)) < 1$ . If all the eigenvalues of  $A$  are positive then  $\rho_q = \rho(q(A)) < 1$  and

$$\rho_q \leq \rho_p^2.$$

*Proof.* If  $\lambda_1 = \lambda_n$  then all the eigenvalues of  $p(A)$  are zero and obviously  $\rho_p < 1$ . If  $\lambda_1 \neq \lambda_n$  then since  $\lambda_1^2$  and  $\lambda_n^2$  are the largest and the smallest eigenvalues of  $A^2$ , we have for  $i = 1, 2, \dots, n$

$$0 < \frac{2\lambda_i^2}{\lambda_1^2 + \lambda_n^2} < 2$$

for  $i = 1, 2, \dots, n$ . Subtracting 1 from the above inequality, we have  $\rho_p < 1$ . If all the eigenvalues of  $A$  are positive, then it is easy to see that  $\mu_n$  is the smallest eigenvalue of the matrix  $S = ((\lambda_1 + \lambda_n)I - A)A$ . Let  $\mu_1$  be the largest eigenvalue of  $S$ . Since  $\mu_1 < \tilde{\mu}_1$ , we have

$$0 < \frac{2\mu_i}{\tilde{\mu}_1 + \mu_n} < 2$$

for each  $i = 1, 2, \dots, n$ . Again subtracting 1 from the above inequality we have  $\rho_q < 1$ . Since the polynomial  $p(x)$  is an increasing function of  $x$  on the interval  $[\lambda_n, \lambda_1]$  we have  $\rho_p = \max\{|p(\lambda_1)|, |p(\lambda_n)|\}$ . But  $|p(\lambda_1)| = |p(\lambda_n)|$  and so  $\rho_p = \left| \frac{\lambda_1^2 - \lambda_n^2}{\lambda_1^2 + \lambda_n^2} \right|$  or

$$\rho_p |\lambda_1^2 + \lambda_n^2| = |\lambda_1^2 - \lambda_n^2| \quad (1)$$

Since all the eigenvalues of  $S$  are positive and since  $-1 < \frac{\mu_i - \mu_n}{\tilde{\mu}_1 + \mu_n} < 1$  for all  $i$ , we obtain

$$\rho_q = \frac{\mu_1 - \mu_n}{\tilde{\mu}_1 + \mu_n}$$

and observing that  $\mu_1 < \tilde{\mu}_1$  we have

$$\rho_q < \frac{\tilde{\mu}_1 - \mu_n}{\tilde{\mu}_1 + \mu_n}$$

Thus it is enough to show that  $\frac{\tilde{\mu}_1 - \mu_n}{\tilde{\mu}_1 + \mu_n} < \rho_p^2$ . To show this, on the right hand side of the above inequality by replaying  $\tilde{\mu}_1$  by  $\frac{(\lambda_1 + \lambda_n)^2}{4}$  and  $\mu_n$  by  $\lambda_1 \lambda_n$ , and simplifying we obtain

$$\rho_q < \frac{(\lambda_1 - \lambda_n)^2}{((\lambda_1 + \lambda_n)^2 + 4\lambda_1 \lambda_n)}.$$

Now multiplying and dividing the right hand side of the above inequality by  $(\lambda_1 + \lambda_n)^2$  and using (1) we get

$$\begin{aligned} \rho_q &< \frac{\rho_p^2(\lambda_1^2 + \lambda_n^2)^2}{(\lambda_1^2 + \lambda_n^2 + 6\lambda_1 \lambda_n)(\lambda_1^2 + \lambda_n^2 + 2\lambda_1 \lambda_n)} \\ &= \frac{\rho_p^2}{\left(1 + \frac{6\lambda_1 \lambda_n}{\lambda_1^2 + \lambda_n^2}\right)\left(1 + \frac{2\lambda_1 \lambda_n}{\lambda_1^2 + \lambda_n^2}\right)}. \end{aligned}$$

Since each of factors in the denominator above is greater than 1 we have  $\rho_q < \rho_p^2$ . Hence the result.  $\square$

**Theorem 2.2.** Let  $\lambda_1 \geq \lambda_2 \geq \dots \lambda_n > 0$  be the absolute values of the eigenvalues of a real matrix  $A$ . Let

$$X_{n+1} = 2X_n - X_n^2 A, \text{ for } n = 1, 2, \dots \quad (2)$$

be the iterates given by the Newton's method where  $X_0$  commutes with  $A$  and  $\rho(X_0 A - I) < 1$ . Then the sequence  $\{X_n\}$  converges to  $A^{-1}$ . In particular let  $X_0 A - I = p(A)$  and if  $A$  has positive eigenvalues let  $X_0 A - I = q(A)$ , where  $p$  and  $q$  are polynomials as defined in the lemma, then the sequence  $\{X_n\}$  converges to  $A^{-1}$ .

**Proof.** Since  $X_0$  and  $A$  commute,  $X_n$  and  $A$  also commute for each  $n$ . Hence multiplying (2) by  $A$  we obtain

$$\begin{aligned} X_{n+1} A - I &= 2X_n A - X_n^2 A^2 - I \\ &= -(X_n A - I)^2. \end{aligned}$$

By induction it follows that

$$X_{n+1} A - I = (-1)(X_0 A - I)^{2^{n+1}}.$$

Since  $\rho(X_0 A - I) < 1$  we can choose  $\epsilon > 0$  such that  $\rho(X_0 A - I) + \epsilon < 1$ . For this  $\epsilon$  there exists a consistent matrix norm  $\|\cdot\|$  (see Stewart [3]) such that

$$\|X_0 A - I\| < \rho(X_0 A - I) + \epsilon.$$



Thus we have

$$\|X_{n+1}A - I\| < (\rho(X_0A - I) + \epsilon)^{2^{n+1}}.$$

Hence the sequence  $\{X_n\}$  converges to  $A^{-1}$ . Since  $\rho(p(A)) < 1$  and  $\rho(q(A)) < 1$  by the above lemma, the second part of the theorem follows.

### 3 Experiments

To generate a matrix with known eigenvalues we use the following method. First a triangular matrix is generated with the prescribed eigenvalues being the diagonal entries and other entries are randomly chosen. In our case the diagonal entries are chosen randomly strictly between 0.1 and 1. A random nonsingular matrix is chosen and the generated triangular matrix is premultiplied and postmultiplied by this nonsingular matrix and its inverse respectively. Let us denote this by  $A$ . Then both methods using  $p(A)$  and  $q(A)$  were applied to matrices of size  $10 \times 10$ ,  $15 \times 15$  assuming that smallest and largest eigenvalues are known. Similarly, the method was applied to diagonally dominant matrices. The diagonally dominant matrices of size  $25 \times 25$ ,  $50 \times 50$ ,  $100 \times 100$  were generated first by generating a random matrix and then replacing the diagonal elements by the largest of the sum of the absolute values of the row and the sum of the absolute values of the column of the element plus 0.1. Although these matrices have complex eigenvalues both methods converged. In the case of complex eigenvalues the smallest and largest real part of the eigenvalues were chosen in the intimal approximates. In all the cases we assumed that they converged when the maximum of the absolute values of the entries of  $X_nA - I$  is less than  $10^{-6}$ . Our method took about half the number of iterations compared to that of Ben-Israel and Pan & Reif. All the simulations were carried out using MATLAB. In the following tables, Method 1 and Method 2 correspond to  $p(A)$  and  $q(A)$  in the Lemma 2.1

| Experiment | $10 \times 10$    |          | $15 \times 15$    |          |
|------------|-------------------|----------|-------------------|----------|
|            | No. of Iterations |          | No. of Iterations |          |
|            | Method 1          | Method 2 | Method 1          | Method 2 |
| 1          | 10                | 5        | 9                 | 5        |
| 2          | 8                 | 4        | 10                | 5        |
| 3          | 9                 | 5        | 11                | 5        |
| 4          | 9                 | 5        | 9                 | 5        |
| 5          | 9                 | 5        | 10                | 6        |

Table 1. Randomly generated matrices with eigenvalues in the interval  $[0.1, 0.9]$

| Experiment | 25 × 25           |          | 50 × 50           |          | 100 × 100         |          |
|------------|-------------------|----------|-------------------|----------|-------------------|----------|
|            | No. of Iterations |          | No. of Iterations |          | No. of Iterations |          |
|            | Method 1          | Method 2 | Method 1          | Method 2 | Method 1          | Method 2 |
| 1          | 5                 | 3        | 5                 | 3        | 5                 | 3        |
| 2          | 5                 | 3        | 5                 | 3        | 5                 | 3        |
| 3          | 5                 | 3        | 5                 | 3        | 5                 | 3        |
| 4          | 5                 | 3        | 5                 | 3        | 5                 | 3        |
| 5          | 5                 | 3        | 5                 | 3        | 5                 | 3        |

Table 2. Randomly Generated Diagonally Dominant Matrices

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*Chapter 10*

## MODIFIED BACK-PROJECTION ALGORITHM ALONG NONLINEAR CURVE AND ITS APPLICATION TO SAR IMAGING

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### Abstract

In this paper a modified back-projection algorithm is proposed to reduce the noise appearing in the ARL BoomSAR image due to the vibration of the boom.

**Key Words:** Back-projection algorithm, SAR, BoomSAR, sidelobe, noise, reconstructed target function, round-trip delay, fast-time bins, point spread function.

**AMS Subject Classification:** Primary 65N06, 78A06.

## 1 Introduction

Synthetic aperture radar (SAR) has been extensively studied during the last three decades for acquiring high resolution radar images. SAR uses a moving antenna platform to synthesise a much larger aperture along the movement trajectory (or flight direction). Army Research Laboratory (ARL) has been investigating the potential of low frequency, ultra-wideband (UWB) synthetic aperture radar to detect targets hidden underneath foliage and objects embedded in the ground (see [3, 4, 5, 6, 7]). Recent development of wide-band high resolution SAR technology has shown that it is possible to detect targets buried close to ground surface over very large open areas [8, 9, 10, 11]. ARL designed a UWB Boom-SAR system which is an impulse radar with an instantaneous frequency that spans from 50

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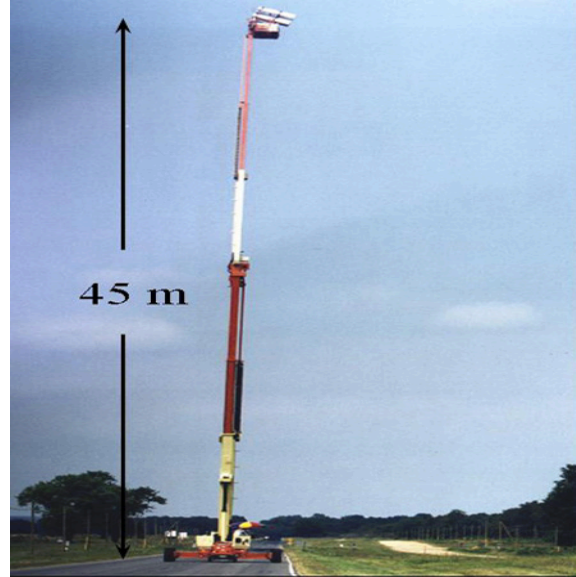


Figure 1: BoomSAR system (from ARL)

to 1100 MHz in a single coherent collection interval, and is mounted on a 45 meters mobile platform that can move at 1 Km/Hr (see Figure 1).

The depression angle of the antennas can be adjusted in  $5^\circ$  increments. The wide bandwidth and wide aperture angle of this radar improve significantly the down-range resolution and cross-range resolution respectively. But new algorithms are still demanded to deal with problems caused by noise.

The inaccuracy of the motion compensation system, the non-ideal motion of the radar and the non-uniform and possibly aliased sampling of the data recorded along the radar path result in artifacts and large side-lobes in the radar imagery. Artifacts and side-lobes limit the radar detection performance on small targets such as mines.

ARL has employed the back-projection (delay-and-sum beam former) technique to form UWB SAR images. The back-projection algorithm is expected to produce higher quality images, provided the original radar motion is available. In the case when the radar aperture is moving on a straight line, the sampling summation over the white noise is zero. Therefore the white noise is eliminated with the back projection method. However, the small vibration, particularly the vibration on the down-range direction, of the boom and basket (that holds the antennas and radar system) produces non-white noise to the radar signal. With the current back projection method, the noise can not be removed because the noise is not white noise, the recorded data is not uniform, and the path is not on a straight line.

In this paper, we propose a new algorithm to reduce the noise and therefore to obtain better quality images.

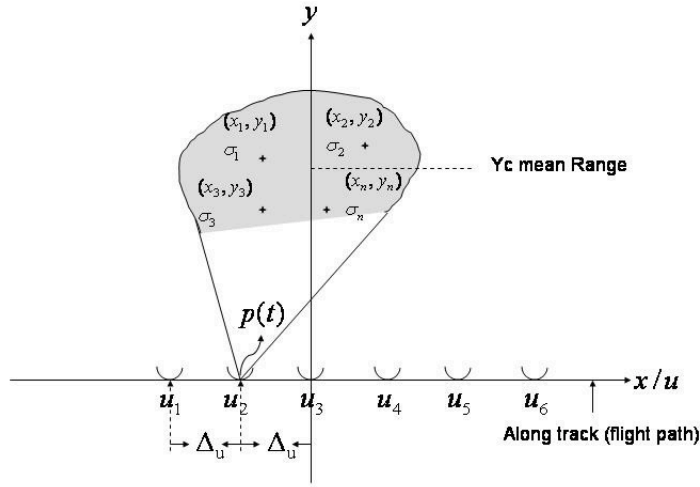


Figure 2: SAR imaging system geometry: broadside target area.

## 2 Mathematical Models and Measurement System

### 2.1 System model

We consider a stationary target region composed of a set of point reflectors with reflectivity  $\sigma_n$  located at the coordinates

$$(x_n, y_n), \quad n = 1, 2, \dots$$

in the spatial  $(x, y)$  domain, where  $x$  is the cross-range domain and  $y$  is the down-range domain. A radar located at  $(u, 0)$  in the spatial domain illuminates the target area with a multifrequency (ultra-wideband) signal  $p(t)$  (see Figure 2). Thus the target function for the radar image and the echoed signal are expressed in the form of

$$V(z) = \sum_n \sigma_n \delta(z - z_n)$$

and

$$s(t, u) = \sum_n \sigma_n p\left(t - \frac{2\sqrt{(x_n - u)^2 + y_n^2}}{c}\right) \quad (2.1)$$

respectively, where  $z = (x, y)$ ,  $z_n = (x_n, y_n)$ , and  $\frac{2\sqrt{(x_n - u)^2 + y_n^2}}{c}$  is the round-trip delay from the radar to the  $n$ th target.

Hereafter the time domain  $t$  is called the fast-time domain and the synthetic aperture domain  $u$  is referred to as the slow-time domain.

### 2.2 Backprojection Algorithm

We denote the fast-time matched-filtered SAR signal via

$$s_M(t, u) = s(t, u) * p^*(-t),$$

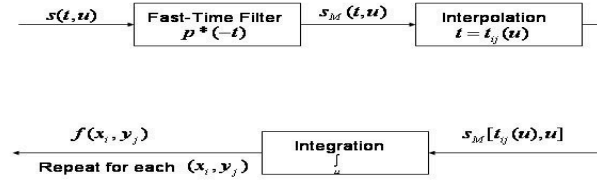


Figure 3: Block diagram of SAR digital reconstruction algorithm via backprojection algorithm.

where  $*$  denotes convolution in the fast-time domain and  $s(t, u)$  is defined by (2.1) and  $p^*$  denotes the complex conjugate of  $p$ . We approximate the target function  $V(z)$  using the following reconstructed target function

$$f(x_i, y_j) = \int_a^b s_M(t_{ij}(u), u) du, \quad (2.2)$$

where  $p(t - t_{ij}(u))$  is the echoed signal from the target located at a given grid point  $(x_i, y_j)$ , and

$$t_{ij}(u) = \frac{2\sqrt{(x_i - u)^2 + y_j^2}}{c} \quad (2.3)$$

is the round-trip delay of the echoed signal recorded by the radar at  $(u, 0)$  from the target at  $(x_i, y_j)$ .

Thus, to form the target function at a given grid point  $(x_i, y_j)$  in the spatial domain, the data  $s_M[t, u]$  for all synthetic aperture locations  $u$  is coherently added up at the fast-time bins  $t_{ij}(u)$ s that correspond to  $(x_i, y_j)$ . A block diagram for the back projection algorithm is shown in Figure 3. To successfully implement this back projection method in practice, accurate  $s_M[t, u]$  at each fast-time bins  $t_{ij}(u)$  is required. Thus,  $s_M(t, u)$  must be obtained through an interpolation method over an available sampling data of fast-time. Therefore, a proper interpolation method is very important.

The backprojection method is used by ARL for the BoomSAR image reconstruction (see [3, 4, 5, 6, 7]). However, the noise can not be removed with the current backprojection method due to the vibration, the non-uniformly recorded data, and the nonlinear path.

### 2.3 Motion Compensation for Backprojection

A nonlinear motion compensation is used for processing the ARL BoomSAR images. The nonlinear motion is caused by the variable speed of the vehicle and the vibration of the Boom.

We denote the vehicle trajectory in the spatial domain as a function of the slow-time via

$$(x_e(u) + u, y_e(u)),$$

where  $x_e(u)$  and  $y_e(u)$ , respectively, are the motion errors in the cross-range and down-range domains. As we mentioned in previous section, the backprojection reconstruction algorithm is based on tracing back the signature of a given reflector in the fast-time domain of the matched-filtered signal  $s_M(t, u)$  at a given slow-time  $u$  and coherently adding the results at the available  $u$  values. The algorithm can be easily modified to incorporate known motion errors.

With the slow-time  $u$ , the actual coordinates of the radar location are  $(x_e(u) + u, y_e(u))$ . For the target at the location  $(x_i, y_j)$ , its signature in the matched-filtered signal can be traced back to the fast-time

$$t_{ij}(u) = \frac{2\sqrt{[x_i - u - x_e(u)]^2 + [y_j - y_e(u)]^2}}{c}.$$

Thus the reconstructed target function  $f(x_i, y_j)$  is approximated by

$$f(x_i, y_j) = \int_a^b s_M\left(\frac{2\sqrt{[x_i - u - x_e(u)]^2 + [y_j - y_e(u)]^2}}{c}, u\right) du. \quad (2.4)$$

ARL BoomSAR system position information is provided by a robotic theodolite that measures angles and distance to an optical beacon located on the antenna array. The position data is sent by radio link to a receiver on the base of the boom. The motion compensation time is non-uniformly distributed due to the limitation of the equipment. Since the vibration in the  $y$  direction has very minor affect on the reconstructed image, here we only deal with the vibration in the  $x$  direction. Moreover, motion errors also cause the actual measured position  $u_l = \bar{u}_l + x_e(\bar{u}_l)$  along the aperture direction unevenly distributed, where  $\bar{u}_l$  stands for the ideal even sample slow-time. Since the unevenly distributed data  $\{s_M[t_{ij}(u_l), u_l]; \quad l = 0, 1, \dots, N\}$  is used to calculate  $f(x_i, y_j)$  in (2.4), the quality of the reconstructed image is reduced. In the ideal case of uniformly distributed sample data, the reconstructed image with the current back-projection algorithm has much lower side lobe level than the case of unevenly distributed data (see Figure 4 and 5). Thus new mathematical algorithm for unevenly sampled data is needed to reduce the side lobe in the reconstructed BoomSAR images.

### 3 Modified Back-Projection Algorithm and Simulation

In this section, we propose a modified back-projection algorithm. In Section 3.1, we describe our modified back-projection algorithm. With an interpolation method, we obtain an approximated continuous path of the moving radar with a new parameter which is used to get an imaginary linear path via a nonlinear transformation and to obtain evenly distributed sampling points of the parameter. We introduce Three-point and Five-point methods to approximate the derivatives for calculating the reconstructed target function. In Section 3.2, we provide the mathematical description of the simulation and some numerical results.



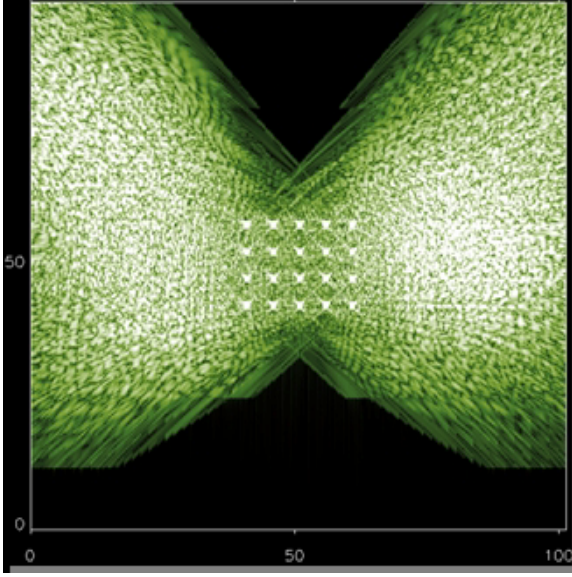


Figure 4: Using motion data from BoomSAR (from ARL)

### 3.1 Interpolation and nonlinear transformation

In the image processing of the ARL Ultra-wideband BoomSAR, the noises mainly come from two sources: one comes from other existing electromagnetic fields in the air, that produces the white noise; and the other comes from the vibration of the Boom, that produces the perturbation noise. Therefore the echoed signal  $S$  may be split into three parts, the original echoed signal  $S_o$  without noise, the random noise  $S_{ran}$  and the perturbation noise  $S_{per}$ ,

$$S(t, u) = S_o(t, u) + S_{ran}(t, u) + S_{per}(t, u).$$

After the matched filtering process, the white noise  $S_{ran}$  is eliminated and the obtained signal is denoted by  $s_M(t, u)$ . Thus  $s_M = s_O + s_{per}$ , where  $s_O$  is the matched-filtered original signal and  $s_{per}$  is the matched-filtered perturbation noise.

**Hypothesis** The perturbation noise has near zero mean, that is,

$$\int_a^b s_{per} du \sim 0.$$

The hypothesis is reasonable because the physical perturbation is often periodic oscillation in the slow time domain.

In theory, we have

$$\begin{aligned} f(x_i, y_j) &= \int_a^b s_M[t_{ij}(u), u] du \\ &= \int_a^b (s_O[t_{ij}(u), u] + s_{per}) du \\ &= \int_a^b s_O[t_{ij}(u), u] du + \int_a^b s_{per} du \\ &\sim \int_a^b s_O[t_{ij}(u), u] du. \end{aligned} \tag{3.1}$$

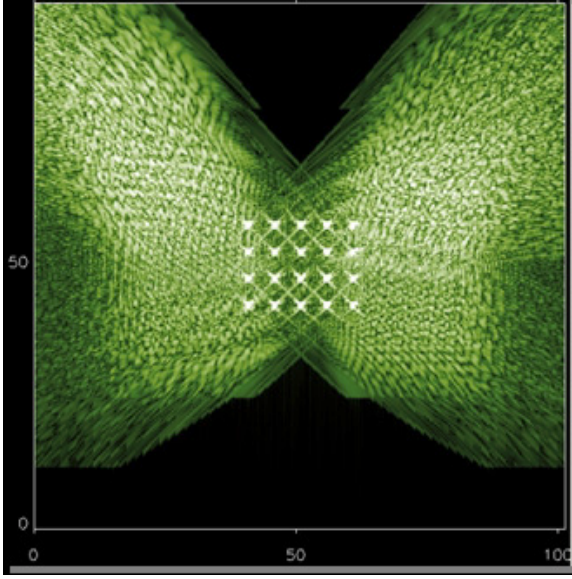


Figure 5: Motion data rearranged to have equal spacing along track. The sidelobe level is significantly reduced (from ARL)

Numerically,  $\int_a^b s_{per} du \sim 0$  in the case of uniformly spaced nodes. However for non-uniform case the numerical integration doesn't approximate 0 in general. Thus to eliminate the perturbation noise, it is helpful to transform non-uniform case to uniform case.

Let  $0 = \tau_1 < \tau_2 < \dots < \tau_n = 1$  be evenly distributed nodes and  $\tilde{u}(\tau)$  be a smooth transformation satisfying  $\tilde{u}(\tau_l) = u_l$ ,  $l = 0, 1, \dots, n$  and

$$\begin{aligned} \tilde{u}'(\tau_l) &= \frac{u_{l+1} - u_{l-1}}{2\Delta\tau}, & 1 \leq l \leq n-1, \\ \tilde{u}'(\tau_0) &= \frac{-3u_0 + 4u_1 - u_2}{2\Delta\tau}, \\ \tilde{u}'(\tau_n) &= \frac{3u_n - 4u_{n-1} + u_{n-2}}{2\Delta\tau}, \end{aligned} \quad (3.2)$$

where  $\Delta\tau = \tau_l - \tau_{l-1}$ ,  $l = 1, 2, \dots, n$ .

The target function is estimated with respect to  $\tau$  as follows

$$f(x_i, y_j) = \int_a^b s_M[t_{ij}(u), u] du \approx \int_0^1 s_M[t_{ij}(\tilde{u}(\tau)), \tilde{u}(\tau)] \tilde{u}'(\tau) d\tau.$$

Thus to calculate the integration, we use equation (3.2), namely Three-point Formulas. Let

$$S(\tau) = s_M[t_{ij}(\tilde{u}(\tau)), \tilde{u}(\tau)] \tilde{u}'(\tau). \quad (3.3)$$

Then we obtain the approximation of the target function  $f(x_i, y_j)$

$$f(x_i, y_j) \approx \frac{\Delta\tau}{3} [S(\tau_0) + S(\tau_n) + 4 \sum_{k=1}^m S(\tau_{2k-1}) + 2 \sum_{k=1}^{m-1} S(\tau_{2k})] \quad (3.4)$$

via the Composite Simpson rule if  $n = 2m$ .

### 3.2 Stationary phase method and the point spread function

We take the following chirp signal as the transmitted signal

$$p(t) = a_\lambda(t) \exp(j\beta t + j\alpha t^2), \quad (3.5)$$

where

$$a_\lambda(t) = \begin{cases} 1, & \text{if } 0 \leq t \leq \frac{2\pi}{\lambda}, \\ 0, & \text{otherwise.} \end{cases}$$

The instantaneous frequency is

$$\omega(t) = \frac{\partial}{\partial t}(\beta t + \alpha t^2) = \beta + 2\alpha t.$$

Thus the domain of the fast time frequency is  $[\beta, \beta + \frac{4\alpha\pi}{\lambda}]$ . We design the emitted signal as follows such that the instantaneous frequency spans from 50 to 1100 MHz,

$$\beta = 2\pi * 50 * 10^6, \quad \beta + \frac{4\alpha\pi}{\lambda} = 2\pi * 1100 * 10^6. \quad (3.6)$$

It follows

$$\alpha/\lambda = 2\pi * 1050 * 10^6 / 4\pi = 5.25 * 10^8.$$

Therefore proper  $\lambda$  will be selected because the signals overlap if  $\lambda$  is not large enough and the hardware is difficult to make if the  $\lambda$  is very large.

Since for the chirp signal

$$T_p = \frac{2\pi}{\lambda},$$

we have the down-range resolution

$$\Delta_x = \frac{\pi c}{2\alpha T_p} = \frac{c}{4\alpha/\lambda} = \frac{3.0}{4 * 5.25} = 0.1428571 \text{ metre},$$

where  $c$  is the speed of light and  $T_p$  is the duration of signal. This down-range resolution is good enough for most applications.

To improve the cross-range resolution, we use the modified back-projection algorithm described in this paper. For the transmitted chirp signal (3.5), the measured echoed signal is

$$s(t, u) = \sum_n \sigma_n p\left[t - \frac{2\sqrt{x_n^2 + (y_n - u)^2}}{c}\right],$$

and the matched-filtered signal is

$$s_M(t, u) = s(t, u) * p^*(-t) = \sum_n \sigma_n \text{psf}_t\left[t - \frac{2\sqrt{x_n^2 + (y_n - u)^2}}{c}\right], \quad (3.7)$$

where the point spread function is

$$\text{psf}_t(t) = \mathcal{F}_\omega^{-1}[|P(\omega)|^2],$$

$\omega$  represents the fast-time frequency domain and  $P(\omega)$  is the Fourier transform of  $p(t)$ .

We use the standard stationary phase method to study the asymptotic behavior of the point spread function  $\text{psf}_t(t)$ . Let  $\phi(t) = t^2 + \frac{\beta - \omega}{\alpha}t$ . Then

$$P(\omega) = \int_{-\infty}^{\infty} p(t)e^{-j\omega t} dt = \int_0^{\frac{2\pi}{\lambda}} e^{j\alpha\phi(t)} dt.$$

and  $t_0 = \frac{\omega - \beta}{2\alpha}$  is the only non-degenerate critical point in  $[0, \frac{2\pi}{\lambda}]$  if  $\beta < \omega < \beta + \frac{4\pi\alpha}{\lambda}$ . By using the stationary phase method we obtain

$$P(\omega) \sim \sqrt{\frac{\pi}{\alpha}} e^{j\frac{\pi}{4}} e^{j\alpha\phi(t_0)} = \sqrt{\frac{\pi}{\alpha}} e^{j[\frac{\pi}{4} - (\frac{\omega - \beta}{2\alpha})^2]}.$$

Thus we obtain

$$|P(\omega)|^2 \sim \begin{cases} \frac{\pi}{\alpha}, & \text{if } \beta \leq \omega \leq \beta + \frac{4\pi\alpha}{\lambda}, \\ 0, & \text{otherwise.} \end{cases}$$

Since  $\text{supp}(\text{psf}_t) \subset [-\frac{2\pi}{\lambda}, \frac{2\pi}{\lambda}]$ , for  $t > 0$ , we have

$$\text{psf}_t(t) \sim \frac{1}{2\pi} a_\lambda(t) \frac{\pi}{\alpha} \int_{\beta}^{\beta + \frac{4\pi\alpha}{\lambda}} e^{j\omega t} d\omega = a_\lambda(t) e^{jt(\beta + \frac{2\pi\alpha}{\lambda})} \frac{\sin(\frac{2\pi\alpha}{\lambda}t)}{\alpha t}. \quad (3.8)$$

### 3.3 Simulation

Let the true targets be four points with the unit reflectivity (see Figure 6). In the simulation, we use the chirp signal (3.5) with the parameters (3.6). Assume that the motion error (that produces perturbation noise) is a sine function as follows

$$x_e(u) = \varepsilon \sin(\omega u),$$

where  $\varepsilon = 0.1$  and  $\omega = 1000$ .

Our algorithm is summarized as follows:

Step 1. Select uniform sampling points  $\{v_1, v_2, \dots, v_n\}$  on the slow-time domain  $u$  ( $n=200$ ), uniform sampling points  $\{\tau_l : 1 \leq l \leq n\}$  on the unit interval  $[0, 1]$ , and grid nodes  $\{(x_i, y_j) : 1 \leq i, j \leq m\}$  ( $m=200$ ) in the square region including the four true targets;

Step 2. Obtain the perturbed nonuniform sampling points  $\{u_1, u_2, \dots, u_n\}$  via the formula  $u_l = v_l + x_e(v_l)$ ,  $1 \leq l \leq n$ ;

Step 3. Compute the fast time bins  $\{t_{ij}(u_l) : 1 \leq i, j \leq m, 1 \leq l \leq n\}$  by using

formula (2.3), the value of matched-filtered signal  $s_M(t_{ij}(u_l), u_l)$  by using (??) and the approximation (3.8) of the point spread function at the sampling points and the grid nodes;

Step 4. Compute the derivatives  $\{\tilde{u}'(\tau_l) : 1 \leq l \leq n\}$  by using the formula (3.2) and the values of  $\{S(\tau_l) : 1 \leq l \leq n\}$  by using (3.3);

Step 5. Compute the values of the reconstructed function  $\{f(x_i, y_j) : 1 \leq i, j \leq m\}$  via the Composite Simpson formula (3.4) and form the image.

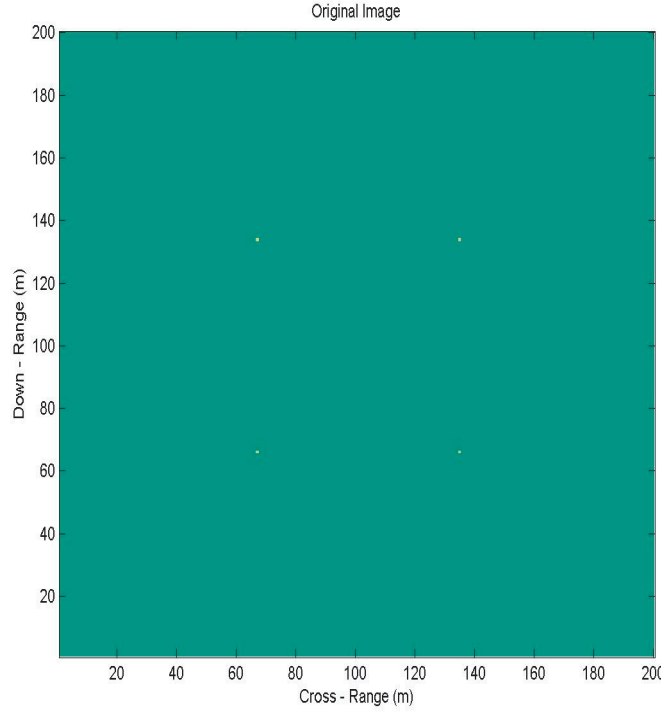


Figure 6: True Targets Image

**Conclusion:** By comparing Figure 7 and 8, we see that our modified back-projection method is more effective for dealing with the nonuniform spacing problem.

## Acknowledgment

The Authors would like to thank Jeffrey Sichina, Lam Nguyen and David Wang from Adelphi Army Research Lab for their lectures given to us at Delaware State University on SAR which lead us to the research done in this paper. The Authors also thank Dr. John Lavery from Army Research Office for his advise and support.

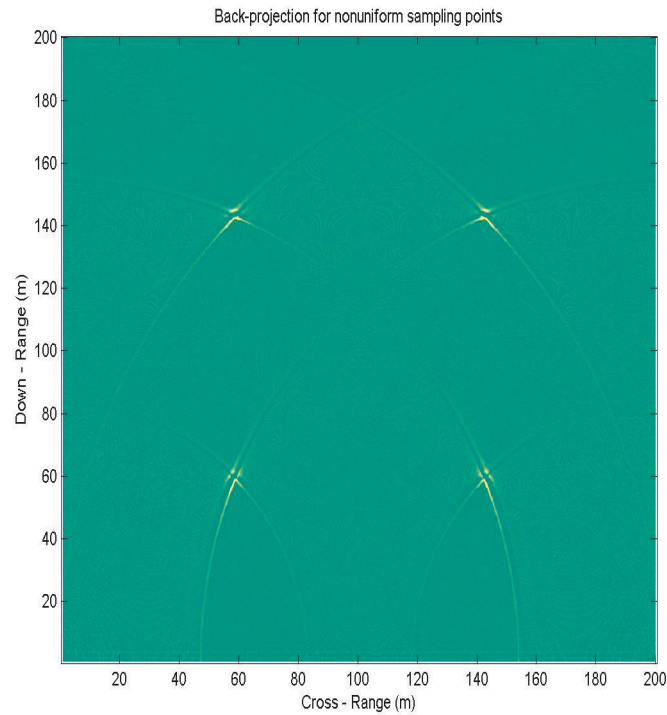


Figure 7: Image formation of nonuniform spacing problem via back-projection method with perturbation noise

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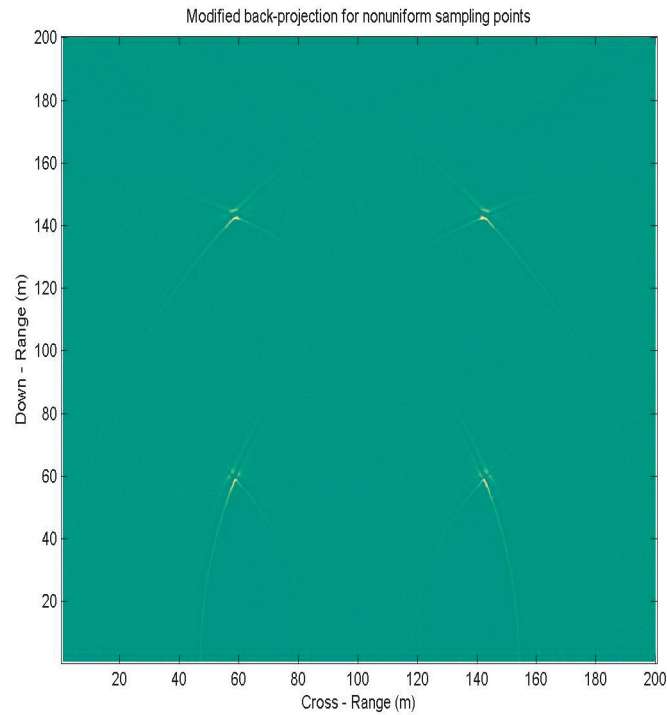


Figure 8: Image formation of nonuniform spacing problem via modified back-projection method with perturbation noise

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*Chapter 11*

## AFFINE TRANSFORMATION METHOD IN AUTOMATIC IMAGE REGISTRATION

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### Abstract

Image registration is a fundamental task in image processing used to match two or more pictures taken, for example, at different times, from different sensors, or from different viewpoints. It arises from widespread scientific fields such as computer vision, medical image analysis, virtual reality, satellite data processing, surface matching, and so on. In this paper, we used affine transformation method to register images automatically.

**Key Words:** surface fitting, B-spline surface patching, Convergent geometric continuity, Polygonal mesh, Quad partition, adaptive ratio compatibility condition.

**AMS Subject Classification:** Primary 68U10, 94A08, 94A12.

## 1 Introduction

Image registration is one of the crucial steps in the analysis of remotely sensed data. A new acquired image must be transformed, using image registration techniques, to match

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the orientation and scale of previous related images. Image registration requires intensive computational effort not only because of its computational complexity, but also due to the continuous increase in image resolution and spectral bands. Thus, the computing techniques for image registration are critically needed.

Regardless of the image data involved and of the particular application, image registration usually consists of the four major steps:

- a Detection of control point candidates (CPCs). CPCs are significant points or structures (edge intersections, objects centroids, significant contour points, etc.) detected automatically or manually.
- b Control point matching. The correspondence between the CPCs in two corresponding images is established. One image is called the reference and the other is called sensed.
- c Estimation of the mapping model. The type of transform between the reference and sensed images are estimated. The type of the mapping function can be global (linear, projective or quadratic transformation) or local (local triangular mapping, radial basis functions, thin-plate splines), depending on the type of the image distortions.
- d Resampling and transformation. The sensed image is transformed over to the reference according to the above mapping model. An appropriate resampling technique is employed to find image values in non-integer coordinates.

## 2 Automatic Image Registration Under Affine Transformation

Automatic image registration is to perform the image registration task without the guidance and intervention of users. The need of automatic image registration comes from widespread applications. For example, one has to use efficiently automatic image registration method to glue together the tremendous amount of satellite images from the Earth Observing System (EOS) program. There are many methods for automatic image registration. Here we will mainly introduce our results of automatic image registration by affine transformation.

We assume that the corresponding parts  $M_1$  and  $M_2$  of two images  $I_1$  and  $I_2$ , can be matched together with an affine transformation. Our task is to find a suitable affine transformation such that we can match images  $M_1$  and  $M_2$  together. An affine transformation is defined by

$$T : X \rightarrow Y, Y = AX + B$$

where  $X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in M_1$ ,  $Y = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \in M_2$ ,  $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$  is a  $2 \times 2$  matrix, and  $B = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$  is a vector.

The mapping  $T : X = AY + B$  is an affine transformation and it seems that our problem is linear, at least in form.

Notes: Both  $A$  and  $B$  are unknown, they have to be determined, and the more serious problem is that the domain of  $T$  is only a part of  $I_1$  and is undetermined. This makes

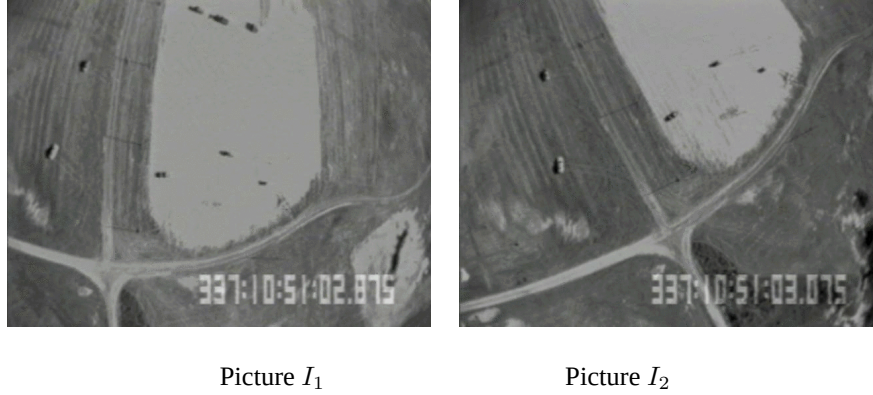


Figure 1:

the image registration to become a non-linear problem(in essential), even we use linear transform.

The main steps of our method to register two images automatically are as follows.

**Step 1 Edge detection.** In this step we use an existing edge detection algorithm. The edges of an image are obtained from its monochrome image. The monochrome image is an image with only colors black and white and is obtained by resetting the grey values of the original image. The grey value is set to be white if the original grey value is great or equal to a given threshold, and set to be black otherwise. The contours of the monochrome image is what we want to obtain. **An observation:** It is remarkable that the edges of the same object in two different pictures can be different. In the following, the two pictures in the last row are the enlarge versions of the regions marked by red rectangles in the first row, respectively. The closed boundaries contained in the rectangles come from the same object. The shape of two closed boundaries are remarkably different.

**What is the meaning of this observation?** We assume that in theory there exists an affine transformation  $T : X = AY + B$  which can match pictures  $M_1$  and  $M_2$  together. We further assume that two corresponding closed boundaries which are respectively determined by the point sets  $\Gamma_1 = \{b_{11}, b_{12}, \dots, b_{1n}\}$  in  $M_1$  and  $\Gamma_2 = \{b_{21}, b_{22}, \dots, b_{2n}\}$  in  $M_2$  are produced from the same object, i.e.,  $b_{1i}$  and  $b_{2i}$  are the boundary points on  $\Gamma_1$  and  $\Gamma_2$  respectively such that

$$b_{2i} = Ab_{1i} + B, \quad 1 \leq i \leq n.$$

But in practice, there always exist computing errors. Therefore, the above equations do not hold exactly. Let  $\varepsilon_i$ ,  $1 \leq i \leq n$  be the computing errors. Then

$$b_{2i} + \varepsilon_i = Ab_{1i} + B, \quad 1 \leq i \leq n.$$

We can not find a suitable affine transformation to match  $M_1$  and  $M_2$  together if  $\varepsilon_i$ ,  $1 \leq i \leq n$  are not small enough. Our observation just shows that these errors may not be small. To overcome this difficult, we employ the centroid of a closed boundary and make the following assumption.

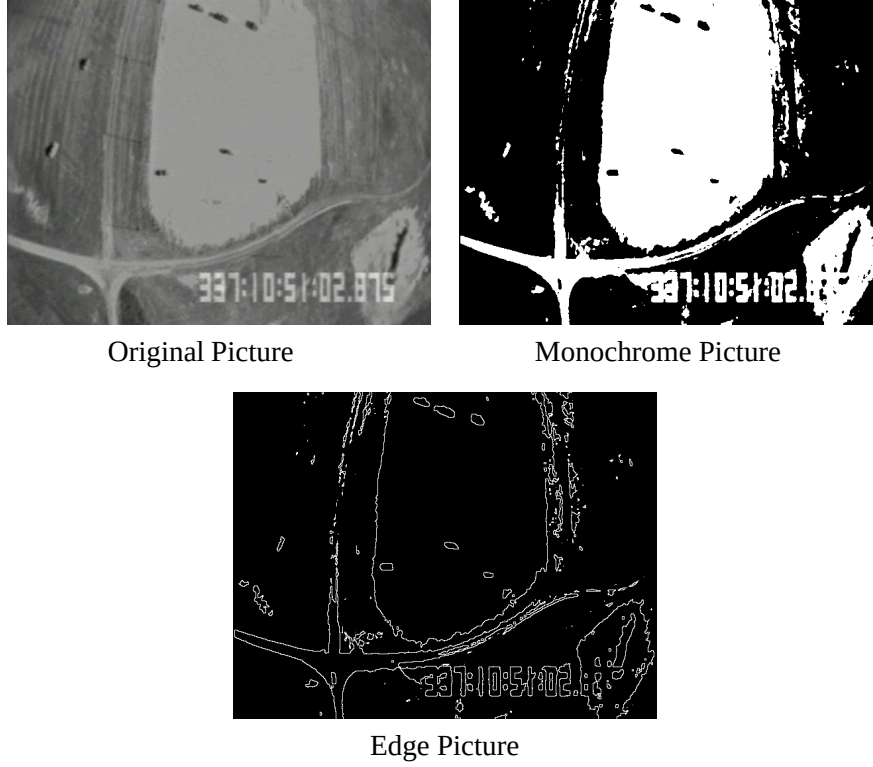


Figure 2:

**Assumption:** There are at least three pairs of closed edges from the same objects in two pictures, respectively. (This assumption is valid for many images)

**Step 2:** For the closed boundary  $\Gamma$  determined by the point set  $\{b_1, b_2, \dots, b_n\}$ , we define  $c = \frac{1}{n} \sum_{i=1}^n b_i$  to be the centroid of  $\Gamma$ . For each picture, we obtain a list of centroids of all the closed boundaries.

If the closed boundaries  $\Gamma_1 = \{b_{11}, b_{12}, \dots, b_{1n}\}$  in  $M_1$  and  $\Gamma_2 = \{b_{21}, b_{22}, \dots, b_{2n}\}$  in  $M_2$  are corresponding boundaries of the same object as described before, then the centroid  $c_1 = \frac{1}{n} \sum_{i=1}^n b_{1i}$  of  $\Gamma_1$  and the centroid  $c_2 = \frac{1}{n} \sum_{i=1}^n b_{2i}$  of  $\Gamma_2$  satisfy the following relationship

$$c_2 = Ac_1 + B.$$

Considering the computing errors, it holds

$$c_2 + \frac{1}{n} \sum_{i=1}^n \varepsilon_i = Ac_1 + B.$$

Above equation shows that, to obtain a suitable transformation, it is much reliable to employ the centroid than the individual boundary points. Another reason of using centroid is based on the fact that the centroid is invariant under affine transformations.

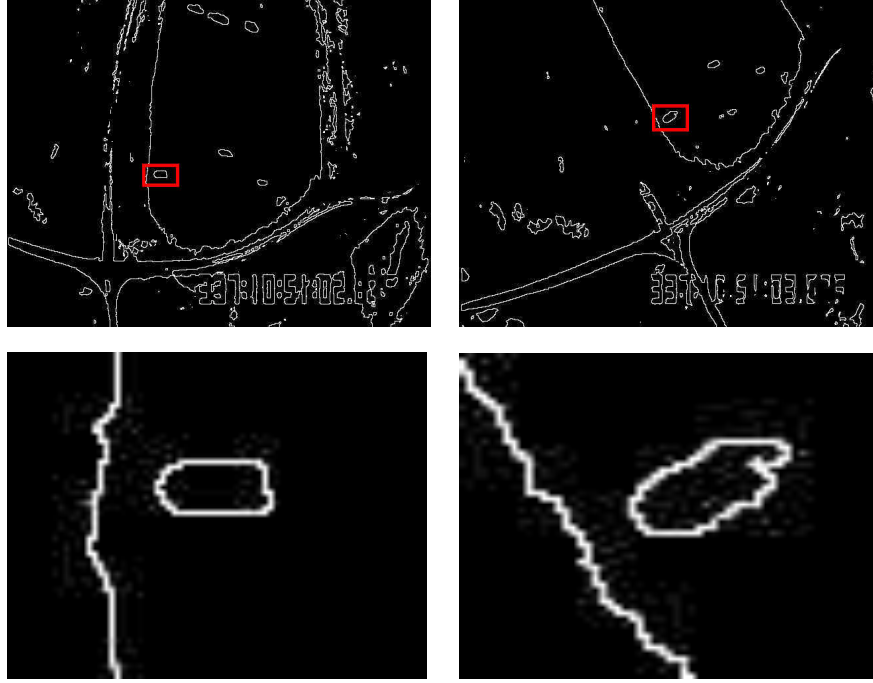


Figure 3:

**Step 3:** Comparing all the combinations of three centroids from the lists of centroids of two pictures  $I_1$  and  $I_2$ , we find the match with the smallest residual error.

Step 3 is very time consuming. To speed up, we used the following techniques:

- 1 Consider only the objects that are both not too small and not too big. Let  $S$  and  $L$  be the area and the length of the boundary of the object, respectively. We require that

$$\begin{aligned} 40 &\leq L \leq 100, \\ 50 &\leq S \leq 300. \end{aligned}$$

This restriction is to be relaxed in the future.

- 2  $\frac{L^2}{S} \leq 30$ . This restriction is also to be relaxed in the future.

The reason of this restriction is based on the fact that for any closed figure, it holds  $\frac{L^2}{S} \geq 4\pi$ . The figure will be a circle if the equality holds. The smaller the value  $\frac{L^2}{S}$  is, the more circle-like the figure is.

- 3 Let  $S_1$  and  $S_2$  be the areas of the triangles formed by the three pairs of the selected centroids, respectively. We restrict that

$$0.81 \leq \frac{S_1}{S_2} \leq \frac{1}{0.81}$$

- 4 Testing results with coarser images of  $I_1$  and  $I_2$ , i.e. reduced resolution of both images before testing for the best match (see Fig 4).

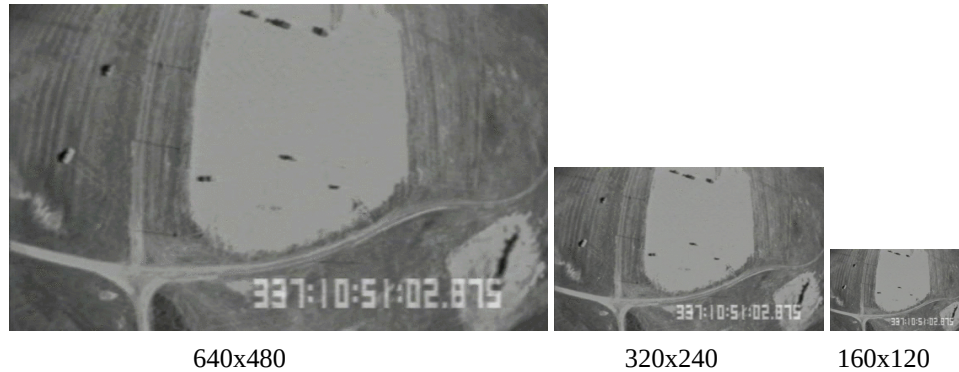


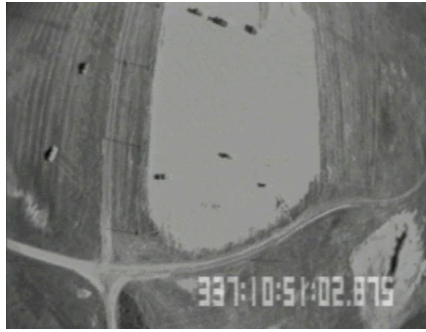
Figure 4:

### 3 The Result Obtained from the Above Processes

The result obtained from the above processes is shown in the Fig 5.

### References

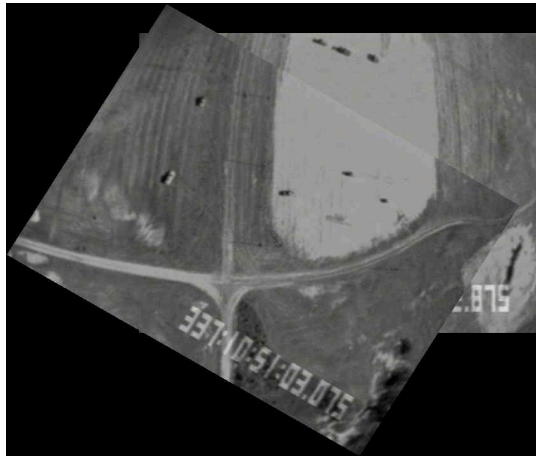
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Original Picture 1



Original Picture 2



The Registered Picture

Figure 5:





*Chapter 12*

## SMOOTHING 2-D IMAGES WITH $B$ -SPLINE FUNCTIONS\*

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### Abstract

To smooth 2-D noisy images from functions of the space  $L^2(\Omega)$  to  $C^2(\Omega)$ , for  $\Omega \subset \mathbb{R}^2$ , we propose a Tikhonov's regularization algorithm in the subspace of bidimensional cubic  $B$ -splines with uniform knots. Considering the tensor product of unidimensional  $B$ -spline functions, we reduce the bidimensional problem to two independent unidimensional problems. The main feature of our algorithm is that its complexity is comparable to that of the bidimensional FFT algorithm.

**Key Words:** smoothing, B-cubic splines, uniform splines, functional of smoothing, parameter of regularization, cross validation, ill posed problem, Tikhonov's regularization.

**AMS Subject Classification:** 65D07, 65D10, 65D15, 65F10, 65K, 92C55.

## 1 Introduction

Splines are piecewise defined curves, where each piece belongs to a certain family (generally polynomial functions). Tensor products of B-splines have been extensively used in computer-aided geometric modeling (CAGM) systems [4, 16, 15, 17]. Least squares spline fitting (LSSF) [10, 6] is used in many areas of scientific computation, computer graphics, optical character recognition (OCR), and signal/image processing in order to generate a

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\*This paper is dedicated to our families for their support

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functional representation of sampled data. The objective of this work is to present a practical algorithm for smoothing 2-D noisy images with the tensor product of cubic  $B$ -splines, which is based on the Tikhonov's regularization method [17, 18]. We consider the model defined as follows: Let  $\Omega = [-1, 1] \times [-1, 1]$ , and  $\Omega_d$  be the observational grid

$$\Omega_d = \{x_i \in [-1, 1] : i = 1, 2, \dots, N_x\} \times \{y_j \in [-1, 1] : j = 1, 2, \dots, N_y\}.$$

The noisy observations are given by

$$u(x_i, y_j) = v(x_i, y_j) + \varepsilon(x_i, y_j), \quad (x_i, y_j) \in \Omega_d, \quad (1.1)$$

where  $v(x, y) \in C^2(\Omega)$  is an unknown function, and  $\varepsilon(x, y)$  is an additive Gaussian noisy process, i.e.,

$$E\{\varepsilon(x, y)\} = 0, \quad E\{\varepsilon(x, y)\varepsilon(r, s)\} = \begin{cases} \sigma^2, & (r, s) = (x, y), \\ 0, & (r, s) \neq (x, y), \end{cases}$$

where  $E$  is the mathematical expectation, and the error variance,  $\sigma^2$ , may be known or unknown.

The bidimensional smoothing problem consists of finding a smooth function  $v(x, y) \in C^2(\Omega)$  in the data set  $u(x_i, y_j), (x_i, y_j) \in \Omega_d$ .

It can be shown that a small modification of the data  $u(x_i, y_j)$ , for instance due to noise, may be propagated in a catastrophic way in the solution function  $v(x, y)$ . In other words, the solution might not depend continuously on the data.

Let  $L^2(\Omega_d)$  be the space of discrete square summable functions in  $\Omega_d$ , and let  $C^s(\Omega)$  be the space of functions with derivatives of order  $s$  continuous in  $\Omega$ . In general the bidimensional smoothing problem can be posed as a typical linear inverse problem, which consists of solving the equation

$$Av = u, \quad v \in C^s(\Omega), \quad u \in L^2(\Omega_d), \quad (1.2)$$

where  $u$  is the data matrix and  $A$  is the linear continuous operator of discretization. However,  $A^{-1}$  is not continuous, and the inverse problem is unstable. This problem is said to be *ill posed in the sense of Hadamard* [17]. In these cases we can find a generalized function or least-squares approximation of minimum norm by some regularization method [1]. Tikhonov's regularization method approaches the ill posed problem by means of a family of nearby well posed problems over the Sobolev space  $W^{2,s}(\Omega)$ . A reasonable way to obtain this regularization is taking the *Tikhonov's stabilizer*,  $R^s(v)$ , which measures the rugosity or instability of  $v$ , and minimizing the energy functional

$$\Phi_\alpha^s(v) = \alpha \|Av - u\|_{L^2(\Omega_d)}^2 + (1 - \alpha)R^s(v), \quad \alpha \in [0, 1], \quad (1.3)$$

on  $S \subset W^{2,s}(\Omega)$ , which is the subspace of the functions  $v$  that are the tensor product of degree  $2s-1$   $B$ -spline functions [4]. The first term in (1.3), being small, assures us that  $v$  is near to the solution of least-squares; the optimal value of the *parameter of regularization or fairness factor*  $\alpha$  must be calculated [3, 17, 18]. In one-dimension, Reinsch and Schoenberg [12, 14] proposed to minimize the functional

$$\frac{\alpha}{n} \sum_{i=1}^n (v(x_i) - u(x_i))^2 + (1 - \alpha) \int_{x_1}^{x_n} (v''(t))^2 dt, \quad \alpha \in [0, 1], \quad (1.4)$$

where  $\alpha$  is the parameter of balance between two conflicting goals: our desire to stay close to the given data and our desire to obtain a smooth function. For a given  $\alpha$ , it is well known that the solution of (1.4) is a spline of order  $2s - 1$  [20]. When  $s = 2$ ,  $v$  is a linear combination of cubic  $B$ -splines.

Craven and Wahba [3] showed how to choose  $\alpha$  objectively, both when the amount of noise associated with the data,  $\sigma^2$  is known, and when it is not known. In the first case, one may minimize the expected mean-square error over the data points, and in the second case, one may minimize the generalized cross validation (GCV), a procedure which is asymptotically the same as minimizing the expected mean-square error. However, in each case, the function to be minimized involves the mean-square residual and the trace of the influence matrix associated with the smoothing spline. The standard way to compute the trace, implementing IMSL [9] involves  $O(n^3)$  operations and  $O(n^2)$  storage, where  $n$  is the number of data points. Hutchinson and de Hoog [8, 5] described an algorithm that uses the rational Cholesky factorization to calculate the trace for a smoothing spline of degree  $2m - 1$ , including the generalized cross validation, in  $O(m^2n)$  operations using  $O(mn)$  storage ( $1 \leq m \ll n$ ).

This paper permits us to show that the problem of 2-D smoothing can be reduced to two independent unidimensional problems by using the tensor product of two  $B$ -splines, providing a great saving in computational complexity essentially compared to the 2-D FFT algorithm.

The outline of this work is as follows. Section 2 provides some basic terminology about splines and cubic  $B$ -splines. Section 3 presents the functional of smoothing and its matrix representation. Section 4 provides the smoothing algorithm and provides numerical results that compares our method with the 2-D FFT method in terms of their complexities. Finally, Section 5 provides conclusions and two examples of smoothing 2-D noisy images with the algorithm in this paper.

## 2 Cubic $B$ -Splines

### Definition 2.1.

- (i) A grid of  $[a, b]$  is a set  $T = \{t_1, \dots, t_n\}$  satisfying

$$a = t_1 < t_2 < \dots < t_n = b;$$

$t_1, \dots, t_n$  are called the knots of  $T$ , and

$$h := \max \{|t_{j+1} - t_j| : j = 1, \dots, n - 1\}$$

is called the mesh size of  $T$ .

- (ii) A spline of order  $k$  (over a grid  $T$ ) is a  $k$  times continuously differentiable function  $S : [a, b] \rightarrow \mathbb{R}$ , such that the  $k$ -th derivative  $S^{(k)}$  is piecewise linear (over  $T$ ).

**Splines of order 2 are called cubic splines.**

Important examples of splines are the so-called  $B$ -Splines. A normalized cubic basis spline ( $B$ -spline), is a cubic spline  $B_l(t)$  that is defined over the extended grid  $t_{-2} < t_{-1} <$

$\dots < t_{n+3}$  and has the property (for some  $l = 0, 1, \dots, n+1$ )  $B_l(t_l) = 1$  and  $B_l(t) = 0$  for  $t \notin (t_{l-2}, t_{l+2})$ .

Given a grid  $T$ , the set of *cubic B-spline* functions can be obtained recursively ([4] p. 131):

$$B_k^{(0)}(t) = \begin{cases} 1, & t_k \leq t < t_{k+1}, \quad 1 \leq k \leq n-1 \\ 0, & \text{otherwise,} \end{cases} \quad (2.1)$$

$$B_k^{(m)}(t) = \frac{t - t_k}{t_{k+m} - t_k} B_k^{(m-1)}(t) + \frac{t_{k+m+1} - t}{t_{k+m+1} - t_{k+1}} B_{k+1}^{(m-1)}(t), \quad (2.2)$$

for  $m = 1, 2, 3; k = 1, \dots, n-m-1$ . These piecewise polynomial functions have  $C^2$  continuity at the knots, and the polynomials involved are cubics. Any  $g(t) \in C^2$  polynomial spline over  $[t_4, t_{n-3}]$  may then be written using the *B-spline* representation

$$g(t) = c^t B = \sum_{k=1}^{n-3} c_k B_k^{(3)}(t). \quad (2.3)$$

Note that this representation is valid *only* when  $t \in [t_4, t_{n-3}]$ . Moreover, the B-spline sequence  $\{B_k^{(3)}\}$  provides a partition of unity on interval  $[t_4, t_{n-3}]$ . In fact,

- (i)  $B_k^{(3)}(t) = 0$  for  $t \notin [t_k, t_{k+4}]$ , for all  $k = 0, \dots, n+1$ .
- (ii)  $B_k^{(3)}(t) > 0$  for  $t \in [t_k, t_{k+4}]$ , for all  $k = 0, \dots, n+1$ .
- (iii)  $\sum_k B_k^{(3)}(t) = \sum_{k=r-3}^{s-1} B_k^{(3)}(t) = 1$ , for all  $t_r < t < t_s$ , and  $4 \leq r < s \leq n-3$ .

## 2.1 Uniform Splines

The curves  $g(t)$  generated by (2.3) above are called *uniform splines* when the basis functions are defined on a uniformly spaced knot sequence  $T = \{t_1 + k\Delta\}_{k=1}^n$ , where  $\Delta = (b-a)/n$ . From the recursive definition (2.1), (2.2) it is easily seen that the uniform *B-splines* defined on  $T$  are shifted versions of each other, i.e.,  $B_{k+1}^{(3)}(t) = B_k^{(3)}(t - \Delta)$ . In other words, the uniform *B-spline* representation (2.3) can be written in terms of the first basis function

$$g(t) = \sum_{k=1}^{n-3} c_k B_1^{(3)}(t - k\Delta). \quad (2.4)$$

In this work we concentrate on the tensor product of two spaces of uniform cubic *B-spline* functions. Let  $\Omega$  be the rectangle  $\Omega = [-1, 1] \times [-1, 1]$ , and  $\tilde{\Omega}_d$  the mesh of knots, also known as 2-D knot-sequence, defined by

$$\tilde{\Omega}_d = \{\tilde{x}_k \in [-1, 1] : k = 1, \dots, M_x\} \times \{\tilde{y}_l \in [-1, 1] : l = 1, \dots, M_y\} \subseteq \Omega_d,$$

where the 1D knot-sequences

$$\{\tilde{x}_k \in [-1, 1] : k = 1, \dots, M_x\} \quad \text{and} \quad \{\tilde{y}_l \in [-1, 1] : l = 1, \dots, M_y\}$$

allow definition of the set *uniform cubic B-Spline functions*

$$\left\{ B_k^{(3)}(x) \right\} \quad \text{and} \quad \left\{ B_l^{(3)}(y) \right\},$$

for  $k = 1, \dots, M_x$  and  $l = 1, \dots, M_y$  respectively.

**Definition 2.2.** Any  $C^2$  function  $v$ , over  $\tilde{\Omega}_d$  can be written using the tensor product B-spline representation, which is called 2-D polynomial spline

$$v(x, y) = \sum_{k=1}^{M_x} \sum_{l=1}^{M_y} C_{kl} B_k^{(3)}(x) B_l^{(3)}(y). \quad (2.5)$$

### 3 Matrix Representation of the Functional $\Phi_\alpha^2$

The following theorem shows that the inverse problem (1.2) can be reduced to an iterative process that finds the minimizer of the functional  $\Phi_\alpha^2$ .

**Theorem 3.1.** Considering the B spline representation of the function  $v$  given by (2.5), and the Tikhonov's stabilizer

$$R^2(v) = \sum_{s_1+s_2=2} \left\| \frac{\partial^{s_1+s_2} v}{\partial x^{s_1} \partial y^{s_2}} \right\|_{L^2(\Omega)}^2, \quad (3.1)$$

the regularization of the inverse problem (1.2), from  $L^2(\Omega_d)$  to  $C^2(\Omega)$ , given by the minimization of the functional (1.3), can be reduced matrixially to an iterative process with initial solution matrix,  $C_{kl}$ , equals to the solution of the least-squares problem.

*Proof.* Using the representation to the function  $v$  given by (2.5), and  $(B_x)_{ki} = B_k^{(3)}(x_i)$ ,  $(B_y)_{jl} = B_l^{(3)}(y_j)$ , the squared norm in the functional (1.3) is expressed by

$$\begin{aligned} \|u - v\|_{L^2(\Omega_d)}^2 &= \\ &= \frac{1}{N_x N_y} \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} \left( u_{ij}^2 - 2u_{ij} \sum_{k=1}^{M_x} \sum_{l=1}^{M_y} C_{kl} (B_x)_{ki} (B_y)_{jl} + \right. \\ &\quad \left. + \sum_{k,k_1=1}^{M_x} \sum_{l,l_1=1}^{M_y} C_{kl} C_{k_1 l_1} (B_x)_{ki} (B_x)_{k_1 i} (B_y)_{jl} (B_y)_{j l_1} \right), \end{aligned} \quad (3.2)$$

and the Tikhonov's stabilizer (3.1) is written by

$$\begin{aligned} R^2(v) &= \\ &= \frac{1}{M_x M_y} \sum_{s_1+s_2=2} \int_0^1 \int_0^1 \left( \sum_{k=1}^{M_x} \sum_{l=1}^{M_y} C_{kl} \frac{d^{s_1} B_k^{(3)}(x)}{dx^{s_1}} \cdot \frac{d^{s_2} B_l^{(3)}(y)}{dy^{s_2}} \right)^2 dx dy \\ &= \frac{1}{M_x M_y} \sum_{s_1+s_2=2} \sum_{k,k_1=1}^{M_x} \sum_{l,l_1=1}^{M_y} F_{k,k_1}^{(s_1)} C_{kl} C_{k_1 l_1} G_{l,l_1}^{(s_2)}, \end{aligned} \quad (3.3)$$

where  $F_{k,k_1}^{(s_1)}, G_{l,l_1}^{(s_2)}$  are defined by

$$\begin{aligned} F_{k,k_1}^{(s_1)} &= \int_0^1 \frac{d^{s_1} B_k^{(3)}(x)}{dx^{s_1}} \cdot \frac{d^{s_1} B_{k_1}^{(3)}(x)}{dx^{s_1}} dx, \\ G_{l,l_1}^{(s_2)} &= \int_0^1 \frac{d^{s_2} B_l^{(3)}(y)}{dy^{s_2}} \cdot \frac{d^{s_2} B_{l_1}^{(3)}(y)}{dy^{s_2}} dy. \end{aligned} \quad (3.4)$$

Substituting (3.3), (3.3) in (1.3) and differentiating with respect to  $C_{kl}$ , we have

$$\begin{aligned} \frac{\partial \Phi_\alpha^2}{\partial C_{kl}} &= \frac{\alpha}{N_x N_y} \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} \left( -2u_{ij} (B_x)_{ki} (B_y)_{jl} + 2 \sum_{k_1=1}^{M_x} \sum_{l_1=1}^{M_y} C_{k_1 l_1} (B_x)_{ki} (B_x)_{k_1 i} (B_y)_{jl} (B_y)_{j l_1} \right) \\ &\quad + \frac{2(1-\alpha)}{M_x M_y} \sum_{s_1+s_2=2} \sum_{k_1=1}^{M_x} \sum_{l_1=1}^{M_y} F_{k,k_1}^{(s_1)} C_{k_1 l_1} G_{l,l_1}^{(s_2)}, \end{aligned} \quad (3.5)$$

or matricially,

$$\frac{\partial \widehat{\Phi}_\alpha^2}{\partial C} = \frac{-2\alpha}{N_x N_y} \left( \widehat{B}_x \widehat{U} \widehat{B}_y^T - \widehat{B}_x \widehat{B}_x^T \widehat{C} \widehat{B}_y \widehat{B}_y^T \right) + \frac{2(1-\alpha)}{M_x M_y} \sum_{s_1+s_2=2} \widehat{F}^{(s_1)} \widehat{C} \widehat{G}^{(s_2)}, \quad (3.6)$$

where

$$\widehat{B}_x = [(B_x)_{ki}] \in \mathbb{R}^{M_x \times N_x}, \quad (3.7)$$

$$\widehat{B}_y = [(B_y)_{lj}] \in \mathbb{R}^{M_y \times N_y}, \quad (3.8)$$

$$\widehat{U} = [u_{ij} = u(x_i, y_j)] \in \mathbb{R}^{N_x \times N_y}. \quad (3.9)$$

Hence, we show that the critical points of the functional,  $\Phi_\alpha^s$ , satisfy the matricial equation

$$\widehat{B}_x \widehat{B}_x^T \widehat{C} \widehat{B}_y \widehat{B}_y^T = \widehat{B}_x \widehat{U} \widehat{B}_y^T + (1 - 1/\alpha) \kappa_{xy} \sum_{s_1+s_2=2} \widehat{F}^{(s_1)} \widehat{C} \widehat{G}^{(s_2)}, \quad (3.10)$$

where  $\kappa_{xy} = \frac{N_x N_y}{M_x M_y}$ . It can be shown that the matrices  $\widehat{B}_x \widehat{B}_x^T$  and  $\widehat{B}_y \widehat{B}_y^T$  are Toeplitz, symmetric, and positive definite [7]. In order to find the matrix of coefficients  $\widehat{C}$  we can use the iterative process

$$\begin{aligned} \widehat{C}^{(n+1)} &= \left( \widehat{B}_x \widehat{B}_x^T \right)^{-1} \left( \widehat{B}_x \widehat{U} \widehat{B}_y^T + \right. \\ &\quad \left. (1 - 1/\alpha) \kappa_{xy} \sum_{s_1+s_2=2} \widehat{F}^{(s_1)} \widehat{C}^{(n)} \widehat{G}^{(s_2)} \right) \left( \widehat{B}_y \widehat{B}_y^T \right)^{-1}, \end{aligned} \quad (3.11)$$

where  $\widehat{C}^{(1)}$  is the solution of the least-squares problem ( $\alpha = 1$ ).  $\square$

*Remark 3.2.* In that way (1.3) and (3.11) give the solution of the smoothing problem in the subspace of cubic splines. The representation of the function as a tensorial product (2.5), allowed us to reduce the 2-D problem to two independent unidimensional problems by separating the variables  $x$  and  $y$ . Furthermore, the matrices in the formula (3.11) are calculated and stored in independent files.

**Corollary 3.3.** *For given values of  $\alpha$ ,  $N_x$ ,  $M_x$ ,  $N_y$ , and  $M_y$  we can represent the expression (3.11) in the compact form*

$$\widehat{C}^{(n+1)} = \widehat{V}_x \widehat{U} \widehat{V}_y + (1 - 1/\alpha) \kappa_{xy} \widehat{P}_x \left( \sum_{s_1+s_2=2} \widehat{F}^{(s_1)} \widehat{C}^{(n)} \widehat{G}^{(s_2)} \right) \widehat{P}_y, \quad (3.12)$$

with  $\widehat{C}^{(1)} = \widehat{V}_x \widehat{U} \widehat{V}_y$ , where  $V_x$  and  $V_y$  are computed by using the thin  $QR$  factorization [7] to the matrices  $\widehat{B}_x^T$  and  $\widehat{B}_y^T$ , and

$$\widehat{P}_x = \left( \widehat{B}_x \widehat{B}_x^T \right)^{-1}, \quad \widehat{P}_y = \left( \widehat{B}_y \widehat{B}_y^T \right)^{-1}. \quad (3.13)$$

*Proof.* Since  $N_x \geq M_x$  and  $N_y \geq M_y$ , and the matrices  $\widehat{B}_x^T$  and  $\widehat{B}_y^T$  have full column rank, their thin  $QR$  factorizations are  $\widehat{B}_x^T = \widehat{Q}_x \widehat{R}_x$ ,  $\widehat{B}_y^T = \widehat{Q}_y \widehat{R}_y$ , where  $\widehat{Q}_x \in \mathbb{R}^{N_x \times M_x}$ ,  $\widehat{R}_x \in \mathbb{R}^{M_x \times M_x}$  and  $\widehat{Q}_y \in \mathbb{R}^{N_y \times M_y}$ ,  $\widehat{R}_y \in \mathbb{R}^{M_y \times M_y}$ . Since the matrices  $Q$  have orthonormal columns and the matrices  $R$  are upper triangular, we can find matrices  $\widehat{V}_x$  and  $\widehat{V}_y$  as the solution of the equations

$$\widehat{R}_x \widehat{V}_x = \widehat{Q}_x^T, \quad \widehat{R}_y \widehat{V}_y = \widehat{Q}_y^T, \quad (3.14)$$

which can be solved by back substitution. Now, we have that

$$\widehat{P}_x = \left( \widehat{B}_x \widehat{B}_x^T \right)^{-1} = \left( \widehat{R}_x^T \widehat{Q}_x^T \widehat{Q}_x \widehat{R}_x \right)^{-1} = \left( \widehat{R}_x^T \widehat{R}_x \right)^{-1},$$

$$\widehat{P}_y = \left( \widehat{B}_y \widehat{B}_y^T \right)^{-1} = \left( \widehat{R}_y^T \widehat{Q}_y^T \widehat{Q}_y \widehat{R}_y \right)^{-1} = \left( \widehat{R}_y^T \widehat{R}_y \right)^{-1}.$$

From (3.14) we get

$$\widehat{P}_x \widehat{B}_x = \widehat{R}_x^{-1} \widehat{Q}_x^T = \widehat{V}_x, \quad \widehat{B}_y^T \widehat{P}_y = \widehat{Q}_y \widehat{R}_y^{-T} = \widehat{V}_y.$$

In the particular case as  $\alpha = 1$ , we have

$$\widehat{C}^{(1)} = \widehat{V}_x \widehat{U} \widehat{V}_y^T. \quad (3.15)$$

Hence, the first approximation to the solution of the smoothing problem is obtained with (2.5), where the coefficients  $C_{kl}$  are given by (3.15)  $\square$

## 4 Smoothing Algorithm

An algorithm based on the tensor product representation (2.5), the iterative process (3.12) above and the stopping criteria of Del Piero [13, 2] for determining the 2-D cubic smoothing spline solution to problem (1.3) now goes as follows:



#### 4.1 Algorithm

- (i) Store the matrix of data with noise,  $\widehat{U}$ .
- (ii) Calculate the matrices  $\widehat{B}_x$  and  $\widehat{B}_y$  by (3.4).
- (iii) Compute a  $QR$  factorization of the matrices  $\widehat{B}_x^T$  and  $\widehat{B}_y^T$ .
- (iv) Solve the equations (3.14) to  $\widehat{V}_x$  and  $\widehat{V}_y^T$ .
- (v) Calculate  $\widehat{C}^{(1)}$  by (3.15).
- (vi) Compute the function  $v(x, y)$  in (2.5), corresponding to  $\widehat{C}^{(1)}$ , the least squares solution ( $\alpha = 1$ ).
- (vii) Compute the  $\alpha$  that minimizes the GCV or the expected mean squared error applying the *Conjugate Gradient Method*, with GCV as stopping criteria [13, 2].
- (viii) Compute the next coefficient matrix by using (3.12).

For convenience in our calculations, we assume equal number of data points in both axes,  $N_x = N_y = N$ , and equal knots in both axes,  $M_x = M_y = M$ . Usually, in practice  $M$  and  $N$  are powers of 2 such that  $2 \leq M \leq 16$  and  $N \gg 1$ . Classical smoothing with 2-D FFT uses  $O((N \log_2 N)^2)$  operations,[19, 11], while a given smoothing parameter  $\alpha$ , the smoothing with cubic  $B$ -splines uses  $O(MN^2)$  operations, and  $O(MN)$  storage providing a great saving of operations. Table 1 shows the computational efficiency of the B-spline for some values of  $M$  and  $N$ .

| $M$ | $N$ | FFT $O((N \log N)^2)$ | B-splines $O(MN) + O(MN^2)$ |
|-----|-----|-----------------------|-----------------------------|
| 4   | 64  | 1.5E5                 | 1.7E4                       |
| 16  | 256 | 4.2E6                 | 2.1E6                       |

Table 1: Computational complexity between the 2-D FTT algorithm and the 2-D B-Spline algorithm.

## 5 Numerical Results and Conclusions

### 5.1 Numerical Results

To test our algorithm, we simulate noisy images by considering a smooth function  $v$ , and generating a random matrix of errors  $\varepsilon$ , with standard normal distribution (Gaussian noise) or uniform distribution (uniform noise). An additive noise over the function  $v$  in each data point is obtained by

$$u_{ij} = v(x_i, y_j) + \varepsilon_{ij}, \quad i = 1, \dots, N_x, j = 1, \dots, N_y, \quad (x_i, y_j) \in \Omega_d.$$

Figure 5.1 on page 10 shows tests of the technique for the functions

$$R(x, y) = \frac{1}{(1 + 25x^2)(1 + 25y^2)}, \text{ the Runge's function}$$

and

$$H(x, y) = e^{-(x^2+y^2)} \sin(5(x^2 + y^2)),$$

over the square  $[-1, 1] \times [-1, 1]$ . Figure 5.1:(a) shows the graph of  $R$  with free noise; (b) shows the graph of  $R$  with a random noise using uniform distribution of 15%; (c) is the first smoothing of  $R$  obtained by the algorithm 4.1. Similarly, graphs (d), (f) and (g) are relative to the function  $H$ .

The Algorithm 4.1 was coded in `Fortran 90` for different 2-D functions. The number of knots considered per axis was  $M = 8$ , and the number of data points per axis was  $N = 64$ . Thus, the domain was discretized by  $64^2 = 4096$  points, where  $R(x, y)$  was evaluated in these points and stored in the vector  $u0$  with dimension 4096. Similarly the noisy function and the smooth function were stored in the vectors  $u$  and  $v$  respectively. The residual  $r$  is the vector  $u - v$ , with  $\|r\| = .7507E - 02$ . Referring to Figure 5.1:(c), on a Pentium with 90 Mhz, the processing time in `Fortran 90` was 2 seconds. Similar results were obtained for the function  $H$ , with corresponding graphs (d), (e) and (f).

## 5.2 Conclusions

In this paper, we provide a solution algorithm based on the tensor product of unidimensional uniform splines for smoothing 2-D noisy images that came from 2-D functions. The algorithm is based in the minimization of the energy functional,  $\Phi_\alpha$ , with respect to the coefficients of the tensor product splines and to  $\alpha$ . The main feature of this algorithm is its great saving in time and computational complexity. The consideration of the cubic splines for a coarser discretization  $\tilde{\Omega}_d$  than the data set on  $\Omega_d$  is a regularization, which provides the complexity advantage to the extent that  $M \ll (\log_2 N)^2$ . There are some limitations for tensor product methods. However, their computational advantages allow their use in many applications. Recently, the tensor product of B-splines surfaces in NURBS form has been studied on reconstruction of smooth biquintic B-spline surfaces over arbitrary topology by X. Shi and his group [16, 15], they considered a constant fairness factor. Our future research work is to implement NURBS in our algorithm. Following the valuable comments of one of the referees, we are considering the comparison of our spline approximation with the local averaging method involving more general type of functions.

## Acknowledgments

The authors wish to thank the referees of the paper, as well as their colleagues Dr. Hector J. Martinez, Dr. Carlos Mejia and Dr. Illia Mikhailov for their encouraging and useful discussions. The first author wishes to thank to Mike Gardebled for his careful reading and comments to improve the paper.

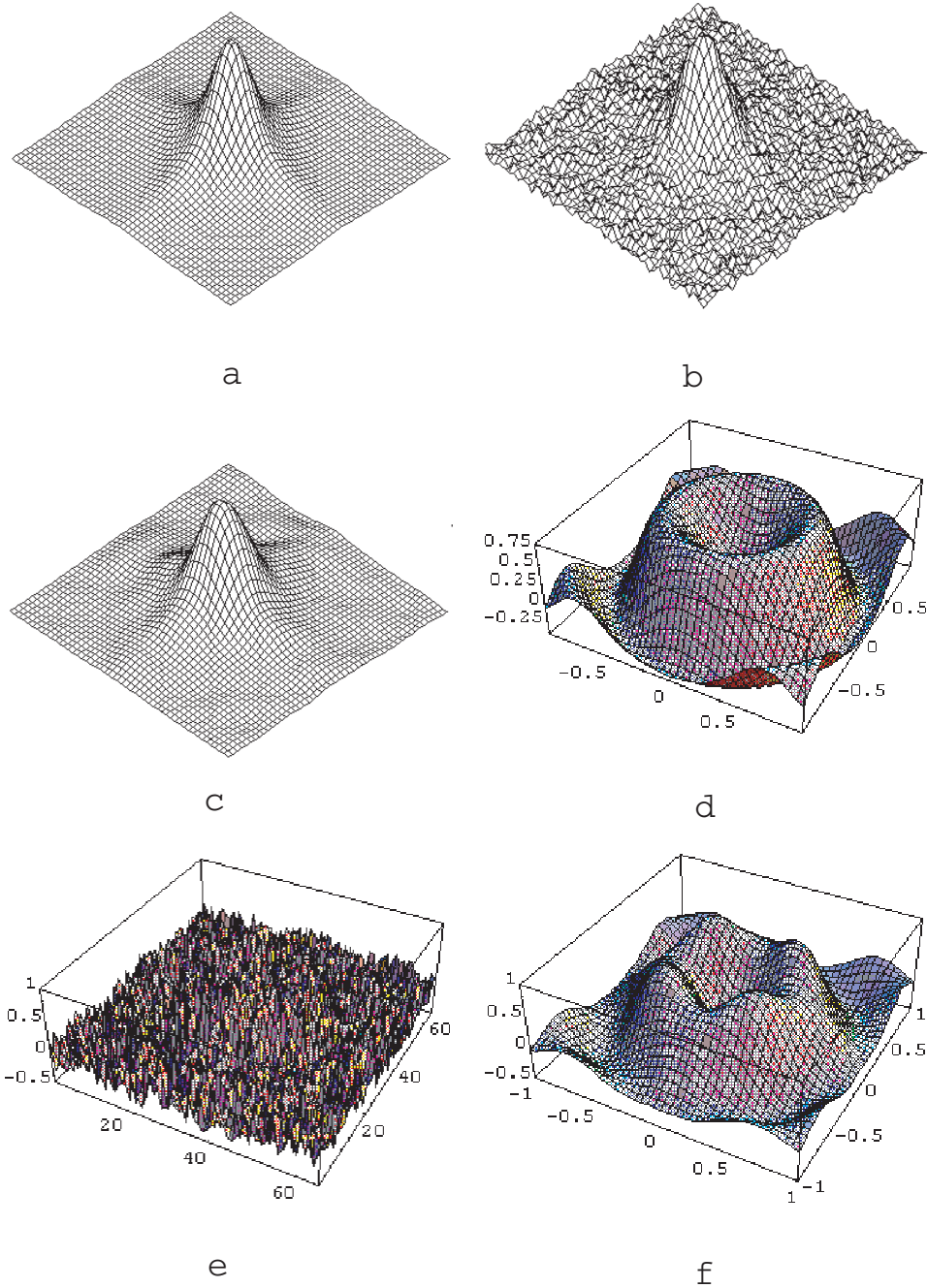


Figure 5.1: (a) Graph of  $R$  without noise. (b) Graph of  $R$  with 15% uniform noise. (c) Reconstruction of  $R$ . (d) Graph of  $H$  with free noise. (e) Graph of  $H$  with 30% uniform noise. (f) Reconstruction of  $H$ .

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*Chapter 13*

## SUPERVISED LEARNING UNDER SAMPLE SELECTION BIAS FROM PROTEIN STRUCTURE DATABASES

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### Abstract

In supervised learning it is commonly assumed that labeled data available for learning a prediction model is a random sample from the underlying distribution. In real-life scenarios, however, this assumption is often violated due to sample selection bias. Consequently, a prediction model learned from such labeled data could be suboptimal and may not generalize well on unseen examples from the same population. In this chapter, we provide a brief review of the sample selection bias problem and existing methods for detecting and correcting such bias. We then describe our contrast classifier framework for addressing this problem and illustrate its effectiveness in several bioinformatics applications related to learning from biased protein structure databases.

**Key Words:** sample selection bias, contrast classifier, supervised learning.

**AMS Subject Classification:** Primary 43A60.

## 1 Introduction

In supervised learning [1, 2], the objective is to learn a prediction model using certain *learning algorithm* from a labeled dataset  $D_L$  with  $N$  examples  $\{(\mathbf{x}_i, y_i)\}_{i=1,2,\dots,N}$  sampled independently from an unknown distribution  $p(\mathbf{x}, y)$ , where  $\mathbf{x}_i = [x_{i1}, x_{i2}, \dots, x_{iK}]$  is the feature vector from  $K$ -dimensional domain  $\mathbb{X}$  and  $y_i$  is the label from domain  $\mathbb{Y}$ . In statistical terminology,  $x_{i1}, x_{i2}, \dots, x_{iK}$  are also called independent variables and  $y_i$  is called dependent variable or outcome. The resulting model should optimally approximate

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\*E-mail address: vucetic@ist.temple.edu; This study was supported in part by NSF grant IIS-0219736 to Z. Obradovic and S. Vucetic and NSF grant CSE-IIS-0196237 to Z. Obradovic and A.K. Dunker.

the unknown relationship between  $\mathbf{x}$  and  $y$ , such that it can predict  $y$  for  $\mathbf{x}$  unseen in  $D_L$  as accurately as possible.

A supervised learning task can either be a *regression* or *classification* task depending on whether label  $y$  is continuous or categorical. For regression tasks, the goal is to approximate the conditional expectation  $E[y|\mathbf{x}]$  as the predicted  $y$  for given  $\mathbf{x}$ . For classification tasks, the goal is to predict the exact class label  $y$  for given  $\mathbf{x}$ . This can be achieved by first estimating the class conditional probability density  $p(y = c|\mathbf{x})$ , where class label  $c = 1, 2, \dots, C$  (number of classes), and then predicting  $y$  as the  $c$  that has maximal  $p(y = c|\mathbf{x})$ .

A basic assumption in supervised learning is that labeled dataset  $D_L$  should be a true random sample from the underlying distribution  $p(\mathbf{x}, y)$  such that the model learned from  $D_L$  is applicable to unseen examples from the same distribution. However, this assumption is often violated in real-life scenarios due to the problem of *sample selection bias*, or *selection bias*, *sampling bias* [3], where  $D_L$  is not representative of the population. Consequently, the resulting prediction model could be suboptimal and may not perform well on unseen examples.

Sample selection bias may be caused by many reasons. The labeled data may be *truncated* or *censored* according to values of  $\mathbf{x}$  and  $y$ . In social study surveys, the participants are often *self-selected* instead of chosen randomly from the general population. A similar situation in direct marketing is the *non-response* problem in which people who do not respond to mailings may have significantly different characteristics from those that do. *Attrition bias* is due to the loss of participants in clinical trial evaluations and other studies. In supervised classification, one type of bias is that labeled data have significantly different class distribution from the population, possibly due to the class-dependent costs to obtain class labels.

Since sample selection bias might significantly deteriorate the learning performance, it is important to detect and correct such bias. However, the underlying bias mechanism is often unknown in real life. Fortunately, quite often an unlabeled dataset  $D_U = \{\mathbf{x}_i\}_{i=1,2,\dots,M}$ ,  $M \gg N$ ,  $\mathbf{x}_i \in \mathbb{X}$ , is also available and can be assumed as representative of the underlying distribution. Under certain conditions, it is possible to model the bias mechanism by comparing the labeled dataset  $D_L$  and the unlabeled dataset  $D_U$ , and to use this knowledge to remedy the problem of bias.

Research on sample selection bias originated from econometrics and related areas in the late 1970s. Since the seminal work by James J. Heckman [3], various statistical methods to address the sample selection bias problem have been proposed in different areas such as econometrics [3, 4], credit scoring [5, 6, 7], social sciences [8], and clinical trials [9]. In machine learning community, specific forms of the sample selection bias problem have been studied for years, such as learning from skewed class distribution [10, 11, 12], learning from labeled and unlabeled data [13], and outlier detection [14]. It was not until recently that studies began to emerge which systematically investigate the bias problem in machine learning context.

Zadrozny [15] formalized the sample selection bias in machine learning terms and classified several popular supervised learning algorithms into two categories, "local" or "global", depending on their sensitivities to a common type of bias. Fan et al. [16] further improved Zadrozny's classification and provided more insights into the effects of sample selection bias in machine learning. Smith and Elkan [17] proposed a Bayesian network

framework as another attempt to formalize sample selection bias in the context of machine learning. In an empirical study by Chawla and Karakoulas [18], the influence of sample selection bias (along with other factors) on performance of semi-supervised learning algorithms was systematically evaluated; they also examined the effectiveness of several bias correction methods in semi-supervised learning scenarios.

The rest of this chapter is organized into two parts. In the first part we provide an overview of the sample selection bias problem as well as existing methods for bias detection and correction. In the second part we present our contrast classifier framework [19] for addressing this problem. After a brief introduction to the contrast classifier, we illustrate its effectiveness on several bioinformatics applications related to protein structure prediction.

## 2 Sample Selection Bias

### 2.1 Formal Definition

Sample selection bias can be formally modeled [15] by associating a binary random variable  $s$  with each possible  $(\mathbf{x}, y) \in X \times Y$ ; an example  $(\mathbf{x}, y)$  may be selected into (or observed in)  $D_L$  if and only if  $s = 1$ . The *sample selection probability*  $p(s = 1|\mathbf{x}, y)$  quantifies the bias toward a given example  $(\mathbf{x}, y)$  and determines how likely it can be sampled besides its original probability density  $p(\mathbf{x}, y)$ . From this perspective, the labeled dataset  $D_L$  is in effect sampled from distribution  $p(\mathbf{x}, y, s = 1) = p(s = 1|\mathbf{x}, y) \cdot p(\mathbf{x}, y)$  instead of the original distribution.

If indicator  $s$  is independent of both  $\mathbf{x}$  and  $y$ , the sample is unbiased. If this is not the case, three types of bias are possible: (I)  $s$  depends on  $\mathbf{x}$  but is independent of  $y$  given  $\mathbf{x}$ , i.e.  $p(s = 1|\mathbf{x}, y) = p(s = 1|\mathbf{x})$  and  $p(y|\mathbf{x}, s = 1) = p(y|\mathbf{x})$ ; (II)  $s$  depends on  $y$  but is independent of  $\mathbf{x}$  given  $y$ , i.e.  $p(s = 1|\mathbf{x}, y) = p(s = 1|y)$  and  $p(\mathbf{x}|y, s = 1) = p(\mathbf{x}|y)$ ; (III)  $s$  depends on both  $\mathbf{x}$  and  $y$ . In supervised learning, Type I or Type II bias can often be assumed. Type III bias is not tractable unless some additional restrictive assumptions can be made.

Note that it is also possible that indicator  $s$  depends on features  $\mathbf{x}_s$  not included in  $\mathbf{x}$ , such that the sample selection probability is specified as  $p(s = 1|\mathbf{x}, \mathbf{x}_s, y)$ . In such case, the sample selection bias cannot be modeled in general. In machine learning, however, the emphasis is on optimizing learning performance from given data rather than on understanding the exact data generating mechanism. Therefore, we do not consider this scenario further.

### 2.2 Effects of Sample Selection Bias on Popular Learning Algorithms

Two recent studies [15, 16] were conducted to investigate the effect of Type I bias on several popular supervised classification algorithms. As claimed by Zadrozny [15], "local" learning algorithms whose outputs depend only on estimated  $p(y|\mathbf{x})$  should not be sensitive to Type I bias, since the actual probability density  $p(y|\mathbf{x}, s = 1)$  estimated from the biased labeled dataset  $D_L$  would be similar to  $p(y|\mathbf{x})$  from the whole population. On the other hand, "global" learning algorithms that rely on both  $p(y|\mathbf{x})$  and  $p(\mathbf{x})$ , or equivalently  $p(\mathbf{x}, y)$ , are more likely to be affected by the bias since  $p(\mathbf{x}, s = 1) \neq p(\mathbf{x})$ . Fan et al. [16] argued that these are not always true. They illustrated that "local" learning algorithms on



particular biased datasets might still be affected by the bias if the estimated  $p(y|\mathbf{x}, s = 1)$  is inaccurate due to incorrect model assumption, while "global" learning algorithms would not be sensitive to the bias if  $y$  and  $s$  depend on two disjoint feature subsets of  $\mathbf{x}$ .

For Type II bias, the sample selection depends only on label  $y$ , i.e.  $p(s = 1|\mathbf{x}, y) = p(s = 1|y)$ . In supervised classification, this implies that  $D_L$  has different class distribution from that of the population but the class conditional distribution  $p(\mathbf{x}|y)$  remains unaltered. One common cause of this type of bias is the class-dependent costs for obtaining class labels ( $y$ ), or equivalently, obtaining examples for certain classes are more difficult than obtaining those for other classes. As illustrated by Weiss and Provost [11], predictor models learned from  $D_L$  with biased class distribution are likely to have, although not always, deteriorated performance on unseen examples from the same population. However, such bias may also be created intentionally to improve the learning performance under certain circumstances, e.g. when the class distribution is extremely imbalanced [10]. In the case of binary classification, this means that one class (minority class) is rare while the other (majority class) is abundant in the population, and the labeled data is assumed to be unbiased. Since many traditional learning algorithms perform poorly under such circumstances, a common strategy is to make a balanced training set by either down-sampling the majority class or up-sampling the minority class [10], and thus creating the bias.

### 2.3 Estimating Sample Selection Probability

It is possible to estimate  $p(s = 1|\mathbf{x}, y)$  if we are given an unlabeled dataset  $D_U = \{\mathbf{x}_i\}_{i=1..M}$ ,  $M \gg N$ , that is sampled from the underlying distribution  $p(\mathbf{x}, y)$ . It is worth noting that, unlike labeled data, unlabeled data is typically easier to obtain in large amount. As we will show later, the sample selection probability  $p(s = 1|\mathbf{x})$  for Type I bias can be estimated indirectly using the contrast classifier approach [19]. For Type II bias, the sample selection probability  $p(s = 1|y)$  can also be obtained indirectly by estimating the class proportion in  $D_U$  using a bootstrapping methodology [12]. For Type III bias, the sample selection probability  $p(s = 1|\mathbf{x}, y)$  cannot be modeled in similar manner as for Type I and II bias due to its dependence on both  $\mathbf{x}$  and  $y$ .

Recently [20], a method for estimating  $p(s = 1|\mathbf{x}, y)$  under Type III bias was proposed which utilizes the equation  $E_D[g(\mathbf{x})] = E_D[s \cdot g(\mathbf{x})/p(s = 1|\mathbf{x}, y)]$ , where  $g(\mathbf{x})$  is a real-valued function,  $E_D[\cdot]$  denotes expectation over  $D$  representing the whole population containing both labeled ( $s = 1$ ) and unlabeled ( $s = 0$ ) examples. Assuming that  $p(s = 1|\mathbf{x}, y)$  is from a parametric family with  $k$  parameters,  $k$  equations can be established using empirical estimates of  $E_D[\cdot]$  for  $k$  different functions  $g_1(\mathbf{x})$ ,  $g_2(\mathbf{x})$ , ...,  $g_k(\mathbf{x})$ . By solving the  $k$  equations, the parameters in  $p(s = 1|\mathbf{x}, y)$  can be determined. However, its practical use might be limited due to the inherent difficulties in choosing appropriate functional forms for both  $p(s = 1|\mathbf{x}, y)$  and  $g(\mathbf{x})$  and solving non-linear equations. Fortunately, under certain assumptions, explicit estimation of  $p(s = 1|\mathbf{x}, y)$  may not be necessary for bias detection and correction as we will see in the next section.

### 2.4 Methods for Sample Selection Bias Detection and Correction

In this section we will review some existing methods that address sample selection bias problem. All of them utilize unlabeled data and model the sample selection probability

explicitly or implicitly.

The famous Heckman's two-step procedure [3] was originally developed for regression analysis in studies of labor force supply. In this procedure, two regression models are built: the *outcome model* is an ordinary linear squares (OLS) model for estimating the outcome ( $y$ ) of interest, while the *selection model* is a probit linear model for modeling the unknown sample selection mechanism. In the first step, the selection model is built using both labeled ( $s = 1$ ) and unlabeled ( $s = 0$ ) data. Using the estimated model parameters, the *inverse Mill's ratio*  $\lambda$ , which is related to the sample selection probability, can be calculated for each  $\mathbf{x}$ . In the second step,  $\lambda$  is included as an additional feature (or independent variable) to build an unbiased outcome model using labeled examples ( $s = 1$ ) only.

An important assumption of the Heckman's procedure, under which  $\lambda$  can be calculated, is that error terms of the two initial models (not including  $\lambda$ ) follow a bivariate normal distribution. If there is no correlation between the two error terms (i.e. a diagonal covariance matrix), the sample selection is independent of  $y$  given  $\mathbf{x}$ , i.e. we have Type I bias; otherwise, we have Type III bias. In the later case, the selection model output may not be regarded as a good estimate for the sample selection probability  $p(s = 1|\mathbf{x}, y)$  since its dependence on  $y$  given  $\mathbf{x}$  is reflected in the error term. In this sense, the Heckman's procedure avoids direct estimation of  $p(s = 1|\mathbf{x}, y)$  but is still able to correct the bias.

The bivariate probit model [6, 7] can be viewed as a variant of the Heckman's procedure that is specific for classification tasks, where the OLS outcome model is replaced by another probit model for binary classification. Like the Heckman's procedure, the bivariate probit model also assumes that error terms of the two models follow a bivariate normal distribution and is applicable to scenarios where Type III bias is present.

In a procedure [21] similar to the Heckman's, Elkan used machine learning algorithms such as Naive Bayes classifiers or decision trees to estimate the selection model. Instead of the inverse Mill's ratio  $\lambda$ , which is no longer applicable, he used the selection model output, or estimated sample selection probability, as the additional feature to build the outcome model. Since no assumption is made about the correlation between the error terms of two models, this procedure implicitly assumes that sample selection depends only on  $\mathbf{x}$  but not on  $y$  and is therefore only applicable to Type I bias. In their empirical study [18], Chawla and Karakoulas further adapted this procedure to solve semi-supervised classification problems.

### 3 Learning from Biased Data with Contrast Classifier

The contrast classifier [19] is a binary classifier which models the distributional difference between the labeled dataset  $D_L$  and the unlabeled dataset  $D_U$ . Its name *contrast* comes from the meaning of its output, which represents a measure of difference, or contrast, in probability density of a given data point  $\mathbf{x}$  between labeled and unlabeled data. Given this property, the contrast classifier could be used in a wide range of important machine learning or data mining applications such as outlier detection, one-class classification, density estimation, and learning from biased data [19]. In the following sections we will show that contrast classifier can be used to estimate the sample selection probability under the Type I bias.

### 3.1 Contrast Classifier

To build a contrast classifier, a new dataset is constructed by assigning class labels 0 and 1 to examples from  $D_L$  and  $D_U$ , respectively. Note that the original label  $y$  is not used and the new label is not related to  $y$  in labeled examples. The contrast classifier is trained as a binary classifier on the constructed dataset using classification algorithms (e.g. feedforward neural networks [1, 2]) which estimate the posterior conditional class probability. An optimally trained contrast classifier would output

$$cc(\mathbf{x}) = \frac{r \cdot u(\mathbf{x})}{(1-r) \cdot l(\mathbf{x}) + r \cdot u(\mathbf{x})}$$

where  $l(\mathbf{x})$  and  $u(\mathbf{x})$  are the probability density functions of  $D_L$  and  $D_U$  respectively, and  $r$  is the fraction of unlabeled examples in training data.

In practice  $D_L$  and  $D_U$  are often *imbalanced* because unlabeled data could be obtained easier and in large amount. Thus, instead of the original  $D_L$  and  $D_U$ , balanced training set with  $r = 0.5$  should be used. For  $r = 0.5$  it follows that

$$\frac{l(\mathbf{x})}{u(\mathbf{x})} = \frac{1 - cc(\mathbf{x})}{cc(\mathbf{x})}$$

Thus, the contrast classifier output  $cc(\mathbf{x})$  is a monotonically decreasing function of the ratio  $l(\mathbf{x})/u(\mathbf{x})$ , which reveals the distributional difference between the two datasets.

If Type I bias can be assumed, i.e.  $p(s = 1|\mathbf{x}, y) = p(s = 1|\mathbf{x})$ , the contrast classifier output  $cc(\mathbf{x})$  is related to the sample selection probability  $p(s = 1|\mathbf{x})$ . While  $l(\mathbf{x})$  approximates  $p(\mathbf{x}, s = 1)$ ,  $u(\mathbf{x})$  may approximate either  $p(\mathbf{x}, s = 0)$  or  $p(\mathbf{x})$  depending on the actual application. For example, in credit scoring problem [6]  $D_L$  contains all accepted applicants ( $s = 1$ ) while  $D_U$  contains all rejected applicants ( $s = 0$ ); thus, it is appropriate to assume  $u(\mathbf{x}) = p(\mathbf{x}, s = 0)$ . In this case

$$cc(\mathbf{x}) = 1 - p(s = 1|\mathbf{x})$$

In some other scenarios it is more reasonable to assume that unlabeled dataset  $D_U$  is representative of the whole population, i.e.  $u(\mathbf{x}) = p(\mathbf{x})$ . In this case

$$cc(\mathbf{x}) = \frac{1}{1 + p(s = 1|\mathbf{x})}$$

In either case,  $cc(\mathbf{x})$  is a decreasing function of  $p(s = 1|\mathbf{x})$ , and could therefore be a useful measure of sample selection bias. In the following discussion, it is assumed that  $u(\mathbf{x}) = p(\mathbf{x})$ .

### 3.2 Selecting Underrepresented Examples

According to the interpretation above, contrast classifier output  $cc(\mathbf{x})$  could be a useful measure of sample selection bias. An example  $\mathbf{x}$  is *underrepresented* if  $cc(\mathbf{x}) > 0.5$ , or *overrepresented* if  $cc(\mathbf{x}) < 0.5$ . Using the  $cc(\mathbf{x})$  distribution for  $D_L$  as a *null distribution*, we can determine if an unlabeled example  $\mathbf{x}$  is likely to come from the same distribution as

$D_L$ , i.e. if  $\mathbf{x}$  is well represented by  $D_L$ . A threshold  $\theta^p$  is first determined such that only 100p% examples in  $D_L$  satisfy  $cc(\mathbf{x}) > \theta^p$ . If an unlabeled example  $\mathbf{x}$  has  $cc(\mathbf{x}) > \theta^p$ , we conclude that it is underrepresented in  $D_L$ .

In a classification problem of  $C(\geq 2)$  classes, the labeled dataset  $D_L$  is inherently heterogeneous, and consequently a single contrast classifier built using the whole  $D_L$  versus  $D_U$  might not detect the sample selection bias effectively. A better approach might be building  $C$  *class-specific* contrast classifiers  $cc_c(\mathbf{x})$ ,  $c = 1, 2, \dots, C$ . For an example  $\mathbf{x}$  to be underrepresented in  $D_L$ , we require that  $cc_c(\mathbf{x}) > \theta_c^p$  holds for all  $c = 1, 2, \dots, C$ . Similar to the discussion above, the threshold  $\theta_c^p$  is chosen for the  $c$ -th contrast classifier such that only 100p% class  $c$  labeled examples in  $D_L$  satisfy the inequality.

### 3.3 Assessing Overall Level of Bias

Contrast classifier output  $cc(\mathbf{x})$  can be used to measure the overall level of bias. One can compare the difference between the two  $cc(\mathbf{x})$  distributions (histograms) of  $D_L$  and  $D_U$ : the bias is negligible if the two distributions largely overlap; it is significant if the two distributions are well separated and/or have different shapes. The two-sample *Kolmogorov-Smirnov* (KS) goodness-of-fit test [22] could be applied to compare the two  $cc(\mathbf{x})$  distributions and the resulting KS statistic could serve as a quantitative measure of the overall bias.

Alternatively, the  $cc(\mathbf{x})$  distribution of  $D_U$  alone could reveal the level of overall bias if  $D_U$  is representative of the population. If this distribution is centered on 0.5,  $l(\mathbf{x})$  should be similar to  $u(\mathbf{x})$  almost everywhere, which indicates that the overall level of bias is negligible. On the other hand, if this distribution is scattered and/or its mean deviates considerably from 0.5, the overall level of bias is significant. We therefore define a quantitative measure  $\Delta$  as

$$\Delta = \sqrt{\frac{|D_U|}{\sum_{i=1}^{|D_U|} (cc(\mathbf{x}_i) - 0.5)^2 / |D_U|}}$$

where  $\mathbf{x}_i$  is  $i$ -th example in  $D_U$ . The value of  $\Delta$  will approach 0 when bias is negligible since all  $cc(\mathbf{x}_i)$  should be close to 0.5. It will be large if the bias is significant and many  $cc(\mathbf{x}_i)$  are far away from 0.5. In the extreme case when all  $cc(\mathbf{x}_i) = 1$ ,  $\Delta$  will be 0.5.

### 3.4 Correcting Sample Selection Bias

It has been shown that resampling the labeled dataset  $D_L$  with weight proportional to  $1/p(s = 1|\mathbf{x})$  could be a useful way to correct sample selection bias [15, 23], given that  $p(s = 1|\mathbf{x})$  is nonzero at any  $\mathbf{x} \in \mathbb{X}$  and can be properly estimated. However, in practice this approach is not always applicable. For example, if  $D_L$  is not large enough and/or highly biased, some regions in the feature space  $\mathbb{X}$  will be highly underrepresented and very few examples in  $D_L$  will come from these regions. Consequently, resampling  $D_L$  alone would be of little use in correcting the bias. Furthermore, the presence of duplicate examples as a result of resampling might lead to overfitting by some learning algorithms. In such a case, it becomes necessary to enlarge  $D_L$  by labeling additional examples.

In general, one can always randomly select unlabeled examples for labeling, and as more examples are labeled the sample selection bias should be eventually corrected. Obviously, this would be inefficient if the labeling cost is significant. Therefore, the unlabeled

examples should be selected in a cost-effective way. The contrast classifier output  $cc(\mathbf{x})$  can serve this purpose since it directly measures the bias toward individual examples. We can first rank all unlabeled examples from  $D_U$  based on their  $cc(\mathbf{x})$  values, and then select those with high values for labeling. Intuitively, this approach would be more effective in bias reduction.

Based on the discussion above, a one-step approach would consist of building a single contrast classifier from the initial  $D_L$  and  $D_U$  and selecting  $G$  underrepresented examples from  $D_U$  based on  $cc(\mathbf{x})$ . In practice it is often difficult to determine an appropriate value for  $G$  such that the labeling cost is minimized. We therefore propose a procedure that iteratively builds contrast classifiers and incrementally selects underrepresented examples [24]. At each iteration, a contrast classifier is built from current  $D_L$  and  $D_U$  and then applied to  $D_U$ . If the  $cc(\mathbf{x})$  distribution for  $D_U$  indicates significant overall bias, i.e. high  $\Delta$  value (Section 3.3), a set of  $B$  ( $B < G$ ) underrepresented examples will be selected and added into  $D_L$  for building the contrast classifier at the next iteration. The whole procedure iterates until the required  $G$  examples have been selected or the sample selection bias becomes negligible.

It should be noted that the actual labeling is not necessary during the procedure since the label information is not used for selection. The labeling for all selected examples can be done after the procedure stops, which is the set difference between the final and initial  $D_L$ . This property is a major difference between the proposed procedure and active learning methods [25, 26]. Another difference is that active sampling methods are primarily designed for supervised classification tasks; they tend to select more examples close to the decision boundary and therefore might introduce more bias. On the other hand, the proposed procedure aims at reducing bias over the entire feature space given a sufficiently large and representative unlabeled dataset. In this sense, it is also applicable to other learning tasks such as nearest-neighbor classification and density estimation [2].

### 3.5 Implementation Issues

Since the unlabeled dataset  $D_U$  is often much larger than  $D_L$ , learning from such imbalanced data would result in a low-quality contrast classifier, while learning time could be prohibitively long. We addressed this by training an ensemble of neural networks on balanced training sets consisting of equal number of labeled (class 0) and unlabeled (class 1) examples randomly sampled from the available data. Similar to bagging [27], we constructed a contrast classifier by aggregating the predictions of these neural networks through averaging. An additional benefit of using an ensemble of neural networks is that the averaging is known as a successful technique for increasing their accuracy by reducing variance while retaining low prediction bias.

## 4 Bioinformatics Applications

In this section, we illustrate how contrast classifiers could be used to detect and correct sample selection bias in protein structure databases and to improve accuracy of protein structure prediction. Before this, we first give a brief introduction to protein structure and objectives of protein structure prediction.

## 4.1 Protein Structure and Protein Structure Prediction

As the "machinery of life", protein is one of the most important macromolecules in living organisms that are essential for the structure and function of all living cells. Proteins constitute more than half of the dry weight of most cell matter and are involved in diverse biological functions such as enzyme action, regulation of cellular functions, signaling, structural support, and transportation and storage of matter. A protein may consist of one or more amino acid sequences (or polypeptide chains), each of which is a linear polymer of amino acids connected by covalent peptide bonds. It is commonly believed that under physiological conditions the amino acid sequence(s) of a protein spontaneously folds into a stable three dimensional (3D) structure, which in turn determines the protein function [28]. Therefore it is of great importance to know the structure of a protein for understanding its functions and the underlying biological mechanisms it is part of.

Protein 3D structure can be determined experimentally mainly by two types of techniques, X-ray diffraction and nuclear magnetic resonance (NMR). For both techniques, the target protein has to be expressed and purified to produce protein solutions of sufficient quantity, purity and concentration. Furthermore, X-ray diffraction requires growing high quality crystals from protein solutions. Although progress has been made in experimental technologies, structure determination is still an expensive and time-consuming process. Consequently, there are only about 30,000 protein structures that have been experimentally determined and deposited in Protein Data Bank (PDB) [29], the main protein structure database. This is only a small subset of the more than 2 million protein sequences deposited in UniProt (Release 6.0) protein database [30]. As more genomes are being completely sequenced, this sequence-structure gap is likely to become even larger. Therefore, computational methods that predict protein structures from amino acid sequence [31] are becoming increasingly attractive.

In a very simplified view, protein structure prediction can be regarded as a supervised learning task where amino acid sequence is represented as feature  $\mathbf{x}$  and the corresponding structure is label  $y$ . Therefore, proteins with known structures form the labeled dataset  $D_L$ , and the goal of learning is to model the sequence-structure relationship. While it is commonly accepted that protein structure is encoded by amino acid sequence [28], predicting 3D structure from sequence is still an open problem since the actual sequence-structure relationship is very complicated and is affected by many factors. Apart from the inherent difficulties, protein structure prediction may also suffer from the sample selection bias problem, i.e. the current protein structure database (PDB) is not a representative sample of all natural proteins.

## 4.2 Characterizing Bias in Database of Disordered Proteins

We first illustrate the effectiveness of the contrast classifier on a typical protein structure prediction problem - prediction of protein disordered regions. Protein disorder [32] refers to an important structural property of some protein regions that do not have a stable 3D structure under physiological conditions. Protein regions that have this property are called *disordered regions*. Proteins that have disordered regions are called *disordered proteins*; otherwise, they are called *ordered proteins*. To develop predictors of protein disorder, a non-redundant database of disordered proteins has been developed [33]. Due to the historical overlooking

of this property, our database is relatively small and is not a good representative of disordered proteins in nature. Thus, in a recent study [19] we applied contrast classifiers to characterize the sample selection bias and identify proteins from large sequence repositories that are statistically underrepresented in our database. Understanding biological properties of these outliers could lead to more effective studies of protein disorder.

In this study, the labeled dataset consisted of 442 proteins, including 152 disordered proteins with disordered regions longer than 30 consecutive residues, and 290 ordered proteins. The unlabeled dataset of 17,676 proteins was constructed as a non-redundant representative subset of the Swiss-Prot (Release 40) database [34]. This unlabeled dataset was assumed to be a representative of all natural proteins. Instead of a single contrast classifier built with all labeled proteins, two class-specific contrast classifiers were trained -  $cc_{disorder}(\mathbf{x})$  as disordered *versus* unlabeled, and  $cc_{order}(\mathbf{x})$  as ordered *versus* unlabeled. For both contrast classifiers, the same knowledge representation as for the VL3 disorder predictor [35, 36] was used.

To select underrepresented proteins, we first filtered the unlabeled dataset and kept only a subset of 6,964 proteins which were 200 to 500 residues long and were not homologous to any labeled proteins. After applying the two contrast classifiers to these proteins, we summarized each protein with two numerical values,  $cc\_avg_{order}$  and  $cc\_avg_{disorder}$ , representing the average predictions of the contrast classifiers over its sequence. Following the method described in Section 3.2, we selected a total of 1,259 proteins that had both high  $cc\_avg_{order}$  and high  $cc\_avg_{disorder}$  values, and were therefore statistically underrepresented by the labeled proteins.

To understand the biological properties of these proteins, we compared the frequencies of 840 protein function keywords listed in the Swiss-Prot database between the selected proteins and 895 Swiss-Prot homologues of the labeled proteins. As the results revealed, keywords Inner Membrane, Membrane, and Transmembrane appear more frequently in the selected proteins. This indicates that many selected proteins belong to a large family of membrane proteins that are known to be structurally and functionally diverse and are highly underrepresented among our labeled order and disorder sequences. It is likely that sample selection bias in our labeled data comes mainly from these membrane proteins. This result is also consistent with the results of our related study [37] where outliers were identified by learning class-conditional distribution models of ordered and disordered protein sequences.

### 4.3 Exploring Bias in Protein Structure Database

As mentioned above, the content of Protein Data Bank (PDB) [29] - the major database of experimental protein structures - is biased in the sense that it does not adequately cover the protein space. As illustrated in [38] the PDB sequences have significantly different amino acid compositions and predicted secondary structures from those in eight complete genomes. It was estimated that current structural information in PDB could cover only about 1/3 of all possible proteins [39, 40]. On the other hand, PDB structures may not be representative of the structure space either. Brenner et al. [41] analyzed the SCOP [42] structural classification of PDB proteins and reported high skewness at all classification levels. Furthermore, the estimated numbers of folds and superfamilies (high-level structural classes) in nature [43] suggest that there are more distinct structures that have not been

represented by known PDB structures.

The sample selection bias in PDB occurs due to several reasons. In general, PDB is positively biased towards proteins that are more amenable to expression, purification and crystallization. Another source of bias is the fact that different research groups usually have different objectives when selecting the target proteins. Understanding the sample selection bias in PDB is important to protein structure prediction since it is the ultimate source of labeled data for building predictors. This is especially true for structure prediction methods based on supervised learning, e.g. secondary structure prediction, transmembrane helix prediction, and protein disorder prediction. Other structure prediction methods could also be affected by the bias. For example, comparative modeling or homology modeling can be viewed as a nearest-neighbor classification method, where the evolutionary distance or sequence similarity measures serve as the distance measure [44]. For these methods, the key to success is a database of representative fold structures that could serve as templates for as many sequences as possible.

In a recent study [45] we applied contrast classifiers to explore the sample selection bias in the Protein Data Bank (PDB). In this application, PDB was considered as labeled data while Swiss-Prot was used as unlabeled data. Swiss-Prot was assumed to be representative of all natural proteins. A contrast classifier was trained on non-redundant subsets of PDB and Swiss-Prot and then applied to the Swiss-Prot sequences to select those underrepresented by PDB. By analyzing the Swiss-Prot keywords and other functional annotations associated with these outlying sequences, the bias in PDB towards different functional protein properties was characterized. The results confirmed some well-known biases, e.g. non-globular regions such as transmembrane, disordered and low complexity regions are significantly underrepresented, while disulfide bonds, metal binding sites, and sites involved in enzyme activity are overrepresented. In addition, this analysis also revealed some less-known bias, e.g. hydroxylation and phosphorylation posttranslational modification sites were found to be underrepresented while acetylation sites were significantly overrepresented [45].

#### 4.4 Prioritizing Protein Targets for Structural Genomics

The ongoing structural genomics projects [46] aim at experimentally determining structures for a carefully selected set of protein targets. Along with those already in PDB, these structures can serve as structural templates to computationally model or predict structures for other proteins with reasonable accuracy. Thus, an important problem, or the *target selection* problem, is how to select protein targets such that the resulting structures can cover as many proteins as possible. Many existing target selection strategies [47] rely on sequence comparison methods [44] to group protein sequences into homologous families, and then select one or several representatives from each family that has no structurally characterized members.

In a recent study (unpublished), we applied contrast classifiers to the target registration database (TargetDB) [48], which contains protein targets selected by different structural genomics projects along with information about their current progress. We used protein targets at different structural determination stages [48], e.g. *selected*, *expressed*, *purified*, *crystallized*, and *structure\_determined*, as the labeled datasets and built a series of contrast



classifiers. As the results revealed, the targets at *selected* stage seem to be indeed unbiased (with respect to our unlabeled dataset from Swiss-Prot), indicating the effectiveness of current target selection strategies. On the other hand, the targets at *structure\_determined* stage exhibit similar bias as in Protein Data Bank [45]. In addition, we observed a clear trend of increasing bias from *selected* stage to *structure\_determined* stage. Further analysis of these results might help understand the underlying mechanism of sample selection bias in current protein structure databases.

While existing target selection strategies might be effective in selecting representative protein targets, the target lists can still be very long due to the vast number of known proteins. Since structural determination is still costly and time-consuming, it is highly desirable to prioritize protein targets in order to *rapidly* achieve a good coverage of the protein space. This target prioritization problem can be cast as a problem of bias correction. As in previous studies, labeled examples are proteins with experimentally determined structures and their homologues, while unlabeled examples are all natural proteins. The goal is to select the most informative unlabeled proteins for labeling (i.e., experimental structure characterization) to rapidly reduce the bias existing in the labeled proteins. As illustrated in another study [24], the iterative procedure for correcting sample selection bias (Section 3.4) could be a suitable choice for this purpose.

We applied the iterative procedure to 28,334 protein sequences (40 amino acids or longer) from a model organism called *Arabidopsis thaliana* (ATH) that is extensively studied in plant biology [49]. ATH is also a major source of structural genomics targets for the Center for Eukaryotic Structural Genomics (CESG). We first clustered the ATH sequences using the CD-HIT program [50] and obtained 14,988 clusters (families). By selecting the longest sequence from each cluster, we obtained a non-redundant set of 14,988 sequences, such that no two sequences have pairwise identity higher than 40%. Out of these sequences 838 were identified to have known structures and thus formed the labeled dataset  $D_L$ , while all of the 14,988 proteins were assigned to the unlabeled dataset  $D_U$ . Note that this process can be viewed as a simplified target selection process and  $D_U$  can be regarded as our target lists.

It was further assumed that at most 500 proteins, i.e.  $G = 500$ , could be selected from  $D_U$  for labeling due to available resources for experimental structure determination. We examined different values of  $B = \{50, 100, 250, 500\}$  and also compared the proposed procedure to simple random selection. As the results showed, the 500 proteins selected by the proposed iterative procedure ( $B = 50$ ) were more effective in correcting sample selection bias than those selected randomly. Furthermore, the iterative procedure with small  $B$  value, e.g.  $B = 50$ , performed considerably better than the one-step approach ( $B = 500$ ) in the sense that it could achieve much lower  $\Delta$  value, or lower level of overall bias, after selected 500 proteins.

At the time of the study, only a small fraction of the 500 proteins were selected as structural genomics targets by the Center for Eukaryotic Structural Genomics and none of them had been solved. We argue that the remaining proteins should also be selected as structural genomics targets and should be given higher priorities. Along with proteins with known structures, these proteins should be very helpful in achieving a rapid and good coverage of the protein space for *Arabidopsis thaliana* genome [49]. We believe that revealing structural properties of such proteins is likely to produce highly significant biological results.

## 5 Discussions and Conclusion

Sample selection bias is a common problem in almost all areas that rely on statistical methods to infer or learn population characteristics from samples. It is therefore of great importance to understand the underlying bias mechanism such that the learning process can be adjusted accordingly to obtain meaningful results. Many statistical methods have been developed for bias detection and correction in various disciplines, although they typically rely on certain restrictive assumptions which may limit their practical use. In machine learning context, systematic studies on sample selection bias are receiving increasing attention. It is expected that more robust learning algorithms would be possible if the sample selection bias is taken into account.

As illustrated in Section 4, the contrast classifier framework provides a simple yet effective way of addressing sample selection bias. However, a major challenge in applying contrast classifiers to learning from biased protein structure data is how to select relevant features. Since the protein sequences have variable length, it is always necessary to map the sequences into certain space ( $\mathbb{X}$ ) with fixed number of features before a contrast classifier can be learned. Only if the features extracted from amino acid sequence are relevant to the underlying bias mechanism can the resulting contrast classifier be effective in bias detection and correction. For example, amino acid compositions and sequence complexity could readily reveal the well-known bias toward non-globular regions in PDB, but are less sensitive to bias toward different structural fold classes [45].

To improve the resulting contrast classifiers, it might be desirable to perform feature selection to remove features that might be irrelevant to the bias. However, existing feature selection techniques [51] may not work well for contrast classifiers, since the labeled dataset could be part of or largely overlapped with the unlabeled dataset. In addition, standard measures of feature relevance for normal binary classifiers may not be suitable for contrast classifiers. It is therefore desirable to modify existing or to develop new feature selection techniques that are more effective for contrast classifier applications.

In current approaches, contrast classifiers are usually implemented as ensembles of neural networks. However, the proper ensemble size and the complexity of component networks (e.g. number of hidden neurons) have to be chosen manually by trial-and-error procedures, which can be inefficient and ineffective especially when the unlabeled dataset is large. Thus, a recently proposed procedure [52] for cost-effective learning of predictors as ensembles of neural networks from arbitrary large datasets will be applied to improve contrast classifier learning. It is expected that the resulting contrast classifiers would utilize the data diversity more efficiently and be adaptive to the inherent complexity of given data.

Methods for incorporating contrast classifiers into the process of protein structure prediction, e.g. protein disorder prediction, will also be investigated to improve the prediction accuracy. A straightforward approach is to assign prediction confidence based on the contrast classifier output  $cc(\mathbf{x})$  to the structure predictors. If a protein produces high  $cc(\mathbf{x})$ , it is very likely that it is underrepresented in the labeled dataset and that the resulting predictor may not generalize well on it. Therefore it might be necessary to report that the prediction confidence is low or even refuse to provide prediction. In this way, the prediction accuracy might be improved, but at the expense of lower coverage.

## 6 Acknowledgements

This study was supported in part by NSF grant IIS-0219736 to Z. Obradovic and S. Vucetic and NSF grant CSE-IIS-0196237 to Z. Obradovic and A.K. Dunker.

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*Chapter 14*

## A NOTE ON NONLINEAR INTEGRABLE HAMILTONIAN SYSTEMS

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### Abstract

This paper presents an approach to determine what kind of canonical Hamiltonian system is integrable and what is nonintegrable. Examples are analyzed to show integrability and nonintegrability. In particular, some new integrable Hamiltonian systems are found and the remarkable peakon dynamical system is a reduction.

## 1 Introduction

The Liouville-Arnold theory has been playing a very important role in the investigation of finite-dimensional integrable system [1]. A motivation for studying integrable systems has been brought by the discovery of soliton equations [5]. New techniques, such as the Lax pair [6] and the spectral curve method [4] etc, were successively involved in soliton theory. For a given Hamiltonian it is usually very difficult to verify whether it is integrable or even to check whether it has additional integrals of motion apart from those simply related to the geometrical symmetries of the potential. The Lax representation is quite an effective way to show the integrability. But, how do we find a Lax pair for a given finite dimensional Hamiltonian system? Calogero [2] proposed a general scheme equation with three different functions to be determined for many-body problem on the line. More details of three different functions  $g(x)$ ,  $\alpha(x)$ ,  $\gamma(x)$  are in Eq. (2.8) or on page 137 of Calogero's book [2].

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In this paper we simplify the Calogero equation (2.8) and reduce it to one single equation (see Eq. (3.18)) only containing a function  $\alpha(x)$  to be determined, and meanwhile the other two  $g(x)$ ,  $\gamma(x)$  are able to be expressed in terms of the function  $\alpha(x)$ . In the paper we provide a procedure to determine what kind of function  $g(x)$  is appropriate and what is inappropriate. Several examples are analyzed to show integrability and nonintegrability. In particular, some new integrable Hamiltonian systems are found and the remarkable peakon dynamical system is a reduction.

## 2 Preliminaries

Let us start from the following two  $N \times N$  matrices:

$$L = \sum_{i,j=1}^N L_{ij} E_{ij}, \quad (2.1)$$

$$M = \sum_{i,j=1}^N M_{ij} E_{ij}, \quad (2.2)$$

where  $\{E_{ij}\}$  is the matrix basis, i.e.  $(E_{ij})_{kl} = \delta_{ik}\delta_{jl}$ ,  $i, j, k, l = 1, \dots, N$ , and

$$L_{ij} = \sqrt{p_i p_j} \alpha(q_i - q_j), \quad (2.3)$$

$$M_{ij} = \sqrt{p_i p_j} \gamma(q_i - q_j), \quad (2.4)$$

where  $\alpha$  and  $\gamma$  are two functions to be determined.

Calogero [2] proved that the Lax equation

$$\dot{L} = [M, L] \quad (2.5)$$

is equivalent to the following canonical Hamiltonian equation

$$(H): \begin{cases} \dot{q}_i = \frac{\partial H}{\partial p_i} = 2 \sum_{j=1}^N p_j g(q_i - q_j), \\ \dot{p}_i = -\frac{\partial H}{\partial q_i} = -2p_i \sum_{j=1}^N p_j g'(q_i - q_j), \end{cases} \quad (2.6)$$

$$\text{with } H = \sum_{i,j=1}^N p_i p_j g(q_i - q_j), \quad (2.7)$$

if and only if the even function  $g(x)$ , together with other two functions  $\alpha(x)$ ,  $\gamma(x)$ , satisfies

$$\begin{aligned} & 2\alpha'(x+y)[g(x) - g(y)] - \alpha(x+y)[g'(x) - g'(y)] \\ &= \alpha(x)\gamma(y) - \alpha(y)\gamma(x), \quad \forall x, y \in \mathbb{R}, \end{aligned} \quad (2.8)$$

where the superscripts denote the corresponding function's derivative with respect to the argument. Eq. (2.8) actually comes from the Ref. [2] (see page 137, equation (\*\*\*)).

**Definition 1** The function  $g(x)$  is said to be **appropriate** for the finite-dimensional integrable system (2.6) if there exist two functions  $\alpha(x)$ ,  $\gamma(x)$  such that Eq. (2.8) holds (i.e. the Lax equation (2.5) is equivalent to the Hamiltonian equation (2.6)). Otherwise, the function  $g(x)$  is said to be **inappropriate**.

Based on the functional equation (2.8), we shall concretely discuss what kind of the function  $g(x)$  is appropriate or inappropriate. We will construct a governing equation of  $\alpha(x)$ , which is available to express  $g(x)$  in terms of  $\alpha(x)$  and as well as easy to make  $g(x)$  appropriate or inappropriate.

### 3 A General Formula for $\alpha(x)$

Since Eq. (2.8) holds for any  $x, y \in \mathbb{R}$ , let us first choose  $y = -x$ . Then Eq. (2.8) reads

$$2\alpha_0 g'(x) = \alpha(-x)\gamma(x) - \alpha(x)\gamma(-x), \quad (3.9)$$

where  $\alpha_0 = \alpha(0)$ . However, setting  $y = 0$  yields

$$2\alpha'(x)[g(x) - g_0] - \alpha(x)[g'(x) + \gamma_0] = -\alpha_0\gamma(x), \quad (3.10)$$

$$2\alpha'(-x)[g(x) - g_0] + \alpha(-x)[g'(x) - \gamma_0] = -\alpha_0\gamma(-x), \quad (3.11)$$

which implies

$$\begin{aligned} & 2[\alpha'(x) + \alpha'(-x)][g(x) - g_0] + [\alpha(-x) - \alpha(x)]g'(x) - \gamma_0[\alpha(x) + \alpha(-x)] \\ &= -\alpha_0[\gamma(x) + \gamma(-x)], \end{aligned} \quad (3.12)$$

and

$$\alpha_0^2 g'(x) = (g(x) - g_0)(\alpha(x)\alpha'(-x) - \alpha(-x)\alpha'(x)) + \alpha(x)\alpha(-x)g'(x), \quad (3.13)$$

where  $\gamma_0 = \gamma(0)$ ,  $g_0 = g(0)$ . Solving Eq. (3.13) yields the relation between  $g(x)$  and  $\alpha(x)$

$$g(x) = c\alpha(x)\alpha(-x) - c\alpha_0^2 + g_0, \quad (3.14)$$

where  $c$  is a non-zero constant and thereafter all  $c$ 's in the examples are non-zero constants.

To get  $\gamma(x)$  in terms of  $\alpha(x)$ , let us take  $y = x + \epsilon\Delta x$  and the derivative  $\frac{d}{d\epsilon}|_{\epsilon=0}$  in both sides of Eq. (2.8). Then we have

$$-2\alpha'(2x)g'(x) + \alpha(2x)g''(x) = \alpha(x)\gamma'(x) - \alpha'(x)\gamma(x),$$

i.e.

$$\frac{d}{dx} \frac{g'(x)}{\alpha(2x)} = \frac{\alpha^2(x)}{\alpha^2(2x)} \cdot \frac{d}{dx} \frac{\gamma(x)}{\alpha(x)}.$$

The integration of this equality directly leads to a relation between  $\gamma(x)$  and  $\alpha(x)$

$$\gamma(x) = c\alpha(x)\Gamma(\alpha(x), \alpha(2x)), \quad (3.15)$$

$$\Gamma(\alpha(x), \alpha(2x)) = \int \frac{\alpha^2(2x)}{\alpha^2(x)} \left( \frac{\alpha'(x)\alpha(-x) - \alpha(x)\alpha'(-x)}{\alpha(2x)} \right)' dx, \quad (3.16)$$

where  $'$  is derivatives with respect to  $x$ . Substituting Eqs. (3.14) and (3.15) into Eq. (2.8) gives the following equation:

$$\frac{1}{2}(A(x, y) - A(y, x)) = \frac{1}{2c} \left( \frac{\gamma(y)}{\alpha(y)} - \frac{\gamma(x)}{\alpha(x)} \right). \quad (3.17)$$

So, we obtain an equation only interfering  $\alpha(x)$ :

$$A(x, y) - A(y, x) = \Gamma(\alpha(y), \alpha(2y)) - \Gamma(\alpha(x), \alpha(2x)), \quad \forall x, y, \quad (3.18)$$

where

$$A(x, y) = \frac{2\alpha'(x+y)\alpha(-x)\alpha(x) - \alpha(x+y)(\alpha'(x)\alpha(-x) - \alpha'(-x)\alpha(x))}{\alpha(x)\alpha(y)}. \quad (3.19)$$

Therefore, we have the following theorem.

**Theorem 1** *If the Lax equation (2.5) has the Hamiltonian canonical form (2.6) with an even function  $g(-x) = g(x)$ , then  $\alpha(x)$  satisfies Eq. (3.18). In addition,  $g(x)$  and  $\gamma(x)$  are given in terms of  $\alpha(x)$  by Eqs. (3.14) and (3.15), respectively.*

This theorem is telling us that the key problem in solving Calogero's equation (2.8) is to find the solution  $\alpha(x)$  of equation (3.18). Let us give some examples as follows.

### Examples

1. Choosing  $\alpha(x) = A \sin \frac{a}{2}x$  ( $A, a$  are two constants, and apparently  $\alpha(x)$  is an odd function) yields

$$g(x) = \lambda \cos ax + \mu, \quad \lambda = -\frac{cA^2}{2}, \quad \mu = g_0 - \lambda. \quad (3.20)$$

By Eqs. (3.16) and (3.15), we take  $\gamma(x) \equiv 0$ . Because this  $\alpha(x)$  satisfies

$$A(x, y) = A(y, x), \quad \forall x, y,$$

the function  $g(x) = \lambda \cos ax + \mu$  is appropriate for the Lax equation (2.5).

2. Choosing  $\alpha(x) = A \operatorname{sgn}(x) \sin \frac{a}{2}x$  ( $A, a$  are two constants, and  $\alpha(-x) = \alpha(x)$ ) yields

$$g(x) = \lambda \cos ax + \mu, \quad \lambda = -\frac{cA^2}{2}, \quad \mu = g_0 - \lambda. \quad (3.21)$$

By Eqs. (3.16) and (3.15), we have

$$\begin{aligned} \Gamma(\alpha(x), \alpha(2x)) &= d, \quad d = \text{constant}, \\ \gamma(x) &= B\alpha(x), \quad B = cd. \end{aligned}$$

Additionally,  $\alpha(x)$  satisfies

$$A(x, y) = A(y, x), \quad \forall x, y.$$

Thus, in this example  $g(x) = \lambda \cos ax + \mu$  is again proved to be appropriate.

The above two examples have same  $g(x)$ , but different  $\alpha(x)$  and  $\gamma(x)$ . This illustrates that the Lax representation is not **unique**. Such  $g(x)$  is a main example of Calogero's book [2].

3. Choosing  $\alpha(x) = x^n$ ,  $n \in \mathbb{R}$ , we get

$$\begin{aligned}\Gamma(\alpha(x), \alpha(2x)) &= n(-1)^n 2^{n+1} x^{n-1}, \\ A(x, y) &= 2n(-1)^{n-1} \frac{x^{n-1}(x+y)^{n-1}}{y^{n-1}},\end{aligned}$$

which imply that Eq. (3.18) holds iff

$$n = 1.$$

Thus,  $\alpha(x) = x$ ,  $\gamma(x) = -4cx$ , and

$$g(x) = -cx^2 + g_0.$$

Therefore, the canonical Hamiltonian system (2.6) with  $g(x) = -cx^2 + g_0$  ( $c \neq 0$ ,  $g_0$  are any constants) is a new integrable system. In particular, we take

$$g(x) = 1 - x^2 \tag{3.22}$$

as an appropriate function for the Lax equation (2.5). However, the following function

$$g(x) = \begin{cases} 1 - x^2, & |x| < 1, \\ 0, & |x| \geq 1, \end{cases} \tag{3.23}$$

is not appropriate, because at  $x = \pm 1$   $g(x)$  does not satisfy the equation (2.8).

4. Choosing  $\alpha(x) = |x|^n$ ,  $n \in \mathbb{R}$ , we have

$$\begin{aligned}\Gamma(\alpha(x), \alpha(2x)) &= \begin{cases} 2^{n+1}n|x|^{n-1}\text{sgn}(x), & n \neq 1, \\ \text{constant}, & n = 1, \end{cases} \\ A(x, y) &= 2n|x|^{n-1}|x+y|^{n-1} \frac{|x|\text{sgn}(x+y) - |x+y|\text{sgn}(x)}{|y|^n},\end{aligned}$$

which imply that Eq. (3.18) holds only for  $n = 1$  because  $A(x, y) = A(y, x)$  and  $\Gamma(\alpha(x), \alpha(2x)) = \Gamma(\alpha(y), \alpha(2y))$  only when  $n = 1$ . Therefore the function  $g(x) = g_0 + cx^2$  ( $g_0$  and  $c \neq 0$  are constants) corresponding  $n = 1$  is appropriate (also see example 3). But when we take  $n = \frac{1}{2}$ ,  $c = -1$ , and  $g_0 = 1$ , the function

$$g(x) = 1 - |x| \tag{3.24}$$

is not appropriate. Therefore,

$$g(x) = \begin{cases} 1 - |x|, & |x| < 1, \\ 0, & |x| \geq 1, \end{cases} \tag{3.25}$$

is not an appropriate function for the Lax equation (2.5).

5. Choosing  $\alpha(x) = e^{-\frac{a}{2}|x|}$  ( $a \in \mathbb{R}$  is a constant), then we have

$$\begin{aligned}\Gamma(\alpha(x), \alpha(2x)) &= -a(1 + \operatorname{sgn}(x)), \\ A(x, y) &= ae^{-\frac{a}{2}(|x+y|+|x|-|y|)} \left( \operatorname{sgn}(x) - \operatorname{sgn}(x+y) \right),\end{aligned}$$

which imply

$$A(x, y) - A(y, x) = \Gamma(\alpha(y), \alpha(2y)) - \Gamma(\alpha(x), \alpha(2x)), \quad \forall x, y.$$

Therefore,  $\alpha(x) = e^{-\frac{a}{2}|x|}$  is a solution of Eq. (3.18). In this case

$$\begin{aligned}g(x) &= ce^{-a|x|} + g_0 - c, \\ \gamma(x) &= -ac e^{-\frac{a}{2}|x|} \operatorname{sgn}(x),\end{aligned}$$

where  $c \neq 0$ ,  $a$ ,  $g_0$  are constants. So,  $g(x) = ce^{-a|x|} + g_0 - c$  is appropriate, and the canonical Hamiltonian system (2.6) with  $g(x) = ce^{-a|x|} + g_0 - c$  ( $a, g_0$  are any constants) is a new integrable system. In particular, this system includes the integrable peakon dynamics [3] as a special reduction with  $c = a = g_0 = 1$  (i.e.  $g(x) = e^{-|x|}$ ).

Two natural questions arise here:

1. If  $g_1(x)$  and  $g_2(x)$  are appropriate, then is their sum function  $g(x) = g_1(x) + g_2(x)$  appropriate? If so, what are the conditions for  $g_1(x)$  and  $g_2(x)$ ?
2. If  $\alpha_1(x)$  and  $\alpha_2(x)$  along with their corresponding functions  $\gamma_1(x)$  and  $\gamma_2(x)$  satisfy Eq. (3.17), do their sum functions  $\alpha(x) = \alpha_1(x) + \alpha_2(x)$  and  $\gamma(x) = \gamma_1(x) + \gamma_2(x)$  still satisfy the equation (3.17)?

Due to the length limit of the paper, we shall discuss the above two problems elsewhere.

## 4 Conclusion

Here we presented a fairly general construction of finite dimensional completely integrable Hamiltonian systems associated with the family of metric function  $g(x)$ . Based on the above discussions, we conclude

1. All canonical Hamiltonian systems (2.6) corresponding to those appropriate functions  $g(x)$  are integrable.
2. All canonical Hamiltonian systems (2.6) corresponding to those inappropriate functions  $g(x)$  are nonintegrable.

Basically, the choice of the metric function  $g(x)$  depends on the  $\alpha(x)$ 's. The latter satisfies the nonlinear integro-differential equation (3.18) which is only respect to  $\alpha(x)$ . An open problem is how to solve equation (3.18) in general. This is really hard. In this paper, we provided some special solutions of  $\alpha(x)$  and gave some theorems and propositions to judge whether the metric function  $g(x)$  is appropriate or not. We also presented some examples to show whether  $g(x)$  is appropriate. However, the more general solution of equation (3.18) is still unknown, which we defer to another time.

## Acknowledgments

The author (Zhijun Qiao) would like to express his sincere thanks to Prof. Fengshan Liu and Prof. Xiquan Shi for their hospitality when he visited DSU in summer 2005.

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*Chapter 14*

## A NOTE ON NONLINEAR INTEGRABLE HAMILTONIAN SYSTEMS

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### Abstract

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## 1 Introduction

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In this paper we simplify the Calogero equation (2.8) and reduce it to one single equation (see Eq. (3.18)) only containing a function  $\alpha(x)$  to be determined, and meanwhile the other two  $g(x)$ ,  $\gamma(x)$  are able to be expressed in terms of the function  $\alpha(x)$ . In the paper we provide a procedure to determine what kind of function  $g(x)$  is appropriate and what is inappropriate. Several examples are analyzed to show integrability and nonintegrability. In particular, some new integrable Hamiltonian systems are found and the remarkable peakon dynamical system is a reduction.

## 2 Preliminaries

Let us start from the following two  $N \times N$  matrices:

$$L = \sum_{i,j=1}^N L_{ij} E_{ij}, \quad (2.1)$$

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where  $\{E_{ij}\}$  is the matrix basis, i.e.  $(E_{ij})_{kl} = \delta_{ik}\delta_{jl}$ ,  $i, j, k, l = 1, \dots, N$ , and

$$L_{ij} = \sqrt{p_i p_j} \alpha(q_i - q_j), \quad (2.3)$$

$$M_{ij} = \sqrt{p_i p_j} \gamma(q_i - q_j), \quad (2.4)$$

where  $\alpha$  and  $\gamma$  are two functions to be determined.

Calogero [2] proved that the Lax equation

$$\dot{L} = [M, L] \quad (2.5)$$

is equivalent to the following canonical Hamiltonian equation

$$(H): \begin{cases} \dot{q}_i = \frac{\partial H}{\partial p_i} = 2 \sum_{j=1}^N p_j g(q_i - q_j), \\ \dot{p}_i = -\frac{\partial H}{\partial q_i} = -2p_i \sum_{j=1}^N p_j g'(q_i - q_j), \end{cases} \quad (2.6)$$

$$\text{with } H = \sum_{i,j=1}^N p_i p_j g(q_i - q_j), \quad (2.7)$$

if and only if the even function  $g(x)$ , together with other two functions  $\alpha(x)$ ,  $\gamma(x)$ , satisfies

$$\begin{aligned} & 2\alpha'(x+y)[g(x) - g(y)] - \alpha(x+y)[g'(x) - g'(y)] \\ &= \alpha(x)\gamma(y) - \alpha(y)\gamma(x), \quad \forall x, y \in \mathbb{R}, \end{aligned} \quad (2.8)$$

where the superscripts denote the corresponding function's derivative with respect to the argument. Eq. (2.8) actually comes from the Ref. [2] (see page 137, equation (\*\*\*)).

**Definition 1** The function  $g(x)$  is said to be **appropriate** for the finite-dimensional integrable system (2.6) if there exist two functions  $\alpha(x)$ ,  $\gamma(x)$  such that Eq. (2.8) holds (i.e. the Lax equation (2.5) is equivalent to the Hamiltonian equation (2.6)). Otherwise, the function  $g(x)$  is said to be **inappropriate**.

Based on the functional equation (2.8), we shall concretely discuss what kind of the function  $g(x)$  is appropriate or inappropriate. We will construct a governing equation of  $\alpha(x)$ , which is available to express  $g(x)$  in terms of  $\alpha(x)$  and as well as easy to make  $g(x)$  appropriate or inappropriate.

### 3 A General Formula for $\alpha(x)$

Since Eq. (2.8) holds for any  $x, y \in \mathbb{R}$ , let us first choose  $y = -x$ . Then Eq. (2.8) reads

$$2\alpha_0 g'(x) = \alpha(-x)\gamma(x) - \alpha(x)\gamma(-x), \quad (3.9)$$

where  $\alpha_0 = \alpha(0)$ . However, setting  $y = 0$  yields

$$2\alpha'(x)[g(x) - g_0] - \alpha(x)[g'(x) + \gamma_0] = -\alpha_0\gamma(x), \quad (3.10)$$

$$2\alpha'(-x)[g(x) - g_0] + \alpha(-x)[g'(x) - \gamma_0] = -\alpha_0\gamma(-x), \quad (3.11)$$

which implies

$$\begin{aligned} & 2[\alpha'(x) + \alpha'(-x)][g(x) - g_0] + [\alpha(-x) - \alpha(x)]g'(x) - \gamma_0[\alpha(x) + \alpha(-x)] \\ &= -\alpha_0[\gamma(x) + \gamma(-x)], \end{aligned} \quad (3.12)$$

and

$$\alpha_0^2 g'(x) = (g(x) - g_0)(\alpha(x)\alpha'(-x) - \alpha(-x)\alpha'(x)) + \alpha(x)\alpha(-x)g'(x), \quad (3.13)$$

where  $\gamma_0 = \gamma(0)$ ,  $g_0 = g(0)$ . Solving Eq. (3.13) yields the relation between  $g(x)$  and  $\alpha(x)$

$$g(x) = c\alpha(x)\alpha(-x) - c\alpha_0^2 + g_0, \quad (3.14)$$

where  $c$  is a non-zero constant and thereafter all  $c$ 's in the examples are non-zero constants.

To get  $\gamma(x)$  in terms of  $\alpha(x)$ , let us take  $y = x + \epsilon\Delta x$  and the derivative  $\frac{d}{d\epsilon}|_{\epsilon=0}$  in both sides of Eq. (2.8). Then we have

$$-2\alpha'(2x)g'(x) + \alpha(2x)g''(x) = \alpha(x)\gamma'(x) - \alpha'(x)\gamma(x),$$

i.e.

$$\frac{d}{dx} \frac{g'(x)}{\alpha(2x)} = \frac{\alpha^2(x)}{\alpha^2(2x)} \cdot \frac{d}{dx} \frac{\gamma(x)}{\alpha(x)}.$$

The integration of this equality directly leads to a relation between  $\gamma(x)$  and  $\alpha(x)$

$$\gamma(x) = c\alpha(x)\Gamma(\alpha(x), \alpha(2x)), \quad (3.15)$$

$$\Gamma(\alpha(x), \alpha(2x)) = \int \frac{\alpha^2(2x)}{\alpha^2(x)} \left( \frac{\alpha'(x)\alpha(-x) - \alpha(x)\alpha'(-x)}{\alpha(2x)} \right)' dx, \quad (3.16)$$

where  $'$  is derivatives with respect to  $x$ . Substituting Eqs. (3.14) and (3.15) into Eq. (2.8) gives the following equation:

$$\frac{1}{2}(A(x, y) - A(y, x)) = \frac{1}{2c} \left( \frac{\gamma(y)}{\alpha(y)} - \frac{\gamma(x)}{\alpha(x)} \right). \quad (3.17)$$

So, we obtain an equation only interfering  $\alpha(x)$ :

$$A(x, y) - A(y, x) = \Gamma(\alpha(y), \alpha(2y)) - \Gamma(\alpha(x), \alpha(2x)), \quad \forall x, y, \quad (3.18)$$

where

$$A(x, y) = \frac{2\alpha'(x+y)\alpha(-x)\alpha(x) - \alpha(x+y)(\alpha'(x)\alpha(-x) - \alpha'(-x)\alpha(x))}{\alpha(x)\alpha(y)}. \quad (3.19)$$

Therefore, we have the following theorem.

**Theorem 1** *If the Lax equation (2.5) has the Hamiltonian canonical form (2.6) with an even function  $g(-x) = g(x)$ , then  $\alpha(x)$  satisfies Eq. (3.18). In addition,  $g(x)$  and  $\gamma(x)$  are given in terms of  $\alpha(x)$  by Eqs. (3.14) and (3.15), respectively.*

This theorem is telling us that the key problem in solving Calogero's equation (2.8) is to find the solution  $\alpha(x)$  of equation (3.18). Let us give some examples as follows.

### Examples

1. Choosing  $\alpha(x) = A \sin \frac{a}{2}x$  ( $A, a$  are two constants, and apparently  $\alpha(x)$  is an odd function) yields

$$g(x) = \lambda \cos ax + \mu, \quad \lambda = -\frac{cA^2}{2}, \quad \mu = g_0 - \lambda. \quad (3.20)$$

By Eqs. (3.16) and (3.15), we take  $\gamma(x) \equiv 0$ . Because this  $\alpha(x)$  satisfies

$$A(x, y) = A(y, x), \quad \forall x, y,$$

the function  $g(x) = \lambda \cos ax + \mu$  is appropriate for the Lax equation (2.5).

2. Choosing  $\alpha(x) = A \operatorname{sgn}(x) \sin \frac{a}{2}x$  ( $A, a$  are two constants, and  $\alpha(-x) = \alpha(x)$ ) yields

$$g(x) = \lambda \cos ax + \mu, \quad \lambda = -\frac{cA^2}{2}, \quad \mu = g_0 - \lambda. \quad (3.21)$$

By Eqs. (3.16) and (3.15), we have

$$\begin{aligned} \Gamma(\alpha(x), \alpha(2x)) &= d, \quad d = \text{constant}, \\ \gamma(x) &= B\alpha(x), \quad B = cd. \end{aligned}$$

Additionally,  $\alpha(x)$  satisfies

$$A(x, y) = A(y, x), \quad \forall x, y.$$

Thus, in this example  $g(x) = \lambda \cos ax + \mu$  is again proved to be appropriate.

The above two examples have same  $g(x)$ , but different  $\alpha(x)$  and  $\gamma(x)$ . This illustrates that the Lax representation is not **unique**. Such  $g(x)$  is a main example of Calogero's book [2].

3. Choosing  $\alpha(x) = x^n$ ,  $n \in \mathbb{R}$ , we get

$$\begin{aligned}\Gamma(\alpha(x), \alpha(2x)) &= n(-1)^n 2^{n+1} x^{n-1}, \\ A(x, y) &= 2n(-1)^{n-1} \frac{x^{n-1}(x+y)^{n-1}}{y^{n-1}},\end{aligned}$$

which imply that Eq. (3.18) holds iff

$$n = 1.$$

Thus,  $\alpha(x) = x$ ,  $\gamma(x) = -4cx$ , and

$$g(x) = -cx^2 + g_0.$$

Therefore, the canonical Hamiltonian system (2.6) with  $g(x) = -cx^2 + g_0$  ( $c \neq 0$ ,  $g_0$  are any constants) is a new integrable system. In particular, we take

$$g(x) = 1 - x^2 \quad (3.22)$$

as an appropriate function for the Lax equation (2.5). However, the following function

$$g(x) = \begin{cases} 1 - x^2, & |x| < 1, \\ 0, & |x| \geq 1, \end{cases} \quad (3.23)$$

is not appropriate, because at  $x = \pm 1$   $g(x)$  does not satisfy the equation (2.8).

4. Choosing  $\alpha(x) = |x|^n$ ,  $n \in \mathbb{R}$ , we have

$$\begin{aligned}\Gamma(\alpha(x), \alpha(2x)) &= \begin{cases} 2^{n+1}n|x|^{n-1}\text{sgn}(x), & n \neq 1, \\ \text{constant}, & n = 1, \end{cases} \\ A(x, y) &= 2n|x|^{n-1}|x+y|^{n-1} \frac{|x|\text{sgn}(x+y) - |x+y|\text{sgn}(x)}{|y|^n},\end{aligned}$$

which imply that Eq. (3.18) holds only for  $n = 1$  because  $A(x, y) = A(y, x)$  and  $\Gamma(\alpha(x), \alpha(2x)) = \Gamma(\alpha(y), \alpha(2y))$  only when  $n = 1$ . Therefore the function  $g(x) = g_0 + cx^2$  ( $g_0$  and  $c \neq 0$  are constants) corresponding  $n = 1$  is appropriate (also see example 3). But when we take  $n = \frac{1}{2}$ ,  $c = -1$ , and  $g_0 = 1$ , the function

$$g(x) = 1 - |x| \quad (3.24)$$

is not appropriate. Therefore,

$$g(x) = \begin{cases} 1 - |x|, & |x| < 1, \\ 0, & |x| \geq 1, \end{cases} \quad (3.25)$$

is not an appropriate function for the Lax equation (2.5).

5. Choosing  $\alpha(x) = e^{-\frac{a}{2}|x|}$  ( $a \in \mathbb{R}$  is a constant), then we have

$$\begin{aligned}\Gamma(\alpha(x), \alpha(2x)) &= -a(1 + \operatorname{sgn}(x)), \\ A(x, y) &= ae^{-\frac{a}{2}(|x+y|+|x|-|y|)} \left( \operatorname{sgn}(x) - \operatorname{sgn}(x+y) \right),\end{aligned}$$

which imply

$$A(x, y) - A(y, x) = \Gamma(\alpha(y), \alpha(2y)) - \Gamma(\alpha(x), \alpha(2x)), \quad \forall x, y.$$

Therefore,  $\alpha(x) = e^{-\frac{a}{2}|x|}$  is a solution of Eq. (3.18). In this case

$$\begin{aligned}g(x) &= ce^{-a|x|} + g_0 - c, \\ \gamma(x) &= -ac e^{-\frac{a}{2}|x|} \operatorname{sgn}(x),\end{aligned}$$

where  $c \neq 0$ ,  $a$ ,  $g_0$  are constants. So,  $g(x) = ce^{-a|x|} + g_0 - c$  is appropriate, and the canonical Hamiltonian system (2.6) with  $g(x) = ce^{-a|x|} + g_0 - c$  ( $a, g_0$  are any constants) is a new integrable system. In particular, this system includes the integrable peakon dynamics [3] as a special reduction with  $c = a = g_0 = 1$  (i.e.  $g(x) = e^{-|x|}$ ).

Two natural questions arise here:

1. If  $g_1(x)$  and  $g_2(x)$  are appropriate, then is their sum function  $g(x) = g_1(x) + g_2(x)$  appropriate? If so, what are the conditions for  $g_1(x)$  and  $g_2(x)$ ?
2. If  $\alpha_1(x)$  and  $\alpha_2(x)$  along with their corresponding functions  $\gamma_1(x)$  and  $\gamma_2(x)$  satisfy Eq. (3.17), do their sum functions  $\alpha(x) = \alpha_1(x) + \alpha_2(x)$  and  $\gamma(x) = \gamma_1(x) + \gamma_2(x)$  still satisfy the equation (3.17)?

Due to the length limit of the paper, we shall discuss the above two problems elsewhere.

## 4 Conclusion

Here we presented a fairly general construction of finite dimensional completely integrable Hamiltonian systems associated with the family of metric function  $g(x)$ . Based on the above discussions, we conclude

1. All canonical Hamiltonian systems (2.6) corresponding to those appropriate functions  $g(x)$  are integrable.
2. All canonical Hamiltonian systems (2.6) corresponding to those inappropriate functions  $g(x)$  are nonintegrable.

Basically, the choice of the metric function  $g(x)$  depends on the  $\alpha(x)$ 's. The latter satisfies the nonlinear integro-differential equation (3.18) which is only respect to  $\alpha(x)$ . An open problem is how to solve equation (3.18) in general. This is really hard. In this paper, we provided some special solutions of  $\alpha(x)$  and gave some theorems and propositions to judge whether the metric function  $g(x)$  is appropriate or not. We also presented some examples to show whether  $g(x)$  is appropriate. However, the more general solution of equation (3.18) is still unknown, which we defer to another time.

## Acknowledgments

The author (Zhijun Qiao) would like to express his sincere thanks to Prof. Fengshan Liu and Prof. Xiquan Shi for their hospitality when he visited DSU in summer 2005.

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*Chapter 15*

## RECONSTRUCTING CONVERGENT $G^1$ B-SPLINE SURFACES FOR ADAPTING THE QUAD PARTITION

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### Abstract

In (Shi et al., 2004), we provided a local scheme of constructing convergent  $G^1$  smooth bicubic B-spline surface patches with single interior knots over a given arbitrary quad partition of a polygonal model. In that paper, like in all the existing literatures, the  $G^1$  conditions do not adapt for the geometric properties of the quad partition, i.e., the conditions do not reflect the different sizes of adjacent B-spline surface patches. In this paper, based on the geometric properties of the quad partition, we provide a new local scheme of constructing convergent  $G^1$  smooth bicubic B-spline surface patches with single interior knots over a given arbitrary quad partition of a polygonal model. Our numerical results show that, for the portion jointed by two distinctly different size B-spline surface patches, the new method improves both the shape and the continuity qualities of the surface model significantly.

**Key Words:** Surface fitting, B-spline surface patching, Convergent geometric continuity, Polygonal mesh, Quad partition, adaptive ratio compatibility condition.

**AMS Subject Classification:** Primary 65D17, 68U07, 68U05, 68W25, 41A15.

## 1 Introduction

When constructing smooth surfaces, in addition to ensure smoothness along the common boundary curve of adjacent surfaces, a very important issue is the shape quality of the surface model. In recent years, the conditions of geometric continuity between two adjacent

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\*E-mail address: xshi@desu.edu; This work is supported by ARO fund (DAAD 19-03-1-0375).

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parametric surface patches have been extensively studied in the literature such as (Du et al., 1990) and the references cited therein. To the authors' knowledge, however, for geometrically continuous conditions between adjacent B-spline surfaces there is no literature directly addressing the problem of how to adapt for the geometric property of the quad partition. The numerical results show that if the continuity conditions are set up by adapting for the geometric construction of the quad partition, both the shape and the continuity qualities of the surface model is improved significantly. The reasons for exploiting B-spline surface representation in smooth surface reconstruction have been clearly addressed in (Eck and Hoppe, 1996; Krishnamurthy and Levoy, 1996; Milroy et al., 1995).

Bicubic B-spline surfaces are the most preferred and acceptable surface representations in CAD/CAM and advanced modeling systems. We define a face-to-face triangulation to be a triangulation with which any two triangles share a common edge, a common vertex, or are disjoint. We define a quad to be a quadrangle surface of face-to-face triangulation. A polygonal model is a continuous surface of face-to-face triangulation, and a quad partition of a polygonal model is a partition of polygonal model into face-to-face quads. Due to the limitation of handling B-spline surfaces with multiple interior knots in some systems, such as Pro/ENGINEER, we focus on bicubic B-spline surfaces with single interior knots. Therefore, in this paper we consider the construction of a local scheme of generating convergent  $G^1$  smooth bicubic B-spline surfaces with single interior knots over an arbitrary quad partition of a polygonal model. Figure 1 shows a polygonal model with an arbitrary quad partition.



Figure 1: A polygonal model with a quad partition.

A scheme for constructing  $G^1$  smooth B-spline surfaces is called *local* (see Shi et al., 2004) if the two kinds of the control points of B-spline surfaces defined below can be determined by the following given order (refer to Figure 2 for an illustration).

- A** The *control points in the neighborhood of a corner* which consist of the corner point, tangent points  $\bullet$ , twist points  $\Delta$ , and curvature points  $\blacksquare$  if necessary.
- B** The *control points in the neighborhood of a boundary curve* which consist of the boundary control points  $\circ$ , and two rows of control points  $\square$  close to the boundary curve.

Reconstruction of smooth B-spline surfaces of arbitrary topological type has been studied in several articles (Eck and Hoppe, 1996; Krishnamurthy and Levoy, 1996; Milroy et al.,

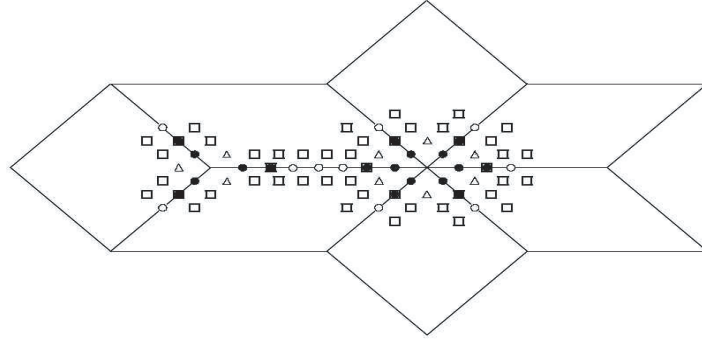


Figure 2: Classification of the control points around corners and edges.

1995, Shi et al., 2004). But the following basic problem related to the shape and continuity of B-spline surfaces are still unsolved:

- How to efficiently set up the geometrically continuous conditions between adjacent B-spline surfaces to produce surface models with better qualities, both in shape and continuity?

The motivation of this paper is to solve above problem. The main contributions of this paper are as follows:

- By adapting for the geometric property of the quad partition, we obtain the  $G^1$  continuity conditions between two adjacent bicubic B-spline surfaces with single interior knots. These conditions are directly represented by the control points of the two B-spline surfaces.
- We provide a new technique for determining the control points in the neighborhood of an  $m$ -patch corner. Determination of the control points preserves the geometric shape at the corner of the model.
- Most importantly, the shape and continuity qualities of the B-spline surface models produced by our algorithm are significantly improved.

The paper is organized as follows. In Section 2, we review relevant previous work. In Section 3, we give the continuity conditions between two bicubic B-spline surfaces with single interior knots. In Section 4, we provide a local scheme of constructing fair and convergent  $G^1$  B-spline surfaces with single interior knots. Finally, in Section 5, we summarize our results and mention some future work.

## 2 Related Work

There is a lot of work on the reconstruction of smooth surfaces of arbitrary topological type using subdivision surfaces. However, subdivision surfaces are not commonly supported by the current modeling systems. Therefore, reconstruction of smooth parametric surfaces, especially the B-spline surfaces widely used in geometric modeling and CAD/CAM, is of

particular important. Many researchers have studied parametric surface fitting and smooth join techniques in the fields of geometric modeling, CAD and approximation theory. We focus here on only the techniques from these areas that are related to fitting smooth B-spline surfaces over any arbitrary quad mesh.

Eck and Hoppe (Eck and Hoppe, 1996) described a method for fitting irregular meshes with a number of automatically placed bicubic Bézier patches, with the continuity of the resulting surface obtained by using Peters' scheme (Peters, 1994). Peters (Peters, 1995) improved Peters' scheme to get another approximately tangent-plane smooth, or  $\epsilon$ - $G^1$  smooth Bézier patches of arbitrary topological type (see (b) on Page 648 in (Peters, 1995)). Krishnamurthy and Levoy (Krishnamurthy and Levoy, 1996) fitted B-spline surfaces of arbitrary topological type with little discussion on the continuity of B-spline surfaces. Milroy et al. (Milroy et al., 1995) provided a scheme of fitting B-spline surfaces, but the scheme is not so good on smoothness. Moreton and Séquin (H. Moreton and C. Séquin, 1992) approximated the nonlinear  $G^1$  constraints by using functional optimization for fair surface design. This approach requires expensive nonlinear optimization and also lacks controlling for the continuity. Peters (Peters, 2000) recently obtained a scheme of constructing  $G^1$  smooth B-spline surfaces with interior double knots from a refined Catmull-Clark subdivision mesh. Obviously the application of the scheme in industry is somewhat limited by current advanced modeling systems.

In contrast, we provided (Shi et al, 2004) a convergent  $G^1$  scheme of the fair B-spline surfaces with single interior knots. The convergent  $G^1$  scheme provided in (Shi et al, 2004) produces fair B-spline surfaces and controls the continuity of the B-spline surfaces within the given tolerance. In this paper, we improve the scheme presented in (Shi et al, 2004) by adapting the  $G^1$  continuity conditions to the geometric properties of the quad partition. Figures 5–7 show that both shape and continuity properties are improved significantly.

### 3 $G^1$ Continuity Conditions of Two Adjacent B-spline Surfaces

In this section, we consider the  $G^1$  continuity conditions of two adjacent bicubic B-spline surfaces. Since these conditions will be used in constructing convergent  $G^1$  smooth B-spline surface models in Section 4, without loss of generality, we assume the number of control points on the common boundary curve of the two B-spline surfaces is not less than 9. All material related to B-spline curves and surfaces can be found in (Piegl and Tiller, 1997).

For general setting, the B-spline surface model is defined as an arbitrary network of tensor product B-spline surface patches. It is usually necessary to assume all B-spline surface patches have the same knot vectors in both directions. Without loss of generality, we assume that the two given bicubic B-spline surfaces are of the form

$$\begin{aligned} S_1(u, v) &= \sum_{i=0}^n \sum_{j=0}^n \mathbf{P}_{i,j} N_{i,3}(u) N_{j,3}(v), \\ S_2(u, v) &= \sum_{i=0}^n \sum_{j=0}^n \mathbf{Q}_{i,j} N_{i,3}(u) N_{j,3}(v), \end{aligned} \tag{3.1}$$

where  $N_{i,3}(u)$  and  $N_{j,3}(v)$  are the basic B-spline functions determined by the  $u$ -directional knot vector  $U = \{0, 0, 0, 0, t_4, \dots, t_n, 1, 1, 1, 1\}$  and the  $v$ -directional knot vector  $V = U$  ( $n \geq 8$ ), and denote their common boundary curve as  $\Gamma(v) = S_1(0, v) = S_2(0, v)$ . For simplicity, we further assume that  $U$  and  $V$  are uniform and only have single interior knots, that is  $h = t_{j+1} - t_j = 1/(n-2)$  ( $3 \leq j \leq n$ ) with  $t_3 = 0$  and  $t_{n+1} = 1$ . However, our method is suitable for general cases of  $U$  and  $V$ .

To deduce the conditions of  $G^1$  continuity between  $S_1(u, v)$  and  $S_2(u, v)$  along their common boundary curve  $\Gamma(v)$ , we begin with defining the following three curves

$$\begin{aligned} C_1(v) &= \left. \frac{\partial S_1(u, v)}{\partial u} \right|_{u=0} = \frac{3}{h} \sum_{j=0}^n \mathbf{P}_j N_{j,3}(v), \\ C_2(v) &= \left. \frac{\partial S_2(u, v)}{\partial u} \right|_{u=0} = \frac{3}{h} \sum_{j=0}^n \mathbf{Q}_j N_{j,3}(v), \\ C_0(v) &= \frac{\partial S_1(0, v)}{\partial v} = 3 \sum_{j=0}^{n-1} \frac{\mathbf{T}_j}{t_{j+4} - t_{j+1}} N_{j,2}(v) \end{aligned} \quad (3.2)$$

where  $\mathbf{P}_j = \mathbf{P}_{1,j} - \mathbf{P}_{0,j}$ ,  $\mathbf{Q}_j = \mathbf{Q}_{1,j} - \mathbf{Q}_{0,j}$ ,  $\mathbf{T}_j = \mathbf{P}_{0,j+1} - \mathbf{P}_{0,j}$ , and  $C_0(v)$  has the knot vector  $V_0 = \{0, 0, 0, t_4, \dots, t_n, 1, 1, 1\}$ .

### 3.1 Decomposition of $C_1(v)$ , $C_2(v)$ and $C_0(v)$

Inserting  $t_4, t_4, \dots, t_n, t_n$  into  $V$  for  $C_1(v)$  and  $C_2(v)$  (refer to Pages 143-144 in (Piegl and Tiller, 1997)), we have

$$(C_1(v), C_2(v)) = \frac{3}{h} \left( \sum_{j=0}^{3(n-2)} \hat{\mathbf{P}}_j N_{j,3}(v), \sum_{j=0}^{3(n-2)} \hat{\mathbf{Q}}_j N_{j,3}(v) \right) \quad (3.3)$$

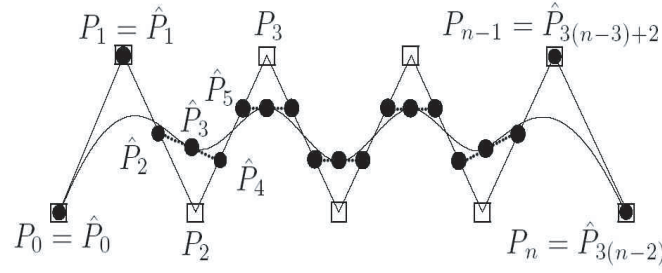
with the knot vector  $\{0, 0, 0, 0, t_4, t_4, t_4, \dots, t_n, t_n, t_n, 1, 1, 1, 1\}$ , where  $\hat{\mathbf{P}}_j$  are determined as follows

$$\begin{aligned} \hat{\mathbf{P}}_0 &= \mathbf{P}_0, \quad \hat{\mathbf{P}}_1 = \mathbf{P}_1, \quad \hat{\mathbf{P}}_2 = \frac{1}{2}(\mathbf{P}_1 + \mathbf{P}_2), \quad \hat{\mathbf{P}}_3 = \frac{1}{12}(3\mathbf{P}_1 + 7\mathbf{P}_2 + 2\mathbf{P}_3), \\ \hat{\mathbf{P}}_{3j} &= \frac{1}{6}(\mathbf{P}_j + 4\mathbf{P}_{j+1} + \mathbf{P}_{j+2}), \quad j = 2, \dots, n-4, \\ \hat{\mathbf{P}}_{3j+1} &= \frac{2}{3}\mathbf{P}_{j+1} + \frac{1}{3}\mathbf{P}_{j+2}, \quad \hat{\mathbf{P}}_{3j+2} = \frac{1}{3}\mathbf{P}_{j+1} + \frac{2}{3}\mathbf{P}_{j+2}, \quad j = 1, \dots, n-4, \\ \hat{\mathbf{P}}_{3(n-3)} &= \frac{1}{12}(2\mathbf{P}_{n-3} + 7\mathbf{P}_{n-2} + 3\mathbf{P}_{n-1}), \\ \hat{\mathbf{P}}_{3(n-3)+1} &= \frac{1}{2}(\mathbf{P}_{n-2} + \mathbf{P}_{n-1}), \quad \hat{\mathbf{P}}_{3(n-3)+2} = \mathbf{P}_{n-1}, \quad \hat{\mathbf{P}}_{3(n-2)} = \mathbf{P}_n, \end{aligned} \quad (3.4)$$

and  $\hat{\mathbf{Q}}_j$  are similarly defined as  $\hat{\mathbf{P}}_j$  in (3.4). In Figure 3, a procedure of decomposing  $C_1(v)$  is illustrated.

Analogously, inserting  $t_4, \dots, t_n$  into  $V_0$  for  $C_0(v)$  yields

$$C_0(v) = \frac{3}{h} \sum_{j=0}^{2(n-2)} \hat{\mathbf{T}}_j N_{j,2}(v) \quad (3.5)$$

Figure 3: Decomposition of  $C_1(v)$ .

with the knot vector  $\{0, 0, 0, t_4, t_4, \dots, t_n, t_n, 1, 1, 1\}$ , and  $\hat{\mathbf{T}}_j$  are given by

$$\begin{aligned} \hat{\mathbf{T}}_0 &= \mathbf{T}_0, \quad \hat{\mathbf{T}}_1 = \frac{1}{2}\mathbf{T}_1, \quad \hat{\mathbf{T}}_2 = \frac{1}{12}(3\mathbf{T}_1 + 2\mathbf{T}_2), \\ \hat{\mathbf{T}}_{2j+1} &= \frac{1}{3}\mathbf{T}_{j+1}, \quad j = 1, \dots, n-4, \\ \hat{\mathbf{T}}_{2j} &= \frac{1}{6}(\mathbf{T}_j + \mathbf{T}_{j+1}), \quad j = 2, \dots, n-4, \\ \hat{\mathbf{T}}_{2(n-3)} &= \frac{1}{12}(2\mathbf{T}_{n-3} + 3\mathbf{T}_{n-2}), \quad \hat{\mathbf{T}}_{2(n-3)+1} = \frac{1}{2}\mathbf{T}_{n-2}, \quad \hat{\mathbf{T}}_{2(n-2)} = \mathbf{T}_{n-1}. \end{aligned} \quad (3.6)$$

### 3.2 $G^1$ conditions between $S_1(u, v)$ and $S_2(u, v)$

Denote respectively  $C_{1,j}(v)$ ,  $C_{2,j}(v)$  and  $C_{0,j}(v)$  to be the restriction of  $C_1(v)$ ,  $C_2(v)$  and  $C_0(v)$  on each interval  $[t_{j+3}, t_{j+4}]$ . Then  $(C_{1,j}(v), C_{2,j}(v), C_{0,j}(v))$  have the following Bézier forms ( $j = 0, \dots, n-3$ ),

$$(C_{1,j}(v), C_{2,j}(v), C_{0,j}(v)) = \frac{3}{h} \left( \sum_{i=0}^3 \hat{\mathbf{P}}_{3j+i} B_{i,3}(\hat{v}), \sum_{i=0}^3 \hat{\mathbf{Q}}_{3j+i} B_{i,3}(\hat{v}), \sum_{i=0}^2 \hat{\mathbf{T}}_{2j+i} B_{i,2}(\hat{v}) \right),$$

where  $v \in [t_j, t_{j+1})$  and  $\hat{v} = (v - t_j)/(t_{j+1} - t_j)$ . Though there are many choices of expressing the  $G^1$  continuity of  $S_1(u, v)$  and  $S_2(u, v)$  along the curve  $\Gamma(v)$  based on  $C_{1,j}(v)$ ,  $C_{2,j}(v)$  and  $C_{0,j}(v)$  (Du et al., 1990), we have opted for the following choice which is most widely used in practice

$$\begin{aligned} \sum_{i=0}^3 \hat{\mathbf{Q}}_{3j+i} B_{i,3}(\hat{v}) &= -a_j \sum_{i=0}^3 \hat{\mathbf{P}}_{3j+i} B_{i,3}(\hat{v}) \\ &+ (b_j(1 - \hat{v}) + c_j \hat{v}) \sum_{i=0}^2 \hat{\mathbf{T}}_{2j+i} B_{i,2}(\hat{v}), \quad j = 0, \dots, n-3 \end{aligned} \quad (3.7)$$

where  $a_j > 0$ ,  $b_j$  and  $c_j$  are constants. To authors knowledge, in all the existing literatures of reconstructing B-spline surface models  $a_j$  is taken as  $a_j = 1$  to produce  $G^1$  conditions. Our results show that, on portions jointed by adjacent B-spline patches with big difference in size, the restriction  $a_j = 1$  reduces the shape and continuity qualities of the surface model. However, in practice this case is inevitable. In this paper, we overcome this problem

by selecting  $a_j$  such that they adapt for the geometric properties of the quad partition. (3.7) yields the  $G^1$  continuity conditions between  $S_1(u, v)$  and  $S_2(u, v)$  as follows

$$\begin{cases} \hat{\mathbf{Q}}_{3j} = -a_j \hat{\mathbf{P}}_{3j} + b_j \hat{\mathbf{T}}_{2j}, \\ 3\hat{\mathbf{Q}}_{3j+1} = -3a_j \hat{\mathbf{P}}_{3j+1} + 2b_j \hat{\mathbf{T}}_{2j+1} + c_j \hat{\mathbf{T}}_{2j}, \\ 3\hat{\mathbf{Q}}_{3j+2} = -3a_j \hat{\mathbf{P}}_{3j+2} + b_j \hat{\mathbf{T}}_{2j+2} + 2c_j \hat{\mathbf{T}}_{2j+1}, \\ \hat{\mathbf{Q}}_{3j+3} = -a_j \hat{\mathbf{P}}_{3j+3} + c_j \hat{\mathbf{T}}_{2j+2} \end{cases} \quad (3.8)$$

for  $j = 0, \dots, n-3$ . By substituting  $\hat{\mathbf{P}}_j$ ,  $\hat{\mathbf{Q}}_j$  and  $\hat{\mathbf{T}}_j$  into (3.8), we obtain that all  $a_j = a$  where  $a$  is defined to be the *adaptive ratio* with respect to the adjacent B-spline patches. The adaptive ratio is very important for the shape and continuity qualities of B-spline surface models. Carefully selecting the adaptive ratios improves the qualities of the surface model significantly. We will discuss its properties and selection next section. In addition to all  $a_j$  being the same, from (3.8) we also obtain the following  $G^1$  continuity conditions directly represented by the control points of  $S_1(u, v)$  and  $S_2(u, v)$

$$\begin{cases} \mathbf{Q}_0 = -a\mathbf{P}_0 + b_0\mathbf{T}_0, \quad 3\mathbf{Q}_1 = -3a\mathbf{P}_1 + b_0\mathbf{T}_1 + c_0\mathbf{T}_0, \\ \mathbf{Q}_2 = -a\mathbf{P}_2 + \frac{c_1}{6}\mathbf{T}_1 + \frac{1}{18}(7c_0 - 2c_1)\mathbf{T}_2 - \frac{c_0}{18}\mathbf{T}_3, \\ \mathbf{Q}_3 = -a\mathbf{P}_3 + \frac{c_2}{9}\mathbf{T}_2 + \frac{1}{18}(7c_1 - 2c_2)\mathbf{T}_3 - \frac{c_1}{18}\mathbf{T}_4, \\ \mathbf{Q}_{j+2} = -a\mathbf{P}_{j+2} + \frac{1}{18}(7c_j - 2c_{j+1})\mathbf{T}_{j+2} + \frac{c_{j+1}}{9}\mathbf{T}_{j+1} - \frac{c_j}{18}\mathbf{T}_{j+3}, \quad j = 2, \dots, n-6, \\ \mathbf{Q}_{n-3} = -a\mathbf{P}_{n-3} + \frac{c_{n-4}}{9}\mathbf{T}_{n-4} + \frac{1}{18}(7c_{n-5} - 2c_{n-4})\mathbf{T}_{n-3} - \frac{c_{n-5}}{12}\mathbf{T}_{n-2}, \\ 3\mathbf{Q}_{n-2} = -3a\mathbf{P}_{n-2} + \frac{c_{n-3}}{3}\mathbf{T}_{n-3} + (2c_{n-4} - \frac{c_{n-3}}{2})\mathbf{T}_{n-2} - c_{n-4}\mathbf{T}_{n-1}, \\ 3\mathbf{Q}_{n-1} = -3a\mathbf{P}_{n-1} + c_{n-4}\mathbf{T}_{n-1} + c_{n-3}\mathbf{T}_{n-2}, \quad \mathbf{Q}_n = -a\mathbf{P}_n + c_{n-3}\mathbf{T}_{n-1}, \end{cases} \quad (3.9)$$

and

$$c_j L_j = 0, \quad j = 0, \dots, n-4, \quad (3.10)$$

where

$$\begin{cases} L_0 = 6\mathbf{T}_0 - 6\mathbf{T}_1 + 3\mathbf{T}_2 - \mathbf{T}_3, \quad L_1 = \frac{3}{2}\mathbf{T}_1 - 3\mathbf{T}_2 + 3\mathbf{T}_3 - \mathbf{T}_4, \\ L_j = \mathbf{T}_j - 3\mathbf{T}_{j+1} + 3\mathbf{T}_{j+2} - \mathbf{T}_{j+3}, \quad j = 2, \dots, n-6, \\ L_{n-5} = \mathbf{T}_{n-5} - 3\mathbf{T}_{n-4} + 3\mathbf{T}_{n-3} - \frac{3}{2}\mathbf{T}_{n-2}, \quad L_{n-4} = \mathbf{T}_{n-4} - 3\mathbf{T}_{n-3} + 6\mathbf{T}_{n-2} - 6\mathbf{T}_{n-1} \end{cases} \quad (3.11)$$

and  $b_j, c_j$  are introduced in (3.7) satisfying

$$\begin{cases} b_{j+1} = c_j, \\ c_j = \frac{n-j-3}{n-2}b_0 + \frac{j+1}{n-2}c_{n-3}, \end{cases} \quad j = 0, \dots, n-4. \quad (3.12)$$

## 4 The determination of the control points of the B-spline surface

In this section, we discuss how to determine the control points of the B-spline surface. We only discuss the determination of tangent points, twist points, and curvature points. For

the determination of other control points please refer (Shi et al., 2004). In this paper, we assume that a B-spline surface is constructed already on each quad of the quad partition of a polygonal model using the method in (Shi et al., 2004).

For a given set of bicubic B-spline surfaces  $\Lambda = \{S_i\}_0^{m-1}$  constructed over quads  $\{Q_i\}_0^{m-1}$ ,  $\Lambda$  is called an  $m$ -patch corner if all  $S_i$  share a common corner point  $\mathbf{P}$ ,  $S_i$  and  $S_{i+1}$  ( $0 \leq i < m-1$ ) join along their common boundary curve  $\Gamma_i$ , and  $\Gamma_i \neq \Gamma_k$  if  $i \neq k$  for  $i, k = 0, \dots, m-1$ . The integer  $m$  is called the *degree* of the corner  $\mathbf{P}$ . If  $S_{m-1}$  and  $S_0$  have a common boundary curve  $\Gamma_{m-1}$ , then the corner is referred to as a *full corner*, otherwise, the corner is referred to as a *partial corner*. A full corner of degree 4 is called *ordinary*, otherwise, the full corner is called *extraordinary*. To simplify the rest of our discussion, the subscripts  $i := (m + i) \bmod m$  will apply throughout.

Based on the  $G^1$  continuity condition (3.9) between two adjacent B-spline surfaces, we discuss how to smoothly stitch a full corner below. The case of a partial corner can be treated similarly. Let  $\mathbf{A}_i$  be the closest control point to  $\mathbf{P}$  of  $\Gamma_i$  (*tangent points*),  $\mathbf{B}_i$  the closest control point to  $\mathbf{P}$  between  $\Gamma_i$  and  $\Gamma_{i+1}$  (*twist points*), and  $\mathbf{D}_i$  the second closest point to  $\mathbf{P}$  of  $\Gamma_i$  (*curvature points*). See Figure 4 for an illustration. Our smooth stitching procedure consists of three successive steps.

**1. Determining the tangent plane at the corner point  $\mathbf{P}$ .** Let  $\mathbf{n}_i(\mathbf{P})$  be the unit normal vector of  $S_i$  at  $\mathbf{P}$  for  $i = 0, \dots, m-1$ , we set the common unit normal vector at  $\mathbf{P}$  as follows:

$$\mathbf{n}(\mathbf{P}) = \sum_{i=0}^{m-1} w_i \mathbf{n}_i(\mathbf{P}) / \left\| \sum_{i=0}^{m-1} w_i \mathbf{n}_i(\mathbf{P}) \right\|$$

where  $w_i > 0$  are the weights such that  $\left\| \sum_{i=0}^{m-1} w_i \mathbf{n}_i(\mathbf{P}) \right\| \neq 0$ . Then the tangent plane  $\Pi(\mathbf{P})$  can be determined by  $\mathbf{P}$  and  $\mathbf{n}(\mathbf{P})$ .

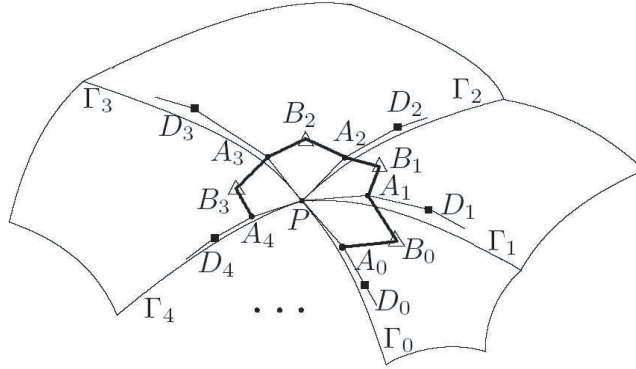


Figure 4: An  $m$ -patch corner.

**2. Determining the tangent points  $\mathbf{A}_i$ .** Project all tangent points  $\mathbf{A}_i$  onto the tangent plane  $\Pi(\mathbf{P})$ . For brevity, we still use  $\mathbf{A}_i$  to denote the projected tangent points.

Let  $\mathbf{R}_i = \mathbf{A}_i - \mathbf{P}$ . It follows from the first equation of (3.9) that there exists constants  $\alpha_i$  (corresponding to  $a$ ) and  $\beta_i$  (corresponding to  $b_0$ ) such that

$$\mathbf{R}_{i+1} + \alpha_i \mathbf{R}_{i-1} = \beta_i \mathbf{R}_i, \quad i = 0, \dots, m-1. \quad (4.1)$$

Let  $\rho_i = \|\mathbf{R}_i\|$ , then by taking the cross product of (4.1) with  $\mathbf{R}_i$ ,

$$\rho_{i+1} \sin \theta_i = \rho_{i-1} \alpha_i \sin \theta_{i-1}, \quad i = 0, \dots, m-1 \quad (4.2)$$

where  $\theta_i$  is the angle formed by  $\mathbf{R}_i$  and  $\mathbf{R}_{i+1}$ , and  $\theta_i < \pi$  is reasonably assumed. According to (4.2), it yields

$$\prod_{i=0}^{m-1} \alpha_i = 1. \quad (4.3)$$

Equation (4.3) is called the *adaptive ratio compatibility (ARC) condition* at the corner  $\mathbf{P}$ . To adapt for the geometric properties of the quads  $\{Q_i\}_{i=0}^{m-1}$ , we set

$$\alpha_i = \frac{\Delta_i}{\Delta_{i-1}}, \quad (4.4)$$

where  $\Delta_i$  is the sum of the areas of all the triangles contained in  $Q_i$ . The ARC condition (4.3) holds automatically if  $\alpha_i$  are determined by (4.4).

Here we give another method to set  $\alpha_i$ . Let  $\{\mathbf{P}_{i,j} | j = 0, 1, 2, 3\}$  be the vertices of the quad  $Q_i$  and  $\bar{\mathbf{P}}_i = 0.25 \sum_{j=0}^3 \mathbf{P}_{i,j}$ , and

$$\alpha_i = \frac{\bar{\Delta}_i}{\bar{\Delta}_{i-1}}, \quad (4.5)$$

where  $\bar{\Delta}_i$  is the sum of the areas of the four triangles with the vertices  $\bar{\mathbf{P}}_i, \mathbf{P}_{i,j}, \mathbf{P}_{i,j+1}, j = 0, 1, 2, 3$  with  $\mathbf{P}_{i,4} = \mathbf{P}_{i,0}$ .

If  $m$  is even, respectively setting  $i = 1, 3, \dots, m-1$  (or  $i = 0, 2, \dots, m-2$ ) in (4.2) yields

$$\prod_{i=0}^{m-1} \sin \theta_{2i} = \prod_{i=0}^{m-1} \alpha_{2i} \sin \theta_{2i+1}. \quad (4.6)$$

Equation (4.6) is referred to as the *compatibility condition* at the corner  $\mathbf{P}$  of even degree. Equation (4.6) is the only restriction for the selection of  $\theta_i$  if  $m$  is even. For odd corner the selection of  $\theta_i$  has no restriction. After selecting  $\theta_i$ , from (4.1) and (4.2) we obtain the following equations to determine the rest parameters in (4.1),

$$\rho_{i+1} = \frac{\sin \theta_{i-1}}{\sin \theta_i} \alpha_i \rho_{i-1} \quad \text{and} \quad \beta_i = \frac{\rho_{i-1} \sin(\theta_{i-1} + \theta_i)}{\rho_i \sin \theta_i}. \quad (4.7)$$

It is easy to show that (4.1) and (4.7) are equivalent.

**3. Determining the twist points  $\mathbf{B}_i$  and curvature points  $\mathbf{D}_i$ .** Denote by  $\mathbf{V}_i = \mathbf{B}_i - \mathbf{P}$  and  $\hat{\mathbf{R}}_i = \mathbf{D}_i - \mathbf{A}_i$ . According to the second equation of (3.9), we obtain

$$\mathbf{V}_i + \alpha_i \mathbf{V}_{i-1} = (1 + \alpha_i + \frac{\gamma_i}{3}) \mathbf{R}_i + \frac{\beta_i}{3} \hat{\mathbf{R}}_i, \quad i = 0, \dots, m-1 \quad (4.8)$$

where  $\alpha_i$  and  $\beta_i$  is defined in (4.7) and  $\gamma_i$  is just the constant  $c_0$  determined by (3.12) with respect to the boundary curve  $\Gamma_i$ .



**Case 1: Ordinary corner.** If the tangent points satisfy the following condition

$$\mathbf{R}_2 = -\alpha_1 \mathbf{R}_0 \quad \text{and} \quad \mathbf{R}_3 = -\alpha_2 \mathbf{R}_1, \quad (4.9)$$

then  $\alpha_0 \alpha_2 = \alpha_1 \alpha_3 = 1$  and  $\beta_i = 0$  ( $i = 0, 1, 2, 3$ ). Therefore, (4.8) is equivalent to

$$\mathbf{V}_i + \alpha_i \mathbf{V}_{i-1} = (1 + \alpha_i + \frac{\gamma_i}{3}) \mathbf{R}_i, \quad i = 0, 1, 2, 3. \quad (4.10)$$

This implies that the curvature points  $\mathbf{D}_i$  can be determined arbitrarily. Since (4.10) is an over-determined system for the fixed  $\mathbf{R}_i$ , each equation in (4.10) may have a residue denoted by  $\epsilon_i$  ( $i = 0, 1, 2, 3$ ), that is,

$$\mathbf{V}_i + \alpha_i \mathbf{V}_{i-1} = (1 + \alpha_i + \frac{\gamma_i}{3}) \mathbf{R}_i + \epsilon_i, \quad i = 0, 1, 2, 3. \quad (4.11)$$

By (4.11),  $L := \sum_{i=0}^3 \bar{\alpha}_i (1 + \alpha_i + \frac{\gamma_i}{3}) \mathbf{R}_i + \sum_{i=0}^3 \bar{\alpha}_i \epsilon_i = 0$ , where  $\bar{\alpha}_i = (-1)^i \prod_{j=i+1}^4 \alpha_j$  and  $\alpha_4 = \alpha_0$ . We define the function  $L_\epsilon$  by the method of *Lagrange multipliers*

$$L_\epsilon = \frac{1}{2} \sum_{i=0}^3 \|\epsilon_i\|^2 + \lambda L \quad (4.12)$$

and set  $\frac{\partial L_\epsilon}{\partial \epsilon_i} = 0$  for  $i = 0, 1, 2, 3$  to get

$$\lambda = \frac{\sum_{i=0}^3 \bar{\alpha}_i (1 + \alpha_i + \frac{\gamma_i}{3}) \mathbf{R}_i}{\sum_{i=0}^3 \bar{\alpha}_i^2} = \frac{(\gamma_0 - \alpha_0 \gamma_2) \mathbf{R}_0 + (\gamma_3 - \alpha_3 \gamma_1) \mathbf{R}_1}{3 \sum_{i=0}^3 \bar{\alpha}_i^2}, \quad \epsilon_i = -\bar{\alpha}_i \lambda. \quad (4.13)$$

Now we can determine the twist points  $\mathbf{B}_i$  by solving (4.11). Since  $|\gamma_k| = O(n^{-1})$  and  $\|\mathbf{R}_k\| = O(n^{-1})$ ,  $\epsilon_i$  satisfy

$$\|\epsilon_i\| = O\left(\sum_{k=0}^3 \|\gamma_k \mathbf{R}_k\|\right) = O(n^{-2}), \quad i = 0, 1, 2, 3. \quad (4.14)$$

If (4.9) does not hold, the case can be included in Case 2 below.

**Case 2: Extraordinary corner.** If  $m$  is odd, (4.8) is an under-determined system. In the case of both  $m = 4$  and (4.9) does not hold, or in the case of  $m > 4$  is even, (4.8) equals

$$\begin{aligned} \mathbf{V}_i + \alpha_i \mathbf{V}_{i-1} &= (1 + \alpha_i + \frac{\gamma_i}{3}) \mathbf{R}_i + \frac{\beta_i}{3} \hat{\mathbf{R}}_i, \quad i = 1, \dots, m-1, \\ \sum_{k=0}^{m-1} \bar{\alpha}_i \beta_i \hat{\mathbf{R}}_k &= - \sum_{k=0}^{m-1} \bar{\alpha}_i (3 + 3\alpha_i + \gamma_k) \mathbf{R}_k, \end{aligned} \quad (4.15)$$

where  $\bar{\alpha}_i = (-1)^i \prod_{j=i+1}^m \alpha_j$  and  $\alpha_m = \alpha_0$ . Equation (4.15) is also an under-determined system. Thus, the twist points  $\mathbf{B}_i$  and curvature points  $\mathbf{D}_i$  can be determined by solving the under-determined systems (4.8) or (4.15) for the relevant cases. These curvature points  $\mathbf{D}_i$  corresponding to  $\beta_i = 0$  can be freely fixed.

**A local scheme of constructing fair and convergent  $G^1$  smooth B-spline surface models.** As described above, for a quad partition of a polygonal mesh, a fair and convergent  $G^1$  smooth bicubic B-spline surface model with adaptive ratios is locally constructed by the following steps.

- Fit a fair bicubic B-spline surface on each quad.
- For each corner point, determine the tangent control points, the twist and curvature points by the method presented above.
- For each boundary curve, adjust the relevant boundary control points by (3.10) if necessary. Then, determine the two row control points close to the boundary curve according to the equations in the linear system (3.9) excluding the first two and last two equations.
- Check the  $G^1$  continuity for the whole model. We specify a maximum angle that should occur between patches, then check all the angles between patches. If the angles are larger than the specified maximum angle, then increase the number of control points of B-spline patches and repeat the above procedure.

**Theorem 4.1.** The B-spline surface model constructed by the above steps is convergent  $G^1$  smooth.

The comparison results of the NURBS models constructed by schemes with and without adaptive ratios are illustrated in Figure 5–7.

## 5 Summary and Future Work

In this paper, based on the geometric properties of the quad partition, we provide a new local scheme of constructing convergent  $G^1$  smooth bicubic B-spline surface patches with single interior knots over a given arbitrary quad partition of a polygonal model. The method is adaptive for the geometric properties of the quad partition, i.e., the conditions reflect the different sizes of adjacent B-spline surface patches. Our numerical results show that, for the portion jointed by two distinctly different-size B-spline surface patches, the new method improves both the shape and the continuity qualities of the surface model significantly.

There are a number of areas for future work. The method of determining the tangent points for an  $m$ -corner can be improved in order to avoid unnecessary geometric distortion around the corner. In addition, there exists a strong interest of studying local schemes for constructing true  $G^1$  smooth B-spline surfaces with single interior knots from both theoretical and practical perspectives. Our results indicate that the biquintic is the smallest degree of existing true  $G^1$  B-spline surface models with single interior knots.

## Acknowledgments

This paper is partly funded by ARO (DAAD19-03-1-0375).

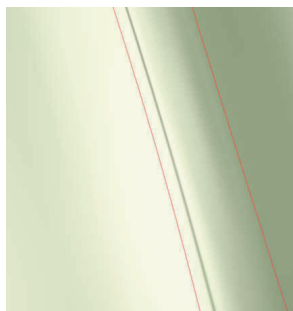
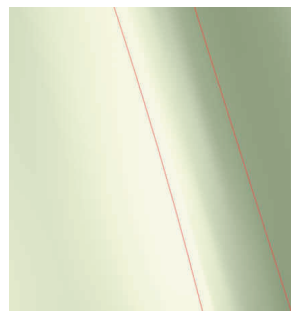
(a)  $20 \times 20, 1.27$ (b)  $20 \times 20, 0.48$ 

Figure 5: (a) is constructed without adaptive ratios and (b) is constructed with adaptive ratios. For each B-spline surface model,  $x \times x, y$  means each patch has  $x \times x$  control points, and the maximum size of dihedral angles of any two adjacent patches is  $y$  degrees.

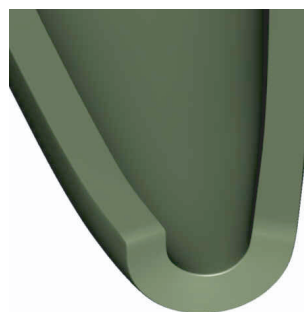
(a)  $20 \times 20, 2.58$ (b)  $20 \times 20, 1.36$ 

Figure 6: (a) is constructed without adaptive ratios and (b) is constructed with adaptive ratios

(a)  $20 \times 20, 3.46$ (b)  $20 \times 20, 1.79$ 

Figure 7: (a) is constructed without adaptive ratios and (b) is constructed with adaptive ratios

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*Chapter 16*

## FEATURE SIZING MODELING FOR PARAMETRIC HUMAN BODY

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### Abstract

A parametric human body model is presented. Contours of human body are acquired from triangle mesh model through resampling, consequently a 3D human body model of quadrangle mesh is obtained. Feature lines, such as waistline and chest girth line, are defined in the human body, then a new 3D human body model with given feature sizing parameters can be created through deforming the contours. The deformation algorithm for contours is an energy-based method and allows local sculpturing, which make it possible to create particular physique immediately.

**Key Words:** Human body model, Feature sizing parameters, Energy-based deformation.

**AMS Subject Classification:** Primary 68U05, 68U20.

## 1 Introduction

3D human body modeling has received much attention recently owing to its importance in many fields of computer graphics, such as garment CAD, online virtual try-on system and 3D game. While the complexity of the structure of human body and the variety of requirements make it very challenging to create realistic human body model. At present, there are mainly wire frame models [1], surface models [2], multi-layered models [3, 4, 5] and anatomy-based model [6]. These models usually aim to produce realistic human body which looks like a human. Whereas in the industrial fields, quantificational description of a human body and modification according to the user input are often required. For example, in

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garment CAD, the designer hope to design clothes according to the sizing parameters, such as chest girth, under bust girth and waist girth. Another example is online try-on system, with which the consumers hope to try on virtually based on their own sizing parameters. Parameterization is a natural solution to these problems.

The automatic modeling of human body from sizing parameters [7] presented by H. Seo and M. Magnenat-Thalmann is one of the most successful parameterized human body models. It is a template-based two-layered model, including skeleton layer and skin layer. The templates are reconstructed from 3D scan data. In this model, producing a new human body with specific sizing parameters is converted into an interpolation problem. Since skeleton-driven deformation is applied, the 3D model is ready to animate. Allen proposed another human body modeling and parameterized method [8] based on 3D scan data. PCA (principal component analysis) is applied to preprocess the large numbers of data primarily, and the parameterized model is produced via finding the relationship between the feature parameters and the principal components. The above two models fall into interpolating methods which required the support of vast scan data. The parameterized model [9] based on resampling presented by K. Qin is similar to our method. The contours of the human body are obtained by resampling from a standard 3D human body model. They use close B-spline curve to fit the contours and then to construct the surface of 3D human body through longitudinal interpolation, but they can only deal with simple deformation such as scaling. Instead, we represent the 3D human body as quadrangle meshes from the contours directly and define feature lines in the model corresponding to the sizing parameters. We will show that human body model with specific sizing parameters can be converted into model deformation with constraints [11, 12, 13], i.e. the deformation of contours. A contour is represented as its intrinsic definition, which make is convenient to apply energy-based deformation. The deformation algorithm for contours allows local sculpturing, which make it possible to create particular physique immediately. The experiments conducted on the female model illustrate the feasibility of our present model.

## 2 The Resampling of 3D Human Body

It is convenient to get a human body model represented as triangle meshes from 3D scanner or 3D software such as Poser. But the model is usually composed of points and the topology, but it seldom includes the semantic information of the real human body, which make it difficult to parameterize it. A useful method to solve this problem is to resample from the model.

In our method, it is required to modify the human body model according to the girths, such as the waist girth and chest girth, so we acquire the horizontal contours through scanning lines [9] (see figure 1 and figure 2). Then a 3D Human body model represented as quadrangle meshes can be constructed directly from the wire frame model composed of contours (see figure 3).

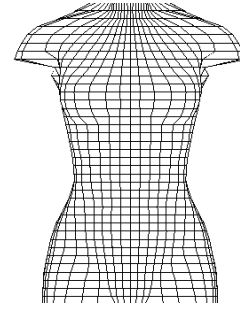
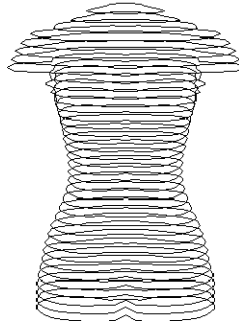
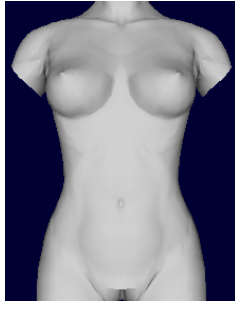


Figure 1: Source human body model

Figure 2: Contours of human body via resampling

Figure 3: Human body model represented as quadrangle meshes

### 3 The Deformation of a Contour

#### 3.1 The intrinsic definition of a planar polyline

A planar polyline (contour) with  $n$  points is represented as its intrinsic definition  $\{Q_1, (\phi_i, l_i)_{i=1}^{n-1}\}$ , where  $Q_1$  is the starting point,  $\phi_i$  is the directional angle, and  $l_i$  is the length of the line segment (see figure 4).

#### 3.2 The deformation energy of the polyline

Energy-based method is an effective deformation technique. “Snakes” (active contour models) [10] is a typical energy-based deformation method for curves. The energy is formulated as

$$V = \frac{1}{2} \int \left[ \alpha(s) \left\| \frac{dv}{ds}(s) \right\|^2 + \beta(s) \left\| \frac{d^2v}{ds^2}(s) \right\|^2 \right] ds$$

where the first and second derivative terms correspond to axial and bending deformations, respectively. The terms  $\alpha(s)$  and  $\beta(s)$  are weighting parameters. Many researchers extended the “snakes” by introducing other energy formulas, but they usually include the two

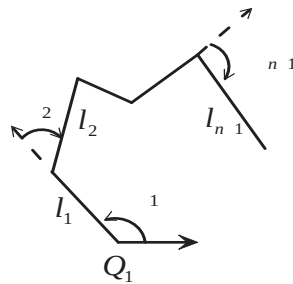


Figure 4: The intrinsic definition of a planar polyline



basic terms: stretching and bending energy terms. In the intrinsic deformation of planar polyline, the length  $l_i$  and the directional angle  $\phi_i$  correspond to the two basic energy terms exactly. In our method, the deformation energy is

$$E = \sum_{i=1}^{n-1} \alpha_i(\phi_i, \phi_{i+1}) \Delta l_i + \sum_{i=1}^{n-1} \beta_i \Delta \phi_i \quad (3.1)$$

where  $\Delta l_i$  and  $\Delta \phi_i$  are the increments of length and directional angle, respectively,  $\alpha_i$  and  $\beta_i$  are weighting parameters. In our implementation,  $\beta_i$  is constant, and  $\alpha_i$  is a variable with respect to the adjacent directional angles.  $\alpha_i$  increases while the directional angles increase, that is because the stretching energy corresponding to axial deformation contributes more in the flatter area.

### 3.3 The constraints

In [14, 15], the mid-point goal deformation is achieved by splitting a polyline into two IK (Inverse Kinematic) chains. Only the directional angles are variables, and the length of every segment and the starting point of the polyline remain constant, which restricts its application. In our method, the polyline is represented as the intrinsic definition, and all the parameters in the intrinsic definition are regarded as variables, including the starting point and the lengths and the directional angles. This extension makes it possible to deal with the displacement constraint of arbitrary point on the polyline and the length constraints simultaneously. The algorithm is described in detail in the rest of this section.

For the  $m$ -th point on the polyline,

$$f(Q_1, \Phi, L) = \mathbf{X}, \quad (3.2)$$

where

$$\begin{aligned} & f(x_1, y_1, \phi_1, \dots, \phi_{m-1}, l_1, \dots, l_{m-1}) \\ &= \begin{bmatrix} x_1 + l_1 \cos \phi_1 + \dots + l_{m-1} \cos(\phi_1 + \dots + \phi_{m-1}) \\ y_1 + l_1 \sin \phi_1 + \dots + l_{m-1} \sin(\phi_1 + \dots + \phi_{m-1}) \end{bmatrix} \end{aligned} \quad (3.3)$$

This is a nonlinear equation. Let  $\mathbf{P} = (x_1, y_1, \phi_1, \dots, \phi_{m-1}, l_1, \dots, l_{m-1})^T$ , then we have

$$\mathbf{J} \Delta \mathbf{P} = \Delta \mathbf{X}, \quad (3.4)$$

where  $\mathbf{J}$  is the Jacobi matrix of  $f$  with respect to  $\mathbf{P}$ .

As for the length constraint of the segment of a polyline, if the length increment of  $i$ -th segment is  $\Delta l_i$ , the constraint formula is

$$\mathbf{J}_{l_i} \Delta \mathbf{P} = \Delta l_i, \quad (3.5)$$

where  $\mathbf{J}_{l_i}$  is a row vector of  $2n$  components, with its  $n + i + 1$ th is 1 and the others are 0. Similarly, If the increment of the length of the whole polyline is  $\Delta l_{sum}$ , the constraint formula is

$$\mathbf{J}_{suml} \Delta \mathbf{P} = \Delta l_{sum}, \quad (3.6)$$

where  $\mathbf{J}_{suml}$  is a row vector of  $2n$  components, with its last  $n - 1$  components are 1 and the others are 0. In general, given the increment of the length of the whole polyline and a probability density function  $f(x)$  of the increment along the polyline, the increment for each line segment can be written as

$$\Delta l_i = \Delta l_{sum} \int_{L_{i-1}}^{L_i} f(x) dx \quad (3.7)$$

where  $L_i = \sum_{j=1}^i l_j$  and define  $L_0 = 0$ . The corresponding constraint equations can be obtained as equation 3.5.

If there are several constraints simultaneously, we can get a linear constraint equation system

$$\mathbf{A} \Delta \mathbf{P} = \mathbf{e}, \quad (3.8)$$

### 3.4 Constrained optimization

In the above sections, we discuss the deformation energy and the constraints respectively, then the deformation of a polyline with constraints is converted into the constraint optimization problem

$$\begin{cases} \min E(\Delta \mathbf{P}) \\ \text{s.t. } \mathbf{A} \Delta \mathbf{P} = \mathbf{e} \end{cases} \quad (3.9)$$

By solving the optimization problem, we can get the increment of the parameter in the intrinsic definition of a polyline, and the deformed polyline consequently. It can be solved by gradient project or feasible direction method.

## 4 Human Body Deformation Based on Feature Sizing Parameters

In order to create 3D human body models according to the feature sizing parameters, we define the corresponding feature lines in the human body model [16]. Since the quadrangle meshes of our human model are obtained from the contours directly, it is easy to define the feature lines (see figure 5, the six feature lines correspond to hip girth, waist girth, under breast girth, breast girth, above breast girth and shoulder width, respectively). According to the user input, we can get the length increment of feature contours, and the increment of the contours between the feature contours can be obtained by interpolation. Then we can get all the deformed contours, and the human body model with specific feature sizing parameters consequently.

Due to the speciality of human body, we must deal with some special part additionally. For example, the length of the feature chest girth in the model is not just the chest girth, so we must convert the chest girth of user input into the length increment of the feature chest girth. Another example, when the waist girth increases, it usually does not grow evenly, with abdomen growing more while back growing less, and the same for decreasing. Accordingly, we should distribute the length increment properly. For example, we want to create a female model with slimmer waist and fatter hip. See figure 6 for boundary curves in both front and side projection views, the contours of waist and hip are shown as figure

7, the probability density function and the length increment distribution of waist girth is shown as figure 8 and figure 9, and the resulting model is shown as figure 10. Figure 11 illustrates an example of increasing waist girth.

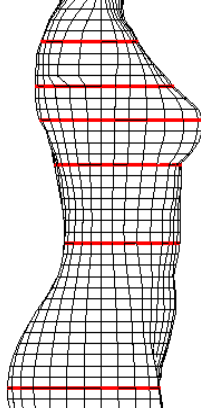


Figure 5: Feature lines in the model

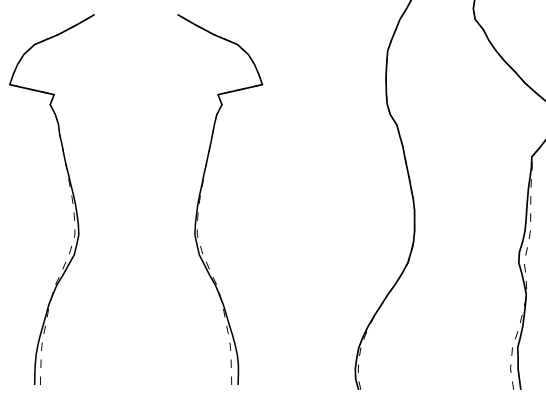


Figure 6: Boundary curves in front and side projection views. Solid line: deformed, dashed line: standard.

Since real human bodies differ from each other greatly, for a special human body model, we can specify the displacement constraints interactively, and the human body model of special physique can be produced by the energy-based deformation discussed in the last section (shown as figure 12 and figure 13).

## 5 Conclusion

In this paper, we present a parametric human body model. Firstly we obtain a human body model represented as quadrangle meshes from the contours. Then feature lines, such as waistline and chest girth line, are defined in the human body, a new 3D human body model of given feature sizing parameters can be created through the contours deformation. The deformation algorithm for contours is an energy-based method and allows local sculpturing, which make it possible to create particular physique immediately. The experiments conducted on female model show that our model works very well.

In the present method, the model is deformed by fitting to girth length parameters, we

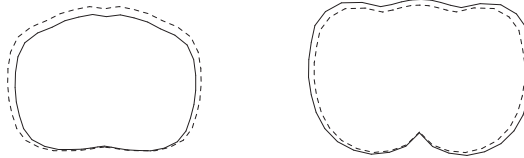


Figure 7: Contours of waist and hip. Solid line: deformed, dashed line: standard.

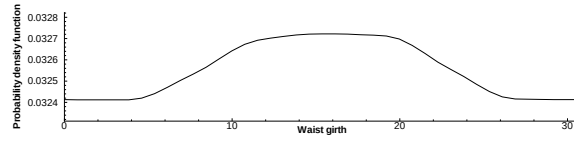


Figure 8: The probability density function of length increment girth.

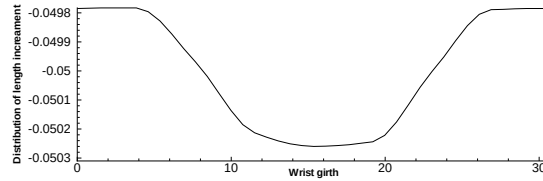


Figure 9: The distribution of length increment of waist girth.

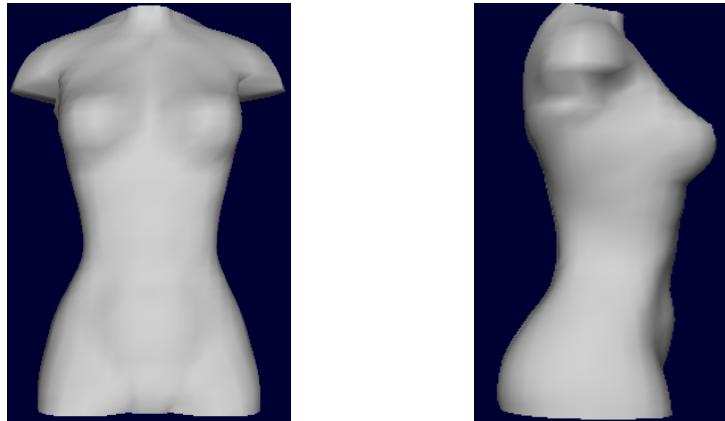


Figure 10: Modeling by decreasing waist girth and increasing hip girth

believe that fitting more feature lines (such as the boundary curves in both front and side projection views) or introducing more constraints (such as the area or volume constraints) will provide more controllable model. Human body is a very complicated object with many special properties. More semantic information of the human body combined into the model will produce more similar models to the real human body. In our method, the user input only include the sizing parameters of a human body, other guide lines of physique, such as WHR (Waist Hip Ratio), should be added into the model. In addition, further work should be done with some knowledge from biological experimental study. For instance, for different part of the body, the probability density function of length increment can be obtained from statistics. For different kinds of people, like African and Asian people, the deformation principles or features are different. For kids, adult people there are different behaviors because of the ages, different templates should be introduced.

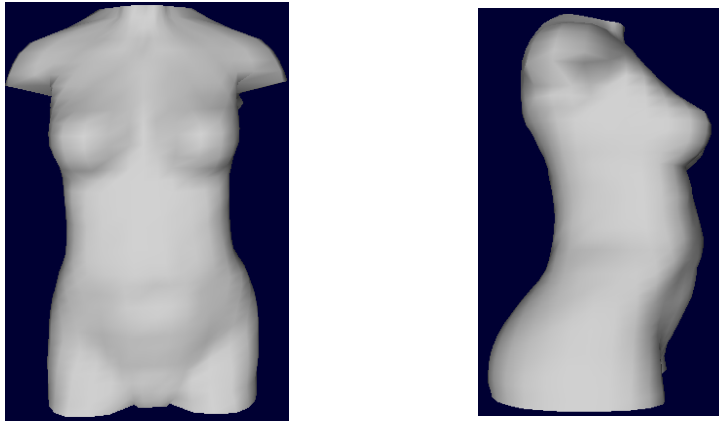


Figure 11: Modeling by increasing waist girth.

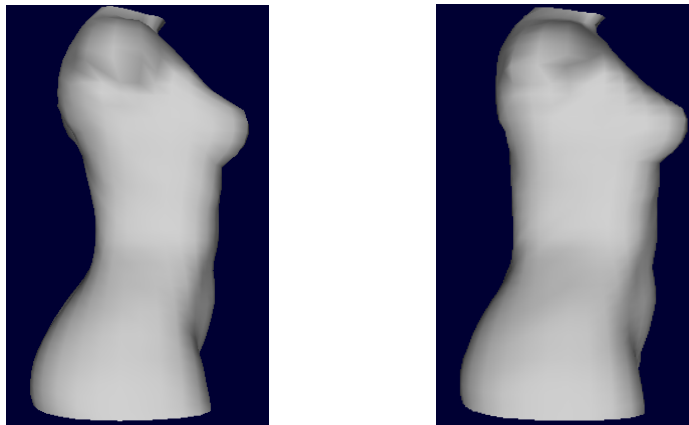


Figure 12: Model of humpbacked woman  
Figure 13: Model of woman with excessively flat back

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*Chapter 17*

# NEW METHOD FOR SIGNAL PROCESSING: DETECTING OF BIFURCATIONS IN TIME-SERIES OF NONLINEAR SYSTEMS

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## Abstract

A new method for detecting of bifurcations in time-series of nonlinear systems is suggested. This method is based on two fundamental nonlinear phenomena: the phenomenon of the pre-bifurcation noise amplification and phenomenon of rise and saturation of the fluctuation correlation time. Theoretical estimates are outlined briefly for both phenomena. It is shown that in saturation regime fluctuation variance is proportional to the square root of external noise variance, whereas fluctuation variance in linear regime is proportional to noise variance. Also it is shown that correlation time saturates at a level inversely proportional to the noise standard deviation. Characteristic features of phenomena are illustrated by the results of numerical simulations. Obtained theoretical estimates are of great practical importance since it can be utilized to design and produce devices which cut off air-engines and rockets in case of detecting in them of the abnormal noise.

**Key Words:** nonlinear dynamics, bifurcations, noise, signal processing.

**AMS Subject Classification:** Primary 43A60.

## 1 Introduction

We introduce a new method for signal possessing which rests on two fundamental nonlinear phenomena. These phenomena could be observed in nonlinear systems near the threshold

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of bifurcation: phenomenon of pre-bifurcation noise amplification and phenomenon of rise and saturation of the fluctuation correlation time.

It is well known a system behavior is qualitatively changes at a bifurcation point. For example such changes as abnormal movements of bridges are life threatening. Hence it is very important to detect an impending bifurcation of a nonlinear system.

First, we consider the phenomenon pre-bifurcation noise amplification. Analysis of pre-bifurcation noise amplification, performed in papers [1]-[4], is based on linear theory, which demonstrates unlimited growth of fluctuations in the immediate vicinity of the bifurcation point. Nonlinear saturation of fluctuations in the vicinity of bifurcation point was studied in the papers [5], [6] for the case of sufficiently slow (quasi-stationary) period doubling bifurcation for nonlinear map and in paper [7] for a pitchfork bifurcation in a nonlinear oscillator.

As was pointed out by Wiesenfeld [1]-[4], the phenomenon of pre-bifurcation noise enhancement as well as weak signal amplification at the bifurcation threshold might be an effective tool for revealing bifurcations in nonlinear systems. Experimental manifestation of pre-bifurcation noise enhancement is described in paper [8] dealing with a period doubling bifurcation in semiconductor laser. Another example of pre-bifurcation noise amplification was given by Garcia-Ojalvo [9] devoted to fluctuation of light in a ring cavity with a nonlinear absorbing medium.

Second, we look at phenomenon of rise and saturation of the correlation time of the fluctuation process.

Correlation time rise near the bifurcation threshold can be anticipated from the relation

$$\tau_c |\lambda| \approx 1$$

which connects the correlation time  $\tau_c$  of fluctuation processes in non-linear systems with a Lyapunov index  $\lambda$ . The last characterizes how close the system is to bifurcation threshold  $|\lambda|=0$  [10], [11]. According to [11], correlation time  $\tau_c$  increases infinitely when  $|\lambda| \rightarrow 0$ .

The phenomenon of pre-bifurcation correlation time rise accompanies the phenomenon of pre-bifurcation increase of fluctuation intensity, and experiences saturation in the vicinity of the bifurcation point [12]. In fact, one can speak about pre-bifurcation rise and subsequent saturation of correlation time.

Notice, that the phenomenon of rise and saturation of the correlation time of the fluctuation processes can be also good “noisy precursor” of bifurcation in nonlinear systems.

The main goal of this paper is to attract the attention of experimenters and engineers to these phenomena and thereby to stimulate searches for specific mechanisms that might be responsible for noise amplification and rise of the fluctuation correlation time in nonlinear systems.

The phenomenon of pre-bifurcation noise amplification is briefly outlined by the example of period doubling bifurcation in a nonlinear map (Section 2) with a special emphasis on the transition from linear regime to the regime of nonlinear saturation of amplification.

The linear theory of correlation time rise and non-linear estimates, evidencing correlation time saturation are presented in Section 3 by the example of a period-doubling bifurcation in a non-linear map.

A new method for signal processing which rests on the theoretical results of Sections 2 and 3 is proposed in Section 4. This method can be utilized to design and produce devices

which cut off air-engines and rockets in case of detecting in them of the abnormal noise.

We summarize the main results of the work in Section 5.

## 2 Pre-bifurcation Noise Amplification at the Threshold of Period Doubling Bifurcation

Let us consider nonlinear noisy map

$$x_{n+1} = F(x_n, \mu) + f_n, \quad (2.1)$$

where  $\mu$  is a control parameter, and  $f_n$  is an external noisy process, which has the zeroth mean value,  $\langle f \rangle = 0$ , and is considered to be  $\delta$ -correlated:  $\langle f_n f_m \rangle = \sigma_f^2 \delta_{mn}$ . If  $\bar{x}$  is a stable point of a map (2.1) in the absence of noise,

$$\bar{x} = F(\bar{x}, \mu), \quad (2.2)$$

then fluctuations  $\xi_n = x_n - \bar{x}$  near steady state obey the equation

$$\xi_{n+1} = \gamma \xi_n + \varepsilon \xi_n^2 + \dots + f_n, \quad (2.3)$$

where

$$\gamma = \frac{dF(\bar{x})}{dx}, \quad \varepsilon = \frac{1}{2} \frac{d^2 F(\bar{x})}{dx^2}.$$

Bifurcation of period doubling arises when modulus of multiplier  $\gamma$  exceeds a unit  $|\gamma| < 1$ .

In frame of linear theory, Eq. (2.3) takes the form

$$\xi_{n+1} = \gamma \xi_n + f_n.$$

The mean squares of the left and right hand parts are equaled to

$$\langle \xi_{n+1}^2 \rangle = \gamma^2 \langle \xi_n^2 \rangle + 2\gamma \langle \xi_n, f_n \rangle + \langle f_n^2 \rangle \quad (2.4)$$

The term  $\langle \xi_n, f_n \rangle$  is to be zero, because  $\xi_n$  and  $f_n$  are statistically independent. Besides, at sufficiently large  $n$ , one has  $\langle \xi_{n+1}^2 \rangle = \langle \xi_n^2 \rangle \equiv \sigma_\xi^2$ , so Eq. (2.4) gives

$$(\sigma_\xi^2)_{lin} = \frac{\sigma_f^2}{1 - \gamma^2} \cong \frac{\sigma_f^2}{2\alpha}. \quad (2.5)$$

Here parameter  $\alpha = 1 - |\gamma|$  characterizes closeness of the nonlinear system to bifurcation threshold  $|\gamma| = 1$ . In frame of linear theory, fluctuation variance  $\sigma_\xi^2$  tends to the infinity at  $\alpha \rightarrow 0$ , which corresponds to the result by Wiesenfeld [1]. Linear theory is valid until the quadrate term in Eq.(2.3) becomes comparable (in statistical sense) with the linear one:

$$(1 - \gamma^2)\sigma_\xi^2 \approx \varepsilon^2 \langle \xi^4 \rangle. \quad (2.6)$$

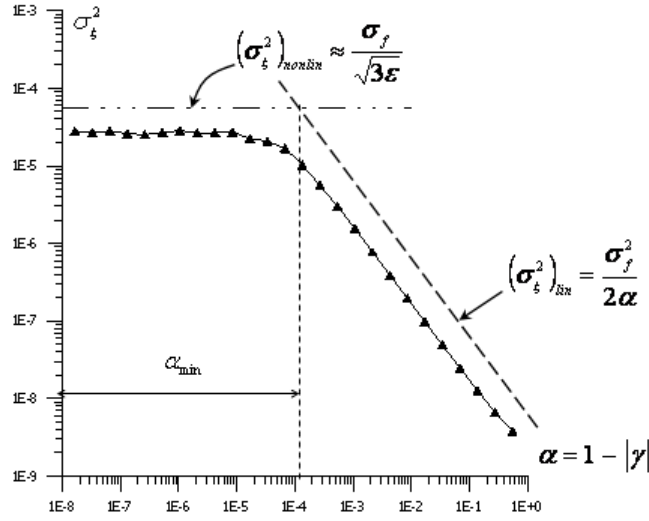


Figure 1: Pre-bifurcation noise amplification for period doubling bifurcation in quadratic map  $F(x) = \mu - x^2$  ( $\sigma_f^2=10^{-8}$ ,  $\varepsilon=1$ ).

According to Eq. (2.3), process  $\xi_n$  is formed under influence of many random values  $f_n$ , so that properties of  $\xi_n$  approach those of Gaussian value. Therefore, one can accept  $\langle \xi_n^4 \rangle \approx 3(\sigma_\xi^2)^2$ , as for Gaussian values, and then Eq. (2.6) takes the form

$$(1 - \gamma^2)\sigma_\xi^2 \cong \varepsilon^2 3(\sigma_\xi^2)^2,$$

or, taking into account (2.5),

$$2\alpha \cong 3\varepsilon^2 \frac{\sigma_f^2}{2\alpha}. \quad (2.7)$$

One can estimate from Eq. (2.7) the minimal value  $\alpha_{min}$  limiting area of linear theory validity:

$$2\alpha_{min} = \sqrt{3}|\varepsilon|\sigma_f. \quad (2.8)$$

Substitution (2.8) into Eq. (2.5) allows to estimate fluctuation variance in a nonlinear regime:

$$(\sigma_\xi^2)_{nonlin} \approx \frac{\sigma_f^2}{\sqrt{3}|\varepsilon|}. \quad (2.9)$$

Dependence of  $\sigma_\xi^2$  on  $\alpha$  is presented in Fig. 1: at  $\alpha > \alpha_{min}$ , the results of the linear theory are valid (dashed line), whereas at  $\alpha < \alpha_{min}$ , nonlinear saturation (2.9) (horizontal pointed line) enters into the play.

These qualitative estimates are in good agreement with the results of numerical simulation, performed in [5] for quadratic map  $F(x, \mu) = \mu - x^2$ . The results are shown at Fig. 1. by a continuous line. Estimate (2.9) turns to be only 30-40% less as compared to numerical data.

Figure 2 illustrates the dependence of the maximal variance of fluctuations  $(\sigma_\xi^2)_{max}$  near the bifurcation threshold on the mean-square fluctuation action  $\sigma_f$ . Points on this plot

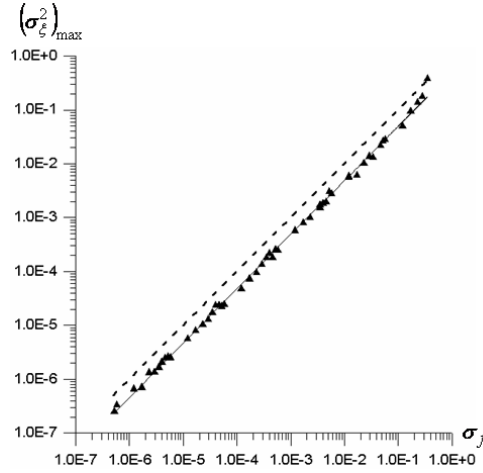


Figure 2: Dependence of the maximal fluctuation variance  $(\sigma_\xi^2)_{\max}$  on the standard deviation of the noise action  $\sigma_f$ . The dashed line corresponds to estimate Eq.(2.9).

can be approximated by the dependence

$$(\sigma_\xi^2)_{\text{nonlin}} \approx 0.5 \cdot \sigma_f,$$

which is in good agreement with the theoretical estimate Eq.(2.9) (the latter is shown in the dashed line in Fig.2). Numerical computations show that Eq.(2.9) can be used not only for the uniform distribution of the fluctuation force  $f_n$ , but also for the normal distribution.

### 3 Phenomenon of Rise and Saturation of the Correlation Time of the Fluctuation Processes

In the framework of the linear theory, (2.3) takes the form:

$$\xi_{n+1} = \gamma \xi_n + f_n. \quad (3.1)$$

Sequential iterations produce:

$$\begin{aligned} \xi_1 &= \gamma \xi_0 + f_0, \\ \xi_2 &= \gamma \xi_1 + f_1 = \gamma^2 \xi_0 + \gamma f_0 + f_1, \\ \xi_3 &= \gamma \xi_2 + f_2 = \gamma^3 \xi_0 + \gamma^2 f_0 + \gamma f_1 + f_2, \\ &\dots \end{aligned}$$

$$\xi_n = \gamma^n \xi_0 + \gamma^{n-1} f_0 + \gamma^{n-2} f_1 + \dots + f_{n-1}. \quad (3.2)$$

Summand with initial value  $\xi_0$  decreases with  $n$  as

$$\gamma^n = (-1)^n (1 - \alpha)^n \approx (-1)^n e^{-n\alpha}. \quad (3.3)$$

Starting with a certain iteration number  $n \gg \frac{1}{\alpha}$ , this term can be neglected. As a result, fluctuations  $\xi$  can be expressed through random forces  $f$  as:

$$\xi_n = \gamma^{n-1} f_0 + \gamma^{n-2} f_1 + \dots + f_{n-1}. \quad (3.4)$$

Correlation function  $C_\xi(k) = \langle \xi_n \xi_{n+k} \rangle$  in linear regime can be found by multiplying  $\xi_n$  from (3.4) by

$$\xi_{n+k} = \gamma^{n+k-1} f_0 + \gamma^{n+k-2} f_1 + \dots + f_{n-1}. \quad (3.5)$$

Taking into account  $\delta$ - correlation of  $f_n$  we get

$$C_\xi(k) = \langle \xi_n \xi_{n+k} \rangle = \sigma_f^2 \gamma^k \frac{1 - \gamma^{2n}}{1 - \gamma^2}, \quad (3.6)$$

or, at large  $n$ , when  $|\gamma|^{2n} \ll 1$ ,

$$C_\xi(k) \approx \sigma_f^2 \gamma^k \frac{1}{1 - \gamma^2} = \sigma_\xi^2 \gamma^k = \sigma_\xi^2 (-1)^k e^{-k\alpha}. \quad (3.7)$$

Here

$$\sigma_\xi^2 = \sigma_f^2 \frac{1}{1 - \gamma^2} \approx \sigma_f^2 \frac{1}{2\alpha} \quad (3.8)$$

is fluctuation  $\xi_n$  variance in frame of linear theory (2.5).

It follows from (3.8), that as far as the linear theory is applicable, the correlation function (3.7) of fluctuations  $\xi_n$  exponentially depends on discrete time  $k$  and the correlation time  $k_c$  increases inverse proportionally to  $\alpha$ :

$$k_c = 1/\alpha, \quad (3.9)$$

what is in agreement with the relation  $\tau_c |\lambda| \approx 1$  [11]. Factor  $(-1)^k$  in Eq. (3.7) describes alternative nature of processes near equilibrium state  $\bar{x}$ .

As shown in Section 2, the linear theory is valid provided

$$\alpha > \alpha_{\min} \propto \sigma_f, \quad (3.10)$$

when the contribution of the quadratic term in (2.3) is negligible compared to that of the linear term. On the contrary, under the opposite condition

$$\alpha < \alpha_{\min} \propto \sigma_f, \quad (3.11)$$

the linear term can be neglected in (2.3). In this case, according to (2.9), fluctuation intensity is estimated as

$$\sigma_\xi^2 = \frac{\sigma_f}{|\varepsilon| \sqrt{3}}. \quad (3.12)$$

It is helpful to merge estimates (3.8) and (3.12):

$$\sigma_\xi^2 = \frac{\sigma_f^2}{2(\alpha + \sigma_f |\varepsilon| \sqrt{3}/2)} \equiv \frac{\sigma_f^2}{2\alpha_{eff}}, \quad (3.13)$$

where

$$\alpha_{eff} = \alpha + \sigma_f |\varepsilon| \sqrt{3}/2. \quad (3.14)$$

Let us estimate correlation time in the immediate vicinity of the bifurcation threshold, substituting in Eq. (3.9) parameter  $\alpha$  with the effective value  $\alpha_{eff}$ :

$$k_c = 1/\alpha_{eff} = \frac{1}{\alpha + \sigma_f |\varepsilon| \sqrt{3}/2}. \quad (3.15)$$

At  $\alpha > \alpha_{\min}$ , when the linear theory is applicable, this formula turns into (3.9), while at  $\alpha < \alpha_{\min}$ , correlation time reaches saturation:

$$k_{c\max} \approx \frac{1}{\alpha_{\min}} \approx \frac{1}{\sigma_f |\varepsilon| \sqrt{3}/2}. \quad (3.16)$$

Estimate (3.15), being in essence an heuristic one, can be substantiated by way of considering the twofold iteration:

$$\begin{aligned} \xi_{n+2} &= \gamma \xi_{n+1} + \varepsilon \xi_{n+1}^2 + f_{n+1} \\ &= \gamma(\gamma \xi_n + \varepsilon \xi_n^2 + f_n) + \varepsilon(\gamma \xi_n + \varepsilon \xi_n^2 + f_n)^2 + f_{n+1} \\ &\approx \xi_n(1 - 2\alpha + \alpha^2 - 2\varepsilon^2 \xi_n^2 + 2\alpha \xi_n^2 \varepsilon^2) + \varepsilon \xi_n^2(\alpha^2 - \alpha) + f_n \alpha + f_{n+1} + \varepsilon f_n^2 \dots \\ &\approx \xi_n(1 - 2\alpha + \alpha^2 - 2\varepsilon^2 \xi_n^2 + 2\alpha \xi_n^2 \varepsilon^2) \\ &\approx \xi_n(1 - \alpha - \varepsilon^2 \xi_n^2)^2. \end{aligned} \quad (3.17)$$

We have neglected here the terms of the third and forth power and substituted  $\gamma$  with  $-(1 - \alpha)$ . Taking into account Eqs. (3.12) and (3.14), one can readily show, that the averaged value  $(1 - \alpha - \varepsilon^2 \xi_n^2)^2$  is close to the value  $\gamma_{eff}^2 = (1 - \alpha_{eff})^2 = (1 - \alpha - \sigma_f |\varepsilon| \sqrt{3}/2)^2$ ,

so that

$$\xi_{n+2} \approx \xi_n \gamma_{eff}^2.$$

In the course of multiple iterations of map (3.17), correlation function assumes a form similar to (3.7), but with  $\alpha_{eff} = \alpha + \sigma_f |\varepsilon| \sqrt{3}/2$  instead of  $\alpha$ :

$$C_\xi(k) = \langle \xi_n \xi_{n+k} \rangle = \sigma_\xi^2 (-1)^k e^{-k\alpha_{eff}} = \sigma_\xi^2 (-1)^k e^{-k(\alpha + \sigma_f |\varepsilon| \sqrt{3}/2)}. \quad (3.18)$$

It is this expression which justifies the heuristic formula (3.15).

As was pointed out in the paper [12], the phenomenon of correlation time rise, considered for period-doubling bifurcations, is of more general nature. It occurs in some other bifurcations, such as pitchfork bifurcation, where non-linear saturation of pre-bifurcation noise amplification also takes place [7]. The phenomenon of correlation time rise can be anticipated also near the generation threshold in various self-oscillating mechanical, electrical, radio physical and optical systems.

Theoretical estimates of pre-bifurcation correlation time rise can be illustrated by numerical modelling by the example of quadratic map

$$x_{n+1} = \mu - x_n^2 + f_n. \quad (3.19)$$

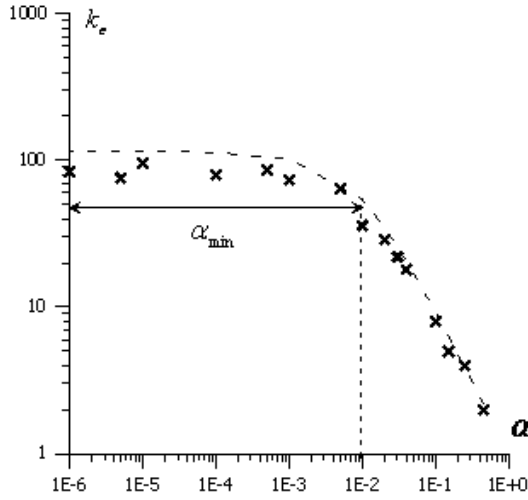


Figure 3: Rise and saturation of correlation time  $k_c$  under  $\alpha \rightarrow 0$  for the noisy quadratic map  $x_{n+1} = \mu - x_n^2 + f_n(|\varepsilon| = 1, \sigma_f = 9.85 \cdot 10^{-3}$ . Dashed line corresponds to the formula (3.14).

During simulation, random value generator produced normally distributed, uncorrelated values of  $f_n$ . Initial values  $x_0$  were chosen equal to the stable point  $\bar{x}$  of map (3.19):  $x_0 = \bar{x} = -\frac{1}{2} + \sqrt{\frac{1}{4} + \mu}$ , which corresponds to  $\xi_0 = 0$ .

In order to obtain 10%-accurate estimates for correlation time  $k_c$ , sampling duration of  $N$  must be chosen to exceed  $N_c = 1/k_c$  not less than 100 times. So, for simulation purposes we took rather large values of noise intensity  $\sigma_f > 10^{-2} - 10^{-1}$ , for which  $N_c \sim 10 - 100$ , and  $N = 10^3 - 10^4$ .

Simulation results are shown in Fig. 3. Crosses denote correlation time values  $k_c$  estimated from condition that normalized correlation function of process  $R_\xi(k) = \frac{C_\xi(k)}{\sigma_\xi^2} = \frac{1}{\sigma_\xi^2} \langle \xi_n \xi_{n+k} \rangle$  falls down to level 0.5. Analytical estimates of correlation time are shown by a dashed line in the figure. As is seen from Fig.3, estimates (3.9), (3.14) and (3.15) are in good agreement with numerical simulation results for both the linear theory domain and saturation region.

#### 4 A New Method for Signal Processing: Detecting of Bifurcations in Time Series of Nonlinear Systems

A noticeable increase in the variation of fluctuations  $\sigma_\xi^2$  and the correlation time of the fluctuation process  $k_c$  near the bifurcation threshold can be used as the basis of a method for detecting bifurcations in time-series of nonlinear systems. Suggested approach is based on three steps.

1. It is the analysis of segments of the time-series with low and high levels of the noise

and the estimate of coefficient of fluctuation amplification

$$K = \frac{\sigma_\xi^2}{\sigma_f^2}.$$

According to Eq.(2.9), the maximum fluctuation “amplification factor”  $K_{max} = (\sigma_\xi^2)_{max}/\sigma_f^2$  is

$$K_{max} = \frac{(\sigma_\xi^2)_{max}}{\sigma_f^2} \sim \frac{1}{\sigma_f}.$$

For example, for variance of noise  $\sigma_f^2=10^{-8}$ , the amplification factor  $K_{max}$  can become as large as  $K \approx 10^4$ .

2. There are estimates of the correlation time of the fluctuation process in segments with high level of the noise. According to Eq.(3.16), the correlation time of the fluctuation process  $k_c$  near bifurcation threshold is inversely proportional to the standard deviation of noise  $\sigma_f$

$$k_{c\max} \approx \frac{1}{\alpha_{\min}} \approx \frac{1}{\sigma_f |\varepsilon| \sqrt{3}/2}.$$

3. The obtained information about two characteristics of the noise amplification in the time series of nonlinear systems might be an effective tool for revealing bifurcations in nonlinear systems. If the value of coefficient of noise amplification is comparable with the value of the correlation time of the fluctuation process, then it possible to draw a conclusion about qualitatively changes of the system’s behavior or about the impending bifurcation of the system.

This approach looks to be prospective for detecting of bifurcations in time-series of different nonlinear systems, for example, for a revealing of dangerous level of noise in movements of bridges.

Moreover obtained theoretical estimates can be used for the making of devices which cut off air-engines and rockets in case of detection in them of the anomaly noise.

## 5 Conclusion

In this paper a new method for signal processing is suggested. This method is of practical interest for detecting of bifurcations in time-series of different nonlinear systems. The method under consideration uses the results of the theoretical estimates of two noise-dependent nonlinear phenomena at the bifurcation threshold. Suggested approach is based on three steps.

i) Conduct analysis of segments of the time-series with low and high levels of the noise and the estimate of coefficient of noise amplification.

ii) Estimate of the correlation time of the fluctuation process in segments with high levels of the noise.

iii) Compare the obtained result from i) with the result from ii). If the value of coefficient of noise amplification is close to the value of the correlation time of the fluctuation process, then it possible to draw a conclusion about qualitatively changes of the system’s behavior or about the impending bifurcation of system.



The proposed a new method might be helpful for detecting of abnormal movements of bridges and dangerous level of noise in air-engines and rockets.

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*Chapter 18*

## COORDINATE ADJUSTMENT BASED ON RANGE AND ANGLE MEASUREMENTS

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### Abstract

This paper demonstrates the procedure and extends the classes of sensors to those that provide inner product measurements.

**Key Words:** Nonlinear Least Square, Weighted Least Square, Distance Observation, Angle Observation, Inner product measurement, pseudorange and GPS.

**AMS Subject Classification:** Primary 43A60.

## 1 Introduction

Sensors using bearing angle and distance information can be combined to find location. Within a system a method to find location is developed based on the given sensors. Sensor packages are evolving and it is worthwhile to have a general approach to fuse bearing angle and distance measurements. This report demonstrates the procedure and extends the classes of sensors to those that provide inner product measurements. The end product of many systems is a location or set of coordinates. Sensor information is combined with known references to determine locations. Typically sensors give range, bearing, or angle measurements to an unknown location. Although these measurements lead to nonlinear relationships between the coordinates and the measurements, least squares methods can be used to minimize the error associated with the set of measurements. This process is referred to as coordinate adjustment as the unknown estimate of location is adjusted from iteration to iteration. The sequel describes the measurements and gives examples of coordinate adjustment. The adjustments are based on the partial derivatives of the measurement with

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respect to each coordinate of interest. Some of the examples will include GPS measurements. Coordinate adjustment is a major technique in surveying see Wolf and Ghilani and geodesy see Strang and Borre. These ideas can be used to enhance navigation.

## 2 Nonlinear Least Squares

First a brief review of least squares. Start with the over determined equation

$$FX = Y,$$

where  $X$  is the vector of unknown coordinates,  $Y$  represents measurements, and  $F$  is a matrix model of a known linear relationship. Then project both sides onto the column space of the matrix  $F$ . Multiply each side by the transpose of  $F$  and then assuming  $X$  is multiplied by a square matrix of full rank multiply each side by the inverse the result will be

$$X = (F^t F)^{-1} F^t Y.$$

This result is discussed in many textbooks. This result will be used as the core of an iterative procedure for nonlinear least squares. Both  $F$  and  $Y$  are known values while  $X$  represents the unknown parameters.

Suppose the functional relationship,  $f$ , between the measurements and the coordinates ( $X$ ) is known. This function is typically nonlinear. To start, the value of  $X$  needs to be guessed or approximated, denote this value as  $X_0$ . The gradient matrix for the measurements with respect to the coordinates will be denoted by  $F$ . If  $X_0$  is reasonably close to  $X$  then the following linearized approximation is valid.

$$f(X) = f(X_0) + F_{X_0}(X - X_0)$$

The left side is the measurement vector while the first term on the right is the evaluation of the functional relationship at the point  $X_0$ . Moving this term to the left side allows the interpretation of the left side as the deviation of the measurements from the approximated coordinates. The gradient evaluated at the point is multiplied by the unknown deviation from the approximation of the location. The previous equation can be written in the form  $Y = FX$  as follows.

$$f(X) - f(X_0) = F_{X_0}(X - X_0)$$

Linear least squares can be used to solve for the location deviation; after the solution is added to the current approximation (and becomes the new current approximation) the process can be repeated until the left side (measurement deviation) gets close to zero. If this is understood what remains is to discuss realizations of  $f$  and  $F$  for different measurements in terms of coordinates. Several rudimentary examples will be given to demonstrate how each type of measurement is used; within the  $F$  matrix different types of measurements can be used. Each measurement will define a row of the matrix.

### 3 Distance Observations

Distance is a scalar mapping of two points and is interpreted as a measure of their proximity. The formula most often used is

$$S_{ij} = ((x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2)^{.5},$$

where  $S_{ij}$  is the distance between the  $i$ th and  $j$ th points. This is referred to as the  $L^2$  or sometimes the  $H^2$  norm. Sometimes issues associated with the measurement process lead to modification of this formula; several of these situations will be discussed.

First consider a two dimensional case where the measurements are two distances. Notice that each measurement contains four coordinates, also each measurement can be expressed as a functional relationship using the four coordinates as unknowns, further, the number of parameters to estimate must be less than or equal to the number of equations; thus, some of the coordinates need to be known. In many situations only one set of coordinates is unknown, and the measurements will be the distance from a known location to an unknown location. Imagine an unknown point whose distance to two known locations is measured. Assume the two known locations are  $(0, 0)$  and  $(10, 10)$ ; the distance measurements are 10 and 6 respectively; further let the first guess of the unknown point,  $X_0$ , be  $(4, 8)$ . First the matrix  $F_{X_0}$  needs to be calculated. Since each measurement is a distance the rows of this matrix will be the partial of the distance with respect to the coordinate corrections. The elements will be of the form

$$\frac{\partial S_{ij}}{\partial x_i} = \frac{x_i - x_j}{((x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2)^{.5}} = \frac{x_i - x_j}{S_{ij}}.$$

Notice that the partial can be interpreted as the direction cosines in three dimensions or as the sine and cosine of the bearing angle in two dimensions. Using  $(0, 0)$  as point  $i$  and  $(4, 8)$  as  $j$ ;  $s = 8.9443$  and the first row of the matrix is just the coordinate difference divided by the distance. The partial matrix for the first step is

$$F_{X_0} = \begin{pmatrix} -0.4472 & -0.8944 \\ 0.9487 & 0.3162 \end{pmatrix}$$

The first value to calculate is the difference between the observations and the distances based on the known locations and the first guess of the unknown coordinates.

$$f(X) - f(X_0) = \begin{pmatrix} 10 \\ 6 \end{pmatrix} - \begin{pmatrix} 8.9443 \\ 6.3246 \end{pmatrix} = \begin{pmatrix} 1.0557 \\ -0.3246 \end{pmatrix}.$$

Using least squares results in an estimate of  $\begin{pmatrix} .0616 \\ -1.2111 \end{pmatrix}$  as the coordinate correction. If this process is repeated the following sequence is generated

$$\begin{pmatrix} 4 \\ 8 \end{pmatrix} \Rightarrow \begin{pmatrix} 3.9384 \\ 9.2111 \end{pmatrix} \Rightarrow \begin{pmatrix} 4.0614 \\ 9.1392 \end{pmatrix} \Rightarrow \begin{pmatrix} 4.0623 \\ 9.1377 \end{pmatrix} \Rightarrow \begin{pmatrix} 4.0623 \\ 9.1377 \end{pmatrix}.$$

The absence of change to four decimal places can be considered a stopping criterion. Using the same procedure with a different starting point yields the following sequence.

$$\begin{pmatrix} 4 \\ -1 \end{pmatrix} \Rightarrow \begin{pmatrix} 9.5308 \\ 3.6426 \end{pmatrix} \Rightarrow \begin{pmatrix} 9.1592 \\ 4.0457 \end{pmatrix} \Rightarrow \begin{pmatrix} 9.1378 \\ 4.0622 \end{pmatrix} \Rightarrow \begin{pmatrix} 9.1377 \\ 4.0623 \end{pmatrix}.$$

For this example each iteration increased the accuracy by a significant digit. The selection of the starting point will determine which local minimum the sequence approaches. This problem could be restated as the intersection of two circles; thus two solutions are expected and can be found analytically. An additional measurement from a different known location would generate a third equation, or row of the matrix, and resolve the ambiguity. This example demonstrates the steps taken for coordinate estimation problems. The observation equations need to be defined then the gradient matrix can be found, and finally a least squares correction is generated. These steps are iterated until the change in the coordinates is deemed insignificant. Also note (see previous equation) that the partials could be expressed as the sine or cosine with respect to the angle measured from one of the axis. In three dimensions the partials could be interpreted as direction cosines. The basic distance observation equation can be modified for certain types of errors. These modifications typically require the addition of a parameter that needs to be estimated. By choosing the proper modification the observation is modeled more realistically and the estimation of the coordinates improves. In some situations, scale errors are introduced. This can be modeled by a simple modification to the distance formula.

$$S_{ij} = (1 - u)((x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2)^{.5}$$

In this case  $u$  is the scale correction factor and needs to be estimated. The gradient matrix will contain an additional column for the derivative with respect to the scale correction factor.

Another modification of the distance formula is referred to as pseudorange or pseudodistance. In this case there is assumed to be a common additive error for each observation; in the literature this error is denoted by  $Z$ . In GPS systems the error due to the receiver clock error can be modelled as a pseudorange. The observation or measurement formula for distance that includes a pseudorange follows.

$$S_{ij} = ((x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2)^{.5} + Z$$

The unknowns need to be augmented to include  $Z$ , this will add a column to the  $F$  matrix. Other modifications of distance are possible and can be developed to incorporate specific types of errors.

## 4 Angle Observations

These equations must establish the relationship between the observed angle and the desired coordinates. It takes three points to define an angle. Some observations assume that the origin is the central point and take one of the axes as a reference for the measure. Azimuth measurements determine the angle to a meridian. This meridian is typically associated with the  $X$ -axis of the coordinate system. Typical examples are the geographic or magnetic meridians. An azimuth measurement is found by taking the arctangent of the slope between two points. Measuring from the  $X$ -axis the formula for the azimuth angle is

$$A_{ij} = \arctan\left(\frac{y_i - y_j}{x_i - x_j}\right)$$

The formula for the partial of this formula is of the form

$$\frac{\partial A_{ij}}{\partial x_i} = \frac{y_i - y_j}{(x_i - x_j)^2 + (y_i - y_j)^2} = \frac{y_i - y_j}{S_{ij}^2} = \frac{\sin(A_{ij})}{S_{ij}}.$$

Comparing this formula to the formula for the partial of the distance, it can be seen that the distance partial could be expressed in terms of the angle and if the whole observation equation was divided by the distance the resulting partials would be similar.

As an example consider a two dimensional space with a known location and an azimuth measurement from two locations. From the point  $(0, 0)$  the azimuth to the unknown location is  $-\pi/8$ . From the point  $(0, -10)$  the azimuth is  $\pi/8$ . Since these angles are equal in magnitude the desired point will be on the perpendicular bisector of the connecting line segment; further, for a 45 degree angle the range and height offsets will be the same. The desired point is  $(5, -5)$ ; however, the goal is to find this point using linearized least squares starting from a guess. Starting with the point  $(6, -7)$  as the initial guess the calculations for the gradient matrix and the deviation of the measurements from the approximated coordinates are as follows.

$$F_{X_0} = \begin{pmatrix} .0824 & .0706 \\ -.0667 & .1333 \end{pmatrix}$$

$$f(X) - f(X_0) = \begin{pmatrix} -\pi/8 \\ \pi/8 \end{pmatrix} - \begin{pmatrix} -.8622 \\ .4636 \end{pmatrix} = \begin{pmatrix} .0768 \\ .3218 \end{pmatrix}.$$

The least squares solution is  $(-.7953, 2.0155)$ . The sequence of corrections is

$$\begin{pmatrix} 6 \\ -7 \end{pmatrix} \Rightarrow \begin{pmatrix} 5.2047 \\ -4.9845 \end{pmatrix} \Rightarrow \begin{pmatrix} 4.9958 \\ -5.0000 \end{pmatrix} \Rightarrow \begin{pmatrix} 5.0000 \\ -5.0000 \end{pmatrix}.$$

Sometimes there is unknown heading angle from which the bearings are measured. In this case the previous azimuth observation can be modified to account for this unknown. The measurement,  $R$ , can be considered as the azimuth diminished by an offset angle,  $Z$ , or

$$A_{ij} - Z_{ij} = R_{ij}.$$

The gradient of this equation is the same as the previous but has an extra element of  $-1$  to account for the offset angle as this angle is the same in each measurement.

## 5 Inner Product Measurements

Next consider the situation when the angle between two line segments is measured. Assume the segments join at a common endpoint. Let the points forming the angle be denoted by  $i$ ,  $j$ , and  $k$ . The inner product can be used to find the cosine of the desired angle. Let  $dx_{ij} = x_i - x_j$  be the  $x$  axis distance between the  $i$ th and  $j$ th locations. The angle is represented as

$$\cos(A) = \frac{dx_{ij}dx_{kj} + dy_{ij}dy_{kj}}{(dx_{ij}^2 + dy_{ij}^2)^{.5}(dx_{kj}^2 + dy_{kj}^2)^{.5}}$$

After taking the arc cosine of each side the partial of the angle can be found with respect to the individual coordinates. Rather than simply stating the partial, several steps used to find it will be shown.

$$N = dx_{ij}dx_{kj} + dy_{ij}dy_{kj}$$

$$D = (dx_{ij}^2 + dy_{ij}^2)^{.5}(dx_{kj}^2 + dy_{kj}^2)^{.5}$$

The procedure for calculating the partial is defined by the following equality.

$$\frac{\partial A}{\partial x_i} = \frac{-1}{\left(1 - \left(\frac{N}{D}\right)^2\right)^{.5}} \frac{N'D - D'N}{D^2} \frac{\partial dx_{ij}}{\partial x_i}$$

$$\frac{\partial N}{\partial x_i} = dx_{kj}$$

$$\frac{\partial D}{\partial x_i} = dx_{ij}(dx_{ij}^2 + dy_{ij}^2)^{-.5}(dx_{kj}^2 + dy_{kj}^2)^{.5} = dx_{ij}(dx_{ij}^2 + dy_{ij}^2)^{-1}D = QD$$

where Q is appropriately defined. Using these expressions the original partial can be written as

$$\frac{\partial A}{\partial x_i} = \frac{-1}{(D^2 - N^2)^{.5}} \frac{N'D - QD}{D} (1) = \frac{Q - N'}{(D^2 - N^2)^{.5}}$$

The denominator can be simplified; the result is

$$(D^2 - N^2)^{.5} = dx_{ij}dy_{kj} - dx_{kj}dy_{ij}$$

Finally in terms of the coordinate differences

$$\frac{\partial A}{\partial x_i} = \frac{dx_{ij}(dx_{ij}^2 + dy_{ij}^2)^{-1} - dx_{kj}}{dx_{ij}dy_{kj} - dx_{kj}dy_{ij}}$$

Using the same steps partials with respect to the other coordinates can be found.

As an example consider the following two dimensional situation. An observer located at an unknown location measures two angles formed with known distant locations. Suppose the observer is located at (10, 1) and angle 1 is formed with points (1, 10) and (-1,-8); angle 2 is formed with points (20, 5) and (25, -10). Further let (9, 3) be the original guess of the location. Iterative estimation yields the following sequence.

$$\begin{pmatrix} 9 \\ 3 \end{pmatrix} \Rightarrow \begin{pmatrix} 10.1624 \\ 1.0750 \end{pmatrix} \Rightarrow \begin{pmatrix} 9.9990 \\ .9979 \end{pmatrix} \Rightarrow \begin{pmatrix} 10.0000 \\ 1.0000 \end{pmatrix}$$

For this example four decimal places of agreement with the true value were obtain after three iterations.

Three dimensional distance formula has already been shown. For azimuth or bearing angles the measurements are restricted to a two dimensional plane. The inner product measure of angle can be augmented to include any number of dimensions, for geolocation three dimensions are needed. The previous procedure is followed with the following modifications.

$$N = dx_{ij}dx_{kj} + dy_{ij}dy_{kj} + dz_{ij}dz_{kj}$$

$$D = (dx_{ij}^2 + dy_{ij}^2 + dz_{ij}^2)^{.5} (dx_{kj}^2 + dy_{kj}^2 + dz_{kj}^2)^{.5}$$

The following relationships hold in the three dimensional case

$$(D^2 - N^2)^{.5} = ((dx_{ij}dy_{kj} - dx_{kj}dy_{ij})^2 + (dx_{ij}dz_{kj} - dx_{kj}dz_{ij})^2 + (dy_{ij}dz_{kj} - dy_{kj}dz_{ij})^2)^{.5}$$

$$\frac{\partial N}{\partial x_i} = dx_{kj}$$

$$\frac{\partial D}{\partial x_i} = dx_{ij}(dx_{ij}^2 + dy_{ij}^2 + dz_{ij}^2)^{-.5} (dx_{kj}^2 + dy_{kj}^2 + dz_{kj}^2)^{.5} = dx_{ij}(dx_{ij}^2 + dy_{ij}^2 + dz_{ij}^2)^{-1} D = QD$$

and finally the form of each derivative is

$$\frac{\partial A}{\partial x_i} = \frac{dx_{ij}(dx_{ij}^2 + dy_{ij}^2 + dz_{ij}^2)^{-1} - dx_{kj}}{((dx_{ij}dy_{kj} - dx_{kj}dy_{ij})^2 + (dx_{ij}dz_{kj} - dx_{kj}dz_{ij})^2 + (dy_{ij}dz_{kj} - dy_{kj}dz_{ij})^2)^{.5}}$$

The numerator of this partial only changes slightly going from two to three dimensions, the denominator is more complex; the extension to a higher dimensional space is straightforward. As an example consider using two angle and two distance measurements to locate a point in a three dimensional world. Let four known locations be  $P1(-2, 10, 9)$ ,  $P2(1, -8, 2)$ ,  $P3(25, -5, -4)$ , and  $P4(19, 12, 7)$ . Let the location of the unknown point be  $P5(10, 2, 3)$ . Using the measurements of the angles  $\angle P1, P5, P2$  and  $\angle P3, P5, P4$ , and the distances  $(P1, P5)$  and  $(P3, P5)$  the rows of the F-matrix can be calculated and the iterative process can be initiated. Following the previous procedure, the following sequence is generated using  $(0, 0, 0)$  as the starting point.

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} 7.9376 \\ 2.0111 \\ -3.5164 \end{pmatrix} \Rightarrow \begin{pmatrix} 9.2479 \\ 3.7065 \\ -0.3460 \end{pmatrix} \Rightarrow \begin{pmatrix} 9.5271 \\ 2.7330 \\ 1.1481 \end{pmatrix} \Rightarrow \begin{pmatrix} 9.7295 \\ 2.4344 \\ 1.9366 \end{pmatrix} \Rightarrow$$

$$\begin{pmatrix} 9.8236 \\ 2.2739 \\ 2.3172 \end{pmatrix} \Rightarrow \begin{pmatrix} 9.8680 \\ 2.1914 \\ 2.5045 \end{pmatrix} \Rightarrow \begin{pmatrix} 9.8937 \\ 2.1369 \\ 2.6207 \end{pmatrix} \Rightarrow \begin{pmatrix} 9.9254 \\ 2.0726 \\ 2.7607 \end{pmatrix} \Rightarrow \begin{pmatrix} 9.9985 \\ 1.1907 \\ 3.0303 \end{pmatrix} \Rightarrow$$

$$\begin{pmatrix} 10.0004 \\ 1.9991 \\ 3.0019 \end{pmatrix} \Rightarrow \begin{pmatrix} 10.0000 \\ 2.0000 \\ 3.0000 \end{pmatrix}$$

This sequence takes eleven iterations to converge. This is a significant increase over the prior two dimensional cases. If more measurements were used convergence should be faster.

## 6 GPS Example

The procedures discussed have been done using Cartesian coordinates. Geographical coordinates (latitude, longitude, height) are used to represent locations on the earth. Typically geographical coordinates are converted to a local Cartesian system prior to coordinate adjustment. The basis of the local coordinate system is formed by starting with the vector



normal to the surface and then adding northing and easting vectors. The vector normal to the surface (or reference ellipsoid) and the northing vector can be found through trigonometric functions of the longitude and latitude coordinates. The easting vector then can be found as the cross product of the normal and northing vectors. GPS uses an earth centered earth fixed (ECEF) coordinate system. The next example will simulate the processing of GPS pseudorange data.

Assume a receiver is located at  $(-2400000, -4700000, 3600000)$  and satellites are located as follows

$$\begin{pmatrix} .7723 & -2.2075 & 1.2556 \\ -2.6089 & -.6510 & .0049 \\ -.5731 & -2.5709 & .1674 \\ -.2757 & -1.5904 & 2.1375 \end{pmatrix} * 10^7.$$

Using the distance formula augmented with pseudorange the gradient matrix can be found. Using the following measurements,  $2.2263e7$ ,  $2.4272e7$ ,  $2.1609e7$ ,  $2.1264e7$ , the iterative estimation process yields the following sequence of estimates of the receiver location and the pseudorange

$$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{pmatrix} -2.9351 \\ -5.6140 \\ 4.3642 \\ 1.5978 \end{pmatrix} \Rightarrow \begin{pmatrix} -2.3733 \\ -4.6475 \\ 3.5615 \\ .2887 \end{pmatrix} \Rightarrow \begin{pmatrix} -2.4005 \\ -4.7009 \\ 3.6007 \\ .2502 \end{pmatrix} \Rightarrow \begin{pmatrix} -2.4000 \\ -4.7000 \\ 3.6000 \\ .2500 \end{pmatrix} * 10^6$$

Starting at the center of the earth this process obtains five-digit accuracy after four iterations. Measures of accuracy based on the location of the satellite and receiver have been developed. These measures are collectively referred to as dilution of precision (DOP) and are functions of the gradient matrix.

## 7 Weighted Least Squares

The methods presented have been based on least squares and can be enhanced by using weighted least squares. This is a straight forward extension and allows the variance of each measurement to influence the final estimate. There are standard methods to compute the variance of most geodesic measurements; however these variance estimates are based on the instrument or sensor used. Models of the variance for each measurement allow the use of weighted least squares. These models can be quite sophisticated and depend on the manufacture of the measurement instrument. Depending on the precision required by the application the extra complexity of weighted least squares might be tolerated.

## 8 Software

A collection of MATLAB functions or toolbox has been developed. These routines can be used to develop iterative procedures to estimate coordinates for specific sensor combinations. These routines can be used as the basis for DSP algorithm development for coordinate estimation subsystems.

## 9 Plane of Best Fit

A problem related to coordinate estimation is to find the plane of best fit for a set of points. These points could be on the surface of the earth, or defined by the trajectory of an object. The problem is easily solved through singular value decomposition (SVD). Principle Component Analysis (PCA) uses SVD to find the dimensions in the data that contain the most variation. It can be conceived of in the following iterative manner; first find the dimension that contains the most variation; remove this dimension from the data and keep repeating. SVD forms an orthogonal set of dimensions that can be used to represent the data. Using SVD to find the plane of best fit, first remove the centroid of the locations and then the first two dimensions or vectors given through an SVD procedure will define the plane of best fit. This is equivalent to ignoring the dimension that contains the least amount of variation. The plane of best fit can be compared to the geodesic local level surface. An artillery trajectory drifts to the right; the rate of change of the plane of best fit could be related to the spin of the projectile.

## 10 Conclusions

Coordinate adjustment techniques can be used to fuse data from different sensors. As the sensors become more accurate high precision techniques result in accurate estimates of location. Many sensors yield inner product information. The mathematics associated with using inner product for coordinate estimation has been developed. Note that both inner product and distance are defined in high dimensional spaces so these ideas can be applied to situations using more than three dimensions. Also the use of angle and distance measurements in coordinate estimation has been reviewed. These techniques will increase the location accuracy of test range information in a post processing situation. Operational flights would benefit if these algorithms can be implemented on a DSP.

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*Chapter 19*

## A GENERALIZED CHINESE REMAINDER THEOREM FOR RESIDUE SETS WITH ERRORS

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### Abstract

Chinese Remainder Theorem (CRT) has been recently generalized from determining a single integer from its remainders to determining multiple integers from their sets (residue sets) of remainders. In this letter, we consider the generalized CRT when the residue sets have errors. We first obtain a sufficient condition on the number of erroneous residue sets so that multiple integers can be still uniquely determined from their residue sets. We then propose a determination algorithm of multiple integers from their residue sets with errors. Finally, we apply the newly proposed algorithm to multiple frequency determination from multiple sensors with low sampling rates and show the effectiveness of the proposed algorithm with considering residue set errors over the one without considering residue set errors.

**Key Words:** Chinese Remainder Theorem, remainder errors, multiple frequency determination, undersampling, sensor networks.

**AMS Subject Classification:** Primary 43A60.

## 1 Introduction

The conventional Chinese Remainder Theorem (CRT) is to determine a single integer from its remainders from a set of modulus. It has tremendous applications in various areas, such as cryptography [11] and digital signal processing [10]. CRT has various generalizations [11]. A different generalization of CRT has been recently proposed in [1, 2, 3] where

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\*E-mail address: xxia@ee.udel.edu; This work was supported in part by the Air Force Office of Scientific Research (AFOSR) under Grant No. FA9550-05-1-0161, and the National Science Foundation under Grant CCR-0097240 and CCR-0325180.

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(instead of a single integer in CRT) multiple integers need to be determined from (not a sequence of remainders but) a sequence of sets, residue sets, of remainders. A residue set consists of the remainders of multiple integers modulo a modulus integer and the residue set is not ordered, i.e., the correspondence between the elements in the residue set and the multiple integers is not specified. The generalized CRT studied in [1] was motivated from the determination of multiple frequencies in a superpositioned signal of multiple sinusoids from its multiple undersampled waveforms. This has applications in a sensor network where multiple sensors have low power and low transmission rates and their sampling rates may be low and much lower than the Nyquist rate of a signal of interest in the field. The generalized CRT has been used in synthetic aperture radar (SAR) imaging of moving targets [4] and polynomial phase signal detection [5].

In the study of the generalized CRT in [1, 2, 3, 6], it is assumed that the residue sets do not have errors, i.e., all remainders are assumed error free. In some applications, such as the multiple frequency determination studied in [1], errors may occur in the remainders. The main goal of this letter is to consider the generalized CRT when some of the remainders in residue sets have errors. We first present a sufficient condition on the number of residue sets with errors so that the multiple integers can be still uniquely determined from the residue sets with errors and the corresponding modulus. We then present a determination algorithm. Finally, we apply the proposed algorithm for the generalized CRT with residue set errors to the multiple frequency determination in a superpositioned signal contaminated by additive noise from its undersampled signals at multiple sensors. Our simulation results show that the error rates of multiple frequencies can be significantly reduced with the proposed algorithm considering residue errors compared to the one in [3] without considering residue set errors. Note that the conventional CRT with remainder errors has been nicely studied in [7, 8, 9].

This letter is organized as follows. In Section 2, we describe the problem. In Section 3, we first present a sufficient condition on the number of residue sets of errors for the unique determination, and then present an algorithm for the unique determination. In Section 4, we apply the proposed algorithm in a sensor network with low sampling rates.

## 2 Problem Formulation

Let  $S = \{N_1, N_2, \dots, N_\rho\}$  be a set of distinct positive integers and  $P = \{p_1, p_2, \dots, p_\gamma\}$  be a set of positive integers that, without loss of generality, are assumed relatively co-prime, i.e., any two of  $p_r$ ,  $1 \leq r \leq \gamma$ , are co-prime, and  $p_1 < p_2 < \dots < p_\gamma$ . The remainder (or residue) of  $N_l$  modulo  $p_r$  is

$$k_{l,r} \equiv N_l \bmod p_r \text{ for } 1 \leq l \leq \rho, 1 \leq r \leq \gamma. \quad (2.1)$$

For  $1 \leq r \leq \gamma$ , define the residue set of  $S$  modulo  $p_r$ :

$$S_r(N_1, N_2, \dots, N_\rho) \triangleq \{k_{l,r} : l = 1, 2, \dots, \rho\}. \quad (2.2)$$

Thus, there are  $\gamma$  residue sets  $S_r(N_1, N_2, \dots, N_\rho)$ ,  $1 \leq r \leq \gamma$ . Furthermore, some of these residue sets may have errors. What we know is  $\hat{S}_r(N_1, N_2, \dots, N_\rho)$ ,  $1 \leq r \leq \gamma$ , that are  $S_r(N_1, N_2, \dots, N_\rho)$ ,  $1 \leq r \leq \gamma$ , contaminated with errors. Suppose the correspondence

between error contaminated residue set  $\tilde{S}_r(N_1, N_2, \dots, N_\rho)$  and  $p_r \in P$  for  $1 \leq r \leq \gamma$  is specified, but the correspondence between  $N_l$  and its remainder  $k_{l,r}$  is not known.

The *problem* is to determine set  $S$  of multiple integers  $N_1, N_2, \dots, N_\rho$  from the  $\gamma$  error contaminated residue sets  $\tilde{S}_r(N_1, N_2, \dots, N_\rho)$  and their corresponding modulus  $p_r$ ,  $1 \leq r \leq \gamma$ .

There are three questions associated with the above problem: 1) what is the dynamic range of these multiple integers  $N_l$  so that they can be uniquely determined? 2) how many errors of the residue sets can be corrected? 3) how can these multiple integers be determined? When  $\rho = 1$  and there is no errors in remainders, CRT provides a complete solution for the above problem. When  $\rho = 1$  but there are errors in remainders, it is the CRT with errors [7]. When  $\rho > 1$  and there is no errors in remainders, it is the generalized CRT studied in [1, 2, 3, 6]. In [1], a dynamic range for the uniqueness of the determination of the multiple integers when the residue sets do not have any errors is given: if

$$\max\{N_1, N_2, \dots, N_\rho\} < \max\{p, p_\gamma\} \quad (2.3)$$

where

$$p = \min_{1 \leq r_1 < r_2 < \dots < r_\eta \leq \gamma} \text{lcm}\{p_{r_1}, p_{r_2}, \dots, p_{r_\eta}\} = p_1 p_2 \dots p_\eta, \quad (2.4)$$

where

$$\gamma = \eta\rho + \theta \quad (2.5)$$

for some  $0 \leq \theta < \rho$ . In [3], based on the above dynamic range, an efficient determination algorithm is proposed. In the following, the results obtained in [1, 3] are generalized to the case when residue sets have errors.

### 3 Uniqueness and A Determination Algorithm

We first have a uniqueness result on the determination of  $N_l$ ,  $1 \leq l \leq \rho$ , from their erroneous residue sets  $\tilde{S}_r(N_1, \dots, N_\rho)$  and  $p_r$ ,  $1 \leq r \leq \gamma$ .

For  $0 < \xi < \eta$  with  $\eta$  defined in (2.5), let

$$p(\xi) \triangleq p_1 p_2 \dots p_\xi, \quad (3.1)$$

and let  $e$  denote the size of the following set (i.e., the number of residue sets with errors):

$$\{r : \tilde{S}_r(N_1, \dots, N_\rho) \neq S_r(N_1, \dots, N_\rho), 1 \leq r \leq \gamma\}. \quad (3.2)$$

Then, we have the following result.

**Theorem 3.1** *If integers  $\{N_l\}$  and the number  $e$  of residue sets with errors satisfy, respectively,*

$$\max\{N_1, \dots, N_\rho\} < \max\{p(\xi), p_\gamma\}, \quad (3.3)$$

$$e \leq \frac{(\eta - \xi)\rho + \theta}{2}, \quad (3.4)$$

where  $\theta$  and  $\eta$  are defined in (2.5), then, integers  $N_l$ ,  $1 \leq l \leq \rho$ , can be uniquely determined from  $\tilde{S}_r(N_1, \dots, N_\rho)$  and  $p_r$ ,  $1 \leq r \leq \gamma$ .

Assume there are two sets of integers  $\{N_1, \dots, N_\rho\}$  and  $\{M_1, \dots, M_\rho\}$  satisfying the above conditions with the same residue sets  $\tilde{S}_r(N_1, \dots, N_\rho)$  and modulus  $p_r$ ,  $1 \leq r \leq \gamma$ . From condition (3.4), we have  $\gamma - 2e \geq \xi\rho$  and thus these two sets of integers share at least  $\xi\rho$  common residue sets without errors. By noticing the dynamic range (3.3), one may replace  $\eta$  in the proof in [1] with  $\xi$  and the rest of the proof to show  $\{N_1, \dots, N_\rho\} = \{M_1, \dots, M_\rho\}$  is the same as the one in [1].

We next generalize the algorithm proposed in [3] to residue sets with errors, which will also confirm the result in Theorem 3.1. Based on the known  $\gamma$  error contaminated residue sets  $\tilde{S}_r(N_1, \dots, N_\rho)$ , we define their product set

$$\tilde{S}(N_1, \dots, N_\rho) = \tilde{S}_1(N_1, \dots, N_\rho) \times \dots \times \tilde{S}_\gamma(N_1, \dots, N_\rho). \quad (3.5)$$

### A Determination Algorithm

**Step 1:** Arbitrarily take a vector  $(k_1, \dots, k_\gamma) \in \tilde{S}(N_1, \dots, N_\rho)$  in (3.5). We know that there are at most  $e$  remainders in the residue vector  $(k_1, \dots, k_\gamma)$  wrong, i.e., there are at least  $\gamma - e$  correct remainders in this residue vector.

**Step 2:** For each  $r$ ,  $1 \leq r \leq \gamma$ , define the coset of  $k_r$ :

$$\mathcal{N}_r = \{k_r + np_r : k_r \leq k_r + np_r < p(\xi), \text{ and integers } n\}, \quad (3.6)$$

where  $p(\xi)$  is defined in (3.1). Note that all the numbers in set  $\mathcal{N}_r$  have the same remainder  $k_r$  modulo  $p_r$ .

**Step 3:** From (3.4), it is easy to check that  $\gamma - e \geq \xi\rho$ , i.e., there are at least  $\xi\rho$  error free residue sets but there are only  $\rho$  different integers  $N_l \in S$ . Thus, there are at least  $\xi$  correct remainders sharing a common integer in  $S$ , i.e., there exist integers  $r_1, \dots, r_\xi$  with  $1 \leq r_1 < \dots < r_\xi \leq \gamma$  such that the remainders  $k_{r_1}, \dots, k_{r_\xi}$  are error free and from a common integer, which means that

$$\mathcal{U} \triangleq \mathcal{N}_{r_1} \cap \mathcal{N}_{r_2} \cap \dots \cap \mathcal{N}_{r_\xi} \neq \emptyset.$$

Based on this observation, Step 3 is to look for indices  $1 \leq r_1 < \dots < r_\xi \leq \gamma$  such that  $\mathcal{U} \neq \emptyset$ . Due to the dynamic range condition (3.3) and (3.6), by the conventional CRT, set  $\mathcal{U} \neq \emptyset$  has only one element, i.e.,  $\mathcal{U} = \{\bar{N}\}$ . Note that, for indices  $1 \leq r_1 < \dots < r_\xi \leq \gamma$  such that  $\mathcal{U} = \{\bar{N}\}$ , its remainders  $k_{r_1}, \dots, k_{r_\xi}$  may not be necessarily remainders of a single integer in  $S$ . If they were remainders of a single integer in  $S$ , we would have  $\bar{N} \in S$ . Next step is to check when  $\bar{N} \in S$ .

**Step 4:** Check whether  $\bar{N}$  is a valid integer: check how many remainders  $\bar{k}_r = \bar{N} \bmod p_r$  belonging to  $\tilde{S}_r(N_1, \dots, N_\rho)$  for  $1 \leq r \leq \gamma$ . Let  $\bar{\gamma}$  be the cardinality of the set  $R \triangleq \{\bar{k}_r \in \tilde{S}_r(N_1, \dots, N_\rho) : 1 \leq r \leq \gamma\}$ . If

$$\bar{\gamma} \geq \gamma - \frac{(\eta - \xi)\rho + \theta}{2} = \frac{\gamma}{2} + \frac{\xi\rho}{2}, \quad (3.7)$$

then,  $\bar{N}$  is a valid integer, i.e.,  $\bar{N} \in S$ .

In fact, when (3.7) holds, due to (3.4) the remainder set  $R$  has at least

$$\bar{\gamma} - \frac{(\eta - \xi)\rho + \theta}{2} \geq \gamma - ((\eta - \xi)\rho + \theta) = \xi\rho$$

correct remainders. Since there are only  $\rho$  integers in  $S$ , there exist at least  $\xi$  remainders in  $R$  sharing a common integer  $N_l$  in  $S$ . With the same argument as in Step 3, we conclude  $\bar{N} = N_l \in S$ .

If  $\bar{N}$  is not a valid integer, take another index set  $\{r_1, r_2, \dots, r_\xi\}$  such that

$$\mathcal{N}_{r_1} \cap \mathcal{N}_{r_2} \cap \dots \cap \mathcal{N}_{r_\xi} \neq \emptyset = \{\bar{N}\}$$

until  $\bar{N}$  is a valid integer. Denote the valid integer as  $N_\rho$ . Note that the existence of such a valid  $\bar{N}$  is ensured by the analysis in Step 3.

**Step 5:** For  $r = 1, 2, \dots, \gamma$ , remove  $k_{\rho,r} = N_\rho \bmod p_r$  from the residue set  $\tilde{S}_r(N_1, \dots, N_\rho)$  to form a new residue set  $\tilde{S}_r(N_1, \dots, N_{\rho-1})$  given in (3.8)

$$\tilde{S}_r(N_1, \dots, N_{\rho-1}) \triangleq \begin{cases} \tilde{S}_r(N_1, \dots, N_\rho) - \{k_{\rho,r}\}, \\ \quad \text{if } k_{\rho,r} \in \tilde{S}_r(N_1, \dots, N_\rho) \text{ and } |\tilde{S}_r(N_1, \dots, N_\rho)| = \rho, \\ \tilde{S}_r(N_1, \dots, N_\rho), \\ \quad \text{if } k_{\rho,r} \in \tilde{S}_r(N_1, \dots, N_\rho) \text{ and } |\tilde{S}_r(N_1, \dots, N_\rho)| < \rho, \end{cases} \quad (3.8)$$

for  $1 \leq r \leq \gamma$ , where  $|\cdot|$  denotes the cardinality of a set. If  $k_{\rho,r} \notin \tilde{S}_r(N_1, \dots, N_\rho)$ , an erroneous residue set is detected, and it is removed from the residue sets for any further consideration and we replace  $\gamma$  by  $\gamma - 1$  and  $e$  by  $e - 1$ , or it is kept unchanged for the next iteration because some of its elements may be correct remainders. We also remove  $N_\rho$  from  $S$  and set  $S = S - \{N_\rho\}$ .

**Step 6:** Go to Step 1 by replacing  $\rho$  with  $\rho - 1$ , and changing the residue product set  $\tilde{S}(N_1, \dots, N_\rho)$  into  $\tilde{S}(N_1, \dots, N_{\rho-1})$ . Repeat this process until  $N_1$  is determined.

**End of the algorithm.**

Note that the above validation of the algorithm also confirms that when the dynamic range (3.3) holds and the number of erroneous residue sets satisfies (3.4), multiple integers can be uniquely determined from their residue sets with errors. Comparing with the generalized CRT without residue set errors obtained in [1, 3], the robustness to the residue set errors studied in this letter comes from the sacrifice of the dynamic range of  $\rho$  uniquely determinable integers  $N_l \in S$ , which is reduced from  $p_1 \cdots p_\eta$  to  $p_1 \cdots p_\xi$  with  $\xi < \eta$ . As a remark, the above algorithm may have high complexity when the parameters are large. Any faster algorithm would be interesting.

As an example, consider two integers, i.e.,  $\rho = 2$  and  $\gamma = 15$  relatively co-prime integers: 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67. Choose  $\xi = 4$ . In this case  $\eta = 7$ . Then, according to the above results, if two positive integers  $N_1, N_2 < 11 \times 13 \times 17 \times 19 = 46189$ , then, these two integers can be uniquely determined from their 15 residue sets even when three of these 15 residue sets are erroneous. Corresponding to the multiple frequency determination application studied in [1], two frequencies as high as 46189Hz in a superpositioned signal can be uniquely determined from 15 undersampled waveforms with the highest sampling rate 67Hz where any three of the undersampled waveforms may be completely damaged. Note that if there is no errors in residue sets, from the results obtained in [1, 2], two uniquely determinable frequencies can go as high as  $11 \times 13 \times 17 \times 19 \times 23 \times 29 \times 31 - 1 = 955049952\text{Hz}$  with the same sampling rates as above.



## 4 Frequency Determination from Multiple Sensors with Low Sampling Rates

In this section, we apply the proposed algorithm in Section 3 to determine multiple frequencies from multiple sensors with undersampling rates. Consider  $\gamma$  sensors with sampling rates  $p_r$  Hz and  $p_r$  satisfy the assumptions made in the beginning of Section 2,  $1 \leq r \leq \gamma$ . Consider  $\rho$  multiple frequencies  $f_1 = N_1$  Hz,  $\dots$ ,  $f_\rho = N_\rho$  Hz in a superpositioned waveform. At the  $r$ th sensor, the received analog signal is of the following form,

$$x_r(t) = \sum_{l=1}^{\rho} A_{l,r} e^{2\pi j f_l t} + w_r(t), \quad (4.1)$$

where  $A_{l,r}$ ,  $1 \leq l \leq \rho$ , are nonzero complex coefficients and  $w_r(t)$  is the additive white noise. The sampled signal at the  $r$ th sensor with sampling rate  $p_r$  Hz is

$$x_r[n] = x_r\left(\frac{n}{p_r}\right) = \sum_{l=1}^{\rho} A_{l,r} e^{2\pi j f_l n/p_r} + w_r\left(\frac{n}{p_r}\right), \quad (4.2)$$

which can be also thought of as a received signal at a base station from the  $r$ th sensor. The problem is to determine the multiple frequencies  $f_l = N_l$ ,  $1 \leq l \leq \rho$ , from the above sampled data  $x_r[n]$ ,  $1 \leq r \leq \gamma$ , where the sampling rates  $p_r$  may be much lower than the signal frequencies  $N_l$ . Note that the above undersampling may be necessary when the transmission rates of multiple sensors are low.

Based on the sampled data  $x_r[n]$  at the  $r$ th sensor, we take  $p_r$ -point DFT and obtain

$$X_r[k] = \text{DFT}_{p_r}(x_r[n]) = \sum_{l=1}^{\rho} \sqrt{p_r} A_{l,r} \delta(k - k_{l,r}) + W_r[k], \quad (4.3)$$

for  $0 \leq k \leq p_r - 1$ , where  $k_{l,r}$  is the remainder of  $N_l$  modulo  $p_r$  the same as in Section 2. Clearly, if the noise power of  $W_r[k]$  is not too high, i.e., the SNR is not too low, the remainder  $k_{l,r}$  can be correctly detected from the DFT coefficients  $X_r[k]$  through (4.3). Otherwise, the remainder  $k_{l,r}$  may have error, which precisely corresponds to the problem studied in Sections 2 and 3. Thus, we may apply the algorithm developed in Section 3 to the above multiple frequency determination problem.

We next see a simulation. In the simulation, consider three frequencies, i.e.,  $\rho = 3$ , and 15 sensors with the sampling rates of primes from 11 to 67 the same as the previous example, and choose  $\xi = 3$ . Thus, the dynamic range in (3.3) is  $p(\xi) = p(3) = 11 \times 13 \times 17 = 2431$ . The SNR is defined as the ratio of the variance of  $A_{l,r}$  over the variance of the additive white Gaussian noise  $w_r(n/p_r)$ . In the simulation, three distinct frequencies  $N_1$ ,  $N_2$ ,  $N_3$  are randomly chosen in the range between 2000 and 2431. The coefficients  $A_{l,r}$  are randomly chosen from zero mean complex Gaussian distributions, independently and identically in terms of  $l$  and  $r$ . From the theory we developed in Section 3, three frequencies can be determined when any three signal waveforms from sensors are damaged or any three residue sets have errors. Two kinds of frequency detection error rates are calculated. One is the error rate of individual frequencies (FrER) and the other is the error rate of a frequency

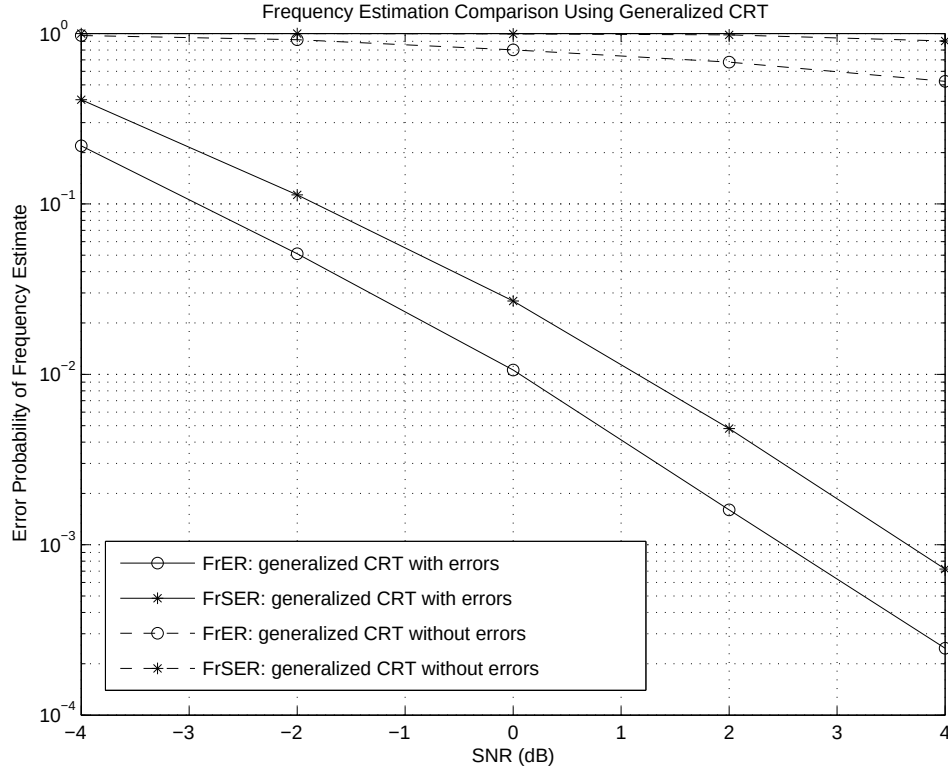


Figure 1: Multiple frequency estimation comparison between generalized CRT with errors proposed in this paper and without errors proposed in [3].

set (FrSER), i.e., when any of the three frequencies in the set is wrong, the set is counted wrong. In the simulation, for convenience, three remainders in  $\hat{S}_r(N_1, N_2, N_3)$  are selected after the  $p_r$ -point DFT from the three largest absolute values of the DFT coefficients for  $1 \leq r \leq 15$ . Also, when no valid integer  $\bar{N}$  can be found in Step 4, a frequency set error and three frequency errors are counted, respectively. It is similarly done for the algorithm in [3]. The proposed algorithm (solid lines in Fig. 1) in Section 3 is compared with the algorithm (dashed lines in Fig. 1) in [3] where residue errors are not considered. From Fig. 1 one can see that the newly proposed algorithm significantly outperforms the one in [3] with a comparable complexity.

## 5 Conclusion

In this letter, we considered the generalized CRT with erroneous residue sets, i.e., the determination of multiple integers from a set of modulus and the corresponding residue sets with errors. We obtained a sufficient condition on the number of erroneous residue sets when a dynamic range of multiple integers is given, so that the multiple integers can be uniquely determined. We then presented an algorithm for the determination. The results in this letter are generalizations of the ones in [1, 3] from error-free residue sets to erroneous residue sets and the robustness to the errors in residue sets comes from the sacrifice of the

dynamic range of multiple uniquely determinable integers. We finally applied the proposed algorithm to determine multiple frequencies from multiple sensors with much lower sampling rates than the Nyquist rate of a signal. Our simulation result shows that the error rates of the detected frequencies can be significantly reduced by using the algorithm proposed in this letter with considering residue errors from the ones using the algorithm proposed in [3] without considering residue errors.

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*Chapter 20*

# LOCAL SMOOTHNESS OF SOLUTIONS FOR NONLINEAR SCHRÖDINGER EQUATIONS WITH POTENTIALS SUPERQUADRATIC

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## Abstract

In this paper we give a strict alternative proof of fractional order derivatives estimate and apply it to investigate the local smoothing property of solutions of the initial value problem for the nonlinear Schrödinger equation with the potential  $V$  superquadratic at infinity, i.e.,  $V$  grows faster than  $C|x|^{2+\varepsilon}$  at infinity for constants  $C > 0$  and  $\varepsilon > 0$ .

**Key Words:** Local smoothness, Nonlinear Schrödinger Equations, Potentials Superquadratic, Fractional order derivatives.

**AMS Subject Classification:** Primary 35G25, 35J10, 35Q40, 46B50.

## 1 Introduction

We consider the initial value problem (IVP) for the nonlinear Schrödinger equation

$$\begin{cases} i\partial_t u = -\Delta u + V(x)u + f(x, u), & (t, x) \in \mathbb{R} \times \mathbb{R}^n, \\ u(x, 0) = u_0(x), & x \in \mathbb{R}^n, \end{cases} \quad (1.1)$$

with potential  $V$  which grows faster than  $C|x|^{2+\varepsilon}$  at infinity, for constants  $C > 0$  and  $\varepsilon > 0$ .

Let  $\langle A \rangle = (1 + |A|^2)^{\frac{1}{2}}$  for a self-adjoint operator  $A$ .

**Assumptions** We assume that the potential  $V(x)$  is a real valued function of  $C^\infty$ -class and that there exists a constant  $R \geq 1$  such that  $V$  satisfies the following conditions for  $|x| \geq R$ :

(V1) For  $k > 2$ ,  $D_1 \langle x \rangle^k \leq V(x) \leq D_2 \langle x \rangle^k$ , where  $0 < D_1 \leq D_2 < \infty$ .

(V2) For any  $\alpha$ , there exists a constant  $C_\alpha$  such that  $|\partial^\alpha V(x)| \leq C_\alpha \langle x \rangle^{k-|\alpha|}$ .

Under the assumption(V1), the operator  $H = -\Delta + V(x)$  on  $C_0^\infty(\mathbb{R}^n)$  is essentially selfadjoint in  $L^2(\mathbb{R}^n)$  and therefore we still denote its unique selfadjoint extension by  $H$ .  $H$  is bounded from below and, without losing generality, we assume  $H \geq 1$ . We define

$$\mathcal{H}^s = \{u \in L^2(\mathbb{R}^n) \mid H^{s/2}u \in L^2(\mathbb{R}^n)\}, \quad \mathcal{H}_{loc}^s = \{u \in L_{loc}^2(\mathbb{R}^n) \mid H^{s/2}u \in L_{loc}^2(\mathbb{R}^n)\}$$

In the previous papers [17] and [18], the authors have studied the smoothing property for the linear Schrödinger equation in one dimension and  $n$  dimension respectively. In this paper we will apply the smoothing properties to study the smoothness of solution of the initial value problem for the related nonlinear Schrödinger equation (1.1).

Currently there are mainly two methods to study the smoothness of solutions. One is to follow Kato and Sjölin's idea (see [4], [5], [10] and [11]). Another is to use the Besov space framework (see [7],[8]). Kato and Sjölin's method is to split the nonlinearity into two parts and then estimate each part independently. This method is difficult to handle the fractional Sobolev space and some of the results derived by it are not optimal (see [11]). Besov space method can be regarded as "micro-local" method, which is suitable to handle the fractional derivatives since Besov space is defined via dyadic decomposition. This method can also give more accurate estimate about the smoothness, but its computation is very complicated. Our method is based on the following fractional order derivatives estimate which makes it possible to handle the fractional derivatives directly during the calculation (see the detail in next section).

$$\| |D|^\alpha F(\vec{u}) \|_{L^r(\mathbb{R}^n)} \leq C \sum_{l=1}^m \|G_l(\vec{u})\|_{L^p(\mathbb{R}^n)} \| |D|^\alpha u_l \|_{L^q(\mathbb{R}^n)}. \quad (1.2)$$

The following local smoothing property has been proved in [18].

**Theorem 1.1** *Let  $V$  satisfy the assumptions (V1) and (V2). Let  $T > 0$  and  $\Psi \in C_0^\infty(\mathbb{R}^n)$ . Then, there exists a constant  $C > 0$  such that*

$$\left( \int_{-T}^T \|\Psi(x) \langle H \rangle^{\frac{1}{2k}} e^{-itH} u_0\|_2^2 dt \right)^{\frac{1}{2}} \leq C \|u_0\|, \quad u_0 \in L^2(\mathbb{R}^n). \quad (1.3)$$

As an application of Theorem 1.1, in [18],[22],[23] and [24] the authors have already systematically studied  $\mathcal{H}^s$  solutions for nonlinear Schrödinger equation (1.1) with such potential  $V$ . We summarize the main results as follows.

Let  $\delta > 0$ ,  $0 < s \leq 1$ ,  $K$  be a compact subset of  $\mathbb{R}^n$ , and  $1 \leq r \leq \frac{kn}{kn-1}$ . We define

$$\begin{cases} Z_{\delta,K} \equiv L^2([-\delta, \delta]; L^{2r}(K)) \cap C([-\delta, \delta]; L^2(\mathbb{R}_x^n)), \\ \tilde{Z}_{\delta,K} \equiv L^2([-\delta, \delta]; L^{2r}(K)) \cap L^\infty([-\delta, \delta]; L^2(\mathbb{R}_x^n)), \\ \|u\|_{Z_{\delta,K}} = \|u\|_{\tilde{Z}_{\delta,K}} = \|u\|_{L^2([-\delta, \delta]; L^{2r}(K))} + \|u\|_{L^\infty([-\delta, \delta]; L^2(\mathbb{R}_x^n))}, \end{cases} \quad (1.4)$$

and

$$\begin{cases} Z_{\delta,K}^s \equiv \{u \in Z_{\delta,K} \mid H^{s/2}u \in Z_{\delta,K}\}, \\ \tilde{Z}_{\delta,K}^s \equiv \{u \in \tilde{Z}_{\delta,K} \mid H^{s/2}u \in \tilde{Z}_{\delta,K}\}, \\ \|u\|_{Z_{\delta,K}^s} = \|u\|_{\tilde{Z}_{\delta,K}^s} = \|u\|_{Z_{\delta,K}} + \|H^{s/2}u\|_{Z_{\delta,K}}. \end{cases} \quad (1.5)$$

**Theorem 1.2** ( $\mathcal{H}^s$  solution) Suppose that  $V$  satisfies the assumptions (V1) and (V2). Let  $1 \leq r \leq \frac{kn}{kn-1}$ ,  $0 < s \leq 1$  and  $L \subset \mathbb{R}^n$  be a compact subset. Suppose that  $f(x, u)$  is of the form  $\lambda(x)F(u)$  and satisfies

$$\lambda(x) = 0, \quad \text{for } x \notin L \quad \text{and} \quad \lambda, \langle D \rangle^s \lambda \in L^\infty(\mathbb{R}^n), \quad (1.6)$$

and

$$|\partial^\gamma F(u)| \leq C_\gamma |u|^{r-|\gamma|}, \quad \text{for } |\gamma| \leq s_*, \quad (1.7)$$

where  $s_*$  is the smallest integer  $\geq s$ ,  $\partial = (\partial_u, \partial_{\bar{u}})$ ,  $\gamma = (\gamma_1, \gamma_2) \in \mathbb{N}^2$  and  $C_\gamma$  are constants. Then, for any compact subset  $K$  such that  $L$  is included in the interior of  $K$  and for any  $u_0 \in \mathcal{H}^s(\mathbb{R}^n)$ , there exists  $\delta > 0$  such that (1.1) admits a unique solution  $u(t, x)$  in  $Z_{\delta,K}^s$ .

**Theorem 1.3** Let  $s > \frac{n}{2}$  and  $V$  satisfy (V1) and (V2). Suppose that  $f(x, u)$  is of the form  $\lambda(x)F(u)$  such that  $\lambda(x)$ ,  $\langle D \rangle^s \lambda(x) \in L^\infty$  and satisfies (1.7). In addition, we assume  $s_* \leq r + 1$  if  $F(u)$  is not a polynomial of  $u$  and  $\bar{u}$ . Then, for any  $u_0 \in \mathcal{H}^s$ , the IVP (1.1) admits a unique solution  $u(t, x)$  in  $C([-\delta, \delta]; \mathcal{H}^s)$ . Furthermore, if  $s_* \leq r$ , then the map  $\mathcal{H}^s \ni u_0 \mapsto u \in C([-\delta, \delta]; \mathcal{H}^s)$  is continuous.

In this paper we will study the local smoothness of the solutions obtained in the aforementioned theorems.

## 2 Estimate of Fractional Order Derivatives

We introduce the estimates of fractional order derivatives which was indicated firstly by Kato ( cf. [5] ). These estimates had ever played an important role in proving the existence of  $\mathcal{H}^s$  solutions for IVP (1.1) (see [22],[23] and [24]). In this section we give an alternative proof of the estimate of fractional order derivatives.

Let  $x = (x_1, \dots, x_n) \in \mathbb{R}^n$ ,  $\vec{u} = (u_1, \dots, u_m) \in \mathbb{C}^m$ ,  $u_l = x_l + iy_l$ ,  $x_l, y_l \in \mathbb{R}$ ,

$$\frac{\partial}{\partial u_l} = \frac{1}{2} \left( \frac{\partial}{\partial x_l} - i \frac{\partial}{\partial y_l} \right), \quad \frac{\partial}{\partial \bar{u}_l} = \frac{1}{2} \left( \frac{\partial}{\partial x_l} + i \frac{\partial}{\partial y_l} \right).$$

$F(\vec{u}) = F(u_1, \dots, u_m) \in C^1(\mathbb{C}^m; \mathbb{C})$  means that

$$\frac{\partial F}{\partial u_l}, \quad \frac{\partial F}{\partial \bar{u}_l} \in C(\mathbb{C}^m; \mathbb{C}), \quad \forall l = 1, \dots, m.$$

Inductively we may define  $F(\vec{u}) = F(u_1, \dots, u_m) \in C^k(\mathbb{C}^m; \mathbb{C})$ ,  $k \in \mathbb{N}$ .

**Assumption 2.1** Let  $F(\vec{\mathbf{u}}) \in C(\mathbb{C}^m; \mathbb{C})$  and for any  $l = 1, \dots, m$ , there exists  $G_l(\xi) \in C(\mathbb{C}^m; \mathbb{R}^+)$  such that

$$|F(\xi) - F(\eta)| \leq \sum_{l=1}^m (G_l(\xi) + G_l(\eta)) |\xi_l - \eta_l| \quad (2.1)$$

holds for all  $\xi, \eta \in \mathbb{C}^m$ .

**Definition 2.2** Let  $\alpha \geq 0$ , one defines the  $\alpha$ -order absolute derivative operator  $|D|^\alpha$  by the Fourier multiplier as follows:

$$|D|^\alpha f(x) = (-\Delta)^{\alpha/2} f(x) = \int_{\mathbb{R}^n} e^{ix \cdot \xi} |\xi|^\alpha \hat{f}(\xi) d\xi, \quad \forall f \in \mathcal{S}(\mathbb{R}^n).$$

By duality we may define  $|D|^\alpha f(x)$  for  $f \in \mathcal{S}'(\mathbb{R}^n)$ .

**Proposition 2.3** Assume that  $F$  satisfies Assumption 2.1. Let  $0 < \alpha < 1$ ,  $1 < p \leq \infty$ ,  $1 < q, r < \infty$  and  $\frac{1}{r} = \frac{1}{p} + \frac{1}{q}$ . If for any  $l = 1, \dots, m$ ,  $|D|^\alpha u_l(x) \in L^q(\mathbb{R}^n)$  and  $G_l(\vec{\mathbf{u}}(x)) \in L^p(\mathbb{R}^n)$ , then  $|D|^\alpha F(\vec{\mathbf{u}}(x)) \in L^r(\mathbb{R}^n)$  and we have the following estimate

$$\| |D|^\alpha F(\vec{\mathbf{u}}) \|_{L^r(\mathbb{R}^n)} \leq C \sum_{l=1}^m \|G_l(\vec{\mathbf{u}})\|_{L^p(\mathbb{R}^n)} \| |D|^\alpha u_l \|_{L^q(\mathbb{R}^n)}. \quad (2.2)$$

Let  $\eta \in C_0^\infty(\mathbb{R}^n)$  be a nonnegative function with support contained in  $\{\xi \in \mathbb{R}^n : \frac{1}{2} < |\xi| < 2\}$  and satisfy

$$\sum_{j=-\infty}^{\infty} \eta(2^{-j}\xi) = 1 \quad \forall \xi \in \mathbb{R}^n \setminus \{0\}.$$

Define the Fourier multiplier operators

$$(Q_j f)^\wedge(\xi) = \eta(2^{-j}\xi) \hat{f}(\xi).$$

**Lemma 2.4** Let  $1 < r < \infty$ ,  $0 < \alpha < 1$ . If  $f \in \mathcal{L}'(\mathbb{R}^n)$ , then

$$\| |D|^\alpha f \|_{L^r(\mathbb{R}^n)} \sim \left\| \left( \sum_{j=-\infty}^{\infty} 2^{2j\alpha} |Q_j f|^2 \right)^{\frac{1}{2}} \right\|_{L^r(\mathbb{R}^n)}.$$

$Q_j$  may be realized as a convolution operator

$$Q_j f = f * \psi_j \quad \text{where} \quad \psi_j(x) = 2^{jn} \hat{\eta}(2^j x) \in \mathcal{S}(\mathbb{R}^n)$$

and satisfies for any  $N$ ,

$$|\psi_j(x)| + 2^{-j} |\nabla_x \psi_j(x)| \leq C_N \Psi_{j,N}(x) \quad (2.3)$$

where  $C_N$  is a constant independent of  $j \in \mathbb{Z}$  and we write

$$\Psi_{j,N}(x) = 2^{jn}(1 + 2^j|x|)^{-N}. \quad (2.4)$$

Note that  $\|\Psi_{j,N}(x)\|_{L^1} \leq C_N$ , with  $j$ -independent constant  $C_N$  and we have

$$\int_{\mathbb{R}^n} \psi_j(x) dx = 0, \quad \forall j \in \mathbb{Z}. \quad (2.5)$$

Constructing also  $\tilde{\eta}$  and  $\tilde{Q}_j$  satisfying the same properties mentioned above as  $\eta$  and  $Q_j$  respectively, but such that  $\tilde{\eta} \cdot \eta = \eta$ . Then we have

$$I = \sum_{j=-\infty}^{\infty} Q_j = \sum_{j=-\infty}^{\infty} \tilde{Q}_j Q_j,$$

where  $I$  means the identity operator on  $\mathcal{L}'(\mathbb{R}^n)$  and  $\tilde{Q}_j$  is realized by convolution with a  $\mathcal{D}_0$  function  $\tilde{\psi}_j$  satisfying (2.3) and (2.5). Let  $M$  denote the maximal function of Hardy-Littlewood:

$$M(f)(x) = \sup_{r>0} \frac{1}{|B_r(0)|} \int_{B_r(0)} |f(x-y)| dy,$$

where  $B_r(0) = \{x \in \mathbb{R}^n : |x| \leq r\}$  and  $|B_r(0)|$  is the volume of the ball  $B_r(0)$ .

**Lemma 2.5** (Stein) *Let  $\psi(x) \in L^1(\mathbb{R}^n)$  and  $\psi$  be a decreasing nonnegative radial function, then we have*

$$|(f * \psi)(x)| \leq \|\psi\|_{L^1(\mathbb{R}^n)} M(f)(x)$$

where  $f \in L^1_{loc}(\mathbb{R}^n)$ .

**Proof** The proof is a standard calculation as same as that in [12] (see pp. 62-63).

**Lemma 2.6** (Fefferman-Stein) *Let  $1 < r, p < \infty$ , then*

$$\|(\sum_{j=-\infty}^{\infty} |M(f_j)|^r)^{1/r}\|_{L^p(\mathbb{R}^n)} \leq C_{p,r} \|(\sum_{j=-\infty}^{\infty} |f_j|^r)^{1/r}\|_{L^p(\mathbb{R}^n)}$$

**Proof** For the proof, we refer the readers to [2].

**Lemma 2.7** *Let  $1 < p \leq \infty$ , then*

$$\|M(f)\|_{L^p(\mathbb{R}^n)} \sim \|f\|_{L^p(\mathbb{R}^n)}.$$

**Lemma 2.8** *Let  $\tilde{Q}_j$  be that mentioned above, for any constant  $0 < c_0 < 1$  there exists a constant  $C_0$  such that for  $\forall j \in \mathbb{Z}$  and  $\forall g \in L^1_{loc}(\mathbb{R}^n)$ , we have*

- (i)  $|\tilde{Q}_j g(x)| \leq C_0 M(g)(x), \quad \forall x \in \mathbb{R}^n$
- (ii)  $|\tilde{Q}_j g(x) - \tilde{Q}_j g(y)| \leq C_0 2^j |x - y| M(g)(x) \quad \text{if } |x - y| \leq c_0 2^{-j}.$



**Proof** For the proof, we refer the readers to [1].

**Proof of Proposition 2.3**

**Step1:** By Lemma 2.4, (2.2) is equivalent to

$$\|(\sum_{j=-\infty}^{\infty} 2^{2j\alpha} |Q_j F(\vec{\mathbf{u}})|^2)^{\frac{1}{2}}\|_{L^r(\mathbb{R}^n)} \leq C \sum_{l=1}^m \|G_l(\vec{\mathbf{u}})\|_{L^p(\mathbb{R}^n)} \|(\sum_{j=-\infty}^{\infty} 2^{2j\alpha} |Q_j u_l|^2)^{\frac{1}{2}}\|_{L^q(\mathbb{R}^n)} \quad (2.6)$$

By (2.5) we have

$$\begin{aligned} Q_j F(\vec{\mathbf{u}})(x) &= \int_{\mathbb{R}^n} F(\vec{\mathbf{u}})(y) \psi_j(x-y) dy \\ &\leq \int_{\mathbb{R}^n} |F(\vec{\mathbf{u}})(y) - F(\vec{\mathbf{u}})(x)| |\psi_j(x-y)| dy \end{aligned}$$

By Assumption 2.1,

$$\begin{aligned} |Q_j F(\vec{\mathbf{u}})(x)| &\leq \sum_{l=1}^m \int_{\mathbb{R}^n} [G_l(\vec{\mathbf{u}})(x) + G_l(\vec{\mathbf{u}})(y)] |u_l(x) \\ &\quad - u_l(y)| |\psi_j(x-y)| dy = I_j^{(1)} + I_j^{(2)}, \end{aligned} \quad (2.7)$$

where

$$\begin{aligned} I_j^{(1)}(x) &= \sum_{l=1}^m G_l(\vec{\mathbf{u}})(x) \int_{\mathbb{R}^n} |u_l(x) - u_l(y)| |\psi_j(x-y)| dy \equiv \sum_{l=1}^m I_{jl}^{(1)}(x) \\ I_j^{(2)}(x) &= \sum_{l=1}^m \int_{\mathbb{R}^n} G_l(\vec{\mathbf{u}})(y) |u_l(x) - u_l(y)| |\psi_j(x-y)| dy \equiv \sum_{l=1}^m I_{jl}^{(2)}(x) \end{aligned}$$

Thus we obtain

$$(\sum_{j=-\infty}^{\infty} 2^{2j\alpha} |Q_j F(\vec{\mathbf{u}})|^2)^{\frac{1}{2}} \leq C \sum_{l=1}^m \{(\sum_{j=-\infty}^{\infty} 2^{2j\alpha} |I_{jl}^{(1)}(x)|^2)^{\frac{1}{2}} + (\sum_{j=-\infty}^{\infty} 2^{2j\alpha} |I_{jl}^{(2)}(x)|^2)^{\frac{1}{2}}\} \quad (2.8)$$

Note that for any  $l = 1, \dots, m$

$$\begin{aligned} I_{jl}^{(1)}(x) &\leq G_l(\vec{\mathbf{u}})(x) \sum_{k \in \mathbb{Z}} \int |\tilde{Q}_k Q_k u_l(x) - \tilde{Q}_k Q_k u_l(y)| |\psi_j(x-y)| dy \\ I_{jl}^{(2)}(x) &\leq \sum_{k \in \mathbb{Z}} \int G_l(\vec{\mathbf{u}})(y) |\tilde{Q}_k Q_k u_l(x) - \tilde{Q}_k Q_k u_l(y)| |\psi_j(x-y)| dy \end{aligned}$$

**Step2:** Estimates of  $I_{jl}^{(1)}(x)$ -terms:

Break the sum over  $k$  into the case  $k < j$  and  $k \geq j$ . Then by Lemma 2.5 and Lemma 2.8

$$\begin{aligned}
& \sum_{k < j} \int_{\mathbb{R}^n} |\tilde{Q}_k Q_k u_l(x) - \tilde{Q}_k Q_k u_l(y)| |\psi_j(x-y)| dy \\
& \leq C_N \sum_{k < j} \int_{|x-y| \leq c_0 2^{-k}} 2^k |x-y| M(Q_k u_l)(x) \Psi_{j,N}(x-y) dy \\
& + C_N \sum_{k < j} \int_{|x-y| > c_0 2^{-k}} [M(Q_k u_l)(x) + M(Q_k u_l)(y)] \Psi_{j,N}(x-y) dy \\
& \leq C_N \sum_{k < j} 2^{k-j} M(Q_k u_l)(x) + C_N \sum_{k < j} 2^{k-j} [M(Q_k u_l)(x) + M^2(Q_k u_l)(x)] \\
& \leq C_N \sum_{k < j} 2^{k-j} M^2(Q_k u_l)(x)
\end{aligned}$$

where  $N > n + 1$  and  $M^2 = M \circ M$ .

Likewise

$$\sum_{k \geq j} \int |\tilde{Q}_k Q_k u_l(x) - \tilde{Q}_k Q_k u_l(y)| |\psi_j(x-y)| dy \leq C_N \sum_{k \geq j} M^2(Q_k u_l)(x).$$

Thus fix a  $N > n + 1$  we have

$$I_{jl}^{(1)}(x) \leq CG_l(\vec{\mathbf{u}}(x)) \left( \sum_{k < j} 2^{k-j} M^2(Q_k u_l)(x) + \sum_{k \geq j} M^2(Q_k u_l)(x) \right)$$

Therefore

$$\begin{aligned}
\left( \sum_{-\infty}^{\infty} 2^{2j\alpha} |I_{jl}^{(1)}(x)|^2 \right)^{\frac{1}{2}} & \leq CG_l(\vec{\mathbf{u}}(x)) \left\{ \sum_{j \in \mathbb{Z}} 2^{2j\alpha} \left( \sum_{k < j} 2^{k-j} M^2(Q_k u_l)(x) \right. \right. \\
& \left. \left. + \sum_{k \geq j} M^2(Q_k u_l)(x) \right)^2 \right\}^{1/2}
\end{aligned} \tag{2.9}$$

Define  $\mathbf{a} = \{a_m\}_{m \in \mathbb{Z}}$  and  $g^{(l)}(x) = \{g_k^{(l)}(x)\}_{k \in \mathbb{Z}}$  as follows:

$$a_m = 2^{-(1-\alpha)m}, \text{ if } m > 0; = 2^{\alpha m}, \text{ if } m \leq 0. \quad g_k^{(l)}(x) = 2^{k\alpha} M^2(Q_k u_l)(x) \tag{2.10}$$

Now we may rewrite the right-hand of (2.10) to

$$CG_l(\vec{\mathbf{u}}(x)) \left\{ \sum_{j \in \mathbb{Z}} \left( \sum_{k \in \mathbb{Z}} a_{j-k} g_k^{(l)}(x) \right)^2 \right\}^{1/2}$$

Note that

$$a_m \leq 2^{-\varepsilon|m|}, \quad \forall m \in \mathbb{Z}$$

where  $\varepsilon = \alpha \wedge (1 - \alpha) > 0$ , then  $\mathbf{a} \in l^1(\mathbb{Z})$ .

Thus Young's inequality implies

$$\left( \sum_{-\infty}^{\infty} 2^{2j\alpha} |I_{jl}^{(1)}(x)|^2 \right)^{\frac{1}{2}} \leq C \|\mathbf{a}\|_{l^1} CG_l(\vec{\mathbf{u}}(x)) \left( \sum_{k \in \mathbb{Z}} [g_k^{(l)}(x)]^2 \right)^{\frac{1}{2}}$$

Therefore by Lemma 2.6 we obtain

$$\begin{aligned} \left\| \left( \sum_{-\infty}^{\infty} 2^{2j\alpha} |I_{jl}^{(1)}(x)|^2 \right)^{\frac{1}{2}} \right\|_{L^r(\mathbb{R}^n)} &\leq C \|G_l(\vec{\mathbf{u}}(x))\|_{L^p(\mathbb{R}^n)} \left\| \left( \sum_{k \in \mathbb{Z}} [M^2(2^{k\alpha} Q_k u_l)(x)]^2 \right)^{\frac{1}{2}} \right\|_{L^q(\mathbb{R}^n)} \\ &\leq C \|G_l(\vec{\mathbf{u}}(x))\|_{L^p(\mathbb{R}^n)} \left\| \left( \sum_{k \in \mathbb{Z}} [(2^{k\alpha} Q_k u_l)(x)]^2 \right)^{\frac{1}{2}} \right\|_{L^q(\mathbb{R}^n)} \end{aligned}$$

**Step3:** Estimates of  $I_{jl}^{(2)}$ -terms: We break up the sum into the cases  $k \leq j$  and  $k > j$  and estimate  $k < j$  as previously

$$\begin{aligned} &\sum_{k < j} \int_{\mathbb{R}^n} G_l(\vec{\mathbf{u}}(y)) |\tilde{Q}_k Q_k u_l(x) - \tilde{Q}_k Q_k u_l(y)| |\psi_j(x-y)| dy \\ &\leq C_N \sum_{k < j} \int_{|x-y| \leq c_0 2^{-k}} 2^k |x-y| G_l(\vec{\mathbf{u}}(y)) M(Q_k u_l)(x) \Psi_{j,N}(x-y) dy \\ &+ C_N \sum_{k < j} \int_{|x-y| > c_0 2^{-k}} G_l(\vec{\mathbf{u}}(y)) [M(Q_k u_l)(x) + M(Q_k u_l)(y)] \Psi_{j,N}(x-y) dy \\ &\leq C_N \sum_{k < j} 2^{k-j} [M(G_l(\vec{\mathbf{u}}))(x) M(Q_k u_l)(x) + M(M(Q_k u_l) G_l(\vec{\mathbf{u}}))(x)] \end{aligned}$$

Likewise,

$$\begin{aligned} &\sum_{k \geq j} \int_{\mathbb{R}^n} G_l(\vec{\mathbf{u}}(y)) |\tilde{Q}_k Q_k u_l(x) - \tilde{Q}_k Q_k u_l(y)| |\psi_j(x-y)| dy \\ &\leq C_N \sum_{k \geq j} [M(G_l(\vec{\mathbf{u}}))(x) M(Q_k u_l)(x) + M(M(Q_k u_l) G_l(\vec{\mathbf{u}}))(x)] \end{aligned}$$

Thus we obtain

$$\begin{aligned} I_{jl}^{(2)}(x) &\leq CM(G_l(\vec{\mathbf{u}}))(x) \left\{ \sum_{j \in \mathbb{Z}} \left( \sum_{k < j} 2^{k-j} M(Q_k u_l)(x) + \sum_{k \geq j} M(Q_k u_l)(x) \right)^2 \right\}^{1/2} \\ &+ C \left\{ \sum_{j \in \mathbb{Z}} \left( \sum_{k < j} 2^{k-j} M(M(Q_k u_l) G_l(\vec{\mathbf{u}}))(x) + \sum_{k \geq j} M(M(Q_k u_l) G_l(\vec{\mathbf{u}}))(x) \right)^2 \right\}^{1/2} \\ &= R'_{jl} + R''_{jl} \end{aligned}$$

Note that

$$M(Q_k u_l)(x) \leq M^2(Q_k u_l)(x), \quad \forall x \in \mathbb{R}^n.$$

Thus

$$\begin{aligned} \left( \sum_{-\infty}^{\infty} 2^{2j\alpha} |R'_{jl}(x)|^2 \right)^{\frac{1}{2}} &\leq CM(G_l(\vec{\mathbf{u}}))(x) \left\{ \sum_{j \in \mathbb{Z}} \left( \sum_{k \in \mathbb{Z}} a_{j-k} g_k^{(l)}(x) \right)^2 \right\}^{1/2} \\ &\leq C \|\mathbf{a}\|_{l^1} M(G_l(\vec{\mathbf{u}}))(x) \left( \sum_{k \in \mathbb{Z}} [g_k^{(l)}(x)]^2 \right)^{\frac{1}{2}} \end{aligned}$$

Similarly by Lemma 2.6 we may obtain

$$\begin{aligned} \left( \sum_{-\infty}^{\infty} 2^{2j\alpha} |R''_{jl}(x)|^2 \right)^{\frac{1}{2}} &\leq C \left\{ \sum_{j \in \mathbb{Z}} \left[ \left( \sum_{k \in \mathbb{Z}} a_{j-k} M(G_l(\vec{u}) g_k^{(l)})(x) \right)^2 \right]^{1/2} \right\} \\ &\leq C \|\mathbf{a}\|_{l^1} \left( \sum_{k \in \mathbb{Z}} [M(G_l(\vec{u}) g_k^{(l)})(x)]^2 \right)^{\frac{1}{2}} \end{aligned}$$

Therefore by Lemma 2.7 and Lemma 2.6 we obtain

$$\begin{aligned} \left\| \left( \sum_{-\infty}^{\infty} 2^{2j\alpha} |I_{jl}^{(2)}(x)|^2 \right)^{\frac{1}{2}} \right\|_{L^r(\mathbb{R}^n)} &\leq C \|M(G_l(\vec{u}))(x)\|_{L^p(\mathbb{R}^n)} \left\| \left( \sum_{k \in \mathbb{Z}} [g_k^{(l)}(x)]^2 \right)^{\frac{1}{2}} \right\|_{L^q(\mathbb{R}^n)} \\ &\quad + C \left\| \left( \sum_{k \in \mathbb{Z}} [M(G_l(\vec{u}) g_k^{(l)})(x)]^2 \right)^{\frac{1}{2}} \right\|_{L^r(\mathbb{R}^n)} \\ &\leq C \|G_l(\vec{u}(x))\|_{L^p(\mathbb{R}^n)} \left\| \left( \sum_{k \in \mathbb{Z}} [(2^{k\alpha} Q_k u_l)(x)]^2 \right)^{\frac{1}{2}} \right\|_{L^q(\mathbb{R}^n)} \\ &\quad + C \left\| \left( \sum_{k \in \mathbb{Z}} [(G_l(\vec{u}) g_k^{(l)})(x)]^2 \right)^{\frac{1}{2}} \right\|_{L^r(\mathbb{R}^n)} \\ &\leq C \|G_l(\vec{u}(x))\|_{L^p(\mathbb{R}^n)} \left\| \left( \sum_{k \in \mathbb{Z}} [(2^{k\alpha} Q_k u_l)(x)]^2 \right)^{\frac{1}{2}} \right\|_{L^q(\mathbb{R}^n)} \end{aligned}$$

Combining the calculations in Step2 and Step3 we obtain (2.6).

The following corollary is used in our study directly.

**Corollary 2.9** *Let  $F \in C^s(\mathbb{C}; \mathbb{C})$  and  $s \in \mathbb{N}$ . Assume that there exists  $k \geq s$  such that*

$$|\partial^\gamma F(\xi)| \leq C_\gamma |\xi|^{k-|\gamma|}, \quad |\gamma| \leq s, \quad (2.11)$$

where  $\partial = (\partial_\xi, \partial_{\bar{\xi}})$ ,  $\gamma = (\gamma_1, \gamma_2) \in \mathbb{N}^2$  and  $C_\gamma$  are constants. If  $0 \leq \alpha \leq s$ ,  $1 < p, r < \infty$ ,  $1 < q \leq \infty$  and  $\frac{1}{r} = \frac{1}{p} + \frac{k-1}{q}$ , then

$$\| |D|^\alpha F(\phi) \|_{L^r(\mathbb{R}^n)} \leq C \|\phi\|_{L^q(\mathbb{R}^n)}^{k-1} \| |D|^\alpha \phi \|_{L^p(\mathbb{R}^n)}. \quad (2.12)$$

### 3 Local Smoothness of Solutions

Before study local smoothing property of solutions by using the estimate of fractional order derivatives, we need the following theorem that has been proved in [18] by using Hörmander's pseudo-differential operators theory.

**Theorem 3.1** *Let  $V(x)$  satisfy the assumptions (V1) and (V2), and  $s \geq 0$ . Then, for any  $1 < p < \infty$ ,*

$$\|H^{s/2}u\|_{L^p} + \|u\|_{L^p} \sim \|\langle D \rangle^s u\|_{L^p} + \|\langle x \rangle^{ks/2} u\|_{L^p}, \quad u \in \mathcal{S}(\mathbb{R}^n), \quad (3.1)$$

and  $H^{s/2}$  extends to a closed operator in  $L^p(\mathbb{R}^n)$ .

Let  $s \geq 0$  and  $s_*$  be the smallest integer  $\geq s$ .

**Theorem 3.2** Suppose that  $V$  satisfies the assumptions (V1) and (V2). Let  $1 \leq r \leq \frac{kn}{kn-1}$ ,  $0 \leq s \leq 1$  and  $L \subset \mathbb{R}^n$  be a compact subset. Suppose that  $f(x, u)$  is of the form  $\lambda(x)F(u)$  and satisfies (1.6) and (1.7). Assume that  $u$  is the unique solution in  $Z_{\delta,K}^s$  to IVP (1.1) obtained in Theorem 1.2 for any given  $u_0 \in \mathcal{H}^s(\mathbb{R}^n)$ . Then we have  $u \in L^2([0, \delta]; \mathcal{H}_{loc}^{s+\frac{1}{k}})$ .

**Proof:** Let  $u$  be the unique solution of (1.1). Then  $u$  satisfies the following equivalent integral equation

$$u(t, x) = e^{-itH}u_0 + i \int_0^t e^{-i(t-\tau)H} [f(x, u(\tau))] d\tau. \quad (3.2)$$

In order to show that  $u \in L^2([0, \delta]; \mathcal{H}_{loc}^{s+\frac{1}{k}})$ , we only need to show that for any  $\phi \in C_0^\infty(\mathbb{R}^n)$ ,

$$\phi(x) H^{\frac{s}{2} + \frac{1}{2k}} u(t) \in L^2([0, \delta]; L^2(\mathbb{R}^n)).$$

By (3.2), We have

$$\phi(x) H^{\frac{s}{2} + \frac{1}{2k}} u(t) = \phi(x) H^{\frac{s}{2} + \frac{1}{2k}} e^{-itH} u_0 - i \int_0^t \phi(x) H^{\frac{s}{2} + \frac{1}{2k}} e^{-i(t-\tau)H} [f(x, u(\tau))] d\tau \quad (3.3)$$

Thus

$$\begin{aligned} \|\phi(x) H^{\frac{s}{2} + \frac{1}{2k}} u(t)\|_{L^2(\mathbb{R}_x^n)} &\leq \|\phi(x) H^{\frac{s}{2} + \frac{1}{2k}} e^{-itH} u_0\|_{L^2(\mathbb{R}_x^n)} \\ &+ \int_0^t \|\phi(x) H^{\frac{s}{2} + \frac{1}{2k}} e^{-i(t-\tau)H} [f(x, u(\tau))]\|_{L^2(\mathbb{R}_x^n)} d\tau \end{aligned}$$

Hence

$$\begin{aligned} \|\phi(x) H^{\frac{s}{2} + \frac{1}{2k}} u(t)\|_{L^2([0, \delta]_t; L^2(\mathbb{R}_x^n))} &\leq \|\phi(x) H^{\frac{s}{2} + \frac{1}{2k}} e^{-itH} u_0\|_{L^2([0, \delta]_t; L^2(\mathbb{R}_x^n))} \\ &+ \int_0^\delta \|\phi(x) H^{\frac{s}{2} + \frac{1}{2k}} e^{-i(t-\tau)H} [f(x, u(\tau))]\|_{L^2([0, \delta]_t; L^2(\mathbb{R}_x^n))} d\tau \end{aligned}$$

By virtue of Theorem 1.1, we obtain

$$\|\phi(x) H^{\frac{s}{2} + \frac{1}{2k}} u(t)\|_{L^2([0, \delta]_t; L^2(\mathbb{R}_x^n))} \leq C \|u_0\|_{\mathcal{H}^s} + C \int_0^\delta \|H^{\frac{s}{2}} f(x, u(\tau))\|_{L^2(\mathbb{R}^n)} d\tau. \quad (3.4)$$

Therefore it only remains to prove that  $f(x, u) \in L^1([0, \delta]; \mathcal{H}^s)$ . It follows from the following lemma.

**Lemma 3.3** Let  $f(x, u) = \lambda(x)F(u)$  satisfy (1.6) and (1.7). Then we have

$$\|H^{s/2} f(x, u(t))\|_{L^1([0, \delta]; L^2(\mathbb{R}^n))} \leq C \|u\|_{\tilde{Z}_{\delta,K}^s}^{r-1} \|u\|_{\tilde{Z}_{\delta,K}^s}. \quad (3.5)$$

**Proof** By Theorem 3.1 we obtain

$$\begin{aligned} & \|H^{s/2}f(x, u(t))\|_{L^1([0, \delta]; L^2(\mathbb{R}^n))} \\ \sim & \|\langle D \rangle^s f(x, u(t))\|_{L^1([0, \delta]; L^2(\mathbb{R}^n))} + \|\langle x \rangle^{ks/2} f(x, u(t))\|_{L^1([0, \delta]; L^2(\mathbb{R}^n))} \end{aligned} \quad (3.6)$$

By (1.6), (1.7) and Hölder inequality, we have

$$\begin{aligned} \|\langle x \rangle^{ks/2} f(x, u(t))\|_{L^1([0, \delta]; L^2(\mathbb{R}^n))} &= \|\langle x \rangle^{ks/2} \lambda(x) F(u)\|_{L^1([0, \delta]; L^2(K))} \\ &\leq C \|F(u)\|_{L^1([0, \delta]; L^2(K))} \\ &\leq C \|u\|_{L^1([0, \delta]; L^{2r}(K))}^r \leq C \|u\|_{Z_{\delta, K}}^r. \end{aligned}$$

We write  $s = [s] + \alpha$ , where  $[s]$  is the integer part of  $s$  and  $0 \leq \alpha < 1$ . Notice that

$$\|\langle D \rangle^s f(x, u(t))\|_{L^2(\mathbb{R}^n)} \sim \sum_{|\gamma| \leq [s]} \|\partial_x^\gamma f(x, u)\|_{L^2(\mathbb{R}^n)} + \sum_{|\gamma| = [s]} \| |D|^\alpha \partial_x^\gamma f(x, u)\|_{L^2(\mathbb{R}^n)}$$

By virtue of (1.6) and (1.7) we may write  $\partial_x^\gamma f(x, u)$  as

$$\partial_x^\gamma f(x, u) = \sum_{|\beta| \leq |\gamma|} R_\beta(x) \partial_x^\beta F(u)$$

where  $R_\beta, \langle D \rangle^{s+|\beta|-|\gamma|} R_\beta \in L^\infty$ . By (2.2) we obtain

$$\|\langle D \rangle^s f(x, u(t))\|_{L^1([0, \delta]; L^2(\mathbb{R}^n))} \leq C \|u\|_{\tilde{Z}_{\delta, K}}^{r-1} \|u\|_{\tilde{Z}_{\delta, K}^s} \quad (3.7)$$

Thus the lemma follows.

**Theorem 3.4** Suppose that  $V$  satisfies the assumptions (V1) and (V2). Let  $s > \frac{n}{2}$ . Suppose that  $f(x, u)$  is of the form  $\lambda(x)F(u)$  such that  $\lambda(x), \langle D \rangle^s \lambda(x) \in L^\infty$  and satisfies (1.7). Assume that  $u$  is the unique solution in  $C([0, \delta]; \mathcal{H}^s)$  to IVP (1.1) for any given  $u_0 \in \mathcal{H}^s(\mathbb{R}^n)$ . Then we have  $u \in L^2([0, \delta]; \mathcal{H}_{loc}^{s+\frac{1}{k}})$ .

**Proof** Following the similar argument with Theorem 3.2, we know that we only need to show that  $f(x, u) \in L^1([0, \delta]; \mathcal{H}^s)$ . Notice that by Sobolev embedding theorem we have  $\|u\|_{L^\infty(\mathbb{R}^n)} \leq C \|u\|_{\mathcal{H}^s}$ ,  $s > n/2$ . By virtue of (3.6) we calculate as follows.

$$\begin{aligned} \|\langle x \rangle^{ks/2} f(x, u(t))\|_{L^1([0, \delta]; L^2(\mathbb{R}^n))} &\leq \|\langle x \rangle^{ks/2} |\lambda(x)| \|u\|^r\|_{L^1([0, \delta]; L^2(\mathbb{R}^n))} \\ &\leq C \|\lambda u\|_{L^1([0, \delta]; L^\infty(\mathbb{R}^n))}^{r-1} \|\langle x \rangle^{ks/2} u\|_{L^1([0, \delta]; L^2(\mathbb{R}^n))} \\ &\leq C \|u\|_{L^\infty([0, \delta]; \mathcal{H}^s)}^r, \end{aligned}$$

and

$$\|\langle D \rangle^s f(x, u(t))\|_{L^1([0, \delta]; L^2(\mathbb{R}^n))} \leq C \|u\|_{L^\infty([0, \delta]; \mathcal{H}^s)}^r \cdot \psi \quad (3.8)$$

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*Chapter 21*

## A MODEL FOR TOTAL ENERGY OF NEMATIC ELASTOMERS WITH NON-UNIFORM PROLATE SPHEROIDS

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### Abstract

We provide a model for the total energy of nematic liquid crystal elastomers whose elastic solid backbones have an average shape of non-uniform prolate spheroids along the director of crosslinked liquid crystalline molecules. This energy is a generalization of the Oseen-Frank free energy for nematic liquid crystals, but incorporates coherently the elastic deformation with changes in the shape and the director of the network. The total free energy we obtain for the elastomer is suitable for the existence of energy minimizers and other mathematical studies of the elastomers and also provides a concrete example that is consistent with both the continuum mechanics theory and the molecular-statistical theory.

**Key Words:** nematic elastomer, modified Oseen-Frank energy, total free energy, existence of energy-minimizers **AMS Subject Classification:** 49J45, 74F20, 82B21.

## 1 Introduction

Nematic elastomers are unusual materials that simultaneously combine the elastic properties of rubbers with the anisotropy of liquid crystals. They consist of polymer networks of

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elastic solid chains (the backbones) formed by the cross-linking of nematic molecules (the mesogens) as the elements of their main-chains and/or pendant side-groups. Because of this structure, any stress on the polymer network influences the nematic order of the elastomer, and, conversely, any change in the nematic order will affect the mechanical shape of the elastomer. The interplay between elastic and orientational changes is responsible for many fascinating properties of such materials that are different from the classical elastic solids and liquid crystals. These materials have been actively studied for both fundamental research and industrial applications; see, e.g., References [2, 4, 7, 8, 9, 11, 12, 14, 16].

The nematic elastomers we shall model in this paper have a common feature that the average shape of backbone nematic polymer is a prolate spheroid with the nematic director as its long axis. Such an elastomer can be formed in two steps: First, in the melt the material is an isotropic elastic rubber with no defined orientation. Under some preferred procedures (via applied fields or temperature change) the nematic chooses an initial shape of a spheroid with a fixed direction  $\mathbf{n}_0$  and *asphericity*  $q_0 \geq 1$  (i.e., the aspect ratio of lengths parallel and perpendicular to the spheroid axis  $\mathbf{n}_0$ ). Then, the network undergoes a deformation  $\mathbf{y}$  from the reference domain to a current domain where the nematic takes a shape of a new shape of prolate spheroid at a deformed point  $z$  with axis  $\tilde{\mathbf{n}}(z)$  and asphericity  $\tilde{q}(z)$ . We assume the elastic deformation is *incompressible* and the shape spheroids are all prolate in the direction  $\tilde{\mathbf{n}}(z)$ ; this is to assume the asphericity  $\tilde{q}(z) \geq 1$  at all points  $z$  in the current configuration.

In this paper we provide a model for the total free energy of the elastomer that consists of three contributions: (1) the nematic elastic energy due to the anisotropy of the elastic solid backbone, (2) the asphericity energy that, to some extent, memorizes the initial spheroid shape of the network, and, (3) the Oseen-Frank penalty energy due to the coupling of non-uniform nematic orders and elastic deformations. This last part of the energy has been usually ignored in the study of nematic elastomers ([4, 7, 9, 11]). We stress that our main contribution here is to provide a mathematically workable and physically sound theory for the total energy of the elastomer that includes the reasonable Oseen-Frank penalty energy for the cross-linking and that, like Ericksen's theory for nematic liquid crystals (see [10, 13, 15]), our model permits the defects in the elastomer network.

We now outline the plan of this paper. In section 2, we introduce the physical quantities that describe the nematic elastomers we want to model. In particular, we introduce the *shape parameters* that combines both the director and the asphericity of the prolate spheroid. In section 3, we build up our mathematical model for the total free energy of the nematic elastomer network that includes three contributions: the nematic elastic energy, the asphericity energy, and the modified Oseen-Frank penalty energy. We remark that our theory is consistent with both the continuum mechanics theory and the molecular-statistical theory. In final two sections of the paper, sections 4 and 5, we discuss the variational properties of our total free energy functional and the existence of energy-minimizers. We omit most of the details here and refer to [5] for proofs and more discussions.

## 2 Physical State Variables and Shape Parameters

We first introduce the physical variables that describe the nematic elastomer and then define some new variables that will be used in our theory.

## 2.1 The state variables

The nematic elastomer modeled here can be described by the following state variables.

(a) *Elastic deformation*  $z = \mathbf{y}(x)$ , where  $x \in \Omega$ , the reference (Lagrangian) domain,  $z \in D$ , the current (Eulerian) domain. The incompressibility of the elastomer network requires that the deformation gradient  $F = \nabla \mathbf{y}$  satisfy

$$\det \nabla \mathbf{y}(x) = 1, \quad \forall x \in \Omega. \quad (2.1)$$

(b) *Nematic director*  $\tilde{\mathbf{n}} = \tilde{\mathbf{n}}(z)$  of the average axis of the crosslinked nematic mesogenic groups (rod-like segments) after deformation; this is a unit vector in  $\mathbf{R}^3$ , that is,

$$|\tilde{\mathbf{n}}(z)| = 1. \quad (2.2)$$

For the nematic elastomers we study here,  $\tilde{\mathbf{n}}$  coincides with the principal axis of the shape spheroid of the elastic solid backbone.

(c) *Current shape* of the spheroidal elastic solid backbone of the nematic elastomer. The *asphericity*  $\tilde{q} = \tilde{q}(z)$  of the spheroidal backbone is the ratio of lengths parallel and perpendicular to the principal axis (which is  $\tilde{\mathbf{n}}(z)$ ) of the current spheroid shape. For prolate spheroids,  $\tilde{q}(z) \geq 1$ . The shape of the spheroid can be described by the so-called *step-length tensor*  $\tilde{L}$ , which is given by matrix

$$\tilde{L} = l_{\perp} I + (l_{\parallel} - l_{\perp}) \tilde{\mathbf{n}} \otimes \tilde{\mathbf{n}} = l_{\perp}(\tilde{q})[I + (\tilde{q} - 1) \tilde{\mathbf{n}} \otimes \tilde{\mathbf{n}}], \quad (2.3)$$

where  $\tilde{q} = l_{\parallel}/l_{\perp}$  is the asphericity, and we have assumed the perpendicular length  $l_{\perp} = l_{\perp}(\tilde{q})$  is a function of asphericity  $\tilde{q}$  only. In this case the current shape is completely described by the step-length tensor; that is,  $\tilde{L} = \tilde{L}(\tilde{q}, \tilde{\mathbf{n}})$ .

(d) *Initial shape* of the nematic elastomer formed at the crosslinking. We assume this shape is constant and given by matrix

$$L_0 = l_0[I + (q_0 - 1)\mathbf{n}_0 \otimes \mathbf{n}_0], \quad (2.4)$$

with *constant* numbers  $l_0 > 0$ ,  $q_0 > 1$  and unit vector  $\mathbf{n}_0$ .

## 2.2 The current shape parameter and material defects

For unit director  $\tilde{\mathbf{n}}$  and prolate asphericity  $\tilde{q} \geq 1$  we introduce the *shape parameter*  $\tilde{\mathbf{d}} = \tilde{\mathbf{d}}(z)$  by

$$\tilde{\mathbf{d}} = \sqrt{\tilde{q} - 1} \tilde{\mathbf{n}}. \quad (2.5)$$

Using this new shape parameter  $\tilde{\mathbf{d}}$  we can write the step-length tensor  $\tilde{L}$  above as

$$\tilde{L} = \mathbf{L}(\tilde{\mathbf{d}}) = a(|\tilde{\mathbf{d}}|)(I + \tilde{\mathbf{d}} \otimes \tilde{\mathbf{d}}), \quad (2.6)$$

where  $a(s) > 0$  is a given function satisfying

$$|\ln a(s)| \leq a_0 s^r + a_1, \quad \forall s \geq 0 \quad (2.7)$$

for some constants  $a_0, a_1$  and  $1 \leq r < 6$ . In [5], we have used  $a(s) = (1 + s^2)^{-1/3}$  to assure the condition  $\det \mathbf{L}(\tilde{\mathbf{d}}) = 1$ ; but we don't need this condition here.

Notice that one can always recover  $\tilde{q}$  from  $\tilde{\mathbf{d}}$  through  $\tilde{q} = 1 + |\tilde{\mathbf{d}}|^2$ . However, one can recover  $\tilde{\mathbf{n}}$  from  $\tilde{\mathbf{d}}$  only when  $\tilde{\mathbf{d}} \neq 0$ , in which case  $\tilde{\mathbf{n}} = \frac{\tilde{\mathbf{d}}}{|\tilde{\mathbf{d}}|}$ . Note that when  $\tilde{\mathbf{d}} = 0$  and hence  $\tilde{q} = 1$ , the backbone shape is a *sphere* and the elastomer is in the isotropic solid phase and has no orientation preference; in this case, we consider the elastomer has a *material defect*. In this aspect, our theory is similar to that of Ericksen's on the nematic liquid crystals with variable degree of orientation (see [10, 15]). Under our theory, the *defect set* of the nematic elastomer is determined by the points of  $z$  where  $\tilde{\mathbf{d}}(z) = 0$ .

### 2.3 The reference shape parameter

Given elastic deformation  $z = \mathbf{y}(x)$  and shape parameter function  $\tilde{\mathbf{d}}(z)$ , define the *reference shape parameter*  $\mathbf{d} = \mathbf{d}(x)$  by

$$\mathbf{d}(x) = \tilde{\mathbf{d}}(\mathbf{y}(x)). \quad (2.8)$$

If  $\mathbf{y}$  is one-to-one, then this equation also uniquely defines  $\tilde{\mathbf{d}}(z)$  from  $\mathbf{d}(x)$ . However, if  $\mathbf{y}$  is not injective, one cannot always recover  $\tilde{\mathbf{d}}$  from  $\mathbf{d}$ ; but this problem can be overcome both mathematically and physically. We intend to address this issue elsewhere. (See also Remarks 3.2 and 3.3 below.)

In the following, we assume the deformation  $\mathbf{y}$  is one-to-one and use  $\mathbf{y}$  and the (independent) reference shape parameter  $\mathbf{d}$  as the state variables for modeling the total energy of the elastomer network. One advantage of using the non-physical  $\mathbf{d}(x)$  instead of the true shape parameter  $\tilde{\mathbf{d}}(z)$  is that  $\mathbf{d}(x)$  and  $\mathbf{y}(x)$  are now both functions on the reference domain  $\Omega$ .

## 3 The Mathematical Model for Total Free Energy

The total free energy of the nematic elastomer consists of three contributions: the nematic elastic energy, the asphericity energy, and finally the Oseen-Frank penalty energy due to the non-uniform distribution of the nematic order. Before we build up these energies, we first introduce some notation, some of which has been already used before.

### 3.1 Notation

Let  $\mathbf{M}^{3 \times 3}$  denote the space of  $3 \times 3$  matrices with standard inner product and Euclidean norm. The components of matrix  $P \in \mathbf{M}^{3 \times 3}$  are written as  $P_{ij}$  with  $1 \leq i, j \leq 3$ . Given vectors  $\mathbf{a}, \mathbf{b} \in \mathbf{R}^3$ ,  $\mathbf{a} \otimes \mathbf{b}$  will denote the (rank-one) matrix in  $\mathbf{M}^{3 \times 3}$  with components  $(\mathbf{a} \otimes \mathbf{b})_{ij} = a_i b_j$ . For  $P \in \mathbf{M}^{3 \times 3}$ , denote by  $\det P$  the determinant of  $P$  and  $\text{adj} P$  the cofactor matrix of  $P$  that satisfies

$$P(\text{adj} P) = (\text{adj} P)P = (\det P)I \quad \forall P \in \mathbf{M}^{3 \times 3}.$$

In particular, if  $P$  is invertible (i.e.,  $\det P \neq 0$ ), we have

$$(\det P)P^{-1} = \text{adj} P. \quad (3.1)$$

Let  $\Omega$  be a bounded domain in  $\mathbf{R}^3$ . For a Sobolev function  $u: \Omega \rightarrow \mathbf{R}^3$  we denote by  $\nabla u(x)$  the Jacobi matrix of  $u$  as a  $3 \times 3$  matrix whose elements are defined almost everywhere in  $x \in \Omega$  by the weak partial derivatives:

$$(\nabla u)_{ij} = \frac{\partial u^i}{\partial x_j}, \quad i, j = 1, 2, 3.$$

### 3.2 The nematic elastic energy

Following [4, 16], the ideal nematic elastic free energy is obtained from a *molecular-statistical theory* and is given by an energy density function that is a simple extension of the classical isotropic Gaussian elasticity. From this, the nematic elastic energy density is given by the so-called *trace formula*:

$$E_{el} = E_{el}(\mathbf{d}, F) = \frac{\mu}{2} \left[ \text{tr}(L_0 F^T L^{-1} F) - \ln \frac{\det L_0}{\det L} - 3 \right], \quad (3.2)$$

where  $\mu > 0$  is an elasticity constant of the network,  $L_0$  and  $L = \mathbf{L}(\mathbf{d})$  are the step-length tensors defined above by (2.4) and (2.6). Note that  $L_0 = \mathbf{L}(\mathbf{d}_0)$ , where  $\mathbf{d}_0 = (q_0 - 1)^{1/2} \mathbf{n}_0 \neq 0$  is a constant vector. The additive constant 3 in the formula is to assure the lowest energy is zero (see below).

Let  $A_0 = L_0^{1/2}$  and  $B(\mathbf{d}) = L^{-1/2} = \mathbf{L}(\mathbf{d})^{-1/2}$ . Then we can write

$$E_{el}(\mathbf{d}, F) = \frac{\mu}{2} [ |B(\mathbf{d}) F A_0|^2 - 2 \ln \det(B(\mathbf{d}) A_0) - 3 ]. \quad (3.3)$$

Therefore, for any given  $\mathbf{d}$ ,  $E_{el}(\mathbf{d}, F)$  is *convex* in  $F$ ; moreover, for all  $F$  with  $\det F = 1$ , one can show that  $E_{el}(\mathbf{d}, F) \geq 0$ , and that  $E_{el}(\mathbf{d}, F) = 0$  if and only if  $B(\mathbf{d}) F A_0 \in SO(3)$ ; that is,

$$F \in \mathcal{U}(\mathbf{d}) \equiv B(\mathbf{d})^{-1} SO(3) A_0^{-1}, \quad (3.4)$$

where  $SO(3)$  is the set of all rotations in  $\mathbf{R}^3$  with determinant 1.  $\mathcal{U}(\mathbf{d})$  is the so-called *energy-well* (see, e.g., [5, 9]).

Suppose now  $\mathbf{y}(x)$  is an elastic deformation which deforms the reference domain  $\Omega$  to the current domain  $D$  and at each point  $z \in D$  the shape parameter  $\tilde{d}(z)$  of the nematic elastomer is given. Throughout the paper we assume the reference domain  $\Omega$  is a bounded open domain with Lipschitz boundary  $\Omega$  in  $\mathbf{R}^3$ . Let  $\mathbf{d}(x) = \tilde{\mathbf{d}}(\mathbf{y}(x))$  be the reference parameter. Then the *nematic elastic energy* is defined by

$$\mathcal{E}_{el} = \mathcal{E}_{el}(\mathbf{d}, \mathbf{y}) = \int_{\Omega} E_{el}(\mathbf{d}(x), \nabla \mathbf{y}(x)) dx. \quad (3.5)$$

### 3.3 The asphericity energy

We postulate that after deformation the nematic elastomer penalizes the change of asphericity except for the isotropic sphere case and remembers only the initial asphericity constant. The total *asphericity energy* is given by

$$\tilde{\mathcal{E}}_{asph} = \tilde{\mathcal{E}}_{asph}(\tilde{\mathbf{d}}) = \int_D \Phi(|\tilde{\mathbf{d}}(z)|) dz, \quad (3.6)$$

where  $\Phi(s)$  is a continuous function satisfying, for some constants  $c_0 > 0$ ,  $c_1 > 0$  and  $1 \leq p < 6$ ,

$$\max\{0, c_0|s|^2 - 1\} \leq \Phi(s) \leq c_1(|s|^p + 1), \quad (3.7)$$

$$\Phi(s) = 0 \iff s = |\mathbf{d}_0| \text{ or } s = 0. \quad (3.8)$$

The growth condition (3.7) and the similar one (2.7) on function  $a(s)$  above will be used in the proof of existence of energy-minimizers later. The usual Ginzburg-Landau type of *double-well* functions can be chosen as such a  $\Phi$ . For instance,

$$\Phi(s) = s^2(s - |\mathbf{d}_0|)^2.$$

### 3.4 The modified Oseen-Frank curvature energy

For a nematic elastomer in the current configuration  $D$ , we use a *modified* Oseen-Frank curvature energy to penalize the spatial change of the shape parameter  $\tilde{\mathbf{d}}$  wherever  $\tilde{\mathbf{d}}(z) \neq 0$ .

#### 3.4.1 The Oseen-Frank curvature energy

In general, in the absence of applied fields, the total *Oseen-Frank curvature energy* for unit vector field  $\tilde{\mathbf{n}}(z)$  is given by the integral

$$\int_D \Psi_{\tilde{\mathbf{n}}}(z) dz,$$

where the energy density  $\Psi_{\tilde{\mathbf{n}}}(z)$  is defined by

$$\Psi_{\tilde{\mathbf{n}}} = k_1(\operatorname{div} \tilde{\mathbf{n}})^2 + k_2(\tilde{\mathbf{n}} \cdot \operatorname{curl} \tilde{\mathbf{n}})^2 + k_3|\tilde{\mathbf{n}} \times \operatorname{curl} \tilde{\mathbf{n}}|^2 + k_4[\operatorname{tr}(\nabla \tilde{\mathbf{n}})^2 - (\operatorname{div} \tilde{\mathbf{n}})^2] \quad (3.9)$$

with the Frank constants  $k_1, \dots, k_4$ . The terms with  $k_1$ ,  $k_2$  and  $k_3$  represent the corresponding *splay*, *twist* and *bend* curvature energies, respectively; the  $k_4$ -term is a *null-Lagrangian* representing the *surface energy* contribution. (See, e.g., [10, 13, 15].)

There is a way to write  $\Psi_{\tilde{\mathbf{n}}}$  as a function of  $\tilde{\mathbf{n}}$  and  $\nabla \tilde{\mathbf{n}}$ . To do so, we define the *Oseen-Frank energy density function*  $\mathcal{W}(\mathbf{n}, P)$  as follows.

$$\mathcal{W}(\mathbf{n}, P) = W_0(\mathbf{n}, P) + k_4 N(P), \quad (3.10)$$

$$W_0(\mathbf{n}, P) = k_1(\operatorname{tr} P)^2 + k_2(\mathbf{n} \cdot \operatorname{ax}(P))^2 + k_3|\mathbf{n} \times \operatorname{ax}(P)|^2, \quad (3.11)$$

$$N(P) = \operatorname{tr}(P^2) - (\operatorname{tr} P)^2, \quad (3.12)$$

where, for each matrix  $P \in \mathbf{M}^{3 \times 3}$ , the vector  $\operatorname{ax}(P) \in \mathbf{R}^3$  denotes the *axial vector* of the matrix  $P$ , which is uniquely defined through the identity

$$(P - P^T)\mathbf{v} = \operatorname{ax}(P) \times \mathbf{v}, \quad \forall \mathbf{v} \in \mathbf{R}^3. \quad (3.13)$$

For a smooth vector field  $\tilde{\mathbf{m}}: D \rightarrow \mathbf{R}^3$ , one can easily check that

$$\operatorname{tr}(\nabla \tilde{\mathbf{m}}) = \operatorname{div} \tilde{\mathbf{m}}, \quad \operatorname{ax}(\nabla \tilde{\mathbf{m}}) = \operatorname{curl} \tilde{\mathbf{m}}.$$

Hence one has

$$\Psi_{\tilde{\mathbf{n}}}(z) = \mathcal{W}(\tilde{\mathbf{n}}(z), \nabla \tilde{\mathbf{n}}(z)).$$

For unit vectors  $\mathbf{n}$ ,  $|\mathbf{n}| = 1$  we have the identity

$$|P|^2 = (\text{tr} P)^2 + (\mathbf{n} \cdot \text{ax}(P))^2 + |\mathbf{n} \times \text{ax}(P)|^2 + N(P). \quad (3.14)$$

Therefore if  $k_2 = k_3$  then  $W_0(\mathbf{n}, P)$  depends only on  $P$ .

Throughout this paper, we assume

$$\kappa = \min\{k_1, k_2, k_3\} > 0. \quad (3.15)$$

It is easily seen that, for  $|\mathbf{n}| = 1$ ,

$$\kappa |P|^2 \leq \mathcal{W}(\mathbf{n}, P) + (\kappa - k_4)N(P). \quad (3.16)$$

If  $k_1 = k_2 = k_3 = k_4 = \kappa > 0$ , one obtains the so-called *one-constant* Oseen-Frank energy formula:

$$\mathcal{W}(\mathbf{n}, P) = \kappa |P|^2. \quad (3.17)$$

### 3.4.2 Extending the Oseen-Frank energy function

We extend the function  $\mathcal{W}(\mathbf{n}, P)$  to arbitrary vectors  $\mathbf{d}$ . We use a simple extension to penalize the directional changes of  $\mathbf{d}$  only; the penalty for  $|\mathbf{d}|$  is already included in the asphericity energy defined above. Define

$$\mathcal{K}(\mathbf{d}, P) = \mathcal{W}(\omega(\mathbf{d}), P); \quad \omega(\mathbf{d}) = \{ \mathbf{d} / |\mathbf{d}| \text{ if } \mathbf{d} \neq 0, \mathbf{n}_0 \text{ if } \mathbf{d} = 0, \quad (3.18)$$

where  $\mathbf{n}_0 = \mathbf{d}_0 / |\mathbf{d}_0|$  is the initial constant director as above. Note that the assumption  $\mathcal{K}(0, P) = \mathcal{W}(\mathbf{n}_0, P)$  reflects a postulation that the elastomer may have some memory of its initial director when the current shape is isotropic.

### 3.4.3 The modified Oseen-Frank energy

We define the modified Oseen-Frank energy for shape parameter  $\tilde{\mathbf{d}}(z)$  by

$$\tilde{\mathcal{E}}_{OF} = \tilde{\mathcal{E}}_{OF}(\tilde{\mathbf{d}}) = \int_D \mathcal{K}(\tilde{\mathbf{d}}(z), \nabla \tilde{\mathbf{d}}(z)) dz \quad (3.19)$$

with the density function  $\mathcal{K}(\mathbf{d}, P)$  defined above.

## 3.5 The total free energy and the reference energy density function

For the total free energy of the nematic elastomer network we simply add the three energies defined above. So the total free energy is

$$\mathcal{E}_{total} = \mathcal{E}_{el} + \tilde{\mathcal{E}}_{asph} + \tilde{\mathcal{E}}_{OF}. \quad (3.20)$$

To derive the total free energy density function in the reference configuration, we need to write

$$\mathcal{E}_{total} = \mathcal{E}_{total}(\mathbf{d}, \mathbf{y}) = \int_{\Omega} f_{\mathbf{d}, \mathbf{y}}(x) dx \quad (3.21)$$



and determine the density  $f_{\mathbf{d}, \mathbf{y}}$  as a function of  $\mathbf{d}, \mathbf{y}, \nabla \mathbf{d}, \nabla \mathbf{y}$ .

As mentioned above, we assume  $\mathbf{y}: \Omega \rightarrow D$  is a *bi-Lipschitz* map. We associate a function  $\tilde{\mathbf{d}}: D \rightarrow \mathbf{R}^3$  to a function  $\mathbf{d}: \Omega \rightarrow \mathbf{R}^3$  by  $\mathbf{d}(x) = \tilde{\mathbf{d}}(\mathbf{y}(x))$  and, *vice versa*, a function  $\mathbf{d}: \Omega \rightarrow \mathbf{R}^3$  to a function  $\tilde{\mathbf{d}}: D \rightarrow \mathbf{R}^3$  through  $\tilde{\mathbf{d}}(z) = \mathbf{d}(\mathbf{y}^{-1}(z))$ . Note that

$$\nabla \mathbf{d}(x) = \nabla \tilde{\mathbf{d}}(\mathbf{y}(x)) \nabla \mathbf{y}(x)$$

and hence

$$\nabla \tilde{\mathbf{d}}(z) = \nabla \mathbf{d}(x) (\nabla \mathbf{y}(x))^{-1}.$$

We can change all the energy integrals on current domain  $D$  to the integrals on the reference configuration  $\Omega$  by the (bi-Lipschitz) coordinate change  $z = \mathbf{y}(x)$ . Since  $\det \nabla \mathbf{y}(x) = 1$ , we have

$$\tilde{\mathcal{E}}_{OF} = \mathcal{E}_{OF}(\mathbf{d}, \mathbf{y}) = \int_{\Omega} \mathcal{K}(\mathbf{d}(x), \nabla \mathbf{d}(x) (\nabla \mathbf{y}(x))^{-1}) dx$$

and

$$\tilde{\mathcal{E}}_{asph} = \mathcal{E}_{asph}(\mathbf{d}, \mathbf{y}) = \int_{\Omega} \Phi(|\mathbf{d}(x)|) dx.$$

Therefore, we have the total energy is given by

$$\mathcal{E}_{total} = \mathcal{E}_{total}(\mathbf{d}, \mathbf{y}) = \int_{\Omega} f_{\mathbf{d}, \mathbf{y}}(x) dx = \int_{\Omega} \psi(\mathbf{d}, \nabla \mathbf{y}, \nabla \mathbf{d}) dx, \quad (3.22)$$

where the density function

$$f_{\mathbf{d}, \mathbf{y}}(x) = \psi(\mathbf{d}(x), \nabla \mathbf{y}(x), \nabla \mathbf{d}(x))$$

is given by *total free energy density function*  $\psi = \psi(\mathbf{d}, F, G)$  defined by

$$\psi(\mathbf{d}, F, G) = E_{el}(\mathbf{d}, F) + \Phi(|\mathbf{d}|) + \mathcal{K}(\mathbf{d}, GF^{-1}), \quad (3.23)$$

with  $E_{el}$ ,  $\Phi$  and  $\mathcal{K}$  defined as above. The state variables  $(\mathbf{d}, F, G)$  for this free energy density function  $\psi$  are

$$\mathbf{d} \in \mathbf{R}^3, \quad F, G \in M^{3 \times 3} \quad \text{with } \det F = 1. \quad (3.24)$$

### 3.6 A few remarks about our total energy formula

**Remark 3.1** We derived the total energy formula (3.22) under the assumption that  $\mathbf{y}: \Omega \rightarrow D$  is a bi-Lipschitz map. However, our final formula (3.22) or (3.23) makes no requirement that the deformation  $\mathbf{y}$  be one-to-one from  $\Omega$  onto  $D = \mathbf{y}(\Omega)$ . So we can use this total energy formula for all  $\mathbf{d}$  and  $\mathbf{y}$ .

**Remark 3.2** Mathematically, the injectivity of the deformations in nonlinear elasticity has been addressed in Ball [3] and Ciarlet & Necas [6] and can be guaranteed either by a pure displacement (Dirichlet) boundary condition with one-to-one boundary data [3] or by imposing an inequality condition [6], which in our case ( $\det \nabla \mathbf{y}(x) = 1$ ) reduces simply to the condition:  $|\Omega| \leq |\mathbf{y}(\Omega)|$ .

**Remark 3.3** *There may be still a way to recover the current shape parameter  $\tilde{\mathbf{d}}(z)$  by restricting the admissible class to a subset of the joint class  $\mathcal{C}$  of  $(\mathbf{d}, \mathbf{y})$  defined by*

$$\mathcal{C} = \{(\mathbf{d}, \mathbf{y}) \mid \mathbf{y} \in W^{1,\infty}(\Omega; \mathbf{R}^3), \mathbf{d}(x) = \tilde{\mathbf{d}}(\mathbf{y}(x)) \exists \tilde{\mathbf{d}} \in W^{1,2}(\mathbf{y}(\Omega); \mathbf{R}^3)\};$$

*this is equivalent to requiring  $\mathbf{d}$  be constant on each level set of  $\mathbf{y}$ .*

**Remark 3.4** *Our density function  $\psi(\mathbf{d}, F, G)$  defined by (3.23), when restricted to the unit vectors  $\mathbf{d} \in S^2$ , satisfies all the frame-indifference and material symmetry properties of the continuum mechanics theory as required in Andersen et al [2] for their density functions (cf. (4.38), (4.49) and (4.52) in [2]):*

$$\psi(R\mathbf{d}, RF, RG) = \psi(\mathbf{d}, F, G) \quad \forall R \in SO(3), \quad (3.25)$$

$$\psi(\mathbf{d}, FQ, GQ) = \psi(\mathbf{d}, F, G), \quad \forall Q \in SO(3) \text{ with } Q\mathbf{d}_0 = \mathbf{d}_0, \quad (3.26)$$

$$\psi(-\mathbf{d}, F, -G) = \psi(\mathbf{d}, F, G). \quad (3.27)$$

*Therefore, our energy density function is also consistent with the continuum mechanics theory.*

**Remark 3.5** *For the one-constant Oseen-Frank energy model, our modified Oseen-Frank energy density reduces to  $\kappa|\nabla\mathbf{d}(\nabla\mathbf{y})^{-1}|^2$  and the model used in [2] reduces to the density  $\kappa|\nabla\mathbf{y}^T\nabla\mathbf{d}|^2$ . In this case, the advantage of our model is that this part of energy density is lower semicontinuous under a weak convergence of the state variables  $(\mathbf{d}, \mathbf{y})$  and so is compatible with the existence of energy-minimizers, while the one used in [2] is not.*

**Remark 3.6** *One can assume the initial shape is an isotropic sphere; that is  $A_0 = I$ . To see this, let  $\det A_0 = \alpha^3$ ,  $\alpha > 0$ . By a linear change of coordinate  $x = A_0\hat{x}$  with  $x \in \Omega$  and  $\hat{x} \in \hat{\Omega}$  and the change of any function  $f(x)$  to  $\hat{f}(\hat{x})$  by  $\alpha\hat{f}(\hat{x}) = f(x)$ , we deduce*

$$\mathcal{E}_{total}(\mathbf{d}, \mathbf{y}) = \hat{\mathcal{E}}_{total}(\hat{\mathbf{d}}, \hat{\mathbf{y}}) = \alpha^3 \int_{\hat{\Omega}} \hat{\psi}(\hat{\mathbf{d}}, \nabla\hat{\mathbf{y}}, \nabla\hat{\mathbf{d}}) d\hat{x}, \quad (3.28)$$

*where the new energy density function  $\hat{\psi}(\hat{\mathbf{d}}, \hat{F}, \hat{G})$  is given by*

$$\hat{\psi}(\hat{\mathbf{d}}, \hat{F}, \hat{G}) = \frac{\mu}{2} \left[ |\hat{B}(\hat{\mathbf{d}})\hat{F}|^2 - 2 \ln \det \hat{B}(\hat{\mathbf{d}}) - 3 \right] + \hat{\Phi}(|\hat{\mathbf{d}}|) + \hat{\mathcal{K}}(\hat{\mathbf{d}}, \hat{G}\hat{F}^{-1}) \quad (3.29)$$

*with the new functions  $\hat{B}$ ,  $\hat{\Phi}$ , and  $\hat{\mathcal{K}}$  given by*

$$\hat{B}(\hat{\mathbf{d}}) = \alpha B(\alpha\hat{\mathbf{d}}), \quad \hat{\Phi}(s) = \Phi(\alpha s), \quad \hat{\mathcal{K}}(\hat{\mathbf{d}}, \hat{P}) = \mathcal{K}(\hat{\mathbf{d}}, \hat{P}),$$

*where  $B(\mathbf{d})$ ,  $\Phi(s)$  and  $\mathcal{K}$  as the same functions as defined above. An interesting fact is that  $\hat{\mathcal{K}}$  remains the same as  $\mathcal{K}$ . Note that the information on  $\mathbf{d}_0$  is still encoded in the asphericity energy function  $\hat{\Phi}(|\hat{\mathbf{d}}|)$  and, of course, also in  $\hat{\mathcal{K}}(\hat{\mathbf{d}}, \hat{P})$ .*

*In the rest of the paper, we shall assume  $A_0 = I$ , so the density function  $\psi(\mathbf{d}, F, G)$  is similar to  $\hat{\psi}$  above.*

## 4 Variational Properties of the Total Energy

By (3.1) above, the free energy density  $\psi$  above can be write

$$\psi(\mathbf{d}, F, G) = E_{el}(\mathbf{d}, F) + \Phi(|\mathbf{d}|) + \mathcal{K}(\mathbf{d}, G \operatorname{adj} F). \quad (4.1)$$

For all  $P \in M^{3 \times 3}$ , one also has

$$N(P) = -2\operatorname{tr}(\operatorname{adj} P). \quad (4.2)$$

Since the map  $\operatorname{ax}: \mathbf{M}^{3 \times 3} \rightarrow \mathbf{R}^3$  defined by (3.13) above is linear, and constants  $k_1, k_2, k_3$  are all positive, the function  $W_0(\mathbf{n}, P)$  (in the definition of  $\mathcal{W}(\mathbf{n}, P)$ ) is convex in  $P$  for any given  $\mathbf{n} \in \mathbf{R}^3$ . Hence, for any  $P, Q \in \mathbf{M}^{3 \times 3}$ ,

$$W_0(\mathbf{n}, P + Q) \geq W_0(\mathbf{n}, P) + \frac{\partial W_0}{\partial P}(\mathbf{n}, P) : Q. \quad (4.3)$$

### 4.1 Compensated compactness and lower semicontinities

The total energy  $\mathcal{E}_{total}(\mathbf{d}, \mathbf{y})$  contains terms like  $\nabla \mathbf{d}(x) \operatorname{adj} \nabla \mathbf{y}(x)$ . Such terms have some compensated compactness property and we list the following two theorems and refer to [5] for the proof and more discussions.

**Theorem 4.1** *Let  $f_\nu \in W^{1,\infty}(\Omega; \mathbf{R}^3)$  and  $g_\nu \in W^{1,2}(\Omega; \mathbf{R}^3)$  satisfy*

$$f_\nu \rightharpoonup \bar{f} \text{ weakly } * \text{ in } W^{1,\infty}, \quad g_\nu \rightharpoonup \bar{g} \text{ weakly in } W^{1,2}.$$

*Then it follows that*

$$\operatorname{adj} \nabla f_\nu \rightharpoonup \operatorname{adj} \nabla \bar{f} \text{ weakly } * \text{ in } L^\infty(\Omega; \mathbf{M}^{3 \times 3}), \quad (4.4)$$

$$\nabla g_\nu \operatorname{adj} \nabla f_\nu \rightharpoonup \nabla \bar{g} \operatorname{adj} \nabla \bar{f} \text{ weakly in } L^2(\Omega; \mathbf{M}^{3 \times 3}), \quad (4.5)$$

*and therefore*

$$\int_\Omega |\nabla \bar{g} \operatorname{adj} \nabla \bar{f}|^2 dx \leq \liminf_{\nu \rightarrow \infty} \int_\Omega |\nabla g_\nu \operatorname{adj} \nabla f_\nu|^2 dx.$$

**Theorem 4.2** *Let  $\mathbf{y}_\nu \in W^{1,\infty}(\Omega; \mathbf{R}^3)$  and  $\mathbf{d}_\nu \in W^{1,2}(\Omega; \mathbf{R}^3)$  satisfy*

$$\mathbf{y}_\nu \rightharpoonup \bar{\mathbf{y}} \text{ weakly } * \text{ in } W^{1,\infty}, \quad \mathbf{d}_\nu \rightharpoonup \bar{\mathbf{d}} \text{ weakly in } W^{1,2}.$$

*Let  $\Phi(|\mathbf{d}|)$ ,  $W_0(\mathbf{n}, P)$  and  $\omega(\mathbf{d})$  be the functions defined above. Then one has*

$$\int_\Omega \Phi(|\bar{\mathbf{d}}|) dx \leq \liminf_{\nu \rightarrow \infty} \int_\Omega \Phi(|\mathbf{d}_\nu|) dx, \quad (4.6)$$

$$\int_\Omega W_0(\omega(\bar{\mathbf{d}}), \nabla \bar{\mathbf{d}} \operatorname{adj} \nabla \bar{\mathbf{y}}) \leq \liminf_{\nu \rightarrow \infty} \int_\Omega W_0(\omega(\mathbf{d}_\nu), \nabla \mathbf{d}_\nu \operatorname{adj} \nabla \mathbf{y}_\nu). \quad (4.7)$$

## 4.2 The null-Lagrangian term and coercivity of the energy

In view of (4.2), the null-Lagrangian term  $N(P)$  in  $\mathcal{W}(\mathbf{n}, P)$  has the following property.

**Theorem 4.3** *Let  $f_1, f_2 \in W^{1,\infty}(\Omega; \mathbf{R}^3)$  and  $g_1, g_2 \in W^{1,2}(\Omega; \mathbf{R}^3)$  with*

$$\det \nabla f_i(x) = 1 \quad \text{a.e. } x \in \Omega, \quad (i = 1, 2)$$

*satisfy  $f_1 = f_2$  and  $g_1 = g_2$  on the boundary  $\partial\Omega$  in the sense of trace. Then*

$$\int_{\Omega} N(\nabla g_1 \text{adj} \nabla f_1) dx = \int_{\Omega} N(\nabla g_2 \text{adj} \nabla f_2) dx. \quad (4.8)$$

From this theorem and (3.16), we easily have the following *coercivity* result.

**Theorem 4.4** *Assume  $\kappa = \min\{k_1, k_2, k_3\} > 0$ . Then*

$$\kappa \int_{\Omega} |\nabla \mathbf{d}(\nabla \mathbf{y})^{-1}|^2 dx \leq \int_{\Omega} \mathcal{K}(\mathbf{d}, \nabla \mathbf{d}(\nabla \mathbf{y})^{-1}) dx + C(\mathbf{d}_1, \mathbf{y}_1) \quad (4.9)$$

*for all  $\mathbf{d} \in W^{1,2}(\Omega; \mathbf{R}^3)$ ,  $\mathbf{y} \in W^{1,\infty}(\Omega; \mathbf{R}^3)$  with  $\det \nabla \mathbf{y}(x) = 1$  a.e. and  $\mathbf{d}|_{\partial\Omega} = \mathbf{d}_1$ ,  $\mathbf{y}|_{\partial\Omega} = \mathbf{y}_1$ , where  $\mathbf{d}_1 \in W^{1,2}(\Omega; \mathbf{R}^3)$  and  $\mathbf{y}_1 \in W^{1,\infty}(\Omega; \mathbf{R}^3)$  with  $\det \nabla \mathbf{y}_1(x) = 1$  a.e. are given functions, and  $C(\mathbf{d}_1, \mathbf{y}_1)$  is a constant only depending on  $\mathbf{d}_1, \mathbf{y}_1$ .*

## 5 Existence of Energy Minimizers

We study the minimization problem for the total nematic elastomer energy given above by

$$\mathcal{E}_{total}(\mathbf{d}, \mathbf{y}) = \int_{\Omega} \psi(\mathbf{d}, \nabla \mathbf{y}, \nabla \mathbf{d}) dx$$

with  $\psi$  defined by (3.23) above or equivalently by (4.1).

### 5.1 Admissible classes

The natural admissible class for the reference shape parameter  $\mathbf{d}$  is the Sobolev space

$$\mathcal{D} = W^{1,2}(\Omega; \mathbf{R}^3).$$

The natural class for deformation  $\mathbf{y}$  is all the *volume-preserving* Lipschitz maps. However since we can not have *a priori* bounds on the Lipschitz constant or the  $W^{1,\infty}$ -norm of  $\mathbf{y}$ , we have to assume such a bound in advance. So we define the admissible class of deformations to be the volume-preserving Lipschitz maps with a given uniform Lipschitz constant  $\Lambda > 0$ . Let

$$\mathcal{Y} = \{\mathbf{y} \in W^{1,\infty}(\Omega; \mathbf{R}^3) \mid \det \nabla \mathbf{y}(x) = 1 \text{ a.e.}\}, \quad (5.1)$$

$$\mathcal{Y}_{\Lambda} = \{\mathbf{y} \in \mathcal{Y} \mid \|\nabla \mathbf{y}\|_{L^{\infty}(\Omega)} \leq \Lambda\}. \quad (5.2)$$

## 5.2 Minimization with given Dirichlet boundary conditions

Let  $\bar{\mathbf{d}} \in \mathcal{D}$ ,  $\bar{\mathbf{y}} \in \mathcal{Y}_\Lambda$  be given. We define the following admissible *Dirichlet classes* for shape and deformation with given boundary anchoring:

$$\mathcal{D}_{\bar{\mathbf{d}}} = \{\mathbf{d} \in \mathcal{D} \mid \mathbf{d}|_{\partial\Omega} = \bar{\mathbf{d}}\}, \quad (5.3)$$

$$\mathcal{Y}_{\Lambda, \bar{\mathbf{y}}} = \{\mathbf{y} \in \mathcal{Y}_\Lambda \mid \mathbf{y}|_\Omega = \bar{\mathbf{y}}\}. \quad (5.4)$$

For simplicity, we denote  $\mathcal{A}_1 = \mathcal{D}_{\bar{\mathbf{d}}} \times \mathcal{Y}_{\Lambda, \bar{\mathbf{y}}}$ . The strong and weak convergences in these Dirichlet classes will be those induced by the same convergence in the Banach space  $W^{1,2} \times W^{1,\infty}$ . We easily see that they are sequentially compact in the weak topology.

We prove the following existence result.

**Theorem 5.1** *Assume  $\mathcal{E}_{total}$  is defined by (3.22) and (3.23). Let  $\Lambda < \infty$ . Then there exists a minimizer  $(\mathbf{d}^*, \mathbf{y}^*) \in \mathcal{A}_1$  such that*

$$\mathcal{E}_{total}(\mathbf{d}^*, \mathbf{y}^*) = \min_{(\mathbf{d}, \mathbf{y}) \in \mathcal{A}_1} \mathcal{E}_{total}(\mathbf{d}, \mathbf{y}).$$

The proof follows a standard direct method of the calculus of variations. Let  $(\mathbf{d}_\nu, \mathbf{y}_\nu) \in \mathcal{A}_1$  be a minimizing sequence; that is,

$$\mathcal{E}_\nu = \mathcal{E}_{total}(\mathbf{d}_\nu, \mathbf{y}_\nu) \rightarrow \mathcal{E}_0 = \inf_{(\mathbf{d}, \mathbf{y}) \in \mathcal{A}_1} \mathcal{E}_{total}(\mathbf{d}, \mathbf{y}).$$

1. Without loss of generality, we assume  $\mathbf{y}_\nu \rightharpoonup \mathbf{y}^*$  weakly \* in  $W^{1,\infty}(\Omega; \mathbf{R}^3)$ . Then  $\mathbf{y}^* \in \mathcal{Y}_{\Lambda, \bar{\mathbf{y}}}$ .

2. We derive a bound on  $\{\mathbf{d}_\nu\}$ . By (4.9), we have  $\{\nabla \mathbf{d}_\nu \text{adj} \nabla \mathbf{y}_\nu\}$  is bounded in  $L^2(\Omega; \mathbf{M}^{3 \times 3})$ . Note that, for  $E, F \in \mathbf{M}^{3 \times 3}$  with  $\det F = 1$  and  $|F| \leq \Lambda$ , it follows that

$$|E| \leq |F| |EF^{-1}| \leq \Lambda |E \text{adj} F|. \quad (5.5)$$

This gives a uniform  $L^2$ -bound on  $\{\nabla \mathbf{d}_\nu\}$ . Since  $\mathbf{d}_\nu \in \mathcal{D}_{\bar{\mathbf{d}}}$ , this implies  $\{\mathbf{d}_\nu\}$  is bounded in  $W^{1,2}(\Omega; \mathbf{R}^3)$ . Without loss of generality we assume that  $\mathbf{d}_\nu \rightharpoonup \mathbf{d}^*$  weakly in  $W^{1,2}(\Omega; \mathbf{R}^3)$  and  $\mathbf{d}^* \in \mathcal{D}_{\bar{\mathbf{d}}}$ .

3. From (3.3) and the growth conditions (2.7) and (3.7), we see that the term  $\psi_1(\mathbf{d}, F) = E_{el}(\mathbf{d}, F) + \Phi(|\mathbf{d}|)$  in the total free energy density function  $\psi$  is a *non-negative convex* function in  $F$  and hence, by a theorem in [1], the corresponding energy is weakly lower semicontinuous in the class  $\mathcal{A}_1$ . By Theorems 4.2 and 4.3, the whole energy  $\mathcal{E}_{total}$  is also weakly lower semicontinuous in this class  $\mathcal{A}_1$ .

4. Finally, by the lower semicontinuity established above, we have

$$\mathcal{E}_{total}(\mathbf{d}^*, \mathbf{y}^*) \leq \liminf_{\nu \rightarrow \infty} \mathcal{E}_{total}(\mathbf{d}_\nu, \mathbf{y}_\nu) = \mathcal{E}_0.$$

This shows that  $(\mathbf{d}^*, \mathbf{y}^*) \in \mathcal{A}_1$  is a minimizer.

### 5.3 The one-constant formula of total energy

In many situations, we need to minimize the total energy with the shape parameter fluctuating freely without anchoring to the boundary. In such cases, we do not have control on the null-Lagrangian term  $N(P)$  in  $\mathcal{K}(\mathbf{d}, P)$  for general choices of the Frank elasticity constants  $k_i$ . In what follows, we will use the one-constant formula for  $\mathcal{K}(\mathbf{d}, P)$ , and therefore, only consider the simplified energy functional

$$\mathcal{I}(\mathbf{d}, \mathbf{y}) = \int_{\Omega} [E_{el}(\mathbf{d}, \nabla \mathbf{y}) + \Phi(|\mathbf{d}|) + \kappa |\nabla \mathbf{d}(\nabla \mathbf{y})^{-1}|^2] dx \quad (5.6)$$

with  $\kappa > 0$ .

#### 5.3.1 Asphericity control

We define, for given constants  $0 \leq \theta_1 \leq \theta_2 \leq \infty$ , the class of asphericity-controlled shape parameters

$$\mathcal{D}_1 = \mathcal{D}_{\theta_1, \theta_2} = \{\mathbf{d} \in W^{1,2}(\Omega; \mathbf{R}^3) \mid \theta_1 \leq |\mathbf{d}(x)| \leq \theta_2 \text{ a.e.}\}.$$

Note that, in this notation,  $\mathcal{D} = \mathcal{D}_{0, \infty}$  and  $\mathcal{D}_{1,1} = W^{1,2}(\Omega; S^2)$ .

#### 5.3.2 Effective elastic response energy

For given  $\mathbf{y}$ , we introduce an *effective elastic response energy* due to the shape fluctuation in  $\mathcal{D}_1$  by

$$\mathcal{J}(\mathbf{y}) = \mathcal{J}_{\theta_1, \theta_2}(\mathbf{y}) = \inf_{\mathbf{d} \in \mathcal{D}_1} \mathcal{I}(\mathbf{d}, \mathbf{y}) = \min_{\mathbf{d} \in \mathcal{D}_1} \mathcal{I}(\mathbf{d}, \mathbf{y}). \quad (5.7)$$

The fact that the minimum in (5.7) is attained can be proved easily by the direct method in a similar way as in Theorem 5.1 above. Due to the property of asphericity energy density function  $\Phi(|\mathbf{d}|)$  assumed before, the *energy-well* of response energy  $\mathcal{J}(\mathbf{y}) = \mathcal{J}_{\theta_1, \theta_2}(\mathbf{y})$  (i.e., the set of  $\mathbf{y}$  with  $\mathcal{J}(\mathbf{y}) = 0$ ) will depend heavily on the values of  $\theta_1$  and  $\theta_2$  (see [5] for more).

#### 5.3.3 Continuities of the response energy

We list the following properties of the effective elastic response energy defined above and refer to [5] for details and more discussions. Note that the results listed below hold for all  $0 \leq \theta_1 \leq \theta_2 \leq \infty$ .

**Theorem 5.2** *The energy  $\mathcal{J}: \mathcal{Y} \rightarrow \mathbf{R}^+$  is continuous in the strong topology and lower semicontinuous in the weak \* topology of  $\mathcal{Y}$ ; namely,*

$$\mathcal{J}(\bar{\mathbf{y}}) \leq \liminf_{\nu \rightarrow \infty} \mathcal{J}(\mathbf{y}_{\nu}) \quad \forall \mathbf{y}_{\nu} \rightharpoonup \bar{\mathbf{y}} \text{ weakly } * \text{ in } \mathcal{Y}, \quad (5.8)$$

$$\mathcal{J}(\bar{\mathbf{y}}) = \lim_{\nu \rightarrow \infty} \mathcal{J}(\mathbf{y}_{\nu}) \quad \forall \mathbf{y}_{\nu} \rightarrow \bar{\mathbf{y}} \text{ strongly in } \mathcal{Y}. \quad (5.9)$$

Moreover, if  $\mathbf{y}_{\nu} \rightarrow \bar{\mathbf{y}}$  strongly in  $\mathcal{Y}$  and let  $\mathbf{d}_{\nu} \in \mathcal{D}_1$  be any minimizer of  $\mathcal{J}(\mathbf{y}_{\nu})$ ; that is,  $\mathcal{J}(\mathbf{y}_{\nu}) = \mathcal{I}(\mathbf{d}_{\nu}, \mathbf{y}_{\nu})$ . Then  $\{\mathbf{d}_{\nu}\}$  has a subsequence  $\{\mathbf{d}_{\nu_j}\}$  strongly converging in  $\mathcal{D}$  to a minimizer  $\bar{\mathbf{d}}$  of  $\mathcal{J}(\bar{\mathbf{y}})$ .

From this result, one easily obtains the following existence theorem.

**Corollary 5.3** *Let  $\mathcal{Y}_1$  be either the class  $\mathcal{Y}_\Lambda$  or a Dirichlet class  $\mathcal{Y}_{\Lambda, \bar{y}}$  with  $\Lambda < \infty$ . Then there exists a  $\mathbf{y}^* \in \mathcal{Y}_1$  such that*

$$\mathcal{J}(\mathbf{y}^*) = \min_{\mathbf{y} \in \mathcal{Y}_1} \mathcal{J}(\mathbf{y}) = \min_{(\mathbf{d}, \mathbf{y}) \in \mathcal{D}_1 \times \mathcal{Y}_1} \mathcal{I}(\mathbf{d}, \mathbf{y}).$$

## Acknowledgement

This paper is based on a joint work initiated during all authors' visit at the Institute of Mathematics and its Applications (IMA) of University of Minnesota in the IMA thematic year 2004-2005. The authors would like to thank the IMA for its hospitality and support. B. Yan is also grateful to Professors F.Liu and X.Shi for organizing a wonderful workshop at the Delaware State University and for providing travel support.

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*Chapter 22*

## SELECTIVE HYPOTHESIS TRACKING IN SURVEILLANCE VIDEOS

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### Abstract

Automatic detection and tracking of moving objects are the fundamental tasks of many video-based surveillance systems. Higher level security assessment and decision making procedures rely upon these essential video analysis tasks. Robust motion detection and object tracking provide the basis for detection of increased activity, entry into a restricted area, detection of objects left behind, tracking of optical flow against established motion patterns, and other similar surveillance requirements. The proposed selective hypothesis tracking method is fundamentally based on the location of spatiotemporal texture motion regions. It uses predicted motion vectors, sub-pixel image registration, and minimum cost estimation using distance, direction, size, and persistence. This method is capable of tracking fast and slow moving objects, objects that disappear and later reappear, and objects that merge and split.

## 1 Introduction

The proposed tracking method utilizes spatiotemporal motion regions, modified image registration technique, and improved minimum cost estimation based on distance, motion

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vectors, direction, and persistence. Selection is made as when to use image registration offsets or estimated velocity vectors. Also, the minimum cost analysis selects the appropriate offsets. There is only one hard limit used in minimum cost estimation, the distance between motion region and any known template. This hard limit is based on the video's frames per second and a fixed threshold, providing a standard time and distance based limit.

Known templates are compared to the motion regions in order to classify the motion regions into either new templates or known templates. Each active template is registered in the current frame using the image alignment technique described in Section X.X. Each template is then associated with a motion region based on image alignment (if template is not merged) or predicted motion (if template is merged). This template association selection is the core of the selective hypothesis tracking method. Finally, minimum cost estimation is computed for each motion region that does not have two or more templates associated with it. If there is exactly one association of template to motion region, it is treated as if motion region had no associated template. This decision allows picking the best template that matches the motion region, as the associated template may not be the best. This applies to the merge-split and disappear-reappear situations.

### 1.1 Image alignment

One of the most widely used image alignment techniques is the Lucas-Kanade algorithm [7]. It has become not only the standard for image alignment but also a standard for optical flow measurement. The basis to image alignment is the gradient descent computation, which is the de facto standard method. The image alignment is just one of three major components of the selective hypothesis tracking algorithm. A brief overview of the modifications made to the Lucas-Kanade algorithm follows.

The goal of the Lucas-Kanade algorithm is to minimize the sum of squared error between the template  $T$  and image  $G$ . Image  $G$  is warped back onto the coordinate frame where template  $T$  resides. This warping requires interpolating the image  $G$  at sub-pixel coordinates relative to the coordinates of the template's frame.

The following error estimation is used to terminate the iterative process

$$E(u, v) = \sum_{x, y \in R} (F(x + u, y + v) - G(x))^2 \quad (1.1)$$

where spatial gradients are  $F_x$  and  $F_y$  and the temporal gradient is  $F_t$ . To minimize  $E(u, v)$ , both gradients are initially set to 0 and vectors  $u$  and  $v$  are computed based on the spatial and temporal gradients(2)

$$\begin{bmatrix} \sum F_x^2 & \sum F_x F_y \\ \sum F_x F_y & \sum F_y^2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = - \begin{bmatrix} \sum F_t F_x \\ \sum F_t F_y \end{bmatrix} \quad (1.2)$$

$E(u, v)$  is checked against the previous values, and the error threshold

$$|E_{old} - E| < \varepsilon \quad (1.3)$$

The algorithm iterates until measured error is small enough and convergence is achieved.

To compute gradients,  $F_x$ ,  $F_y$ , and  $F_t$ , the image is blurred and convoluted with spatial derivative filters. The  $F_t$  is computed using weighted averaging kernel where emphasis is on the center. Both  $F_x$  and  $F_y$  are computed using standard deviation where emphasis is on the spread. The selection of Gaussian kernel derivatives to compute the gradients is based on mathematical convenience and efficiency. They also provide localization of the derivative and low signal to noise ratio.

Sub-pixel image interpolation is needed for the warping part of the registration algorithm, as image pixels will not be located on an integral grid. Simple bilinear interpolation is used to warp frame  $G$ , since the assumption is that the image is locally bilinear.

## 2 Selective Hypothesis Tracking

### 2.1 Template data

Each known template has the following information associated with it:

$N$  Number of tracked objects

$T^i$   $i^{th}$  tracked objects

$T_f^i$  tracked object's last registered frame number

$T_t^i$  tracked object's time to live value

$T_m^i$  tracked object's merged flag

$T_a^i$  tracked object's assigned flag

$T_C^i$  tracked object's centroid location

$T_R^i$  tracked object's bounding rectangle

$T_v^i$  tracked object's registration offset vector

$T_s^i$  tracked object's motion speed vector tracked

$T_M^i$  object's motion bounding rectangle

### 2.2 Motion region data

Each new motion region in frame  $f$  has the following information associated with it:

$K$  Number of motion regions in frame  $f$

$M^j$   $j^{th}$  motion region in a frame  $f$

$M_C^j$   $j^{th}$  motion region's centroid location

$M_R^j$   $j^{th}$  motion region's rectangle

$\vec{M}_T^j$   $j^{th}$  motion region's vector of associated templates

### 2.3 Position estimation

Selective hypothesis tracking algorithm processes each frame, aligning and computing the minimum cost between known templates and current motion regions. If there are no templates created and there is motion in the current frame, then templates are created and new next frame is processed. Each active template is first aligned to the current frame  $G$  only if its time to live value is set to the maximum threshold.

$$\begin{bmatrix} u_e \\ v_e \end{bmatrix} = \begin{bmatrix} \sum F_x^2(T_R^i) & \sum F_x F_y(T_R^i) \\ \sum F_x F_y(T_R^i) & \sum F_y^2(T_R^i) \end{bmatrix}^{-1} \cdot \begin{bmatrix} \sum F_t F_x(T_R^i) \\ \sum F_t F_y(T_R^i) \end{bmatrix} \quad (2.1)$$

The position estimation is based single or merged status of the template. The template alignment provides the  $(u, v)$  offset from last know template's frame to the current frame  $G$  when the template's status is active-single. However, when the template's status is hidden-merged, the template predicted velocity is used.

$$T_s^i = S \cdot T_s^i + (1 - S) \cdot (M_C^j - T_C^i) \quad (2.2)$$

To compute a template to motion region association,  $P$ , of active-single templates, the image alignment offsets are used along with motion region:

$$P \leftarrow (T_C^i + T_v^i) \in M_R^j \quad (2.3)$$

To compute a template to motion region association,  $P$ , of hidden-merged templates, the predicted velocity offsets are used along with motion region:

$$P \leftarrow (T_C^i + T_s^i) \in M_R^j \quad (2.4)$$

If association  $P$  with new motion region  $M$  is true then the its template vector is incremented.

$$\vec{M}_T^j = \vec{M}_T^j + T^i \quad (2.5)$$

## 2.4 Selective decay of unassociated templates

Each unassociated template has its time-to-live factor decremented based on the presence of any new motion region within its vicinity. The vicinity is defined as

$$\left| M_C^j - T_C^i \right| < P_{MAX} \quad (2.6)$$

and template's time-to-live is decremented as follows when (9) is true

$$T_t^i = T_t^i - 1 \quad (2.7)$$

otherwise it is decremented by a factor of  $\delta > 1$

$$T_t^i = T_t^i - \delta \quad (2.8)$$

The penalty given to each template that is not associated with any new motion region, and not within a vicinity of any motion region allows for faster decay of template's usability in the tracking process.

## 2.5 Selective position update of associated templates

Motion regions that have more than one template, such as during merged condition, provide a bounding rectangle to the predicted new position of the merged templates. Each template's centroid and rectangle is updated using the predicted velocity (5), however it is bound by the motion rectangle.

$$T_C^i = T_C^i + T_s^i \quad (2.9)$$

$$T_R^i = T_R^i + T_s^i \quad (2.10)$$

where  $T_C^i$  and  $T_R^i$  are bound by  $M_R^j$

Motion regions that have one or no associated template are treated the same. The single association between a template and the motion region may not be the best, as in split situations. Therefore, a minimum cost estimation is computed for each unselected and active template based on distance, direction, size, and persistence [4].

The cost based on distance is estimated as

$$\Delta p = \left| M_C^j - T_C^i \right| \quad (2.11)$$

the cost based on size is estimated as

$$\Delta s = \left| M_R^j - T_R^i \right| / (M_R^j + T_R^i) \quad (2.12)$$

the cost based on direction is estimated as

$$\Delta d = \left| \arctan(T_s^i) - \arctan(M_C^j - T_C^i) \right| \quad (2.13)$$

and the cost based on persistence is estimated as

$$\Delta t = (TTL_{MAX} - T_t^i) / TTL_{MAX} \quad (2.14)$$

The total cost between a motion region and all templates is then estimated as

$$C = w_p \Delta p + w_d \Delta d + w_s \Delta s + \Delta t \quad (2.15)$$

where

$w_p$  Cost weight factor for position and distance offset

$w_d$  Cost weight factor for direction difference

$w_s$  Cost weight factor for size difference

$TTL_{MAX}$  Maximum time to live

$P_{MAX}$  Maximum distance as a function of block size and

$$w = w_p + w_d + w_s = 1 \quad (2.16)$$

Each new frame containing motion regions is evaluated against know templates. New template is only created if a motion region has no associated template based on image alignment, predicted position, or minimum cost computation. New template's characteristics are only based on an unassociated motion region. Once the association is established, template's predicted velocity is still computed despite the image alignment calculation, as it may be necessary to use predicated position when templates merge.

Templates that are not within any motion region have their time to live  $T_t^i$  decreased. Once this value reaches zero, the template will no longer be used during the association with the motion region. This may provide false results when an object disappears and reappears beyond the time to live timeframe. A higher level template matching method may be used to solve this problem.

### 3 Results

Two methods are used to evaluate selective hypothesis tracking results: the visual inspection of identified tracking objects and comparison of tracking centroids to independent ground truth data. The simplest visual evaluation involves tracking a single object that appears and disappears in the field of view, without any obstruction or merging with other objects. More challenging scenarios involve several objects merging and splitting where two or more objects cross paths as observed by the camera and some objects become hidden during the merging. The tracking algorithm must predict the possible location of each individual object despite the fact that the motion detection only provides a single motion rectangle. In this scenario, the image registration technique will not work as possibly only

one of the merged objects is in the foreground. The known velocity of each object before the merge occurred is used to update the predicted position of each object. The predicted position is then bound by the observed motion rectangle, limiting the objects position to within the motion rectangle.

Another challenging tracking sequence involves an object that appears and disappears within the field of view. This may occur when object disappears behind a tree, building, or a parked car and reappears within a few seconds later. This scenario has one major difference from the objects that merge: there is no motion rectangle presents. When object that is tracked disappears, the corresponding motion rectangle is not present. The algorithm keeps tracking templates for some period of time in case this template reappears later.

There is one limitation to this scheme. An object that disappears and reappears much later than is allowed by the algorithm, a new template is created instead of matching to templates already seen by the system. The splitting of single template into multiple objects is a difficult case of label assignment. The single label of the object before the split must now be assigned to one of the motion regions after the split.

An example of two objects crossing the same path in the field of view is shown in Figure 5. In frame 863, object 3 (van) and object 4 (group of people) approach each other. In frame 898 the merge occurs with object 3 is in the foreground. Image registration of object 4 is impossible as it is partially visible. The predicted position based on last know motion velocity along with bounding motion rectangle provide a sufficient location of individual objects (frame 923). In frame 963 both objects split and continue in their respective directions while maintaining the correct labels.

Single object split is shown in Figure 3. Single object 4 (group of people) is approaching parked cars in frame 1043. In frame 1099 a person left one of the parked cars while object 4 passed next to it. In this situation there is a single motion blob corresponding to tracking object 4. While this motion region is expanding due to the fact that a single person is walking in the opposite direction to the object 4, there is still single object being tracked. The split occurs in frame 1141, where the single person is assigned new tracking label 6 while object 4 continues along its course maintaining it own label, as seen in frame 1214.

Example of an object that disappears and later reappears is shown in Figure 4. Infra 3 is a thermal infrared video sequence showing single person walking behind two trees. In frame 233 object 1 approaches the first tree and becomes invisible to the camera in frame 252. It reappears in frame 263 as the tracking algorithm keeps the same object label and does not create a new template. Object 1 is tracked continuously until frame 431 when it disappears again. Later it reappears again as object 1 in frame 461 before leaving the field of view.

### 3.1 Ground truth data evaluation

Independent ground truth data was used not only to test the proposed motion detection method but also to verify the selective hypothesis tracking algorithm. Split 1 video along with its ground truth data is used to evaluate the proposed tracking method. Once each tracking object is identified, their centroids were compared to the ground truth data. Figure 1 displays the projection of all ground truth data onto a single frame, this includes individual object projection and group projection [6]. A close up of the projection were two objects



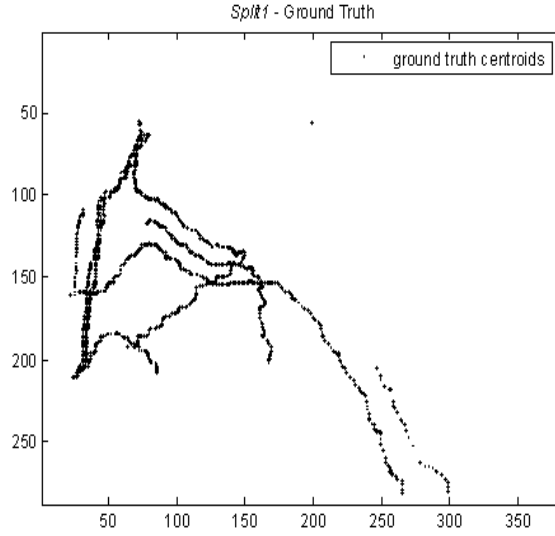


Figure 1: Split 1 video ground truth data

merged and split is also shown, Figure 2. On average, the tracking centroid distance from ground truth data was 5.2 pixels with standard deviation of 2.6 pixels for Split 1 video, while the motion block size was 4x4 pixels.

## 4 Spatiotemporal Texture Vectors

The selective hypothesis tracking is based on spatiotemporal texture motion regions presented by [5].

Videos are represented as three-dimensional (3D) arrays of monochromatic (infrared or graylevel) pixel values  $g_{i,j,t}$  at a time instant  $t$  and a pixel location  $(i, j)$ . A video is characterized by temporal dimension  $Z$  corresponding to the number of frames, and by two spatial dimensions, characterizing number of pixels in horizontal and vertical direction of each frame. Each image is divided in a video sequence into disjoint  $N_{BLOCK} \times N_{BLOCK}$  squares (e.g., 8x8 squares) that cover the whole image. Spatiotemporal 3D blocks are obtained by combining squares in consecutive frames at the same video plane location. All experiments reported here use 8x8x3 blocks that are disjoint in space but overlap in time, i.e., two blocks at the same spatial location at times  $t$  and  $t+1$  have two squares in common. The fact that the 3D blocks overlap in time allows us to perform successful motion detection in videos with very low frame rate, e.g., in experimental results, videos with 2 to 30 frames a second are included. The obtained 3D blocks are represented as 192-dimensional (8x8x3) vectors of monochromatic pixel values [5].

In general the blocks are represented by N-dimensional vectors  $b_{I,J,t}$ , specified by spatial indexes  $(I, J)$  and time instant  $t$ . Vectors  $b_{I,J,t}$  contain all graylevel values  $g_{i,j,t}$  of pixels in the corresponding 3D block. To reduce dimensionality of  $b_{I,J,t}$  while preserving information to the maximal possible extent, we compute a projection of the normalized

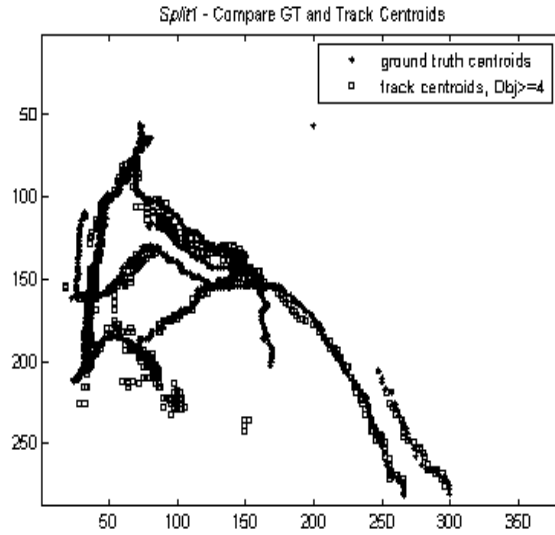


Figure 2: Split 1 compare tracking to ground truth data.

block vector to a vector of a significantly lower length  $K = N$  using a PCA [3] projection matrix  $P_{I,J}^K$  computed for all  $b_{I,J,t}$  at video plane location  $(I, J)$ . The resulting spatiotemporal texture vectors.  $b_{I,J,t}^* = P_{I,J}^K \bullet b_{I,J,t}$  provide a joint representation of texture and motion patterns in videos and are used as input of algorithms for detection of motion and objects tracking.

Value of  $K=10$  is used in all experiments and value of  $K=3$  is used for simplicity in creating motion orbit graphs. The obtained  $K$ -dimensional vectors form a compact spatiotemporal texture representation for each block. It is important to notice that a different projection matrix  $P_{I,J}^K$  is used for each video plain location. This assures that the obtained texture vectors are able to optimally distinguish different textures that appear in a given block. The initial projection matrix is trained on the first  $t_0$  frames under the assumption that only background is present in all block locations. The projection matrices are then updated during the time periods in which no motion is detected in a given block location [1,2].

## 5 Acknowledgments

D. Pokrajac has been partially supported by NIH-funded Delaware IDeA Network of Biomedical Research Excellence(INBRE)Grant, NSF infrastructure Grant(award # 0320991) and NSF grant "Seeds of Success: Comprehensive Program for the Retention, Quality Training, and Advancement of STEM student" (award # HRD-0310163).

Jingsi Gao and D. Pokrajac are partially supported by DoD HBCU/MI Infrastructure Support Program(45395-MA-ISP Department of Army).

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Figure 3: Campus 1 single split tracking. Example of single object, splitting into two distinct objects, when the single object retains most of its previous shape and direction. Object 4 approaches parked cars when a person appears inside object 4 and walks in the opposite direction. Object 4 splits from and continues while new object 6 is created. Finally object 4 and 6 continue in the opposite directions. Campus 1 video sequence 1043-1214.

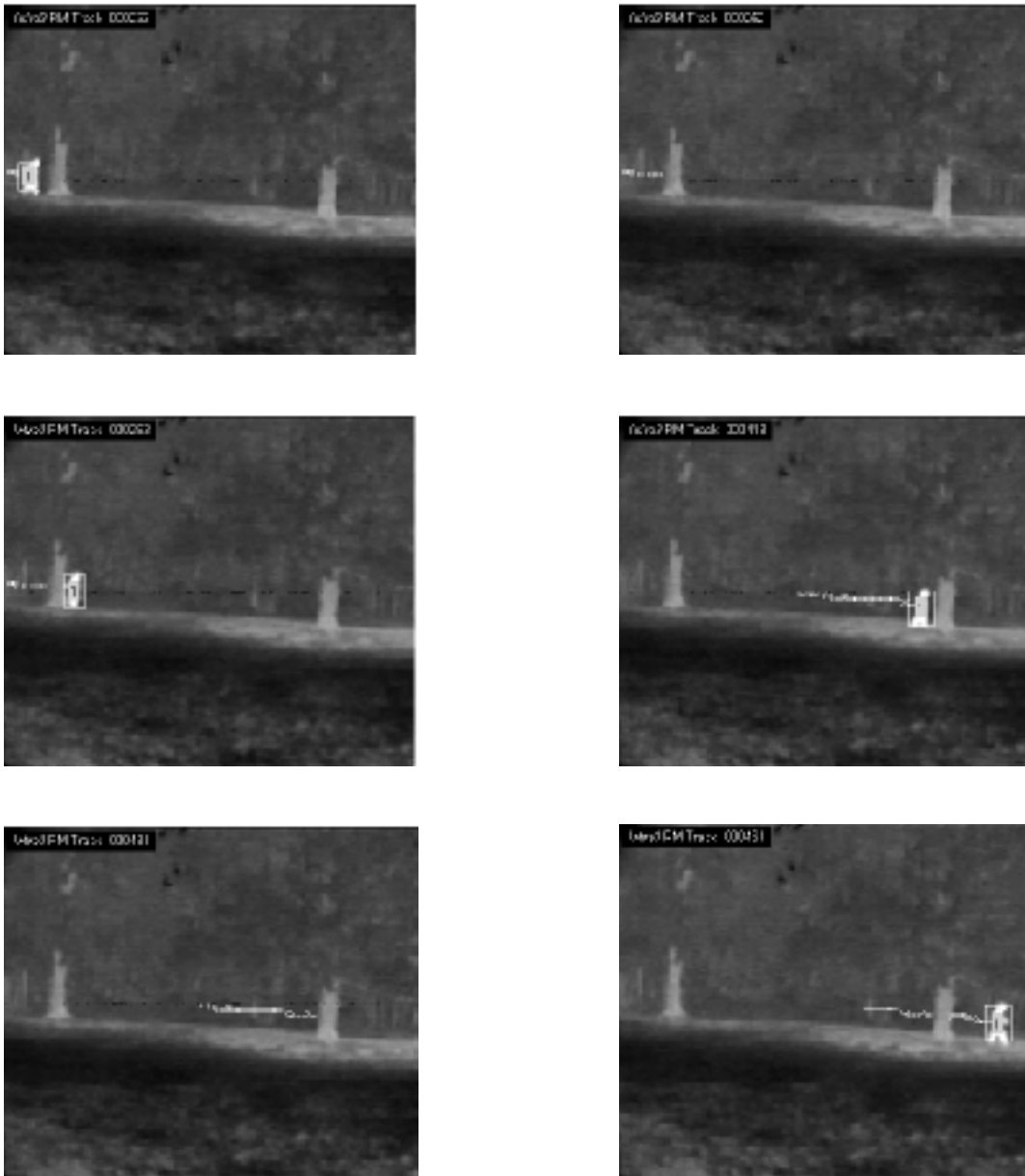


Figure 4: Infra 3 disappear-reappear tracking. Example of tracking an object that disappears from the field of view. Person appears before and after each tree while label is maintained. Person is hidden behind a tree, as no active object is being tracked. Infra 3 video sequence 233-461.



Figure 5: Campus 1 cross-tracking. Example of two objects crossing paths, when one becomes obscured by the foreground objects. Objects 3 and 4 approach each other. Object 4 is less visible as object 3 is in the foreground, and finally objects 3 and 4 continue in opposite directions. Campus 1 video sequence 863-963.



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